

Schneider Electric

C-Gate Manual

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Table of Contents

Section	Page
1 Introduction	17
2 About C-Gate	18
2.1 What is C-Gate?	18
2.2 Why use C-Gate?	18
2.3 How does C-Gate work with C-Bus?	18
2.4 When to use C-Gate	19
2.5 When not to use C-Gate	19
2.6 Limitations	20
3 Quick Start Guide	21
3.1 Start Here	21
3.2 Concepts	21
3.3 C-Bus Network	22
3.4 System Requirements	22
3.5 Installing C-Gate	23
3.6 Running C-Gate (Windows Service)	24
3.7 Running C-Gate (Console Application)	24
3.8 Connect To C-Gate	25
3.9 Enter Commands	27
3.10 Working with Projects	27
3.11 Starting a Project or Network	28
3.12 Explore the Network	29
3.13 Operation	30
3.14 Events	32
3.15 Getting Help for Commands	32
3.16 Online Resources	32
4 Reference	34
4.1 C-Gate Launcher (Windows)	34
4.1.1 C-Gate Launcher Configuration	34
4.1.2 New vs Old Launcher	35
4.1.3 C-Gate Launcher Diagnostics	35
4.1.4 C-Gate As A Service	36
4.2 C-Gate Jar	36
4.2.1 Zip Distribution	36
4.2.2 Java Runtime Environment	37
4.2.3 Manual Launch	37
4.3 Command and Monitoring Interfaces	38
4.3.1 Command Interface	39
4.3.1.1 Console Connection Details	39
4.3.1.2 TCP/IP Connection Details	40

4.3.1.3	Commands and Replies	40
4.3.1.4	Commands	41
	Terminology.....	41
	Objects and Addresses.....	41
	Array Filtering.....	44
	Unique Command IDs.....	46
	Adding IDs to commands	46
	Background processing.....	46
	Verbose output.....	47
	Responses.....	47
	Example of Command ID.....	47
4.3.1.5	C-Gate Response Codes.....	47
	General Response Codes.....	49
	Pre-Release Debugging Information.....	50
	Successful Completion Messages.....	50
	Security Messages	50
	Object Information Messages.....	51
	Client Side Command Errors.....	52
	Macro Errors	52
	Access Control Errors	52
	Method Failures.....	53
	Internal Errors.....	53
	Network Errors.....	54
	Continuation Confirmation.....	54
4.3.2	Event Interface	54
4.3.2.1	Event Buffer Overflow.....	55
4.3.2.2	Setting the event level.....	55
4.3.2.3	Event Table	56
	Information Event Codes	56
	Energy Related.....	56
	Command Events.....	57
	Network Information.....	57
	PCI Synchronisation.....	58
	Scene and License	58
	Unit Configuration Warnings.....	59
	Debugging Information.....	59
	Command Related.....	59
	Security Related.....	60
	Non Critical Network Errors.....	60
	Critical Alarms	61
	Connected Network Alarms.....	62
	Interface Errors.....	62
	File errors	63
	Licensing Errors.....	63
	Unit Specification Errors.....	63
	Scene Errors.....	64
	Access Control.....	64
4.3.3	Status Change Port	64
4.3.4	Config Change Port	65
4.3.5	Session ID and Command ID	65
4.3.6	SSL Connection	65
4.4	Logging	65

4.4.1	C-Gate logs	66
4.4.2	New vs old C-Gate log	67
4.4.3	Installer logs	67
4.4.4	Launcher logs	68
4.5	Command Descriptions	68
4.5.1	#	68
4.5.2	//	69
4.5.3	AIRCON	69
4.5.4	AIRCON REFRESH	70
4.5.5	AIRCON SET_HUMIDITY_SETBACK_LIMIT	71
4.5.6	AIRCON SET_HUMIDITY_LOWER_GUARD_LIMIT	72
4.5.7	AIRCON SET_HUMIDITY_UPPER_GUARD_LIMIT	73
4.5.8	AIRCON SET_HVAC_LOWER_GUARD_LIMIT	74
4.5.9	AIRCON SET_HVAC_SETBACK_LIMIT	75
4.5.10	AIRCON SET_HVAC_UPPER_GUARD_LIMIT	76
4.5.11	AIRCON SET_WARD_OFF	77
4.5.12	AIRCON SET_WARD_ON	77
4.5.13	AIRCON SET_ZONE_HUMIDITY_MODE	78
4.5.14	AIRCON SET_ZONE_HVAC_MODE	79
4.5.15	APIVER	81
4.5.16	AUDIO	82
4.5.17	AUDIO CURRENT_FEED	83
4.5.18	AUDIO DYNAMIC_1	84
4.5.19	AUDIO DYNAMIC_2	85
4.5.20	AUDIO HIGH_PRIORITY	86
4.5.21	AUDIO MUTE	86
4.5.22	AUDIO NEXT_FEED	87
4.5.23	AUDIO NEXT_LANGUAGE	88
4.5.24	AUDIO OFF	89
4.5.25	AUDIO ON	90
4.5.26	AUDIO OUTPUT_COMMON_CONTROL	90
4.5.27	AUDIO OUTPUT_DEVICE_STATUS_REQUEST	91
4.5.28	AUDIO OUTPUT_ERROR_CODE	92
4.5.29	AUDIO PREVIOUS_FEED	93
4.5.30	AUDIO RAMP	93
4.5.31	AUDIO REQUEST_CURRENT_FEED	94
4.5.32	AUDIO SET_FEED	95
4.5.33	AUDIO TERMINATERAMP	96
4.5.34	AUDIO ZONE_DESCRIPTOR_REQUEST	97
4.5.35	AUDIO ZONE_FEED_LABEL_REQUEST	98
4.5.36	BROADCAST_EVENT	98
4.5.37	CGL	99
4.5.38	CGL EXPORT	100
4.5.39	CGL IMPORT	101
4.5.40	CLOCK	102
4.5.41	CLOCK DATE	102

4.5.42	CLOCK REQUEST_REFRESH	103
4.5.43	CLOCK TIME	104
4.5.44	CONFIG	105
4.5.45	CONFIG INI SET	105
4.5.46	CONFIG INI GET	106
4.5.47	CONFIG GET	107
4.5.48	CONFIG INFO	107
4.5.49	CONFIG LOAD	108
4.5.50	CONFIG SAVE	109
4.5.51	CONFIG SET	110
4.5.52	CONFIG OBGET	111
4.5.53	CONFIG OBSET	112
4.5.54	CONFIG OBRESET	112
4.5.55	CONFIRM	113
4.5.56	DBADD	114
4.5.57	DBADDSAFE	115
4.5.58	DBCOPY	116
4.5.59	DBCOPYSAFE	117
4.5.60	DBCREATE	119
4.5.61	DBCREATENET	119
4.5.62	DBDELETE	121
4.5.63	DBGET	121
4.5.64	DBGETXML	122
4.5.65	DBLOAD	123
4.5.66	DBNETWORKPATH	124
4.5.67	DBNEW	125
4.5.68	DBRENAMENET	126
4.5.69	DBRENAMENETSAFE	127
4.5.70	DBSAVE	128
4.5.71	DBSET	128
4.5.72	DBSETSAFE	129
4.5.73	DBSETXML	130
4.5.74	DBTAGLIST	131
4.5.75	DBUPDATE	132
4.5.76	DBVALIDATE	133
4.5.77	DBVERIFY	133
4.5.78	DO	134
4.5.79	ENABLE	135
4.5.80	ENABLE LABEL	135
4.5.81	ENABLE REMOVE	137
4.5.82	ENABLE SET	138
4.5.83	EREPORT	138
4.5.84	EREPORT MESSAGE	139
4.5.85	EVENT	140
4.5.86	GET	141
4.5.87	GETSTATE	142

4.5.88	HELP	143
4.5.89	LIGHTING	144
4.5.90	LIGHTING LABEL	145
4.5.91	LIGHTING UNICODELABEL	147
4.5.92	LIGHTING OFF	148
4.5.93	LIGHTING ON	148
4.5.94	LIGHTING RAMP	148
4.5.95	LIGHTING TERMINATERAMP	148
4.5.96	LOCK	148
4.5.97	LOGIN	149
4.5.98	LOGOUT	149
4.5.99	MEASUREMENT	150
4.5.100	MEASUREMENT DATA	151
4.5.101	MEDIATRANSPORT	151
4.5.102	MEDIATRANSPORT CATEGORY_NAME	152
4.5.103	MEDIATRANSPORT ENUMERATE	153
4.5.104	MEDIATRANSPORT ENUMERATION_SIZE	154
4.5.105	MEDIATRANSPORT FORWARD	154
4.5.106	MEDIATRANSPORT NEXT_CATEGORY	155
4.5.107	MEDIATRANSPORT NEXT_SELECTION	156
4.5.108	MEDIATRANSPORT NEXT_TRACK	157
4.5.109	MEDIATRANSPORT PAUSE	157
4.5.110	MEDIATRANSPORT PLAY	158
4.5.111	MEDIATRANSPORT REPEAT	159
4.5.112	MEDIATRANSPORT REWIND	159
4.5.113	MEDIATRANSPORT SELECTION_NAME	160
4.5.114	MEDIATRANSPORT SET_CATEGORY	161
4.5.115	MEDIATRANSPORT SET_SELECTION	162
4.5.116	MEDIATRANSPORT SET_TRACK	163
4.5.117	MEDIATRANSPORT SHUFFLE	163
4.5.118	MEDIATRANSPORT SOURCE_POWER	164
4.5.119	MEDIATRANSPORT STATUS_REQUEST	165
4.5.120	MEDIATRANSPORT STOP	165
4.5.121	MEDIATRANSPORT TOTAL_TRACKS	166
4.5.122	MEDIATRANSPORT TRACK_NAME	167
4.5.123	NET	167
4.5.124	NET CHECKUNIT	168
4.5.125	NET CLOCKS	170
4.5.126	NET CLOSE	171
4.5.127	NET CREATE	172
4.5.128	NET DELETE	173
4.5.129	NET FLUSH	174
4.5.130	NET LEARN	174
4.5.131	NET LIST	175
4.5.132	NET LIST_ALL	176
4.5.133	NET LOAD	177

4.5.134	NET OPEN	178
4.5.135	NET PINGU	179
4.5.136	NET PROJECT_IDENTIFY	179
4.5.137	NET RENAME	180
4.5.138	NET SAVE	181
4.5.139	NET SET_PROJECT_IDENTIFY	182
4.5.140	NET SYNC	183
4.5.141	NET SYNCNEW	184
4.5.142	NET UNRAVELUNIT	186
4.5.143	NET UNRAVEL	187
4.5.144	NETWORK	189
4.5.145	NETWORK LOCATE	189
4.5.146	NEW	190
4.5.147	NOOP	191
4.5.148	OFF	191
4.5.149	OID	192
4.5.150	ON	193
4.5.151	PORT	193
4.5.152	PORT CNISCAN	194
4.5.153	PORT CNISCAN2	195
4.5.154	PORT IFLIST	197
4.5.155	PORT LIST	197
4.5.156	PORT PROBE	198
4.5.157	PORT REFRESH	199
4.5.158	PROJECT	200
4.5.159	PROJECT CLOSE	201
4.5.160	PROJECT COPY	201
4.5.161	PROJECT DELETE	202
4.5.162	PROJECT DIR	203
4.5.163	PROJECT LIST	204
4.5.164	PROJECT LOAD	205
4.5.165	PROJECT NEW	206
4.5.166	PROJECT RENAME	206
4.5.167	PROJECT REPAIR	207
4.5.168	PROJECT RESTORE	208
4.5.169	PROJECT SAVE	209
4.5.170	PROJECT START	210
4.5.171	PROJECT STOP	210
4.5.172	PROJECT USE	211
4.5.173	QUIT	212
4.5.174	RAMP	212
4.5.175	REPORT	214
4.5.176	RUN	214
4.5.177	SCENE	215
4.5.178	SECURITY	215
4.5.179	SECURITY ARM	216

4.5.180	SECURITY_DISPLAY_MESSAGE	217
4.5.181	SECURITY_EMULATE_KEYPAD	217
4.5.182	SECURITY_RAISE_ALARM	218
4.5.183	SECURITY_REQUEST_ZONE_NAME	219
4.5.184	SECURITY_STATUS_REQUEST	219
4.5.185	SECURITY_TAMPER	220
4.5.186	SESSION_ID	221
4.5.187	SESSION_ID ALL	221
4.5.188	SESSION_ID TAG	222
4.5.189	SET	223
4.5.190	SHOW	224
4.5.191	SHUTDOWN	224
4.5.192	SHORTMESSAGE	225
4.5.193	SHORTMESSAGE_REFRESH	225
4.5.194	SHORTMESSAGE_SEND	226
4.5.195	STOP	227
4.5.196	TELEPHONY	228
4.5.197	TELEPHONY_CLEAR_DIVERSION	229
4.5.198	TELEPHONY_DIVERT	229
4.5.199	TELEPHONY_ISOLATE_SECONDARY_OUTLET	230
4.5.200	TELEPHONY_RECALL_LAST_NUMBER_REQUEST	231
4.5.201	TELEPHONY_REJECT_INCOMING_CALL	232
4.5.202	TEMPERATURE	232
4.5.203	TEMPERATURE_BROADCAST	233
4.5.204	TERMINATERAMP	233
4.5.205	TEST_SPAM	234
4.5.206	TEST_SPAM_EREPORT	235
4.5.207	TEST_SPAM_LIGHTING	236
4.5.208	TEST_SPAM_LIST	237
4.5.209	TEST_SPAM_STOP	237
4.5.210	TOPOLOGY	238
4.5.211	TOPOLOGY_EXPLORE	239
4.5.212	TREE	239
4.5.213	TREEXML	241
4.5.214	TREEXMLDETAIL	243
4.5.215	TRIGGER	246
4.5.216	TRIGGER_EVENT	246
4.5.217	TRIGGER_LABEL	247
4.5.218	TRIGGER_UNICODELABEL	248
4.5.219	UNLOCK	250
4.5.220	CONVERTUNIT_CHECK	250
4.5.221	CONVERTUNIT_CONVERT	251
4.5.222	NAC_ROUTING	253
4.5.223	DBGETJSON	254
4.5.224	NAC_TAGMAP	254
4.5.225	NAC_OBJECTS_LIST	256

4.5.226	LOG EXTRACT	257
4.6	Configuration	258
4.6.1	Configuration File	258
4.6.2	Scope and Scope Objects	259
4.6.3	Configuration Commands	259
4.6.4	Configuration Parameters	260
4.6.4.1	accept-connections-from.....	260
4.6.4.2	access-control-file.....	260
4.6.4.3	allow-fast-start	261
4.6.4.4	allow-v3-pci	261
4.6.4.5	application.catalog.filename.....	262
4.6.4.6	application.catalog.directory.....	262
4.6.4.7	auto-reopen	263
4.6.4.8	cbus-application.....	263
4.6.4.9	cbus-tx-delay	264
4.6.4.10	cbus.tx-cache	264
4.6.4.11	ccp.display-oids.....	265
4.6.4.12	ccp.display-state.....	265
4.6.4.13	cgate-name	266
4.6.4.14	clock.master	266
4.6.4.15	clock.mastermode.....	267
4.6.4.16	clock.update-interval.....	267
4.6.4.17	cggroups-file	268
4.6.4.18	command-local-address.....	268
4.6.4.19	command-port	269
4.6.4.20	command.encoding.....	269
4.6.4.21	comms-debug	269
4.6.4.22	config-change-port.....	270
4.6.4.23	config-path	270
4.6.4.24	console.enable-commands.....	271
4.6.4.25	default-tag-db	271
4.6.4.26	enable.save-state.....	272
4.6.4.27	enable-xml-to-sql-background-job.....	272
4.6.4.28	event-filename	273
4.6.4.29	event-file.event-level.....	273
4.6.4.30	event-file.keep-days.....	274
4.6.4.31	event-file.split	274
4.6.4.32	event-file.split-count.....	275
4.6.4.33	event-file.split-size.....	275
4.6.4.34	event-host	275
4.6.4.35	event-millis	276
4.6.4.36	event-mode	276
4.6.4.37	event-port	277
4.6.4.38	event-printer	277
4.6.4.39	event.display-oids.....	278
4.6.4.40	file.base	278

4.6.4.41	global-event-level.....	279
4.6.4.42	heartbeat-time	279
4.6.4.43	hide-project-names	280
4.6.4.44	instance.lock-file.....	280
4.6.4.45	instance.lock-timeout.....	281
4.6.4.46	lighting.learn-update.....	281
4.6.4.47	load-change-port.....	281
4.6.4.48	local-flow-control.....	282
4.6.4.49	macro-path	282
4.6.4.50	memory-report.....	283
4.6.4.51	network.application-connect.....	283
4.6.4.52	network.error.commands-failed.....	284
4.6.4.53	network.error.units-failed.....	284
4.6.4.54	network.error.units-failed-hysteresis.....	285
4.6.4.55	network.pci.poll-interval.....	285
4.6.4.56	network.retries	286
4.6.4.57	network.retries.pci-check.....	286
4.6.4.58	network.source.....	287
4.6.4.59	network.state-interval.....	287
4.6.4.60	networks-file	287
4.6.4.61	patch.archive-file.....	288
4.6.4.62	pci-flow-control.....	288
4.6.4.63	pci.local-sal	289
4.6.4.64	pp.spec-base-directory.....	289
4.6.4.65	project.default	290
4.6.4.66	project.default.archive-dir.....	290
4.6.4.67	project.default.dir.....	291
4.6.4.68	project.start	291
4.6.4.69	reopen-delay	291
4.6.4.70	report-new-objects.....	292
4.6.4.71	response-delay.....	292
4.6.4.72	scene-base	293
4.6.4.73	secure.bind-address.....	293
4.6.4.74	secure.port-base.....	294
4.6.4.75	serial.fixbaud	294
4.6.4.76	speed-write	295
4.6.4.77	startup-delay	295
4.6.4.78	sweep-timeout	296
4.6.4.79	sync-fast-pci	296
4.6.4.80	sync-time	297
4.6.4.81	sync.free-periods.....	297
4.6.4.82	sync.global-pool-size.....	298
4.6.4.83	sync.padding.enabled.....	299
4.6.4.84	sync.padding.sync-time-factor.....	299
4.6.4.85	sync.gateway-pool-size.....	300
4.6.4.86	tag-autosave	300

4.6.4.87	tag-name-output.....	301
4.6.4.88	tag-use-zip	301
4.6.4.89	tag-validate-db	302
4.6.4.90	unit-auto-update-db.....	302
4.6.4.91	unit-auto-delete.....	303
4.6.4.92	unitcatalog.directory.....	303
4.6.4.93	unitcatalog.filename.....	303
4.6.4.94	use-1.0-addressing.....	304
4.6.4.95	use-cgroups	304
4.6.4.96	use-config-change-port.....	305
4.6.4.97	use-event-file	305
4.6.4.98	use-load-change-port.....	306
4.6.4.99	use-queue-sweeper	306
4.6.4.100	use-scenes	307
4.6.4.101	use-tags	307
4.7	C-Bus Applications	307
4.7.1	Specifying Applications	308
4.7.1.1	applications.xml.....	308
4.7.1.2	applications.spe override.....	311
4.7.2	Airconditioning Application	312
4.7.2.1	Airconditioning Application Overview.....	312
4.7.2.2	Objects	313
4.7.2.3	Events	313
	Status Events.....	313
	Humidity Schedule Entry	313
	HVAC Schedule Entry.....	314
	Zone Humidity Plant Status	315
	Zone HVAC Plant Status.....	317
	Zone Humidity	318
	Zone Temperature	318
	Command Events.....	319
	Refresh.....	319
	Set Ward On	320
	Set Ward Off	320
	Set Zone HVAC Mode.....	321
	Set HVAC Upper Guard Limit.....	321
	Set HVAC Low er Guard Limit.....	322
	Set HVAC Setback Limit.....	322
	Set Zone Humidity Mode.....	323
	Set Humidity Upper Guard Limit.....	323
	Set Humidity Low er Guard Limit.....	324
	Set Humidity Setback Limit.....	325
4.7.2.4	Commands	325
4.7.2.5	Specification	326
4.7.3	Audio Application	326
4.7.3.1	Audio Application Overview.....	326
4.7.3.2	Objects	327
4.7.3.3	Events	327
	Current Feed.....	327

Dynamic 1.....	328
Dynamic 2.....	328
High Priority.....	329
Mute	329
Next Feed.....	330
Next Language.....	331
Off	331
On	332
Output Common Control.....	332
Output Device Status Request.....	333
Output Error Code.....	333
Previous Feed.....	334
Ramp	334
Request Current Feed.....	335
Set Feed	336
Terminate Ramp.....	336
Zone Descriptor Request.....	337
Zone Feed Label Request.....	337
4.7.3.4 Commands	338
4.7.3.5 Specification	338
4.7.4 Clock and Timekeeping Application	338
4.7.4.1 Application Overview.....	338
4.7.4.2 Objects supported.....	339
4.7.4.3 Methods	339
4.7.4.4 Commands	340
4.7.4.5 Events issued	340
4.7.4.6 Status Change Port.....	340
4.7.4.7 Network time	340
4.7.4.8 Configuration properties.....	340
4.7.4.9 Usage Notes	340
4.7.4.10 Specification	340
4.7.5 Enable Control Application	341
4.7.5.1 Enable Introduction.....	341
4.7.5.2 Enable Application Overview.....	341
4.7.5.3 Persistent storage.....	341
4.7.5.4 Objects supported.....	341
Enable Control Application Object.....	341
Enable Network Variables.....	342
4.7.5.5 Commands supported.....	342
4.7.5.6 Events issued	342
4.7.5.7 Status Change Port Reporting.....	342
4.7.5.8 Project database.....	342
4.7.5.9 Usage Notes	343
4.7.5.10 Enable Specification.....	343
4.7.6 Error Reporting Application	343
4.7.6.1 Application Overview.....	343
4.7.6.2 Objects Supported.....	344
4.7.6.3 Send Event	344
4.7.7 Measurement Application	344

4.7.7.1	Specification	344
4.7.7.2	Application Overview.....	345
4.7.7.3	Objects Supported.....	345
	Measurement Application Object.....	345
	Commands Supported.....	346
	MEASUREMENT ?.....	346
	Events Supported.....	346
	Object Parameters	346
	Measurement Device Object.....	347
	Commands Supported.....	347
	Events Supported.....	347
	Object Parameters	347
	Channel Object.....	348
	Commands Supported.....	348
	MEASUREMENT DATA.....	348
	Events Supported.....	349
	Measurement Data.....	349
	Object Parameters	349
4.7.7.4	Events Issued	350
	Data	350
4.7.7.5	Unit Code Table.....	351
4.7.8	Media Transport Control Application	352
4.7.8.1	Application Overview.....	352
4.7.8.2	Objects Supported.....	353
4.7.8.3	Events Issued	353
	Category Name.....	354
	Enumerate.....	354
	Enumeration Size.....	355
	Forward	355
	Next Selection.....	356
	Next Category	357
	Next Track.....	357
	Pause	358
	Play	358
	Repeat	359
	Rewind	359
	Selection Name.....	360
	Set Selection.....	361
	Set Category	361
	Set Track.....	362
	Shuffle	362
	Source Power Control.....	363
	Status Request.....	363
	Stop	364
	Track Name.....	364
4.7.9	Security Application	365
4.7.9.1	Security Application Overview.....	365
4.7.9.2	Objects Supported.....	366
	Security Application Object.....	366
	Commands Supported.....	367
	Events Supported.....	367
	System Armed/Disarmed	367

System Disarmed.....	367
Exit Delay Started.....	368
Entry Delay Started.....	368
Alarm On.....	369
Alarm Off	369
Tamper Off.....	370
Tamper On.....	370
Current Alarm Type.....	370
Panic Cleared.....	371
Panic Activated.....	371
Low Battery Corrected.....	372
Low Battery Detected.....	372
Battery Charging.....	373
Status Report 1.....	373
Status Report 2.....	374
Password Entry Status.....	375
Mains Failure.....	376
Mains Restored.....	376
Arm Ready.....	376
Line Cut Alarm Raised.....	377
Line Cut Alarm Cleared	377
Fire Alarm Raised.....	378
Fire Alarm Cleared	378
Arm Failed Raised.....	379
Arm Failed Cleared.....	379
Gas Alarm Raised.....	379
Gas Alarm Cleared.....	380
Other Alarm Raised.....	380
Other Alarm Cleared.....	381
Object Parameters	381
Security Zone Object.....	383
Commands Supported.....	383
Events Supported	383
Zone Unsealed.....	383
Zone Sealed.....	384
Zone Open.....	384
Zone Short.....	384
Zone Isolated	385
Zone Name.....	385
Arm Not Ready.....	386
Object Parameters	386
4.7.9.3 Specification	387
4.7.10 Short Message Application	387
4.7.10.1 Short Message Overview.....	387
4.7.10.2 Events Issued	388
Short Message Refresh.....	388
Short Message Send.....	389
4.7.11 Telephony Application	390
4.7.11.1 Telephony Application Overview.....	390
4.7.11.2 Objects supported.....	390
Telephony Application Object.....	390
4.7.11.3 Commands supported.....	391
4.7.11.4 Events supported.....	391

Line On Hook.....	391
Line Off Hook.....	391
Dial Out Failure.....	392
Dial In Failure.....	392
Ringin g	392
Recall Last Number Response.....	393
Internet Connection Request Made.....	393
Isolate Secondary Outlet.....	393
Recall Last Number Request.....	394
Reject Incoming Call.....	394
Divert	394
Clear Diversion.....	395
Configuration properties.....	395
4.7.11.5 Usage Notes	395
4.7.11.6 Specification	395
4.7.12 Temperature Broadcast Application	395
4.7.12.1 Application Overview.....	395
4.7.12.2 Objects Supported.....	396
Temperature Broadcast Application Object.....	396
Commands Supported.....	397
Events Supported	397
Object Parameters	397
Temperature Group Object.....	397
Commands Supported.....	398
Events Supported	398
Temperature Broadcast.....	398
Object Parameters	398
4.7.12.3 Specification	399
4.7.13 Trigger Control Application	399
4.7.13.1 Trigger Overview.....	399
4.7.13.2 Trigger Objects Supported.....	399
Trigger Application Object.....	399
Trigger Group Object.....	399
4.7.13.3 Trigger Events.....	400
Trigger Commands	400
4.7.13.4 Status Change Port.....	400
4.7.13.5 Configuration	400
4.7.13.6 Project Database Issues.....	400
4.7.13.7 Usage Notes	401
4.7.13.8 Trigger Specification.....	401
4.8 Projects	401
4.8.1 Concept of Project	401
4.8.2 Projects And Tags	401
4.8.3 Project Names	401
4.8.4 Projects And Networks	401
4.8.5 Addressing	401
4.8.6 Default Project	402
4.8.7 Current Project	402
4.8.8 Project Properties	402
4.8.9 Access Control Issues	402

4.9	Networks	402
4.9.1	Opening a Network	403
4.9.1.1	Network Interface State Diagram	404
4.9.2	Network State	404
4.9.2.1	Network State Diagram	406
4.9.3	Networks In Error	407
4.9.4	Network Syncing	407
4.9.4.1	Calculation of NextSyncTime	409
4.9.4.2	Sync Padding	409
4.9.4.3	Sync-Free Periods	410
4.9.5	Unit Syncing	411
4.9.5.1	Unit State Diagram	412
4.9.6	Command Retries	412
4.9.7	C-Bus Clock Recovery	413
4.9.8	PCI Connection Check	413
4.9.9	PCI Polling	414
4.9.10	Unit Online Status	415
4.9.10.1	Unit Online Status Diagram	417
4.10	Access Control	418
4.10.1	Access Control Basics	418
4.11	CGL Format	419
4.11.1	CGL Specification v1.0	419
4.11.1.1	CGL Specification v1.0 Example	419
4.11.1.2	CGL Specification v1.0 Constraints	420
4.12	Object Overview	423
4.12.1	Working with Objects	423
4.12.2	All Objects	425
4.12.3	CGate Object	426
4.12.4	Projects	427
4.12.5	CBusNetworkManager	427
4.12.6	CBusNetwork	427
4.12.7	CBusUnit	434
4.12.8	CBus Output Units	435
4.12.9	CBus Output Terminal	436
4.12.10	DIN and Pro Dimmers and Relays	436
4.12.11	CBus Din/Pro Terminal	437
4.12.12	CBus Temperature Sensor	437
4.12.13	CBus PE Cell / Light Level Sensor	438
4.12.14	CBusApplication	439
4.12.15	CBusGroup	439
4.13	Building Clients for C-Gate	441
4.13.1	Communicating with C-Gate	441
4.13.2	Controlling Communications with C-Gate	441
4.13.3	Multiple Clients	442
4.13.4	Logging with C-Gate	442

5	Version History	443
5.1	v2.11.6	443
5.2	v2.11.5	443
5.3	v2.11.4	443
5.4	v2.11.3	443
5.5	v2.11.2	443
5.6	v2.11.1	443
5.7	v2.11.0	443
5.8	v2.10.6_3169	444
5.9	v2.10.6	444
5.10	v2.10.5	444
5.11	v2.10.2	444
5.12	v2.10.0	444
5.13	v2.9.8	445
5.14	v2.9.7	445
5.15	v2.9.5	445
5.16	v2.9.4	445
5.17	v2.9.2	446
5.18	v2.9.0	446
5.19	Upgrade from 1.5 to 2.5	446
5.19.1	Major changes from 1.5 to 2.x	446
5.19.2	Setting 1.5 compatibility	447
6	Not Used	449
6.1	Set Plant HVAC Level	449
6.2	Set Plant Humidity Level	449
Index		451

1 Introduction

The C-Gate Manual gives basic information about installing and operating the C-Gate server. This is the complete reference documentation for working with C-Gate.

Documentation History

This manual was newly released with version 2.5 of the C-Gate server. It combines the previously released C-Gate Server Reference and C-Gate User Guide, plus newer material like the [Quick Start Guide](#) into a new C-Gate Manual.

For deep detail, the [Reference](#) section is the place to begin.

Or, for an overview, keep reading into [About C-Gate](#).

If you are building client software to integrate with C-Gate, please see the [Building Clients for C-Gate](#) section.

Media

This manual is available as a Adobe® Acrobat® PDF or Microsoft® HTML help. Always get the latest version from the Schneider Electric website <http://www2.clipsal.com/cis/technical/downloads>

Version

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2 About C-Gate

2.1 What is C-Gate?

C-Gate is a software package that monitors and controls the components of the Clipsal C-Bus wiring system.

C-Gate runs on a PC or server platform to provide high-speed monitoring and control of multiple C-Bus networks.

The C-Gate software is either run on a separate server or in the background on a PC, to allow other C-Bus front end software or building management systems to have high-speed, high level control and monitoring of C-Bus.

C-Gate uses industry-standard TCP/IP interfaces to support:

- **Multiple C-Bus networks** - connected to a TCP/IP backbone network using TCP/IP terminal servers
- **Multiple connections** - from one or more front-end or building management systems using TCP/IP sockets
- **Simple connection** - to web servers for Internet-based C-Bus control and monitoring.

A lot of flexibility comes from using the TCP/IP standard, opening up the ability to control, connect to, monitor and Internet-enable C-Bus.

Internally, C-Gate builds an object model of the C-Bus Networks that it is controlling and monitoring.

It uses the smart monitoring features of C-Bus to fully simulate the operation of C-Bus input and output devices.

So, C-Gate can rapidly give up-to-date information about the status of C-Bus without having to slowly poll individual devices.

2.2 Why use C-Gate?

C-Gate allows you to connect one or more networks with C-Bus devices to other software or systems for control or monitoring.

C-Gate's high level Command and Event Interfaces mean that you don't need to go into the details of the C-Bus protocol in order to interface to C-Bus.

C-Gate allows you to connect to C-Bus networks across a TCP/IP backbone network as well as through a local RS232 interface or via a modem or other mechanisms.

A number of C-Bus networks can be managed in parallel at high speed, meaning that control is rapid and monitoring is accurate - even where a number of networks are involved.

2.3 How does C-Gate work with C-Bus?

The C-Gate server software opens a connection to each C-Bus network it is going to manage.

To connect to a C-Bus network, C-Gate tries to open a connection to a C-Bus Network through one of the available connection methods, which include:

- C-Bus PC Interface (model 5100PC)
- C-Bus Network Interface (model 5500CN)
- via a modem to a C-Bus Telephone Interface
- via a TCP/IP socket connection to a terminal server

Once connected, C-Gate scans each C-Bus network and builds an object model of the network and the network's C-Bus units. This model is the base that C-Gate uses for control and monitoring.

Once this initial model is established, C-Gate listens to each network, receiving monitoring events from the network that will update the model with the latest state of the network.

An event is received by C-Gate every time a button is pushed or every time a C-Bus Group Level changes as a result of a button push or activity from a sensor or other C-Bus device.

To ensure the C-Gate model of the network is up to date, the network is re-scanned on a continuous basis. This also ensures that the network is still connected and communicating with the C-Gate server.

Applications or building management systems connect to C-Gate's Command Interface and can issue commands (such as ON, OFF and RAMP and commands associated with other C-Bus Applications) and query the status of any object (using the GET and SET commands) in C-Gate's model of each network.

C-Gate also provides several event, status and configuration interfaces to allow applications or building management systems to receive a continuous real-time list of events that have occurred on the C-Bus networks that C-Gate is managing.

2.4 When to use C-Gate

Use the C-Gate server when:

- You need to accurately monitor and control one or more C-Bus networks
- You need high performance control of one or more C-Bus networks
- You need to control the C-Bus network from another Building management or control system
- You want to use a custom front-end or user interface to C-Bus, such as C-Lution or Citect
- You want to provide an Internet or TCP/IP interface to control and monitor C-Bus networks
- Control based on a C-Bus network connected by bridges is too slow
- You want to monitor status of the C-Bus devices on a C-Bus network.

2.5 When not to use C-Gate

Don't use the C-Gate server when:

- You are implementing a small, stand-alone C-Bus network that has **no need** for any external control or monitoring or any interface to a computer or other devices.

2.6 Limitations

The C-Gate server has the following limitations that should be considered when using C-Gate to control and monitor C-Bus networks:

- There is no absolute maximum number of networks that C-Gate can interface to. A reasonable operating maximum is 64 C-Bus networks.
- Above 64 networks, server configuration will have to be carefully arranged to ensure C-Gate will perform reasonably. Please contact Schneider Electric. for this configuration information

3 Quick Start Guide

Start Here...

To quickly get C-Gate up and running for you, start reading here. We'll guide you through connecting to and controlling C-Bus from C-Gate in just a few minutes.

Proceed to [Start Here](#)

3.1 Start Here

Here are your basic steps to start working with C-Bus and C-Gate:

1. Understand some basic [Concepts](#) about C-Bus and C-Gate
2. Find a [C-Bus Network](#) to work with
3. [Install C-Gate](#), if not already installed
4. [Run C-Gate](#), if not installed as a Windows Service
5. [Connect to C-Gate](#)
6. [Enter Commands](#)
7. [Working with Projects](#)
8. [Starting a Project or Network](#)
9. [Explore the Network](#)
10. [Operation](#)
11. [Events](#)
12. [Getting Help for Commands](#)
13. [Online Resources](#)

Then, dive into the [Reference](#) section for the details of C-Gate's commands, events, operation, C-Bus applications and many more things.

3.2 Concepts

There are a few concepts we need to explain to get started.

When we talk about **C-Bus Devices** and **C-Bus Units**, we mean hardware modules that are designed to connect to other **C-Bus Devices** using C-Bus cable.

When a group of C-Bus Devices (like switches, dimmers and other devices) are wired together with C-Bus cable, we call this a **C-Bus Network**.

Two C-Bus Networks can be connected together using a C-Bus Bridge. A bridge has connections to two separate C-Bus Networks.

A Project is a group of C-Bus Networks grouped together, and operated together by C-Gate or C-Bus Toolkit. Think of a Project as the set of C-Bus Networks used in, for example, a house. The C-Bus Networks in the house work together to automate the house, so they are grouped together as a project. Find out more about Projects in the Projects section of the Reference.

Read on to find out more about [C-Bus Networks](#).

3.3 C-Bus Network

In order to start working with C-Gate, you will need a [C-Bus Network](#) to control.

You can connect C-Gate to the C-Bus network in two common ways:

1. Using a PC Interface (PCI) connected to your computer's serial port

Connect your C-Bus network's PCI to a serial port on your computer, or via a USB to RS232 Serial converter. You should have a cable. It is best to know the name of the serial port you are connecting to. The name will be something like COMx where x is a number. Typically it is COM1, COM2, COM3, or even something like COM7 or COM12.

2. Using a C-Bus Network Interface (CNI) connected to an Ethernet network that your computer is connected to

You'll need to know the IP address of the CNI you want to connect to. Refer to the CNI documentation and software to find out or set the IP address of your CNI. You can also try and discover your CNI's IP address using C-Gate commands, which will be explained in [Running C-Gate](#).

3.4 System Requirements

To install and operate C-Gate, your system should preferably have the following as a minimum:

Hardware

- Pentium 4 processor, 2 GHz OR Core 2 Duo processor E6300 or better.
- 512 MB of RAM.
- 5 GB of free disk space.
- CD-ROM drive (for software installation from CD only).
- Mouse.
- Colour display capable of 800 x 600 resolution.
- USB port for use with a USB/serial converter OR a DB9 standard serial port.
- Network adaptor 10 Mbps or better (optional).

Software

- Microsoft Windows XP, Service Pack 2 or later.
- Adobe Acrobat Reader v4.0 (or higher) required for viewing and printing the Online Documentation.

- Windows Media Player v7 or higher, required for viewing the Video Presentation included with the Online Documentation.
- Anti-virus software.

System RAM, CPU and network resource requirements are increased as more C-Bus networks are opened. The requirements quoted here are for small-scale C-Gate systems accessing 8 or less networks.

3.5 Installing C-Gate Installation

The easiest way to get C-Gate installed on Windows is to install the C-Bus Toolkit software. C-Bus Toolkit requires C-Gate to communicate with C-Bus networks, and includes C-Gate in the installer. If you already have C-Bus Toolkit installed, you'll have C-Gate installed already.

C-Gate requires Sun's Java Runtime Environment 8 or later for operation. A recent stable version of Java is also included in the installer (of either C-Bus Toolkit or C-Gate).

Have a look

Look for the \Schneider Electric\C-Gate 3 directory on your default hard disk. If the directory is there, you already have C-Gate installed. Go ahead to [Running C-Gate](#).

Installing C-Bus Toolkit and C-Gate

1. Download C-Bus Toolkit from the CIS website at <http://www2.clipsal.com/cis/technical/downloads/software>
2. Run the installer, and install C-Bus Toolkit using the standard defaults. This will also install C-Gate.

Installing C-Gate alone

If installing C-Gate independently of C-Bus Toolkit:

1. Download C-Gate from the CIS website at <http://www2.clipsal.com/cis/technical/downloads/software>
2. Run the installer, and install C-Gate using the standard defaults.

C-Gate as a Windows Service

Note that during C-Gate installation the Components page now has an option called "CGate as a Windows Service".

C-Gate as a Windows Service

If you choose to install C-Gate as a Windows Service, it will automatically start running after installation. See [Running C-Gate \(Windows Service\)](#) for more information on starting and stopping

the C-Gate Service.

C-Gate as a Console Application

If you choose NOT to install C-Gate as a Windows Service, it will install in the traditional manner, as a console application. After installation is complete, you will need to start C-Gate yourself. Proceed to [Running C-Gate](#) to learn more.

Note that the best way to switch between service and console mode is to re-run the C-Gate installer.

3.6 Running C-Gate (Windows Service)

Windows platforms:

When C-Gate is installed as a Windows service, it will be started automatically after installation, and also after the computer boots into Windows.

There are several ways to stop and start C-Gate:

1. You can stop and start C-Gate from the Service Properties dialog as above.
2. You can also control the C-Gate service from the Start Menu:

Start / All Programs / Clipsal / C-Gate / Start the C-Gate service

Start / All Programs / Clipsal / C-Gate / Stop the C-Gate service

3. You can send the [SHUTDOWN](#) command to instruct C-Gate to shutdown. When C-Gate is shut down, the service will automatically stop.

Note that when C-Gate is running as a service, you will not see the C-Gate console window. See [Connect To C-Gate](#) for how to issue commands to CGate.

3.7 Running C-Gate (Console Application)

(For Windows platforms)

If C-Gate is not installed as a Windows Service, or the service is stopped or uninstalled, then C-Gate needs to be run as a console application.

Some applications such as C-Bus Toolkit will automatically start C-Gate in console mode if it is not already running. If you are using these applications you should allow them to start C-Gate where possible.

In some situations you may want to start and stop C-Gate yourself.

To run C-Gate, select the following from the Start Menu:

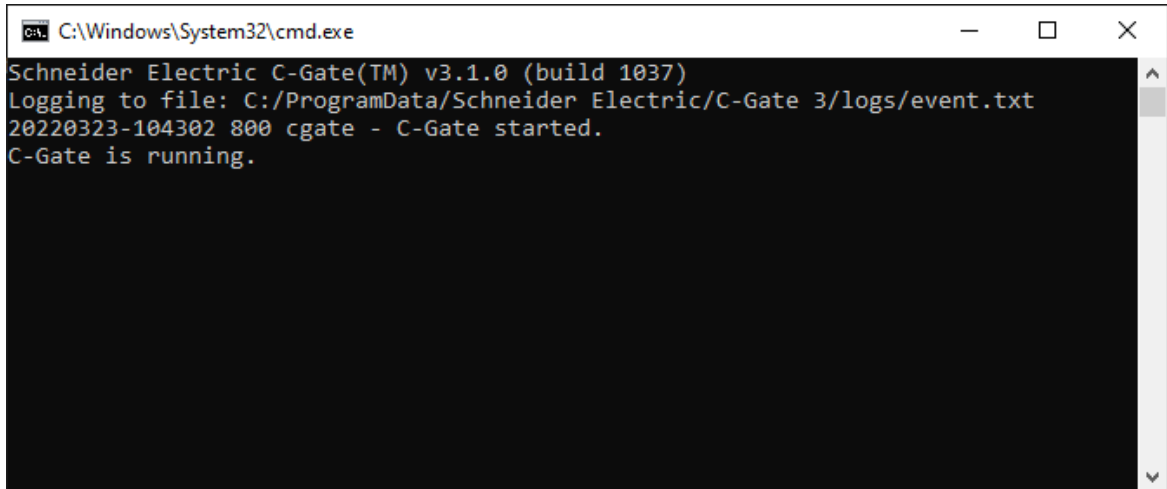
Start / All Programs / Clipsal / C-Gate / C-Gate

Alternatively, run the cgate.exe executable located within the C-Gate directory:

C:\Program Files\Schneider Electric\C-Gate 3\cgate.exe

This will open a Windows console window, running C-Gate inside of it. You can view the stream of events indicating what C-Gate is up to.

Here is an example of what you should see when you start the console version of C-Gate:

A screenshot of a Windows command prompt window titled 'C:\Windows\System32\cmd.exe'. The window has a black background with white text. The text displayed is: 'Schneider Electric C-Gate(TM) v3.1.0 (build 1037)', 'Logging to file: C:/ProgramData/Schneider Electric/C-Gate 3/logs/event.txt', '20220323-104302 800 cgate - C-Gate started.', and 'C-Gate is running.'.

```
C:\Windows\System32\cmd.exe
Schneider Electric C-Gate(TM) v3.1.0 (build 1037)
Logging to file: C:/ProgramData/Schneider Electric/C-Gate 3/logs/event.txt
20220323-104302 800 cgate - C-Gate started.
C-Gate is running.
```

Note: we do not recommend running the cgate.jar Java binary directly. This is because cgate.exe takes care of things such as allocating the correct amount of memory for C-Gate.

To find out more about customising the launch configuration of cgate.exe, see [Launch Configuration](#).

To stop C-Gate:

1. Use the [SHUTDOWN](#) command in the console window, or
2. Close the console window C-Gate is running in, or
3. Type Ctrl-C in the console window.

To use C-Gate:

You can enter commands directly into the console window, but it is easier to open a separate window for commands to avoid confusion.

So, let's [Connect To C-Gate](#) and then try out some commands.

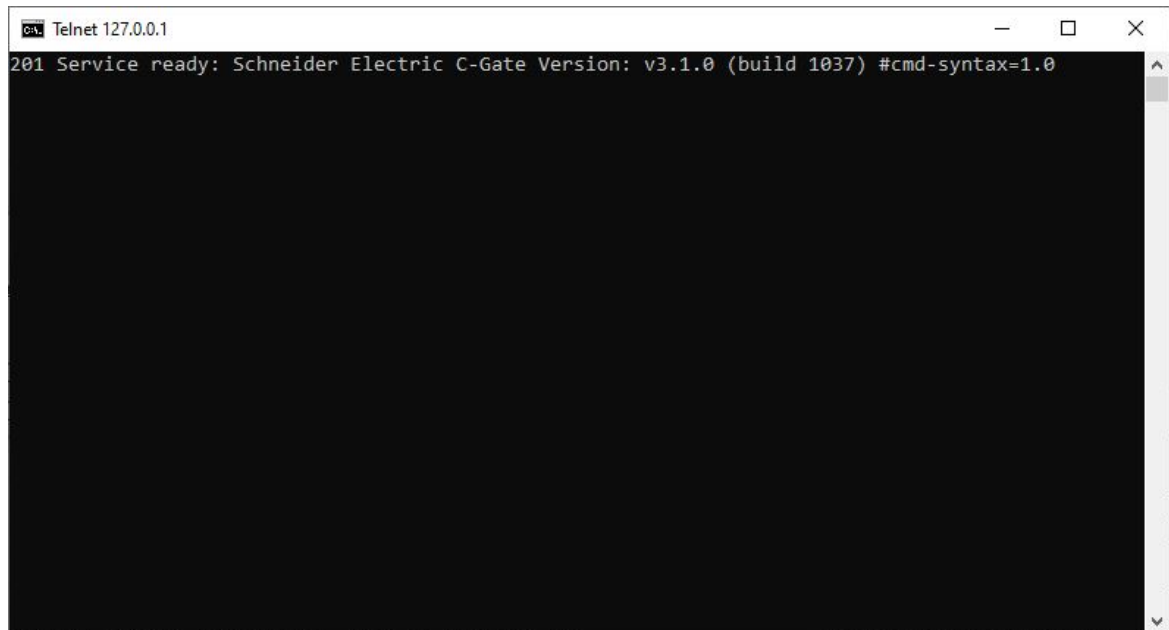
3.8 Connect To C-Gate

Windows platforms:

To connect to C-Gate, you are going to open a command-line session using the Telnet program that comes with Windows. The C-Gate installation makes it easy to make a connection from the start menu:

Start / All Programs / Clipsal / C-Gate / Connect to C-Gate

A new window will open similar to this one:



If the C-Gate server is not running, then the window will close almost immediately. Otherwise you are ready to start

The line:

```
201 Service ready: Clipsal C-Gate Version: v2.6.1 (build 2245)
```

shows C-Gate is ready to accept commands.

See [Enter Commands](#) to take the next step.

Telnet Client on Windows Vista, Windows 7 and Windows 8:

Since Windows Vista, the Windows Telnet client is turned off by default. You need to enable the telnet client feature in Windows before using the 'Connect to C-Gate' start menu item. To enable telnet client on Windows Vista, Windows 7 and Windows 8:

- Open Control Panel from the Windows Start menu.
- Select Programs.
- Under Programs and Features, click Turn Windows features on or off.
- From the list that appears, select the Telnet Client checkbox to enable telnet.

Other platforms:

You'll need to find and run the telnet client for your platform. In most cases, issuing a command like this will open a command session in the current window.

```
telnet localhost 20023
```

If this succeeds, you should see a welcome message like "201 Service ready: Clipsal C-Gate Version: v2.6.1 (build 2245)" which indicates C-Gate is ready to receive commands.

See [Enter Commands](#) to take the next step.

3.9 Enter Commands

C-Gate is controlled by a series of commands typed on the C-Gate command line. Each command starts at the beginning of the line and is ended by pressing the Enter or Return key. C-Gate will send a response back for each command, indicating whether the command succeeded or failed and any other information requested in the command.

The rhythm of this is enter a command, wait for a response, enter a command, wait for a response, enter a command, wait for a response.

For example, the simplest command is `noop`, which just returns an 'OK' response:

type the word:

```
noop
```

then press Return or Enter

C-Gate will respond with:

```
200 OK.
```

In this quick start guide, commands we type are in **bold** while C-Gate's responses are not.

Try it yourself.

```
noop
```

```
200 OK.
```

Each C-Gate response starts with a three-digit number, in this case 200, followed by some text that gives you an idea what happened. The three digit number is there to make it easy for client programs connected to C-Gate to understand C-Gate responses. The text is there to make it easy for you to understand what is going on.

The `noop` command does nothing else but return an OK response. It is a good way of making sure that you are talking to C-Gate's command interface.

3.10 Working with Projects

When you configure or program a network with the C-Bus Toolkit software, it relies on C-Gate to create the project and perform the programming. Once the project has been created, you can load and use this project to control and monitor your C-Bus networks.

Available projects

Projects that are in C-Gate's memory are ready to be used immediately. To get a list of these projects, use the [PROJECT LIST](#) command.

project list

```
123 project=HOME state=stopped
```

As you can see, there is one loaded project, and it is stopped. We can use this project by entering a [PROJECT USE](#) command. This makes sure that this command session is working with the project we want.

project use HOME

```
200 OK.
```

Projects that are stored on disk can be loaded into C-Gate and then used.

Now, you can move on to [Starting a Project or Network](#)

Loading a project from disk into C-Gate

If your project doesn't show up in the results of the [PROJECT LIST](#) command, you can load it from disk.

To get a list of the projects on disk, use [PROJECT DIR](#):

project dir

```
123-project=CIS_PROJ
123-project=CIS_TEST
123 project=HOME
```

The project we want, HOME , can be loaded with the [PROJECT LOAD](#) command:

project load HOME

```
20060206-144513 761 cmd1 - Command: project load HOME
```

and then set up this command session to use it:

project use HOME

```
200 OK.
```

Now, you can move on to [Starting a Project or Network](#)

3.11 Starting a Project or Network

To start C-Gate working with a C-Bus network, the network has to be opened. Opening a network means that C-Gate opens communication with the network and starts to build a model of the devices and operation of the network so you can use commands to manipulate the devices on the network. For example, to type in commands to turn on or off some relays controlling lights, the C-Bus network needs to be opened.

To open just one network use the NET OPEN command:

```
net open 254
120-initializing
120-opening port
120-starting network threads
120-pci reset
120-open complete
200 OK: //HOME/254
```

To open all the networks in your project, use the PROJECT START command.

```
project start
200 OK.
```

For more information and advanced usage see: [Opening a Network](#).

Once opened the network will begin to sync which may take a minute or two. For more information on this see: [Network Syncing](#).

Once the network is synced you can [Explore the Network](#).

3.12 Explore the Network

Tree

Once a network has been opened, you can start to explore C-Gate's model of the network using some commands. The [TREE](#) command lists a lot of information about a network and the C-Bus Devices on the network.

```
tree 254
320- Network name:254 type=cni address=localhost:14000 state=ok
320- Unit count=12
320- Units:
320-//HOME/254/p/0 ($0) type=PC_INT_1 app=255($ff),255($ff) state=ok
groups=
320-//HOME/254/p/1 ($1) type=DIMDN8 app=56($38),255($ff) state=ok
groups=13,14,15
320-//HOME/254/p/2 ($2) type=KEYM2 app=56($38),255($ff) state=ok
groups=13,14,35,36,37,38,39,40,5
320-//HOME/254/p/3 ($3) type=KEYM4 app=56($38),255($ff) state=ok
groups=13,14,15,38,39,40
320-//HOME/254/p/4 ($4) type=KEYM8 app=56($38),255($ff) state=ok
groups=13,2,3,4,5,6,7,8
320-//HOME/254/p/5 ($5) type=SCNCTL5 app=56($38),202($ca) state=ok
groups=
320-//HOME/254/p/6 ($6) type=RELDN12 app=56($38),255($ff) state=ok
groups=0,2,3,4,5,6,7,8,9,10,11,12,5,6
320-//HOME/254/p/9 ($9) type=PC_INT_1 app=255($ff),255($ff) state=ok
groups=
320-//HOME/254/p/14 ($e) type=KEYBL5 app=56($38),136($88) state=ok
groups=1,44,10,2
320-//HOME/254/p/15 ($f) type=PC_CTA app=255($ff),255($ff) state=ok
groups=
```

```

320-//HOME/254/p/19 ($13) type=PCLOCAL4 app=255($ff),255($ff) state=ok
groups=
320-//HOME/254/p/249 ($f9) type=GATEWLSN app=56($38),49($31) state=ok
groups=
320- //HOME/254/56/10 ($a) level=0 state=ok units=6,14
320- //HOME/254/56/11 ($b) level=0 state=ok units=6
320- //HOME/254/56/12 ($c) level=0 state=ok units=6
320- //HOME/254/56/13 ($d) level=255 state=ok units=1,2,3,4
320- //HOME/254/56/14 ($e) level=255 state=ok units=1,2,3
320- //HOME/254/56/15 ($f) level=0 state=ok units=1,3
320- //HOME/254/56/35 ($23) level=255 state=ok units=2
320- //HOME/254/56/36 ($24) level=255 state=ok units=2
320- //HOME/254/56/37 ($25) level=255 state=ok units=2
320- //HOME/254/56/38 ($26) level=255 state=ok units=2,3
320- //HOME/254/56/39 ($27) level=255 state=ok units=2,3
320- //HOME/254/56/40 ($28) level=255 state=ok units=2,3
320- //HOME/254/56/44 ($2c) level=255 state=ok units=14
320- Application 202 ($ca) [trigger]
320- Net Vars:
320- //HOME/254/202/255 ($ff) level=0 state=ok
320 -end-

```

The tree command is pretty much designed for human reading. Client applications that want this kind of information can use [TREEXML](#) to get a structured output that is much easier to parse.

So, what does the TREE output contain:

- The name, state and interface address of the network
- The number of C-Bus Units on the network
- A list of all the units on the network, with their Unit Type string, the applications they support, the unit state, and any Groups that are used.
- A list of applications in use on the network, with a listing of Groups or Network Variables, depending on the applications.

Applications, Groups and Network variables are used to control the C-Bus Devices. Find out how to use them in the next section on [Operation](#).

3.13 Operation

Once a network is up and running, we can use commands to control some of the C-Bus devices on the network. For now, let's talk about controlling lighting devices as these are the most common you'll encounter on C-Bus.

Lighting is one of the Applications that C-Bus supports. Applications group a set of commands together for a broad purpose. Lighting Application commands are commands like ON, OFF and RAMP. The Lighting Application has an application number, which is 56.

In the lighting application, turning on and off and dimming lights occurs by changing the values of some network variables in the Lighting Application. Just in the lighting application, these network variables are called Groups. There are 255 Groups, with addresses starting at 0 and going up to 254. And each group has a value between 0 and 255.

C-Bus Units are programmed using the C-Bus Toolkit software to associate Groups with keypresses on Switches, and control of Dimmers and Relays to turn light on or of as well as dim them. Normally a group is set to 0 to turn it off, and 255 to turn it on, with the values in between allowing dimming.

There are three main commands for manipulating group values, [ON](#), [OFF](#) and [RAMP](#).

ON sets a group value to 255

OFF sets a group value to 0

RAMP can set a value to any level between 0 and 255, with optional rate of change control.

So let's find some groups on our network. Remember the application number is 56. Let's have a look at some of the output from the TREE command again.

```
...
320-Applications:
320- Application 56 ($38) [lighting]
320- Groups:
320- //HOME/254/56/0 ($0) level=0 state=ok units=6
320- //HOME/254/56/1 ($1) level=255 state=ok units=14
320- //HOME/254/56/2 ($2) level=0 state=ok units=4,6,14
320- //HOME/254/56/3 ($3) level=0 state=ok units=4,6
320- //HOME/254/56/4 ($4) level=0 state=ok units=4,6
320- //HOME/254/56/5 ($5) level=0 state=ok units=2,4,6
320- //HOME/254/56/6 ($6) level=0 state=ok units=4,6
320- //HOME/254/56/7 ($7) level=0 state=ok units=4,6
320- //HOME/254/56/8 ($8) level=228 state=ok units=4,6
320- //HOME/254/56/9 ($9) level=0 state=ok units=6
...
```

You can see that groups 0 through 9 are defined for lighting application 56 on our network 254 in our project HOME. Group 0 is at level 0, and group 1 is at level 255. So 0 is off, and 1 is on.

The //HOME/254/56/0 gives the fully qualified address of group 0. We can refer to the group in commands as:

//HOME/254/56/0

or given we have previously run the PROJECT USE HOME command, we can also use:

254/56/0

because the project part is assumed from the PROJECT USE command.

So, let's turn on group 0:

on 254/56/0

200 OK: //HOME/254/56/0

Now, let's check the level has changed using the [GET](#) command.

get 254/56/0 level

300 //HOME/254/56/0: level=255

And let's turn that off again.

off 254/56/0

200 OK: //HOME/254/56/0

And now, let's set it to a mid level like, say, 100, but do it slowly:

ramp 254/56/0 100 2m


```
200 OK: //HOME/254/56/0
```

This will take approximately two minutes to reach level 100.

Whereas:

```
ramp 254/56/0 100
```

```
200 OK: //HOME/254/56/0
```

will cause the level change to take place immediately.

As you can see, this series of commands allow control of lighting devices from the command line. However, in addition to this control, C-Gate produces a lot of event information that allows client programs to subscribe to a range of events that occur on the C-Bus networks. Read on to find out about [Events](#).

3.14 Events

C-Gate can provide a wide range of events to client programs to allow them to keep up-to-the-second information as to the state of the network.

Events are all delivered as line-by-line text, where the latest information is presented as a single line indicating the event.

To gain an understanding of events, please read about [Command and Monitoring Interfaces](#) in the Reference section.

To learn about events which are logged on disk, refer to the [Logging](#) section.

3.15 Getting Help for Commands

C-Gate has a HELP command, and you can use that to get some help for commands that you want to work with.

Type help and the command name, and C-Gate will give you some help with the command if there is some available.

To find documentation for available commands, look in the [Commands](#) section of the C-Gate Reference.

3.16 Online Resources

Here are a few selected online resources to help with C-Bus and C-Gate.

C-Bus Forums

<http://www.cbusforums.com>

Several thousand people using C-Bus swap tips and ideas at the C-Bus Forums. Go here for general advice or esoteric tips and tricks, plus information about the latest software releases.

Schneider Electric Website

<http://www.clipsal.com/cis>

Consult the official Schneider Electric website for technical support and software downloads.

4 Reference

This section is the reference guide to C-Gate.

4.1 C-Gate Launcher (Windows)

The Windows installer includes everything needed to get started out of the box. This includes:

- a launcher executable (cgate.exe), and
- a private Windows 32-bit Java Runtime Environment (/jre8 subdirectory).

The launcher executable serves multiple functions:

1. It can register C-Gate as a Windows Service that runs automatically as long as the computer is turned on.
2. It can run C-Gate in console mode.
3. It can find the included JRE and pass parameters to it as needed.

These operations are controlled with specific [configurations](#).

4.1.1 C-Gate Launcher Configuration

The C-Gate Launcher uses a default configuration which works for most users. For certain scenarios a custom configuration may be needed.

The C-Gate Launcher takes its configuration from the following methods in order of priority:

1) Launcher parameters

cgate.exe accepts the following command-line parameters:

-install	Install as a Windows service
-uninstall	Uninstall the Windows service
-start	Start the Windows service if CGateService is not disabled
-stop	Stop the Windows service
-console	Start C-Gate as console application
-restart	Restart C-Gate, or start it if it was not previously running
-silent	Suppresses pop-up error dialogs (most of them). Can be used with any other parameter.

If no parameter is supplied the executable first attempts to start the Windows service and failing that launches C-Gate as a console application.

2) INI config file options

cgate.exe can use an optional cgate.ini file in the same directory which supports the following options:

JAVA_DIR=<value> Use the Java binary at <value>/bin/java.exe to launch C-Gate. Useful when there are multiple versions of Java on the machine.

If not specified, defaults to: <C-Gate root directory>/jre

3) Registry options

cgate.exe will look in the following registry location for an optional key that determines the full string of parameters to be passed to the Java VM.

HKEY_LOCAL_MACHINE\SOFTWARE\Schneider Electric\C-Gate\CurrentVersion\
String value: "JavaParameters"

Example value: "-Xms64M -Xmx256M"

4) Registry options set by C-Bus Toolkit

The following options are able to be changed using C-Bus Toolkit's Preferences dialog.

cgate.exe will look in the following registry location for an optional key that determines the Maximum memory amount to be allocated to the Java Virtual Machine.

HKEY_CURRENT_USER\SOFTWARE\Schneider Electric\C-Gate\CurrentVersion\
String value: "JavaHeapMax"

Note: This option does not have any effect if the JavaParameters registry key is defined.

5) YAJSW Configuration

As the launcher utilises the open-source YAJSW wrapper it is possible for experienced users to customise the configuration files in the /yajsw directory. These configurations are made at the user's own risk and are not supported by Clipsal technical support.

4.1.2 New vs Old Launcher

C-Gate 2.11.0 introduced a new cgate.exe launcher to replace the old one.

It offers improved stability and logging, while retaining the same behaviours and command-line parameters as the old launcher.

There are some small differences:

- Memory options are now respected in Service mode. Whereas previously Service mode would always use 512MB, the maximum memory amount specified in the Toolkit preferences will now apply to the Service mode as well as to the Console mode.
- The -silent parameter no longer applies as the new launcher does not produce any popup dialogs.

The new launcher produces far more detailed logging, which it writes to two new log files on disk, see [Launcher logs](#).

It's no longer necessary to run C-Gate in console mode to see the text of unhandled exceptions, and messages are no longer sent to the Windows Event Viewer.

4.1.3 C-Gate Launcher Diagnostics

As well as cgate.jar's own logging, the cgate.exe launcher writes to two log files:

- logs/launcher.txt – this captures logging specific to cgate.exe that was previously sent to Windows Event Viewer.
- logs/wrapper.txt – this captures the stdout/stderr output that was previously only visible in

Console mode.

4.1.4 C-Gate As A Service

When C-Gate is installed as a Windows service, it will be started automatically after installation, and also after the computer boots into Windows.

You can confirm that it is working via the Services dialog in Windows. The easiest way to access this is as follows:

- From the Windows Start Menu, select Start / Run..
- Type 'services.msc' and press ENTER.
- The Services dialog will open.
- Find and double-click on the service called 'C-Gate Service' to open the Service Properties dialog.
- Confirm that the Startup Type is 'automatic'.
- Confirm that the service status is 'Started'.

4.2 C-Gate Jar

C-Gate at its core is a Java application (cgate.jar) and it can run on a range of platforms as long as there is an appropriate Java Runtime Environment.

For users on Windows platforms it is recommended to use the [C-Gate Launcher](#) as it provides a greatly simplified C-Gate experience.

For users on other platforms it is recommended to use the C-Gate zip distribution and you will need to supply the appropriate JRE and a launch mechanism.

4.2.1 Zip Distribution

The zip distribution of C-Gate is recommended for non-Windows users.

It is functionally identical to the installation by the C-Bus Toolkit installer, with the following differences:

1. A private JRE is not included. It is up to you to provide a Java Runtime Environment on the system.
2. A simple cgate.sh launcher is included to launch from a Linux shell.
3. C-Gate is not installed as a service or daemon. It is up to you to configure C-Gate to launch as and when desired.
4. Windows USB drivers are not included. The first-time you install on a Windows system you should use the C-Bus Toolkit installer.
5. Built-in support for serial ports on Java 32-bit for Windows, Linux and Mac OSX. Note that you can install a 32-bit Java on a 64-bit operating system.
6. Built-in support for serial ports on Java 64-bit for Linux.
7. Configurable support for serial ports on Java for Windows 64-bit and Java for MacOS X 64-bit by moving and renaming files (see \\lib\\serial-readme.txt).

Installation is straight-forward: unzip to the directory of your choice.

WARNING: Please don't contact Technical Support for problems with the zip distribution. You should have moderate familiarity with your target OS, with platform scripting and with executing Java applications.

4.2.2 Java Runtime Environment

C-Gate is compiled with the following version of Java compiler:

Java Compiler: Oracle Java 8 (8u202) 64-bit JDK for Windows

Java Runtime Environment (JRE) is included in the Windows installer and the [C-Gate Launcher](#) start the application with the JRE. The C-Gate zip distribution does not include a JRE.

JRE: Adoptium OpenJDK 11 (11.0.24+8) 64-bit JRE for Windows

It is possible to use your own JRE. For best results use one that is the same JRE major version (listed above).

Bear in mind that some libraries included with C-Gate may not work if the JRE:

- is not in the JRE major version listed above,
- or is built for different architecture (eg. ARM-based processor),
- or is for a different platform (e.g. Linux, Mac).

There may be other limitations not yet documented.

4.2.3 Manual Launch

JRE Location

First you should confirm the availability and location of your JRE. For example you can check to see if it is on your path:

```
$ java -version
java version "1.8.0_91"
Java(TM) SE Runtime Environment (build 1.8.0_91-b15)
Java HotSpot(TM) 64-Bit Server VM (build 25.91-b15, mixed mode)
```

JRE Parameters

In order to reproduce the configuration used by C-Bus Toolkit and the C-Gate Launcher you should pass the following parameters to your JRE:

```
-Xms64M           # don't need to change this.
-Xmx256M          # increase to 512 etc as needed.
```

Not doing this may cause C-Gate to run out of memory when working with large project files.

Manual Launch

Putting all the above together, to launch C-Gate in a standard shell:

```
$ cd ~/cgate2
$ java -jar cgate.jar -Xms64M -Xmx256M
```

This will start C-Gate in the current shell.

C-Gate will display a message like this:

```
Clipsal C-Gate(TM) v2.6.1 (build 2245)
```

You can enter commands directly into the console session but it is easier to open a separate window for commands to avoid confusion.

For more information see the [Command and Monitoring Interfaces](#).

Automatic Launch

It is up to the user to configure daemons or scheduled tasks for their operating system to automatically launch C-Gate.

4.3 Command and Monitoring Interfaces

Once C-Gate is running, it makes available a number of interfaces to allow command to be issued to the C-Gate server and to allow monitoring and event information to be received from C-Gate. These interfaces are available as telnet ports, with optional SSL encryption with client authorisation. [Access control](#) also applies to these connections. In addition, the [console](#) can be used for the delivery of event and issuing commands.

For all of these interfaces, multiple clients can connect at once, allowing multiple operations and control clinets to talk to C-Gate and control and monitor connected C-Bus Networks.

All of these interfaces are telnet-style TCP/IP sockets, which can be accessed directly using a telnet client program to allow for manual use. The SSL interfaces require a SSL telnet client or similar.

The interfaces are:

Command Interface

The [command interface](#) is used to issue commands and get responses back from C-Gate. This is the prime method used by client applications that wish to interface to C-Gate. There is a rich command set available for this interface allowing all full configuration and operation of C-Gate from a remote location.

The command interface is normally found at TCP/IP port 20023, or port 20123 for a secured SSL connection.

Note: using the [EVENT](#) command it is possible to flexibly enable the delivery of all sorts of events on the Command Interface.

Event Interface

The [event interface](#) is used to provide a continuous stream of events as they occur on the server. Clients can listen to the event interface and receive the stream of events as they occur. The events have date, time and object information attached to them, and the list of events is well defined.

The event interface is normally found at TCP/IP port 20024, or port 20124 for a secured SSL connection.

Status Change Port (SCP)

The [Status Change Port](#) allows listening clients to receive an ongoing series of events that result from status changes in C-Bus Applications. Status changes include lighting application on, off and ramp messages along with messages for other C-Bus applications.

The output of this port is rendered as a series of command interface commands and comments, which allows their recording for client macro facilities.

The Status Change Port is normally found at TCP/IP port 20025, or port 20125 for a secured SSL connection.

Config Change Port (CCP)

The [Config Change Port](#) allows listening clients to see any configuration change in the C-Bus networks connected to C-Gate. These configuration changes include addition or deletion of C-Bus Units and unit synchronisation states.

The Config Change Port is normally found at TCP/IP port 20026, or port 20126 for a secured SSL connection.

4.3.1 Command Interface

Commands can be sent to C-Gate and responses and events received via the [console \(startup window\)](#) or via a TCP/IP interface, with or without [SSL encryption](#).

The command interface supports a wide range of [commands](#) used to configure, control and monitor C-Bus networks and the C-Bus server itself.

4.3.1.1 Console Connection Details

When the C-Gate server starts, a console window is normally opened. This console window can be used to view events and submit commands to the C-Gate server.

The ability to type commands can be controlled by the [console.enable-commands](#) configuration option. It is enabled by default.

The console acts as a special command session. It is started without a 201 Service Ready message and is started with event output already turned on. The console starts up in event mode e +s0c0. The event mode can be changed by the [EVENT](#) command.

The console starts up with access control level Program. This can not be changed.

4.3.1.2 TCP/IP Connection Details

The command interface for the C-Gate server is accessed as a TCP/IP socket interface using a series of text commands and responses to allow both human and automated use.

To access the C-Gate server to type commands and view status:

1. Start the telnet program.
2. Connect to the hostname or IP address of the C-Gate server, using port 20023, or the port number set in the cgate system parameter command-port.
3. The C-Gate server will respond with the 201 Service Ready response if the connection is established successfully.

Automatic systems wishing to access the C-Gate server should open a TCP/IP socket to the IP address of the C-Gate server; port 20023. The 201 Service Ready message should be received in response to the connection.

8 bit bytes are supported over the socket interface. All command and response data is 8 bit USASCII characters. The high-order bit is ignored in commands.

Note: if the C-Gate server's cgate parameter accept-connections-from is set to a list of IP addresses, then connections will only be accepted from hosts with those IP addresses. See section 0 for further details of this parameter.

4.3.1.3 Commands and Replies

The command interface is based on the issuing of a command and a response following the execution of the command by the C-Gate server. All commands are one line only, terminated by a CRLF pair. Each response is one line long.

The figure shows the opening of a connection to the C-Gate server and the ongoing conversation. Once the controlling system opens the connection, the C-Gate server responds with a 201 Service Ready message. This response is to the opening of the connection and shows the controlling system that the IG is ready to receive commands.

When the controlling system issues the NOOP command, CGate responds with a 200 OK message, a single response message for this command.

When the controlling system sends a GET command, a reply is sent.

Controlling system (open connection)	open connection	C-Gate server
	----->	
		201 Service ready (connected)
	<-----	
(issue command:)	NOOP	
	----->	(process command and respond)
	200 OK	
(receive response)	<-----	
(issue command:)	GET 2/56/1 level	
	----->	(process command and respond)
	300 2/56/1: level=102	
(receive response)	<-----	
(issue next command:)	...	

----->

A command is a line of ASCII text terminated by a CRLF pair. Each command line begins with an action verb and contains one or more parameters following the action verb. White space (ASCII space and tab characters) are used to separate commands and parameters.

Command syntax is described here, and details of individual commands are described in the Commands section.

command = action-verb [parameters] CRLF

action-verb = name ; See the list of commands below.

4.3.1.4 Commands

This section describes the commands supported by the C-Gate Command Interface.

4.3.1.4.1 Terminology

Augmented BNF

The descriptions of C-Gate commands uses an augmented Backus-Naur Form (ABNF) to describe the grammar of commands, responses and events. Reference should be made to the [IETF's RFC2234](http://www.ietf.org/rfc/rfc2234.txt) (<http://www.ietf.org/rfc/rfc2234.txt>) which fully defines augmented ABNF as used in this specification.

This additional definition is used widely in the ABNF used in this document:

NAME = ALPHA [* <CHAR except CTRL, whitespace, "\$", "/", "~", "-", ",", ":", or "." >]

4.3.1.4.2 Objects and Addresses

Object Approach in C-Gate

The C-Gate server adopts an object-oriented approach to controlled devices. All controlled devices and many internal components of the C-Gate server are exposed to the command and event interfaces as objects that can be accessed with commands.

In general, every object has:

- An object identifier that allows the object to be referred to. Object identifiers may be network addresses, symbolic names, or reserved system names.
- Parameters that may be set or viewed, using the [SET](#) and [GET](#) commands
- Methods which may be executed with the [DO](#) command

Objects are either created by the system on startup, created as a C-Bus network is scanned, or are created with the [NEW](#) command.

System objects provide parameters and methods to allow configuration and operation of the C-Gate server itself. Key system objects include the cgate object and defined C-Group objects. Parameters for these objects are described in

Addressing Controlled Components and Objects

Both commands and events reference the components and objects in networks attached to the C-Gate and contained within it. This section defines the format of addresses for network devices used in the gateway and for object identifiers, that can reference addressed components on networks, but can also reference local system objects.

Addresses

Addresses are constructed from one or more address parts separated by the "/" (slash) character. Address parts get more specific to the right hand side of an address. Address parts except for network names, are all expressed as positive decimal integers, in the range from 0 to 32767 or as hexadecimal integers proceeded by the '\$' character.

Object Identifiers

An object identifier can contain:

- An address, as described above, or
- an object name that is known to the C-Gate and is part of a flat namespace typically used for system objects and CGroups.

C-Bus address components

The table shows the components of a C-Bus address that would be used to control and monitor C-Bus networks connected to the C-Gate.

<u>Address type</u>	<u>Description</u>
Project Name	The name of a project. Generally project names shall not start with a number, and will be 8 or less uppercase characters. E.g. HOME or CIS.
C-Bus Network name	This is a name or number describing a C-Bus network controlled by the C-Gate server. Important notes: Network names are case-sensitive. Network A42 and network a42 are different networks. When using C-Gate with Clipsal C-Lution, you must use numbers for the C-Bus network name, and the numbers must be in the range 1 through 255. Network name 0 is reserved for system use.
C-Bus Unit number	Identifies a single physical C-Bus unit on a C-Bus network. Unit numbers are unique on a C-Bus network. C-Bus unit numbers range from 0 to 255. Unit 255 is the initialisation address for new C-Bus units that have not been allocated a device address, but otherwise operates as a normal unit address.
C-Bus terminal number	This is the number of a unique relay terminal, dimmer circuit, or key switch pushbutton on a C-Bus unit. C-Bus relays and dimmers typically have from 1 to 12 terminals or circuits and this part of an address allows them to be individually addressed. There is no terminal 0 (zero) on any C-Bus unit.
C-Bus Application Number	This is the number of a C-Bus application. There are 255 possible applications, ranging in number from 0 through 254. Application number 255 indicates the unassigned application in a C-Bus unit.

C-Bus Group or Area number	C-Bus Group and area numbers are used in sending commands and monitoring commands sent to the C-Bus network. In the programming of a C-Bus network, C-Bus Units are programmed to send or act upon commands that specify a Group or an Area for a specific application. C-Bus Group and Area numbers range from 0 to 255 in any C-Bus network. Groups share the numbering space with Area numbers.
-----------------------------------	--

Representing Addresses

The augmented BNF forms below show how addresses are represented. There are two main types of addressing:

- physical addressing by network components or physical devices, and
- group and application addressing which addresses the network by C-Bus applications and C-Bus groups (used in C-Bus for command and status information).
-

Physical addresses are indicated by the addition of the "p/" path element to the beginning of the address.

A wildcard (match all objects) address can be indicated by using a "*" symbol in the last part of an address.

```
numeric-address = 1*DIGIT | "$" 1*HEX_DIGIT | "*"
```

```
terminal-number = numeric-address
```

```
unit-number = numeric-address
```

```
network-name = 1*DIGIT | NAME
```

```
application-number = numeric-address | "~"
```

```
group-number = numeric-address
```

```
project-name = 8*CHAR ;; should be uppercase
```

```
project-prefix = "/" + project-name + "/"
```

```
physical-address = [project-prefix | "/"] "p" "/" network-name "/" unit-number  
[ "/" terminal-number ]
```

```
application-address = [project-prefix | "/"] network-name ("/" application-  
number ) | "/"
```

```
group-address = application-address "/" group-number
```

```
network-address = [project-prefix | "/"] network-name
```

```
project-address = "/" + project-name
```

```
system-address = NAME
```

Representing Object Identifiers

The augmented BNF below shows the form of an object identifier. Note that an object identifier full encompasses the representation of an address.

```
object-identifier = NAME | network-address | physical-address | application-address  
| group-address | project-address
```

Addressing Examples

Some example addresses are shown below:

To address network 254 in project HOME, the address would be:

254

or

//HOME/254

To address physical C-Bus unit 15 on network 57 the address would be constructed as:

p/57/15

or

//HOME/p/57/15

To address all the units on network 57 the address would be constructed as:

p/57/*

To address terminal 2 on the same unit:

p/57/15/2

To address C-Bus group 12 on C-Bus network 2 on application 56, the address would be:

2/56/12

To address group 5 in the default application of network 11

11//5

or

11/~ /5

4.3.1.4.3 Array Filtering

Array Filtering is a syntax permitted in certain C-Gate commands that allows you to reference an object in an Array not by its unique index but by filtering on a field of the object.

Where you would normally use the array index in brackets e.g. **[3]** you would use a **[name=value]** syntax.

This is particularly useful with the database commands such as DBGET.

Examples

This is the conventional method to reference an array object. The index [3] has no meaning to the user other than as a unique index, so it is not easy to find the object you want.

```
dbget 254/Unit[3]
342-254/Unit[3]/OID=b31a30f0-0736-1033-a235-9e41e9666ddf
342-254/Unit[3]/TagName=NEWUNIT
342-254/Unit[3]/Address=207
342-254/Unit[3]/Description=
...
```

Instead we can use an array filter to locate a Unit object by its C-Bus address:

```
dbget 254/Unit[Address=207]
342-254/Unit[3]/OID=b31a30f0-0736-1033-a235-9e41e9666ddf
342-254/Unit[3]/TagName=NEWUNIT
```

```
342-254/Unit[3]/Address=207
342-254/Unit[3]/Description=
...
```

We can add a second array filter to retrieve a known field from the PP array:

```
dbget 254/Unit[Address=207]/PP[Name=ClockGenEnable]
342-254/Unit[3]/PP[9]/Name=ClockGenEnable
342 254/Unit[3]/PP[9]/Value=1
```

Depending on the unit type the PP fields are found at different indexes or may not even exist. We can query all the units at once:

```
dbget 254/Unit/PP[Name=ClockGenEnable]
342-254/Unit[1]/PP[14]/Name=ClockGenEnable
342-254/Unit[1]/PP[14]/Value=1
342-254/Unit[2]/PP[9]/Name=ClockGenEnable
342-254/Unit[2]/PP[9]/Value=0
342-254/Unit[3]/PP[9]/Name=ClockGenEnable
342 254/Unit[3]/PP[9]/Value=1
```

A further refinement to just retrieve the Values:

```
dbget 254/Unit/PP[Name=ClockGenEnable]/Value
342-254/Unit[1]/PP[14]/Value=1
342-254/Unit[2]/PP[9]/Value=0
342 254/Unit[3]/PP[9]/Value=1
```

Filtering by blank fields:

```
dbget 254/Unit/Description
342-254/Unit[1]/Description=woh
342-254/Unit[2]/Description=
342 254/Unit[3]/Description=

dbget 254/Unit[Description=]/Description
342-254/Unit[2]/Description=
342 254/Unit[3]/Description=
```

Known Limitations

1) You can't use a space in your filter value:

```
# 8 characters long inc space at end
dbget 254/Unit/PP[Value=NEWUNIT ]
400 Syntax Error: Too many parameters
```

```
# array of integers
dbget 254/Unit/PP[Value=0x0 0x0]
400 Syntax Error: Too many parameters
```

2) Even if you are filtering on what you believe to be a unique identifier there is no guarantee that you'll only get one object in response. You should validate the contents of the response including the index or indexes.

History

- C-Gate [v2.11.0](#) implemented the Array Filtering feature.

4.3.1.4.4 Unique Command IDs

Each command to the C-Gate server can have a [unique ID](#) attached to it. When an id is added to a command, all the responses for that command are prefixed with the id given for the command. This makes it easier for client programs to connect commands and responses.

Commands can also be set to run in the [background](#), allowing the command to run in a separate thread and complete while other commands are executed.

[Verbose output](#) can be enabled as well, resulting in additional responses from some commands.

4.3.1.4.4.1 Adding IDs to commands

An ID prefix can be added to any command, and then all responses to the command are prefixed with the same prefix.

The ID prefix has the following syntax:

```
['*'] [ ['&' [ priority-digit ] ] '[' id-string ']' ] command
```

where:

command is any existing C-Gate command, including any required parameters.

priority-digit is a digit in the range 0 through 9 giving the thread priority of this command, and 4 is the regular (normal) priority.

id-string is a string provided by the client program connecting to C-Gate. This ID string should conform to the following rules:

1. can not contain whitespace characters, control characters, or NUL
2. can not contain left or right square brackets ('[' or ']')
3. can not contain left and right angle brackets ('<' or '>')

In addition, it is useful to make the *id-string unique for each command sent*. If this is done, responses can be related to command easily. Otherwise, there is little point in adding IDs to commands.

4.3.1.4.4.2 Background processing

If it is desired that the command be processed in the background so that another command can be entered immediately, adding a '&' to the front of the command line, and an optional single digit priority will run the given command in a separate thread allowing another command to be entered immediately.

The standard priority for the new thread is 4. A priority of 0 is the lowest possible, while 9 is the highest possible.

USE PRIORITIES WITH CARE, and note that they may behave very differently on different platforms.

DO NOT habitually use really high or really low priorities.

4.3.1.4.4.3 Verbose output

Some commands may provide additional verbose output. This may be useful for debugging purposes. Verbose output can be turned on by making the first character of the command line a '*' (star).

4.3.1.4.4.4 Responses

When an id string is added to a C-Gate server command, then all responses will have the following syntax:

```
'[' id-string ']' response
```

where id-string is the id-string given in the command that this response is from, and response is a standard C-Gate response, including a response code and a description.

Note: even when XML snippets are being delivered as responses (without leading response code values) there will still be an [id] added to the commencement of each line of XML.

4.3.1.4.4.5 Example of Command ID

The following shows an example session using these ID numbers:

```
[1] noop
[1] 200 OK.
&1[bgsync] do 1 sync
&[2] get cgate networks
[ab1134] tree 1
[2] 300 cgate: version=v1.6.0 pre-alpha (build 1604)
[ab1134] 320- Network name:1 type=Cni address=10.1.1.25:10001 state=new
[bgsync] 202 Done.
[ab1134] 320- Unit count=0
[ab1134] 320- Units:
[ab1134] 320-Applications:
[ab1134] 320 -end-

# example of stopping a command
[33] net sync myNet
stop 33
[33] 207 Stopped.
```

4.3.1.5 C-Gate Response Codes

The C-Gate Response Codes may be broken into several separate categories. These are described in the following sections.

Every command results in a response. Responses can be one or more lines of text. Each response line is terminated with a CRLF pair. If event output is enabled using the [EVENT](#) command, then responses can include events that are not in response to commands.

The BNF form of all valid responses are shown below. Note that responses to commands can come from backgrounded commands with id strings, and events can be included with or without prefix codes.

```
response = ( command-response|event-response|buffer-overflow-indicator ) CRLF
```

```
command-response = [command-id] command-data
```

```
command-id = '[' id-string ']'
```

```
command-data = response-code ( continuation-mark | " " ) text CRLF
```

```
continuation-mark = "-" ; this indicates another line follows
```



```
response-code = 3*DIGIT;
```

```
event-response = [event-id] event-data
event-id = '#' e|s|c '#' SP
event-data = event-time event-code object-identifier event-info
;; event-id is not given when the event mode e+ is used.
```

```
buffer-overflow-indicator = "####!!!Event buffer overflow. Events have been
missed.!!!###"
```

The digits in the response code are coded to make dealing with the response easy. The table explains the general use of the three digit response codes. Note that the 7xx, 8xx and 9xx responses are used to define events in the event interface.

<u>First DIGIT of response-code</u>	<u>General meaning</u>	<u>Use</u>
0xx	Not used	Not used.
1yz Informational	Informational response only, to indicate useful information	Generally for human users, help and suggestions. Generally ignored by automatic systems.
2yz Successful	Operation successful and complete	Usual in response to successful SET commands.
3yz Object Status	Returns status of objects or other requested information. May be one or more response lines of object information which need to be decoded.	Generally 3xx codes are issued in response to a SHOW command
4yz Client Error	Incorrect or malformed command, address, object id or other client-side problem has caused the problem to fail	Example: 400 Syntax Error
5yz Server Error	Failure of the command due to some failure in the connected controlled devices or in the IG itself.	
6yz Continuation	Issued when a following command is required to complete the specified operation	For example, a restart command requires another confirming command. 6xx is issued to ask for confirmation before performing a restart.
7yz	Reserved for events	
8yz	Reserved for events	
9yz	Reserved for events	

The second and third digits give increased granularity to the error codes. Refer to the full response code table for details of these codes.

4.3.1.5.1 General Response Codes

Code	Meaning	Syntax of Response
10x	Help Information	
101	Help information	(typically multiple lines such as) 101-help-text 101-help-text 101-help-text
110	Starting macro file	"Starting macro file:" filename
111	Ending macro file	"Ending macro file:" filename
112	Indicating a macro command from an executing macro file	"Macro command:" command-line
120	Additional information from executing method or command	method-name":" information
121	Names of parameter programming session	"121 name=" name "address=" address
122	Indicates that no parameter programming session exists	"no open sessions"
123	List of projects (from PROJECT DIR command)	"123 project=" project-name
124	Indicates that no projects exist in directory	"124 no projects found"
125	Port info from PORT LIST command	"125 port=" port-name "status=" status
126	No ports found from PORT LIST command	"126 no ports found"
127	List of interfaces from PORT IFLIST command	"127 address=" ip-address "interface=" interface-name
128	No interfaces found from PORT IFLIST command	"128 no interfaces found"
129	List of CNIs from PORT CNISCAN command	"129 found CNI ip-address=" ip-address "status=" status "port=" port-number
130	No CNIs found from PORT CNISCAN command	"130 no CNIs found"
131	List of defined networks from NET LIST command	"131 network=" network-address "state=" status "InterfaceState=" interface-state
132	No network found from NET LIST or NET LIST_ALL command	"132 no networks found"
133	List of catalog numbers and descriptions	"133 catalogNumber=" "description=" description
134	Calculator results	"134-result:" "FAILED" "OK" "134-current_supply (mA)=" value "134-current_consumption (mA)=" value "134-impedance (ohms)=" value "134-units_calculated=" unit-count "134-units_not_calculated=" unit-count

135	Results from NET LIST_ALL command	"135 project=" project-name "address=" net-address "OID=" net-oid "interfaceType=" interface-type "interfaceAddress=" interface-address "state=" object-state "interfaceState=" interface-state
136	Results from DBNETWORKPATH command, compact form	"136" hex-bytes
137	Results from DBNETWORKPATH command, OID form	"137" oids
138	api version results from APIVER command	"138" api=version

4.3.1.5.2 Pre-Release Debugging Information

Code	Meaning	Syntax of response
19x	Pre-release debugging information	
191	Pre-release debugging information	Unstructured information (not for automated use)

4.3.1.5.3 Successful Completion Messages

Code	Meaning	Syntax of response
20x	Successful completion	
200	Completed successfully	"OK"
201	Service ready	"C-Gate-name=" NAME "Service ready on "C-Gate-name"
202	Method successfully executed	"Done:" object-id
203	Macro complete	"Macro:" filename "complete."
204	Closing connection	"Closing connection"
205	Restart confirmed	"Restart confirmed."
206	Shutdown confirmed	"Shutdown confirmed."
207	Command Stopped by STOP command	"Stopped"
208	Queued for sending via a backgrounded command	"Queued for sending"

4.3.1.5.4 Security Messages

Code	Meaning	Syntax of response
2xy	Security	
210	Indicating the current access level	"Access level" access-level-name
211	Indicating the new access level	"Access level set to" access-level-name
225	Indicating a successful lock operation	object-identifier ":" "Locked."
226	Indicating a successful unlock operation	object identifier ":" "Unlocked."
230	Network detected from PORT PROBE command	"Probe succeeded: C-Bus network detected"
231	Indication of network time	"231 Time set to:" hh:mm:ss ds hh = 00..23 hours

		mm= 00..59 minutes ss= 00..59 seconds ds= 1 0 for daylight savings time on or off respectively
232	Indication of network date	"Date set to:" yyyy-mm-dd dow yyyy= Year as four digits (eg 2003) mm=01..12 months dd=01..31 days dow= 0..6 day of week (eg 0=Monday, 3= Thursday etc)

4.3.1.5.5 Object Information Messages

Code	Meaning	Syntax of response
30x	Object information	
300	Object information	parameter-name = token parameter-value = token *DIGIT quoted-string one or more lines of: object-identifier ":" *(parameter-name "=" parameter-value) For example: 300 1/56/1: level=200
301	Object ID information (results from a DBADD command where an OID is created)	"OID=" oid-value for example: 301 OID=df7c9feb-5919-11d7-a52d-004854d15455
302	Unit list from network (from NET PINGU command)	"302" "Units=" unit-list for example: 302 Units= 1,2,3,4,255
303	Type Version Serial info from NET SYNCNEW command	"303" "Unit found: address=" unit-address "stype=" type "version=" version "serial=" serial-number
304	Directory information from FILE DIR command	"304" "directory=" dir-name "files=" file-count
305	File information from FILE DIR command	"305" "name=" filename "size=" size "modified=" modified-dates
306	Event mode information from EVENT command	"306" event-mode event-mode = "e" (DIGIT '+') "s" DIGIT "c" DIGIT
315	Parameter programming , parameter information returned from a PP command.	parameter-name = token parameter-value = token *DIGIT quoted-string parameter-name "=" parameter-value
316	Return of raw data	"RawData=" hex-byte-pairs (note that bytes with no valid values are replaced with ??)
320	Tree information. Returned from the tree command.	The tree command returns a complex set of lines designed for human consumption.
321	Network serial number from topology explore	"321" "Network Serial" net-name serial-number
322	Alternative path found in Topology	"322" "Alternative Path Found" net-name port-path
323	Network found in topology scan	"323" "Network Found" net-name port-path
324	Bridge found in topology scan	"324" "Bridges Found " name number-of-bridges
325	Tunnelling Found in topology explore	"Tunnelling Found" Network to Network "ALL" or "app, app"
326	Application connect found in	"Application Connect Found" net type "ALL" or "app,

	topology explore	app"
327	Patch information line	"327" "Version=" patch-version "Name=" patch-file-name "Special=" ("true" "false") "Description=" quote description quote quote = ""
343	Indicates the start of an XML snippet	"Begin XML Snippet"
344	Indicates the end of an XML snippet	"End XML Snippet"
345	Indicates start of a file download	"Start file download for file: " filename
346	Indicates end of a file download	"End file download"
347	An XML data line	XML data

4.3.1.5.6 Client Side Command Errors

Code	Meaning	Syntax of response
40x	Client side command error	
400	Syntax error	"Syntax error."
401	Bad object or device ID	bad-object-id "=" object-identifier or "Bad object address" or "Field not found" or various others
402	This object can't do the requested operation (see the C-Gate Command Interface Guide to see what is supported by each object)	"Operation not supported by:" object-identifier
403	Unsupported ramp level	"Unsupported Ramp Level:" ramp-level
404	Unsupported ramp time	"Unsupported Ramp Time:" ramp-time
405	Parameter out of range	bad-parameter "=" parameter-name "Parameter out of range: bad-parameter"
406	Error creating object	"NEW command error: details"
407	Error processing scene. A scene could not be played or recorded due to an error.	"Error processing scene."
408	Indicates that a SET, GET or other method failed for a given object	"Operation failed:" details or "OID field can not be changed"

4.3.1.5.7 Macro Errors

Code	Meaning	Syntax of response
41x	Macro errors	
410	Macro command error	"Macro command error at:" filename line-number line-number = 1*DIGIT
411	Enable command not allowed in macro	"Enable not allowed in macro at:" filename line-number
412	Macro loop detected	"Macro loop detected at:" filename line-number
413	Macro file not found	"Macro file not found:" filename

4.3.1.5.8 Access Control Errors

Code	Meaning	Syntax of response
42x	Access control errors	

420	Access denied	"Access denied."
421	A command connection was refused.	"Connection refused."
422	Bad username and password in a login command	"Username and Password do not match."
425	Can't perform a lock command on something that is already locked	"Already locked by:" session-name
426	Can't perform an unlock command	"Unlock failed."
427	Can't perform a cancel lock command	"Cancel lock failed."

4.3.1.5.9 Method Failures

Code	Meaning	Syntax of response
4xy	Method failures	
430	Parameter sync failed	"Parameter sync failed."
431	PORT PROBE failed because port is in use	"431 Probe failed: Port in use."
432	PORT PROBE failed because there is no network	"432 Probe failed: No -Bus network detected."
442	Antimatter injectors failed to shut down	"442 Core breach imminent"
446	Unable to set a DB entry	"Unable to set: " error-message
444	XML snippet could not be produced	"XML creation failed:" reason
460	Programming parameter name not found	"Parameter not found."
461	Array index out of bounds	"Array index out of bounds"
462	Parameter value is out of range	"Parameter value out of range"
463	A bad parameter value was supplied	"Bad parameter value"
464	Project already loaded	"Project already loaded."
465	The project could not be found	"Project not found."
466	Can't save the project	"Can not save the project."
467	Tried to close a project without shutting it down first.	"Can not close running project."
468	A network can't be deleted while it is still open	"Can not delete open network"
469	Can't create or load a network	"Can not create network:" reason
470	Bad interface specification	"Bad interface specification for " net-name " interface-spec
471	Cannot create network	"Can not create network" net-name interface-spec
472	Cannot open network	"Can not open network" net-name

4.3.1.5.10 Internal Errors

Code	Meaning	Syntax of response
50x	Internal errors	
500	Internal error of some kind	"Internal Error." debug-information

	debug-information = text describing error
--	---

4.3.1.5.11 Network Errors

Code	Meaning	Syntax of response
52x	Network errors	
520	Network unavailable	"Network" object-identifier "unavailable"
521	Unit unavailable	"Unit" object-identifier "unavailable"
522	Unit error	"Unit error:" unit-error-information unit-error-information = text describing what the error is
523	Interface error	"Interface error on:" interface-name interface-name = NAME
524	Error sending command to network	"Command send error"

4.3.1.5.12 Continuation Confirmation

Code	Meaning	Syntax of response
60x	Continuation confirmation	
600	Confirm that you want to proceed by typing the word "confirm"	"Critical operation. Type CONFIRM to continue."

4.3.2 Event Interface

Network Connection

The event interface operates on a similar basis to the way responses are provided on the command interface. However, the event interface uses a separate TCP/IP connection to provide event information.

On startup of the C-Gate server, the C-Gate server will attempt to open a TCP/IP connection to the host IP address and port number specified in the sys-event object parameters. Once this connection is established, the C-Gate server will send events on a line-by-line basis, one event per line. The level and type of events that are delivered is controlled by the parameters of the sys-event object. Parameters to this can be shown and set via the command interface and stored in the system configuration file for startup.

Structure of Individual Events

The structure of an event line is shown in the augmented BNF form below:

```
event-response = event-time event-code object-identifier event-info
```

```
event-time = YYYYMMDD-HHMMSS [ "."mmm]
; YYYY = 4 digit year code
; MM = 2 digit month code (01-12)
; DD = 2 digit day code (01-31)
; HH = 2 digit hour (00-23)
; MM = 2 digit minute code (00-59)
; SS = 2 digit second code (00-59)
```

; mmm = 3 digit millisecond code (optional if enabled with [event-millis](#))

event-code = ("7" | "8" | "9") 2*DIGIT ; three digit event code
 ; 7 series for status reports
 ; 8 series for medium priority
 ; 9 series for alarms (high priority)

event-info = <see event table for details of event information>

Millisecond timing can be added to events by setting the config parameter **event-millis** to **yes**. This adds a dot and a three digit millisecond count to the event time. It is disabled by default.

The event-time is based on the internal clock used by the C-Gate server. It is set to the system time at startup.

4.3.2.1 Event Buffer Overflow

Client programs that connect to command or event streams must frequently read any events sent to them. Failure to read often enough can result in events being lost as the C-Gate server can only buffer approximately 200 events before buffers overflow.

Any event stream that displays the message:

```
###!!!Event buffer overflow. Events have been missed.!!!###
```

has lost some events. If this occurs, the client software should be redesigned to read events more often.

4.3.2.2 Setting the event level

As a guide to setting the correct event level, the following table gives an overview of the levels and the types of events that occur at that level. As an example, setting the event level to 5 means that all events with a **lower** or **equal** event level will be sent to the event outputs.

Event level	Events at this level
1	Critical errors, such as syntax errors in c-gateconfig , cgroups , networks files. Internal errors in cgate Operational failure messages
2	Network errors Interface errors Discovery of unknown devices
3	Startup and shutdown advice messages Network sync failures Object information (from 701, GETSTATE command) Power usage (720)
4	Recoverable C-Bus communication errors: checksum failures, PCI busy errors
5	Heartbeat events (700) Network open/close advice Command interface open/close advice, Enable on/off advice
6	Network synchronisation messages
7	Reserved
8	Network device information (73x) Device configuration Reports from sync operations
9	Individual send/receive commands for each interface

In typical operation, event level 5 provides a good level of events. At a minimum, event level 3 allows the receipt of object information and critical errors. Event level 9 is principally used for diagnostic purposes.

The **global-event-level** parameter of the **cgate** system object sets the event-level used by all new objects when they are created. Setting this parameter to the right level at startup, via the **cgate** config file, will ensure that all objects report at the appropriate level. In addition, every object has a **event-level** parameter of its own, and this can be set from the command interface or in one of the other startup files to allow more detailed monitoring of a particular aspect of system operation.

4.3.2.3 Event Table

This section shows the event table information for C-Gate.

4.3.2.3.1 Information Event Codes

70x	Event Level	Information Code Message	Message
700	5	Heartbeat to indicate C-Gate server is operating.	"Heartbeat"
701	3	Object parameter information. This is an event that does nothing but report the state of a pre-defined object. These events can be triggered by the GETSTATE command.	parameter-name "=" value
702	8	Application information event - the text of this event varies with the particular C-Bus application that is using it. (This is also used by network variables in applications for similar reasons)	'[application-name]' application-information
703	3	Send an event cause by a broadcast_event command.	'broadcast_event' event-class [event-text]
704	3	Network state information	"interfaceType=" interface-type "interfaceAddress=" interface-address "state=" object-state "interfaceState=" interface-state

4.3.2.3.2 Energy Related

72x	Event Level	Energy Use Message	Message
720	3	Power usage report	An event indicating that the calculated power usage for the loads connected to the object (and sub-objects) has changed. This calculation requires that individual sub-unit definitions for wattage of loads are set. These are set on the load-power parameter for each unit and sub-unit.
			power-usage = *DIGIT
			"power=" power-usage "W"

4.3.2.3.3 Command Events

73x	Event Level	Command events	Message
730	7	New level advice for group	"New level=" *DIGIT "sourceunit=" *DIGIT "ramptime=" *DIGIT ["sessionId=" string "commandID=" strings
731	8	Unit advice	"unit type:" type "version:" version
732	8	Group advice	"group at" *DIGIT
733	9	Unit type advice	"address:" unit-address "unit type:" TOKEN "version:" version
734	9	Response line:	"response:" STRING
735	9	Command send advice	"send cmd" ["(fastpci)"]:" command-string
736	8	Application advice	"applications:" application-number "," application-number
737	8	Area advice	"area: " *DIGIT
738	7	New level advice from sync	"SyncUpdate: new level=" *DIGIT

4.3.2.3.4 Network Information

74x and 75x	Event Level	Network Information	Message
740	5	Opened C-Bus network	"Opened cbus network:" network-name
741	5	Closed C-Bus network	"Closed cbus network:" network-name
742	5	Network created	"Network created type=" type "address=" address
743	5	Network deleted	"Network deleted"
744	7	Unit online status has changed. This is used to indicate to clients that some units have changed their online statuses, e.g. online, offline, error, etc.	"Unit online status changed: " online_status_name=online_status_unit_list
751	5	Tag information has changed. This is used to indicate to clients that information has changed at this level or deeper, in the tag database.	"Tag information changed at tag address:" tag-address "oldtag:" old-TagName "newtag:" new-TagName
752	5	Indicates that as a part of a DBUPDATE operation, a unit element in the tag database has not been matched and will be moved aside (have its address set to null) and replaced with a new unit.	"Unmatched Unit element moved aside, OID=" oid
753	7	Provides detailed network sync information from background or interactive sync operations	"Net Sync:" details
754	7	Indicates attempted network state changes. Only appears if the state actually changed.	net-addr oid "State="state "Previous="state "Reason="

			text
755	7	Indicates network interface state change	"State=" state
756	7	Indicates network synchronisation state change.	"SyncState=" sync-state sync-state = "idle" "init" "mmi" "pci" "bridge" "units" "app"
757	9	Indicates attempted network state changes. Only appears if the state didn't change value.	net-addr oid "State="state "Previous=(unchanged) Reason=" text
758	9	Provides detailed network parameters	"Network parameters: " + parameter list
759	9	Gives detailed network information about failures, errors and retries	• "command-fail-count=" + cmdFailCount + " unit=" + unitAddress + " reason=" + reason • "Checking PCI before retrying command (" + original command + ")" • "command timed out: " + original command + " after " + response delay + "ms, beginning retry " + retries + " of " + max retries • "command timed out: " + original command + " after " + response delay + "ms, retry limit of " + max retries + " reached" • "confirm-retry-count=" + retries + " (in packetConfirm)"

4.3.2.3.5 PCI Synchronisation

76x	Event Level	PCI Synchronisation Event	Message
761	9	Command detail	"Command: " + command-line
762	6	Network synchronisation has succeeded	"Network sync ok"
763	6	Version 3 or later PCI detected (allowing fast communications)	"C-Bus PCI V3 detected at unit:" unit-address
765	8	Indication of confirm packet for sent command (with V3 PCI)	"got packet confirm:" CHAR CHAR

4.3.2.3.6 Scene and License

77x	Event Level	Scene or License Event	Message
770	6	Scene playing notification	"Playing scene:" scene-name
771	6	Scene recording notification	"Recording scene:" scene-name

772	6	Reloading scene notification	"Reloading scene:" scene-name
775	8	Licence check	"Checking license"

4.3.2.3.7 Unit Configuration Warnings

78x	<u>Event Level</u>	<u>Unit configuration warnings (non-critical)</u>	<u>Message</u>
781	8	The expected count of terminals for this output unit was not found in the unit	"Terminal count does not match for unit:" unit-number
782	8	The expected count of minimum values for this output unit was not found	"Min level terminal count does not match for unit:" unit-number
783	8	The expected count of group values was not found for this unit	"Group Value count does not match for unit:" unit-number
784	8	The expected set of current sensors does not match for this unit	"Current sense does not match for unit:" unit-number
785	8	The extendo-dignostic summary for this unit is missing or bad.	"Bad extendo-diagnostic summary for unit:" unit-number
786	8	Automatic database update failed	"Database update for unit" unit-address "failed (" message ")"
787	7	Warning of a unsigned unit specification in use.	"Unsigned unit specification: " filename " is being used"

4.3.2.3.8 Debugging Information

79x	<u>Event Level</u>	<u>Information</u>	<u>Message</u>
79x	9	No currently defined codes for this series	No messages yet

4.3.2.3.9 Command Related

80x	<u>Event Level</u>	<u>Command Message Meaning</u>	<u>Message</u>
800	5	Startup	"C-Gate started"
801	3	Shutdown	"C-Gate shutdown"
802	3	Restart	"C-Gate will restart"
803	5	Command interface or monitoring interface opened	• "Host:" hostname "/" ip-address "opened command interface from port: " port-numbers • "Host:" hostname "/" ip-address "opened event interface from port: " port-numbers • "Host:" hostname "/" ip-address "opened load change interface from

			port: " port-numbers • "Host:" hostname "/" ip-address "opened config change interface from port: " port-numbers
804	5	Command interface or monitoring interface closed	• "Host:" hostname "/" ip-address "closed command interface from port: " port-number • "Host:" hostname "/" ip-address "closed event interface from port: " port- number • "Host:" hostname "/" ip-address "closed load change interface from port: " port-number • "Host:" hostname "/" ip-address "closed config change interface from port: " port-number
805	5	Command connection refused	"Host:" ip-address "command connection refused."
806	5	Connection refused by access control	"Access control refused" [connection- type] "connection from" STRING
807	?	Reserved for future audit commands	
808	?	Reserved for future audit commands	

4.3.2.3.10 Security Related

81x	<u>Event Level</u>	<u>Security Message</u>	<u>Message</u>
810	5	Enable succeeded	"Command interface:" interface-no "user:" username enabled
811	5	Enable failed	"Command interface:" interface-no "user:" username failed enable
812	5	End enable	"Command interface:" interface-no "user:" unenabled
813	5	Lock cancelled	"Lock cancelled name=" lock-name "network=" network- address "command_session=" command-session-id

4.3.2.3.11 Non Critical Network Errors

82x	<u>Event Level</u>	<u>Network Error (Non Critical)</u>	<u>Message</u>
820	2	Network unavailable	"Network" object-identifier "unavailable"
821	2	Unit unavailable	"Unit unavailable:" additional-details
822	2	Unit error	"Unit error:" unit-error information
823	3	PCI busy	"PCI busy indication"
824	2	Unknown cbus response	"unknown cbus response:" STRING
825	2	Response for unknown group	"response for unknown group:" STRING
826	4	C-Bus receive error	"cbus receive exception:" additional- information
827	2	C-Bus send error	"cbus send error" ["(fast pci)"] ":" additional-information

828	2	Checksum error	"receive checksum error"
829	3	New C-Bus unit not defined for this network	"new cbus unit: address=" unit-address "type=" unit-type "version=" version "not defined for this network."
830	3	New C-Bus group found on this application	"new group: address=" group-address "not defined for this application."
831	3	Error creating a unit as a C-Gate object	"error creating cbus unit: address="unit-address "type=" unit-type "version=" version
832	4	Response for unknown application	"Response for unknown application address=" application-address
833	3	New C-Bus area not defined for this application	"new cbus area: address=" unit-address "not defined for this application."
834	2	The transmitter thread has been restarted after being paused and not restarted due to communication issues	"Restarting stuck transmitter"
835	2	Failed to send command to C-Bus – timeout after retries	"C-Bus Send Error: command timed out, failed"
838	2	Unit is not responding and has been deleted	"Unit not responding, will be deleted"
839	5	Unit has new serial number and therefore appears to have been replaced	"Unit replaced: new serial=" serial-number
840	2	The PCI appears not to be responding, either because it isn't physically connected or it doesn't have power	"PCI not responding"
841	2	A network port is not available	"Network port not available"
842	2	The terminal adapter, providing RS232 connections to C-Bus PCs, is not available	"Terminal adapter not available"
843	3	Synchronisation of groups on this application has failed.	"Group sync failed"
844	3	Synchronisation of units on this network has failed.	"Unit sync failed"
845	2	Network synchronization has failed	"Network sync failed"
850	5	Can't execute scene action	"Can't execute scene action for scene:" scene-name "address:" address
851	5	Application warning event generated by a C-Bus application	"[' application-name ']' application-warning-info
82x	Event Level	Network Error (Non Critical)	Message
852	3	Network powerup detected from PCI PUN message	"C-Bus Network Powerup Detected"
853	3	PCI parameter change notification received	"PCI parameter change detected"

4.3.2.3.12 Critical Alarms

90x	Event	Critical Error Condition	Message
------------	--------------	---------------------------------	----------------

	Level		
900	1	Internal error message	"Internal error: " debug-information
902	1	The unit specification file could not be found, so unit types can not be established	"Missing unit specification file" file-name
903	1	An exception was noted and information has been saved to a file.	"Critical Server Exception: refer exception.info for details"
904	1	The unit specification file (cbusunit.spe) is missing, so unit types can't be established.	"Missing unit specification file" filename

4.3.2.3.13 Connected Network Alarms

91x	Event Level	Connected network and interface alarms	Message
910	2	Can't open C-Bus network interface	"Can't open event interface at:" ip-address "port:" port-number
911	2	The expected count of minimum values for this output unit was not found	"Can't open:" debugging-information
912	2	Can't open command interface	"Can't open command interface on port:" *DIGIT
913	2	Can't start event server socket	"Can't start event server socket on port:" *DIGIT
914	2	Unable to open event file	"Unable to open event file:" file-name
915	2	Can't start config change port	"Can't start config change port socket on port:" *DIGIT
916	1	Unable to open event printer	"Unable to open event printer:" file-name
917	2	Can't start load change port	"Can't start load change port socket on port:" *DIGIT

4.3.2.3.14 Interface Errors

92x	Event Level	Interface errors	Message
920	1	A checksum error was noted in a received message	"Receive checksum error:"
921	1	Bad or unknown interface type	"Unknown/unimplemented CBus interface type" interface-type
922	1	Unable to open C-Bus Network	debugging-information
923	1	Error closing C-Bus Network	error-information

4.3.2.3.15 File errors

93x	Event Level	File Error	Message
931	1	Can't open networks file	"Can't open networks file:" file-name
932	1	Bad token in networks file	"bad token in networks file: " additional-detail
933	1	Can't open Cgroup file	"Can't open Cgroups file:" file-name
934	1	Bad token in networks file line	"Bad token in networks file line:" line-name
935	1	Syntax error in configuration parameter	"Syntax error:" additonal-details
936	1	C-Gate could not start the given class as a network interface	"Unable to start network class:" class "(" additional-details ")"
937	1	Unknown object in cgroup	"Unknown object " object "
938	8	Deprecated option in C-GateConfig.txt is not using the default value or the option is obsoleted	• "config warning: deprecated option has non-default value: " name=value • "config warning: obsolete option will be ignored: " name

4.3.2.3.16 Licensing Errors

94x	Event Level	Licensing error	Message
940	1	Hardware key not found	"Protection key not found" error-details
941	1	Network count exceeded	"Network license exceeded: (" *DIGIT "in use," *DIGIT "licensed.)"
942	1	C-Gate has shut down because there are more networks connected than are allowed for by the license	"Cgate Shutdown – licence exceeded"
945	1	Error event issued by a C-Bus application	[' application-name ']' application-error

4.3.2.3.17 Unit Specification Errors

95x	Event Level	Unit Specification Errors	Message
950	8	The expected count of terminals for this output unit was not found in the unit	"Bad class in unit specification type: " type "Class:" class-name "not found" "(" additional-information ")"
951	8	Unable to load the class indicated in the class specification file due to a error in the constructor	"Error in constructor for Type: "type "Class:" class-name "(" additional-information ")"
952	8	Error reading unit specification	"Error reading unit specification:" file-name
953	8	Syntax error in unit specification file	"Syntax error in unit specification file:" name "after type:" details
954	8	Unable to start class to represent unit	"Unable to start unit class" "(" additional-

			detail ")")
955	8	No specification available for unit	"No specification for unit address:" unit-number "type:" type "version:" version "-using base CBusUnit"

4.3.2.3.18 Scene Errors

96x	<u>Event Level</u>	<u>Scene Error</u>	<u>Message</u>
961	1	"Unknown notify from scene control group. Scene" scene-name "group" group-number)	"Unknown notify from scene control group. Scene" scene-name "group" group-number)
962	1	Missing scene file	"Missing scene file" file-name
963	1	Error reading scene file	"Error reading scene file" file-name
964	1	Syntax error in scene file	"Syntax error in scene file:" file-name "last token:" TOKEN
965	1	Bad address in scene	"Bad address" group-address "in scene:" scene-name
966	1	Error writing scene	"Error writing recorded scene:" scene-name "error:" additional-information
967	1	Scene directory not found	"Scene directory not found:" directory-name

4.3.2.3.19 Access Control

97x	<u>Event Level</u>	<u>Access Control</u>	<u>Message</u>
970	1	Missing access control file	"Missing access control file" file-name
971	1	Error reading access control file	"Error reading access control file:" filename
972	1	Syntax error in access control file	"Syntax error in access control file:" file-name "after word:" TOKEN
974	1	Can't resolve address in access control file	"Can not resolve address: " address "on line" sp TOKEN sp "in access control file:" file-name
975	1	Bad access level in access control file	"Bad AccessLevel in access control file, setting to None"

4.3.3 Status Change Port

The Status Change Port or SCP as it is also known, allows listening clients to receive an ongoing series of events that result from status changes in [C-Bus Applications](#). Status changes include lighting application on, off and ramp messages along with messages for other C-Bus Applications.

The output of this port is rendered as a series of command interface commands and comments,

which allows their recording for client macro facilities.

The Status Change Port is normally found at TCP/IP port 20025, or port 20125 for a secured SSL connection.

4.3.4 Config Change Port

The Config Change Port allows listening clients to see any configuration change in the C-Bus networks connected to C-Gate. These configuration changes include addition or deletion of C-Bus Units and unit synchronisation states.

The Config Change Port is normally found at TCP/IP port 20026, or port 20126 for a secured SSL connection.

4.3.5 Session ID and Command ID

All application events visible in the Status Change Port and the Event port can include additional sessionId and commandId information if the command resulting in the event was initiated from a C-Gate command session (and not from a unit on the network). This is in addition to the specified events as shown in the individual application's documentation.

Format

The additional information is a text string as shown below appended as the last item to the event:

```
"sessionID=" session-id "commandID=" command-id
```

where

session-id is a string that defines the individual instance of the command interface used to send the command. Typically this is something like cmd*n* where *n* is the count of command sessions that have been opened.

command-id is a unique id given to the command as explained in the [Unique Command ID](#) section of this reference, or is given as {none} if no session was used to issue the command.

Note this is only added when a command is issued from a command interface.

The session id for a command interface can be determined by the [SESSION ID](#) command.

4.3.6 SSL Connection

C-Gate supports SSL connections for the Command, Event, Config Change and Status Change interfaces when enabled..

The SSL interfaces are always enabled.

4.4 Logging

A summary of all logs in the system follows:

- [C-Gate logs](#) in \logs\event*.txt
- [Installer logs](#) in \logs\install\Setup Log*.txt
- [Launcher logs](#) in \logs\launcher.txt and \logs\wrapper.txt

History

- C-Gate [v2.11.0](#) added a new launcher with new logs. See [New vs Old Launcher](#).
- C-Gate [v2.10.0](#) introduced a new logging system. See [New vs old C-Gate log](#).

4.4.1 C-Gate logs

C-Gate implements logging by directing event messages to a file.

Log files are stored in the /logs subdirectory.

Log files are named as follows: `event-yyyymmdd-x.txt`

For example the first two logs for Christmas Day 2012 would be:

`event-20121225-0.txt`

`event-20121225-1.txt`

Default Logging Configuration

By default C-Gate logs at event level 9 and the log files roll over daily or when reaching a size of 5 megabytes. The file-name includes the time of creation. Logs are kept for seven days with older logs being discarded.

This default configuration is defined by the following configuration variables:

[use-event-file](#) is set to yes.

[event-file.event-level](#) is set to 9. See [Setting the event level](#) for more information.

[event-file.keep-days](#) is set to 7.

[event-file.split-size](#) is set to 5000000 (4.768 MB)

Increase the number of logs kept (recommended)

If storage space permits, please increase the amount of logs kept to retain more historical information. For example:

[event-file.keep-days](#) can be set to 28 to keep logs for four weeks instead of 7 days.

Restart C-Gate to make sure the changes take effect.

Adjust file sizes

The maximum size of individual files can be adjusted but this won't affect the total size of logs kept. For example:

[event-file.split-size](#) can be set to 20000000 to keep 20 megabyte files instead of 5.

Restart C-Gate to make sure the changes take effect.

Please note that log files greater than 5 MB are problematic to send via e-mail unless they are zipped beforehand.

Decrease the detail of logging

At the default event level of 9 the event file captures everything, including but not limited to commands, responses, status and config change events, C-Bus messages, and debugging information.

[event-file.event-level](#) can be set to 5 to reduce the level of logging but this may be insufficient for diagnostics.

4.4.2 New vs old C-Gate log New logging system

The new logging system introduced in C-Gate [v2.10.0](#) is different to the old system in that it:

- saves log files in a dedicated sub-directory.
- defaults to the maximum (level 9) amount of logging.
- the maximum size threshold is now defined in terms of days not megabytes. Care should be taken to ensure there is sufficient storage space for the logs.
- is asynchronous and should not block other threads.

The following configuration variable is still relevant in C-Gate but it no longer affects the logging system:

[global-event-level](#). This should remain at level 5. See [Setting the event level](#) for more information.

These configuration variables are not relevant to the new logging system and will eventually be deprecated:

[event-filename](#)
[event-file.split](#)
[event-file.split-count](#)

Revert to the old Logging System

It is possible to revert to the old logging system used in C-Gate v2.9 and earlier with the following configuration:

[use-event-file](#) is set to old.
[global-event-level](#) is set to 5. See [Setting the event level](#) for details on how to do this.
[event-file.split-size](#) is set to 5000000.
[event-filename](#) is set to event.log.
[event-file.split](#) is set to yes.
[event-file.split-count](#) is set to 50.

Restart C-Gate to make sure the changes take effect.

4.4.3 Installer logs

The C-Gate installer writes a log to the /logs/install subdirectory. The filename is in the format:

Setup Log YYYY-MM-DD #NNN.txt

The log is written here once the installation is successfully completed. If the installation is failing without being completed, you can find the installer log in this directory instead:

C:\Users\YOUR_ACCOUNT\AppData\Local\Temp

4.4.4 Launcher logs

The cgate.exe launcher logs to two files in the /logs subdirectory.

launcher.txt

Captures all output from the YAJSW library. This includes all the shell commands being executed and the versions of Java being used.

wrapper.txt

Captures the stdout and stderr output from cgate.jar. This includes any unhandled exceptions.

4.5 Command Descriptions

4.5.1 # Command

#

Access Level

None

Syntax

[any-text]

Use

Inserts a comment into the command stream. Comments are ignored by the command processor.

//' can also be used as a comment marker.

Example

turn on these lights

Successful Response

No response is given to the command

Failure response

No response is given to the command

See Also

//, NOOP

4.5.2 **// Command**

//

Access Level

Any

Syntax

// [any-text]

Use

Inserts a comment into the command stream. Comments are ignored by the command processor.

Because // is also used in specifying a fully-qualified address including a project, it is only a comment when it is the first token on the line.

'#' can also be used as a comment marker.

Example

// try out the lights

Successful Response

No response is given to the command

Failure response

No response is given to the command

See Also

#, NOOP

4.5.3 **AIRCON Command**

AIRCON

Access Level

Operate

Syntax

AIRCON [?]

Use

Lists the AIRCON sub-commands:

AIRCON REFRESH - Sends a refresh request to an air-conditioning ward.

AIRCON SET_HUMIDITY_SETBACK_LIMIT - Sets the error allowed in the set humidity for zones.

AIRCON SET_HUMIDITY_LOWER_GUARD_LIMIT - Sets the absolute minimum humidity allowed in zones.

AIRCON SET_HUMIDITY_UPPER_GUARD_LIMIT - Sets the absolute maximum humidity allowed in zones.

AIRCON SET_HVAC_LOWER_GUARD_LIMIT - Sets the absolute minimum temperature allowed in zones.

AIRCON SET_HVAC_SETBACK_LIMIT - Sets the error allowed in the set temperature for zones.

AIRCON SET_HVAC_UPPER_GUARD_LIMIT - Sets the absolute maximum temperature allowed in zones.

AIRCON SET_WARD_OFF - Switches off all plant in all the zones in the specified ward.

AIRCON SET_WARD_ON - Returns an air-conditioning ward to its previous operational state.

AIRCON SET_ZONE_HUMIDITY_MODE - Broadcasts Humidity mode and level required for zones.

AIRCON SET_ZONE_HVAC_MODE - Broadcasts HVAC mode and level required for zones.

Success Response

A series of 101 Help lines giving a list of the AIRCON commands.

Failure Responses

400 Syntax Error

420 Access denied

4.5.4 AIRCON REFRESH Command

AIRCON REFRESH

Access Level

Operate

Syntax

AIRCON REFRESH app ward

app = air-conditioning application address (net/\$AC)

ward = air-conditioning ward number (0..255)

Use

Sends a refresh request to an air-conditioning ward.

Example

aircon refresh 254/172 1

Success Response

200 OK.

Failure Responses

400 Syntax error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.5 AIRCON SET_HUMIDITY_SETBACK_LIMIT Command

AIRCON SET_HUMIDITY_SETBACK_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HUMIDITY_SETBACK_LIMIT app ward zone-list level mode raw-flag

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0..255)
zone-list = comma-separated list of zone numbers (0..6) (0 is the 'unswitched' zone)
level = required setback limit
mode = required Humidity mode
raw-flag = 0 if level is a humidity percentage, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = humidify only
 - 2 = dehumidify only
 - 3 = humidity control
- humidity level is 0% to 100%
- raw humidity level can be a fraction of plant capacity eg 50% or -10%

Use

Sets the error allowed in the set humidity for zones.

Example

aircon set_humidity_setback_limit 254/172 1 3,4 15 1 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.6 AIRCON SET_HUMIDITY_LOWER_GUARD_LIMIT Command

AIRCON SET_HUMIDITY_LOWER_GUARD_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HUMIDITY_LOWER_GUARD_LIMIT app ward zone-list level mode raw-flag

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0..255)
zone-list = comma-separated list of zone addresses (0..6) (0 is the 'unswitched' zone)
level = required lower humidity limit
mode = required Humidity mode
raw-flag = 0 if level is a humidity percentage, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = humidify only
 - 2 = dehumidify only
 - 3 = humidity control
- humidity level is 0% to 100%
- raw humidity level can be a fraction of plant capacity eg 50% or -10%

Use

Sets the absolute minimum humidity allowed in zones.

Example

aircon set_humidity_lower_guard_limit 254/172 1 0,1,2 20 3 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
 401 Bad object or device ID:
 405 Parameter out of range:
 408 Operation failed

4.5.7 AIRCON SET_HUMIDITY_UPPER_GUARD_LIMIT Command

AIRCON SET_HUMIDITY_UPPER_GUARD_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HUMIDITY_UPPER_GUARD_LIMIT app ward zone-list level mode raw-flag

app = air-conditioning application address (net/\$AC)
 ward = air-conditioning ward number (0..255)
 zone-list = comma-separated list of zone addresses (0..6) (0 is the 'unswitched' zone)
 level = required lower humidity limit
 mode = required Humidity mode
 raw-flag = 0 if level is a humidity percentage, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = humidify only
 - 2 = dehumidify only
 - 3 = humidity control
- humidity level is 0% to 100%
- raw humidity level can be a fraction of plant capacity eg 50% or -10%

Use

Sets the absolute maximum humidity allowed in zones.

Example

aircon set_humidity_lower_guard_limit 254/172 1 1,2 70 3 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.8 AIRCON SET_HVAC_LOWER_GUARD_LIMIT Command

AIRCON SET_HVAC_LOWER_GUARD_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HVAC_LOWER_GUARD_LIMIT app ward zone-list level mode raw-flag

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0..255)
zone-list = comma-separated list of zone addresses (0..6) (0 is the 'unswitched' zone)
level = required lower temperature limit
mode = required HVAC mode
raw-flag = 0 if level is a temperature value, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = heat only
 - 2 = cool only
 - 3 = heat & cool
 - 4 = vent/fan only
- temperature level is in degrees centigrade
- raw temperature level can be a fraction of plant capacity, eg 50% or -10%

Use

Sets the absolute minimum temperature allowed in zones.

Example

aircon set_hvac_lower_guard_limit 254/172 1 0,1,2 18 3 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:

408 Operation failed

4.5.9 AIRCON SET_HVAC_SETBACK_LIMIT Command

AIRCON SET_HVAC_SETBACK_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HVAC_SETBACK_LIMIT app ward zone-list limit mode raw-flag

app = air-conditioning application address (net/\$AC)

ward = air-conditioning ward number (0..255)

zone-list = comma-separated list of zone addresses (0..6) (0 is the 'unswitched' zone)

limit = required setback limit

mode = required HVAC mode

raw-flag = 0 if level is a temperature value, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = heat only
 - 2 = cool only
 - 3 = heat & cool
 - 4 = vent/fan only
- temperature level is in degrees centigrade
- raw temperature level can be a fraction of plant capacity, eg 50% or -10%

Use

Sets the error allowed in the set temperature for zones.

Example

aircon set_hvac_setback_limit 254/172 1 0,1,2 2 3 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:

401 Bad object or device ID:

405 Parameter out of range:

408 Operation failed

4.5.10 AIRCON SET_HVAC_UPPER_GUARD_LIMIT Command

AIRCON SET_HVAC_UPPER_GUARD_LIMIT

Access Level

Operate

Syntax

AIRCON SET_HVAC_UPPER_GUARD_LIMIT app ward zone-list level mode raw-flag

app = air-conditioning application address (net/\$AC)

ward = air-conditioning ward number (0..255)

zone-list = comma-separated list of zone addresses (0..6) (0 is the 'unswitched' zone)

level = required lower temperature limit

mode = required HVAC mode

raw-flag = 0 if level is a temperature value, 1 if level is a raw value

- mode:
 - 0 = off
 - 1 = heat only
 - 2 = cool only
 - 3 = heat & cool
 - 4 = vent/fan only
- temperature level is in degrees centigrade
- raw temperature level can be a fraction of plant capacity, eg 50% or -10%

Use

Sets the absolute maximum temperature allowed in zones..

Example

```
aircon set_hvac_lower_guard_limit 254/172 1 1,2 27 3 0
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:

401 Bad object or device ID:

405 Parameter out of range:

408 Operation failed

4.5.11 AIRCON SET_WARD_OFF Command

AIRCON SET_WARD_OFF

Access Level

Operate

Syntax

AIRCON SET_WARD_OFF app ward

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0..255)

Use

Switches off all plant in all of the zones in the specified ward.

Example

```
aircon set_ward_off 254/172 1
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.12 AIRCON SET_WARD_ON Command

AIRCON SET_WARD_ON

Access Level

Operate

Syntax

AIRCON SET_WARD_ON app ward

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0-255)

Use

Returns an air-conditioning ward to its previous operational state.

Example

```
aircon set_ward_on 254/172 1
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed
```

4.5.13 AIRCON SET_ZONE_HUMIDITY_MODE Command

```
AIRCON SET_ZONE_HUMIDITY_MODE
```

Access Level

Operate

Syntax

```
AIRCON SET_ZONE_HUMIDITY_MODE app ward zone-list mode raw-flag setback-enabled
guard-enabled use-aux-level type level aux-level
```

```
app = air-conditioning application address (net/$AC)
ward = air-conditioning ward number (0..255)
zone-list = comma-separated list of zone numbers (0..6) (0 is the 'unswitched'
zone)
mode = required Humidity mode
raw-flag = 0 if level is a humidity percentage, 1 if level is a raw value
setback-enabled = 1 for setback enabled
guard-enabled = 1 for guard enabled
use-aux-level = 1 if aux-level is used, else 0 for automatic operation
type = type of the Humidity plant
level = humidity % value or raw level
aux-level = auxiliary level value
```

- mode:
 - 0 = off
 - 1 = humidify only
 - 2 = dehumidify only
 - 3 = humidity control
- type:
 - 0 = none
 - 1 = evaporative

2 = refrigerative
3 = both

- humidity level is 0% to 100%
 - raw humidity level can be a fraction of plant capacity eg 50% or -10%
 - aux-level is flags & fan mode combined into one byte: RMmmmmmm where...
 - R = reserved, always 0
 - M = 0 for automatic fan, 1 for continuous fan
 - fan speed mmmmm: 0 = default speed, 1..63 = speed setting (plant dependent)
- For example, binary 01000011 = decimal 67, which is: fan on, speed = 3.
binary 00000010 = decimal 2, which is: fan automatic, speed = 2.

Use

Broadcasts humidity mode and level required for zones.

Example

```
aircon set_zone_humidity_mode 254/172 1 0,1,2 3 0 0 0 1 2 40 64
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.14 AIRCON SET_ZONE_HVAC_MODE Command

AIRCON SET_ZONE_HVAC_MODE

Access Level

Operate

Syntax

AIRCON SET_ZONE_HVAC_MODE app ward zone-list mode raw-flag setback-enabled guard-enabled use-aux-level type level aux-level

app = air-conditioning application address (net/\$AC)
ward = air-conditioning ward number (0..255)
zone-list = comma-separated list of zone numbers (0..6) (0 is the 'unswitched' zone)
mode = required HVAC mode
raw-flag = 0 if level is a temperature value, 1 if level is a raw value
setback-enabled = 1 for setback enabled

guard-enabled = 1 for guard enabled
 use-aux-level = 1 if aux-level is used, else 0 for automatic operation
 type = type of the HVAC plant
 level = temperature or raw level
 aux-level = auxiliary level value (eg fan speed and mode)

- mode:
 - 0 = off
 - 1 = heat only
 - 2 = cool only
 - 3 = heat & cool
 - 4 = vent/fan only
- type:
 - 0 = none
 - 1 = furnace
 - 2 = evaporative
 - 3 = reverse-cycle
 - 4 = heat-pump-heating
 - 5 = heat-pump-cooling
 - 6 = furnace/evap
 - 7 = furnace/heat-pump-cooling
 - 8 = hydronic
 - 9 = hydronic/heat-pump-cooling
 - 10 = hydronic/evap
 - 11-254 = reserved (don't use)
 - 255 = any (type 255 allows service to select a plant type from those available)
- temperature is in degrees centigrade
- raw temperature level can be a fraction of plant capacity, eg 50% or -10%
- aux-level is flags & fan mode combined into one byte: RMmmmmmm where...
 - R = reserved, always 0
 - M = 0 for automatic fan, 1 for continuous fan
 - fan speed mmmmm: 0 = default speed, 1..63 = speed setting (plant dependent)
 For example, binary 01000011 = decimal 67, which is: fan on, speed = 3.
 binary 00000010 = decimal 2, which is: fan automatic, speed = 2.

Use

Broadcasts HVAC mode and level required for zones.

Example

```
aircon set_zone_hvac_mode 254/172 1 0,1,2 3 0 1 0 1 255 23 64
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
 401 Bad object or device ID:
 405 Parameter out of range:

408 Operation failed

4.5.15 APIVER Command

APIVER

Access Level

Monitor

Syntax

APIVER [details]

Use

Lists API versions for C-Gate commands, modules and othe components.

Add the 'details' option to give the details of the last API change.

Example

```
apiver
138-event=1.0
138-ccp=1.0
138-scp=1.0
138-schema-tag=1.1
138-schema-unitspec=1.0
138-schema-patch=1.0
138-schema-cbusunits=1.0
138-schema-transform=1.0
138-schema-applications=1.0
138-obj-application=1.2
138-obj-unit=1.4
138-obj-network=1.0
138-cmd-syntax=1.0
138-cmd-project=1.1
138-cmd-pp=1.1
138-cmd-net=1.0
138-cmd-port=1.0
138-cmd-file=1.0
138-cmd-general=1.0
138-cmd-cbus=1.0
138-cmd-db=1.1
138-cmd-transform=1.1
138-topology=1.0
138-calculator=1.0
138-config=1.0
138-access=1.0
138 scene=1.0
```

```

apiver details
138-event=1.0 LastChange="initial version"
138-ccp=1.0 LastChange="initial version"
138-scp=1.0 LastChange="initial version"
138-schema-tag=1.1 LastChange="DBVersion updated to 2.1"
138-schema-unitspec=1.0 LastChange="initial version"
138-schema-patch=1.0 LastChange="initial version"
138-schema-cbusunits=1.0 LastChange="initial version"
138-schema-transform=1.0 LastChange="initial version"
138-schema-applications=1.0 LastChange="initial version"
138-obj-application=1.2 LastChange="added ereport application"
138-obj-unit=1.4 LastChange="new class and added params for DALI gatetway"
138-obj-network=1.0 LastChange="initial version"
138-cmd-syntax=1.0 LastChange="initial version"
138-cmd-project=1.1 LastChange="additional responses to indicate wrong project
DBVersion"
138-cmd-pp=1.1 LastChange="fixed bugs in bit field processing (4879)"
138-cmd-net=1.0 LastChange="initial version"
138-cmd-port=1.0 LastChange="initial version"
138-cmd-file=1.0 LastChange="initial version"
138-cmd-general=1.0 LastChange="initial version"
138-cmd-cbus=1.0 LastChange="initial version"
138-cmd-db=1.1 LastChange="fixed bug in hashtable creation stopping dbset working
properly"
138-cmd-transform=1.1 LastChange="minor command response syntax changes"
138-topology=1.0 LastChange="initial version"
138-calculator=1.0 LastChange="initial version"
138-config=1.0 LastChange="initial version"
138-access=1.0 LastChange="initial version"
138 scene=1.0 LastChange="initial version"

```

Success Response

A series of 138 messages. See samples above.

Failure Responses

400 Syntax Error
420 Access denied

4.5.16 AUDIO Command

AUDIO

Access Level

Operate

Syntax

AUDIO [?]

Use

Lists the AUDIO sub-commands:

AUDIO CURRENT_FEED - Reports the current feed for the given zone.
 AUDIO DYNAMIC_1 - Request a matrix switcher to send a Dynamic 1 operation in the given zone.
 AUDIO DYNAMIC_2 - Request a matrix switcher to send a Dynamic 2 operation in the given zone.
 AUDIO HIGH_PRIORITY - Request an output device to turn on and go to a set output level and feed.
 AUDIO MUTE - Set the mute mode of an amplifier.
 AUDIO NEXT_FEED - Set the next feed for the given zone.
 AUDIO NEXT_LANGUAGE - Set the next language for the given zone.
 AUDIO OFF - Send an off operation.
 AUDIO ON - Send an on operation.
 AUDIO OUTPUT_COMMON_CONTROL - Request all devices to perform a function.
 AUDIO OUTPUT_DEVICE_STATUS_REQUEST - Request status of all devices.
 AUDIO OUTPUT_ERROR_CODE - Send an error code notification.
 AUDIO PREVIOUS_FEED - Set the previous feed for the given zone.
 AUDIO RAMP - Send a ramp operation.
 AUDIO REQUEST_CURRENT_FEED - Request the current feed of a zone.
 AUDIO SET_FEED - Set the current feed of a zone.
 AUDIO TERMINATERAMP - Terminate a ramp in progress.
 AUDIO ZONE_DESCRIPTOR_REQUEST - Send feed description to DLT labelling devices.
 AUDIO ZONE_FEED_LABEL_REQUEST - Send feed description and dynamic labels to DLT labelling devices.

Success Response

A series of 101 Help lines giving a list of the AUDIO commands.

Failure Responses

400 Syntax Error
 420 Access denied

4.5.17 AUDIO CURRENT_FEED Command

AUDIO CURRENT_FEED

Access Level

Operate

Syntax

AUDIO CURRENT_FEED app multiplexer zone feed gain
 AUDIO CURRENT_FEED app Z function gain

app = audio application address (net/\$CD)

```
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
feed = feed number (0..7)
function = zone function (0..255)
gain = amount of gain (0..4)
```

Use

Reports the current feed for the given zone.

Example

```
audio current_feed 254/205 2 7 7 4
audio current_feed 254/205 Z 192 4
```

Success Response

200 OK.

Failure Responses

```
400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed
```

4.5.18 AUDIO DYNAMIC_1 Command

AUDIO DYNAMIC_1

Access Level

Operate

Syntax

```
AUDIO DYNAMIC_1 app multiplexer zone
AUDIO DYNAMIC_1 app Z function
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)
```

Use

Instruct the matrix switcher in the given zone to send a feed-specific Dynamic Function 2 command to the internal NIRT / C-Bus.

Example

```
audio dynamic_1 254/205 1 2
audio dynamic_1 254/205 Z 192
```

Success Response

200 OK.

Failure Responses

```
400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed
```

**4.5.19 AUDIO DYNAMIC_2
Command**

AUDIO DYNAMIC_2

Access Level

Operate

Syntax

```
AUDIO DYNAMIC_2 app multiplexer zone
AUDIO DYNAMIC_2 app Z function
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)
```

Use

Instruct the matrix switcher in the given zone to send a feed-specific Dynamic Function 1 command to the internal NIRT / C-Bus.

Example

```
audio dynamic_2 254/205 1 2
audio dynamic_2 254/205 Z 192
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.20 AUDIO HIGH_PRIORITY Command

AUDIO HIGH_PRIORITY

Access Level

Operate

Syntax

AUDIO HIGH_PRIORITY app multiplexer level feed

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
level = level (0..255)
feed = feed number (0..7)

Use

Request an output device in a zone to turn on and go to a set output level and feed, or if the level is zero to return to the prior state.

Example

audio high_priority 254/205 2 192 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.21 AUDIO MUTE Command

AUDIO MUTE

Access Level

Operate

Syntax

AUDIO MUTE app multiplexer zone mode
AUDIO MUTE app Z function mode

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)
mode = mode (0, 2, 5, 7, 255)
 0 = turn amplifier off
 2 = amplifier on, volume normal, speakers off
 5 = amplifier on, volume preset, speakers off
 7 = amplifier on, volume preset, speakers on
 255 = amplifier on, volume normal, speakers on

Use

Set the mute mode of an amplifier.

Example

audio MUTE 254/205 2 1 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.22 AUDIO NEXT_FEED Command

AUDIO NEXT_FEED

Access Level

Operate

Syntax

AUDIO NEXT_FEED app multiplexer zone
AUDIO NEXT_FEED app Z function

app = audio application address (net/\$CD)

multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Sets the next feed for the given zone.

Example

```
audio next_feed 254/205 2 7
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.23 AUDIO NEXT_LANGUAGE Command

AUDIO NEXT_LANGUAGE

Access Level

Operate

Syntax

AUDIO NEXT_LANGUAGE app multiplexer zone
AUDIO NEXT_LANGUAGE app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Sets the next language for the given zone.

Example

```
audio next_language 254/205 2 7
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.24 AUDIO OFF Command

AUDIO OFF

Access Level

Operate

Syntax

AUDIO OFF app multiplexer zone code
AUDIO OFF app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = zone code (0..7)
function = zone function (0..255)

Use

Set the value of the Function Code referenced by the Zone Function Variable to 0.

Example

audio off 254/205 2 4 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.25 AUDIO ON Command

AUDIO ON

Access Level

Operate

Syntax

AUDIO ON app multiplexer zone code
AUDIO ON app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = zone code (0..7)
function = zone function (0..255)

Use

Set the value of the Function Code referenced by the Zone Function Variable to 255.

Example

audio on 254/205 2 4 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.26 AUDIO OUTPUT_COMMON_CONTROL Command

AUDIO OUTPUT_COMMON_CONTROL

Access Level

Operate

Syntax

AUDIO OUTPUT_COMMON_CONTROL app control-code

app = audio application address (net/\$CD)
control-code = control code (0)
0 = turn off

Use

All output devices are to act upon this command according to the value of <Control Code>.

Example

audio output_common_control 254/205 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.27 AUDIO OUTPUT_DEVICE_STATUS_REQUEST Command

AUDIO OUTPUT_DEVICE_STATUS_REQUEST

Access Level

Operate

Syntax

AUDIO OUTPUT_DEVICE_STATUS_REQUEST app parameter

app = audio application address (net/\$CD)
parameter = parameter (0)

Use

Requests that all output devices return their status.

Example

audio output_device_status_request 254/205 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.28 AUDIO_OUTPUT_ERROR_CODE Command

AUDIO_OUTPUT_ERROR_CODE

Access Level

Operate

Syntax

AUDIO_OUTPUT_ERROR_CODE app multiplexer zone code

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = error code (0..1)
 0 = overheating
 1 = low voltage

Use

A device is reporting an error condition.

Example

audio_output_error_code 254/205 1 2 0

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.29 AUDIO PREVIOUS_FEED Command

AUDIO PREVIOUS_FEED

Access Level

Operate

Syntax

AUDIO PREVIOUS_FEED app multiplexer zone
AUDIO PREVIOUS_FEED app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Sets the previous feed for the given zone.

Example

audio previous_feed 254/205 2 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.30 AUDIO RAMP Command

AUDIO RAMP

Access Level

Operate

Syntax

AUDIO RAMP app multiplexer zone code level rate

AUDIO RAMP app Z function level rate

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = zone code (0..7)
function = zone function (0..255)
level = level to ramp to (0..255)
rate = time taken to perform the ramp (0..15)
 0 = instantaneous
 1 = 4 seconds
 2 = 8 seconds
 3 = 12 seconds
 4 = 20 seconds
 5 = 30 seconds
 6 = 40 seconds
 7 = 1 minute
 8 = 90 seconds
 9 = 2 minutes
 10 = 3 minutes
 11 = 5 minutes
 12 = 7 minutes
 13 = 10 minutes
 14 = 15 minutes
 15 = 17 minutes

Use

Ramp the zone function to the specified level at the given rate.

Example

```
audio ramp 254/205 2 4 7 255 1
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.31 AUDIO REQUEST_CURRENT_FEED Command

AUDIO REQUEST_CURRENT_FEED

Access Level

Operate

Syntax

AUDIO REQUEST_CURRENT_FEED app multiplexer zone
AUDIO REQUEST_CURRENT_FEED app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Request the current feed for the given zone.

Example

audio request_current_feed 254/205 2 7

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.32 AUDIO SET_FEED Command

AUDIO SET_FEED

Access Level

Operate

Syntax

AUDIO SET_FEED app multiplexer zone feed option
AUDIO SET_FEED app Z function option

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
feed = feed number (0..7)
function = zone function (0..255)


```
option = option (0..1)
    0 = allow labels and annunciation on feed change
    1 = disable labels and annunciation on feed change
```

Use

Sets the current feed for the given device.

Example

```
audio set_feed 254/205 2 7 7 4
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed
```

4.5.33 AUDIO TERMINATERAMP Command

AUDIO TERMINATERAMP

Access Level

Operate

Syntax

```
AUDIO TERMINATERAMP app multiplexer zone code
AUDIO TERMINATERAMP app Z function
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = zone code (0..7)
function = zone function (0..255)
```

Use

Terminate a ramp operation in progress on the given zone.

Example

```
audio terminatoramp 254/205 2 4 7
```

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.34 AUDIO_ZONE_DESCRIPTOR_REQUEST Command

AUDIO_ZONE_DESCRIPTOR_REQUEST

Access Level

Operate

Syntax

AUDIO_ZONE_DESCRIPTOR_REQUEST app multiplexer zone
AUDIO_ZONE_DESCRIPTOR_REQUEST app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Instruct the matrix switcher in the given zone to send the feed description to DLT labelling devices.

Example

audio_zone_descriptor_request 254/205 1 2

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.35 AUDIO_ZONE_FEED_LABEL_REQUEST Command

AUDIO_ZONE_FEED_LABEL_REQUEST

Access Level

Operate

Syntax

AUDIO_ZONE_FEED_LABEL_REQUEST app multiplexer zone
AUDIO_ZONE_FEED_LABEL_REQUEST app Z function

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Use

Instruct the matrix switcher in the given zone to send the feed description, Dynamic 1 and 2 labels to DLT labelling devices.

Example

audio zone_feed_label_request 254/205 1 2

Success Response

200 OK.

Failure Responses

400 Syntax Error:
401 Bad object or device ID:
405 Parameter out of range:
408 Operation failed

4.5.36 BROADCAST_EVENT Command

BROADCAST_EVENT

Access Level

Operate

Syntax

BROADCAST_EVENT event-class [event-text]

event-class = A string (ending at the first whitespace) that gives the class of this event.

There are no restriction on values of the class. This is determined by the using application.

event-text = (optional) Further descriptive string to the end of the line.

Use

Broadcasts an event on the event port and status change port with the given *event-class* and *event-text*.

Example

```
broadcast_event ShutDown C-Gate will be shutdown in 10 minutes
200 OK.
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error
420 Access denied
```

4.5.37 CGL Command

CGL

Access Level

Admin

Syntax

CGL [?]

Use

Lists the CGL sub-commands:

CGL EXPORT - Exports a network to CGL format.

CGL IMPORT - Imports a network in CGL format to the given project.

Success response

A series of lines listing these commands in the form:

```
101-Help: help-information
```

4.5.38 CGL EXPORT Command

CGL EXPORT

Access Level

Monitor

Syntax

CGL EXPORT project-name [network-list [application-list]]

Use

Gets a part of the current Tag Database as a snippet in [CGL Format](#).

The db-address given is any object in the XML database, and can be specified as either a path from the base of the tag database (ie /Installation/InstallationDetail/Installer), or a C-Bus network address, of a tag name.

This command returns:

- a 343-Begin XML snippet response
- The XML snippet requested, as one or more lines starting with "347-". This starts with a <?xml ... ?> line. Note that the XML is not pretty-printed.
- a 344 End XML snippet response

If a single element is selected by address, for example an element that simply contains a string, then an XML snippet similar to the following will be returned just giving the raw type of the field:

```
347-<?xml version="1.0"?>
347-<string>2001-11-01T13:05:34.153</string>
```

Examples

```
DBGETXML Installation/Version
DBGETXML p/1/20
```

Successful Response (example)

```
343-Begin XML snippet
347-<?xml version="1.0"?>
347-<Unit xmlns="http://www.clipsal.com/cis/schema/2001/cbus.xsd">
347-<TagName>n1_u20</TagName><Address>20</Address><UnitType>KEYM8</
UnitType><UnitName>NEWUNIT </UnitName><FirmwareVersion>1.3.06</
FirmwareVersion></Unit>
344 End XML snippet
```

Failure response

```
401 Bad object or device ID
440 There is no tag database to perform this operation on.
```

444 XML creation failed

See also

[CGL IMPORT](#)

4.5.39 CGL IMPORT Command

CGL IMPORT

Access Level

Admin

Syntax

CGL IMPORT project-name << end-tag

Use

Replaces all or part of the current Tag Database with a snippet in [CGL Format](#) given after the command. The project-name must be a project that has already been loaded in C-Gate.

The CGL required for this command is provided as a here document, meaning is it provided in one or more lines following the command, with the command terminated by the end-tag given on the command line after the << characters, on a line by itself.

This command returns:

- a series of 380 messages for each item that was added or modified.
- a 200 OK message when concluded.
- one or more errors if other conditions occur.

Examples

```
CGL IMPORT home << EOF
.. one or more lines of CGL data ...
EOF
```

Successful Response

```
301 OID=oid-value
This response returns the OID of the created element
```

Failure response

```
401 Bad object or device ID: No project specified
401 Bad object or device ID: Project not found
400 Syntax Error: No CGL data supplied
408 CGL validation failed:
408 CGL import failed:
```

See also[CGL EXPORT](#)**4.5.40 CLOCK
Command**

CLOCK

Access Level

Operate

Syntax

CLOCK [?]

Use

Lists the CLOCK sub-commands:

CLOCK DATE - Gets or sets the date.

CLOCK REQUEST REFRESH - Request a time-master to broadcast the date and time.

CLOCK TIME - Gets or sets the time.

Success response

A series of lines giving help for these commands in the form: 101 Help: help-information

Failure responses

400 Syntax Error

420 Access denied

**4.5.41 CLOCK DATE
Command**

CLOCK DATE

Access Level

Operate

Syntax

CLOCK DATE app [date | "system"]

app = clock application address (net/223)

date = yyyy-mm-dd

- yyyy = 4 year date (eg. 2008)
- mm = months (01..12)
- dd = days (01-31)

"system" -if this is given instead of yyyy-mm-dd, C-Gate's current date is used

Use

Gets or sets the date.

CLOCK DATE app with no parameters will return the current date. Adding a date will set the network date by sending a clock date command to the network.

Success response

232 Date set to: *yyyy-mm-dd dow*

- yyyy = 4 year date (eg. 2008)
- mm = months (01..12)
- dd = days (01-31)
- dow = day of week (0..6 where 0 = Monday)

Failure responses

401 Bad object or device ID
402 Operation not supported by: ...
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.42 CLOCK REQUEST_REFRESH Command

CLOCK REQUEST_REFRESH

Access Level

Operate

Syntax

CLOCK REQUEST_REFRESH app

app = clock application address (net/223)

Use

Broadcasts a request_refresh message, which will cause a time-master device to broadcast the current date and time.

Example

clock request_refresh 254/223

Success response

200 OK.

Failure responses

400 Syntax Error

4.5.43 CLOCK TIME Command

CLOCK TIME

Access Level

Operate

Syntax

CLOCK TIME app [time | "system" [daylight-flag]]

app = clock application address (net/223)

time = hh:mm:ss

-if "system" is given instead, C-Gate's current time is used

- hh = 00..23
- mm = 00..59
- ss = 00..59

daylight-flag:

- 0 = no daylight saving
- 1 = this time includes 1 hr advance
- 255 = daylight saving offset unknown

- If the daylight-flag is omitted, it defaults to a value of 255 (\$FF).

Use

Gets or sets the time.

CLOCK TIME app with no parameters will return the current time as understood by this application. Adding a time and possibly a daylight saving flag will set the network time by sending a clock time command to the network.

Success response

231 Time set to: *hh:mm:ss daylight-flag*

- hh = 00..23
- mm = 00..59
- ss = 00..59
- daylight-flag = 0 | 1 | 255

Failure responses

401 Bad object or device ID

402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.44 **CONFIG Command**

CONFIG or CONFIG ?

Access Level

Admin

Syntax

CONFIG [?]

Use

Lists the CONFIG sub-commands:

CONFIG GET - Get the value of a configuration parameter
CONFIG INFO - Get information about a configuration parameter
CONFIG LOAD - Load config parameters from either a project db or a file or both
CONFIG SAVE - Save the current config parameters to either a project db or a file
CONFIG SET - Set the value of a configuration parameter
CONFIG OBGET - Get the value of a configuration parameter for a scope object
CONFIG OBSET - Set the value of a configuration parameter for a scope object
CONFIG INI - Used to get and set values in the cgate.ini file. It provides two commands: CONFIG INI SET and CONFIG INI GET.

Success Response

A series of 101 Help lines giving details of the config commands

Failure Responses

400 Syntax Error
420 Access denied

4.5.45 **CONFIG INI SET Command**

CONFIG INI SET

Access Level

Admin

Syntax

CONFIG INI SET `JAVAHEAPMAX` [value]

value = The value for the parameter must be a positive integer

Use

Updates the JAVAHEAPMAX parameter in the cgate.ini file to the specified value. This command is used to configure the maximum JVM heap size for C-Gate. The change will take effect after restarting C-Gate

Example

```
CONFIG INI SET JAVAHEAPMAX 100
200 OK.
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error
420 Access denied
```

4.5.46 CONFIG INI GET Command

```
CONFIG INI GET
```

Access Level

Admin

Syntax

```
CONFIG INI GET JAVAHEAPMAX
```

Use

Retrieves the current value of the JAVAHEAPMAX parameter from the cgate.ini file. This command is used to view the configured maximum JVM heap size

Example

```
CONFIG INI GET JAVAHEAPMAX
303 JAVAHEAPMAX=100
```

Success Response

```
303 JAVAHEAPMAX "=" value
```

Failure Responses

```
400 Syntax Error
420 Access denied
```

4.5.47 CONFIG GET Command

CONFIG GET

Access Level

Admin

Syntax

CONFIG GET config-parameter

config-parameter = config parameter name or "*" to get all config parameters

Use

Returns the value of a configuration parameter, or values of all known configuration parameters if the parameter is given as *.

See [Configuration](#) for more information.

Example

```
config get sync-time  
303 sync-time=200
```

Success Response

303 config-parameter "=" value

Failure Responses

```
400 Syntax Error  
408 Operation failed: config parameter not found  
420 Access denied
```

4.5.48 CONFIG INFO Command

CONFIG INFO

Access Level

Admin

Syntax

CONFIG INFO config-parameter

config-parameter = config parameter name or "*" to get all config parameters

Use

Returns full information over several lines about a configuration parameter, or values of all known configuration parameters if the parameter is given as `*`.

Example

```
config info config-path
304-parameter=config-path
304-value=config
304-description=Path to directory where configuration files are held
304-defaultValue=config
304-scope=global
304 effective=restart
```

Success Response

Six lines per parameter in the following form and order:

```
304-parameter= name of the parameter
304-value= current value of the parameter
304-description= description of this parameter
304-defaultValue= the default value of this parameter
304-scope= the scope of this parameter (what part of the server is it active over)
304 effective= when do changes of this parameter come into effect
```

See the [Configuration](#) section in this guide to decode the meanings of these lines.

Failure Responses

```
400 Syntax Error
408 Operation failed: config parameter not found
420 Access denied
```

4.5.49 CONFIG LOAD Command

CONFIG LOAD

Access Level

Admin

Syntax

```
CONFIG LOAD config-type [filename]
```

```
config-type = "project" | "global" | "all"
```

Use

Loads configuration parameters into the server from either the current project database (as defined by the PROJECT USE command and the PROJECT commands) or from the global configuration file, or

both.

Use this command to load an existing on-disk configuration into the server. Note that there may be some immediate impact from the loading of some parameters.

Example

```
config load global
200 OK.
```

or

```
config load project
200 OK.
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error
408 Operation failed: No project in use
408 Operation failed: Global config load from filename failed: further-details
420 Access denied
```

4.5.50 CONFIG SAVE Command

CONFIG SAVE

Access Level

Admin

Syntax

```
CONFIG SAVE config-type [filename]
```

```
config-type = "project" | "global" | "all"
```

Use

Saves configuration parameters from the server to either the current project database "project" (as defined by the PROJECT USE command and the PROJECT commands) or to the global configuration file "global", or both "all".

Use this command to save a current configuration for later use. This has no impact on the currently operating parameters in the server.

Example

```
config save global
```

200 OK.

or

config save all
200 OK.

Success Response

200 OK.

Failure Responses

400 Syntax Error
408 Operation failed: No project in use
408 Operation failed: Global config save to *filename* failed: *further-details*
420 Access denied

4.5.51 CONFIG SET Command

CONFIG SET

Access Level

Admin

Syntax

CONFIG SET config-parameter [value]

config-parameter = config parameter name
value = new value for the parameter, can include spaces

Use

Sets the value of a configuration parameter to the given value. If the value is not given, the parameter takes on the null value. The value given can have embedded spaces and these will be included in the value.

See [Configuration](#) for more information on setting config parameters, their scope and when changes are effective.

Example

config set sync-time 100
200 OK.

Success Response

200 OK.

Failure Responses

400 Syntax Error
408 Operation failed: unknown parameter *parameter-name*
420 Access denied

4.5.52 CONFIG OBGET Command

CONFIG OBGET

Access Level

Admin

Syntax

CONFIG OBGET object config-parameter

object = the global, project or network scope object from which to get the config parameter

config-parameter = config parameter name or "*" to get all config parameters

Use

Returns the value of a configuration parameter of the object, or values of all known configuration parameters of the object if the parameter is given as *.

See [Configuration](#) for more information. See [Scope and Scope Objects](#) for more information on scope objects.

Example

```
config obget global global-event-level
303 global-event-level=5
```

```
config obget project network.retries
303 network.retries=2
```

```
config obget 254 sync-time
303 sync-time=200
```

Success Response

```
303 config-parameter "=" value
```

Failure Responses

400 Syntax Error
408 Operation failed: config parameter not found
420 Access denied

4.5.53 CONFIG OBSET Command

CONFIG OBSET

Access Level

Admin

Syntax

CONFIG OBSET object config-parameter [value]

object = the global, project or network scope object on which to set the config parameter

config-parameter = config parameter name

value = new value for the parameter, can include spaces

Use

Sets the value of a configuration parameter of an object to the given value. if the value is not given, the parameter takes on the null value. The value given can have embedded spaces and these will be included in the value.

See [Configuration](#) for more information. See [Scope and Scope Objects](#) for more information on scope objects.

Example

```
config obset sync-time 100
200 OK.
```

Success Response

200 OK.

Failure Responses

400 Syntax Error

408 Operation failed: unknown parameter *parameter-name*

420 Access denied

4.5.54 CONFIG OBRESET Command

CONFIG OBRESET

Access Level

Admin

Syntax

CONFIG OBRESET object [config-parameter]

object = the project or network scope object on which to reset the config parameter to the global value

config-parameter = config parameter name

Use

Resets the value of a configuration parameter of an object to the global value. if the parameter is not given, all parameters are reset.

See [Configuration](#) for more information. See [Scope and Scope Objects](#) for more information on scope objects.

Example

```
config obreset sync-time
200 OK.
```

Success Response

```
200 OK.
```

Failure Responses

```
400 Syntax Error
408 Operation failed: unknown parameter parameter-name
420 Access denied
```

4.5.55 CONFIRM Command

CONFIRM

Access Level

Admin

Syntax

CONFIRM

Use

Used when a 600 message is received, to confirm the execution of a [shutdown](#) or restart before it proceeds. This is meant to stop people shutting down the server by having their fingers typing on automatic.

Examples

#in response to 600 message from shutdown or restart

CONFIRM

Successful Response

206 Shutdown confirmed.

Failure response

400, 420

See Also

[SHUTDOWN](#)

4.5.56 DBADD Command

DBADD

Access Level

Operate

Syntax

DBADD db-address element-type

db-address = address of an existing element under which to add the new element

element-type = type of the new object to add to the database

Suitable types include: Project, Network, Unit, Application, Group, Level.

Use

Adds an object to a database.

Examples

DBADD Installation InstallationDetail

Successful Response

200 OK.

(where no OID is used for this element), or

301 *OID=oid-value*

(where an OID is applicable to the element that has been created)

Failure response

400 Syntax Error.

401 Bad object or device ID:

440 There is no tag database to perform this operation on.

4.5.57 DBADDSAFE Command

DBADDSAFE

Access Level

Operate

Syntax

DBADDSAFE parent-address element-type [element-address] [tag-name]

parent-address = address of an existing element under which to add the new element

element-type = type of the new object to add to the database

element-address = address of the new object, it is mandatory if the object has an Address field

tag-name = tag name of the new object, it is mandatory if the object has a TagName field

Allowed types include but not limited to: Project, Network, Unit, Application, Group, Level, TagDLT.

Use

Adds an object to a database and apply business rules check when the object has Address and TagName fields.

The rules for element-address:

- it can not be null or blank
- if element-type is Network, Unit, Application, Group or Level, it must be the string representation of decimal number (i.e. only 0~9 is allowed in the element-address parameter)
- if element-type is Network, Unit, Application, Group or Level, it must be within the range of [0, 255]
- it can not be the same as the address of an existing object under parent-address

The rules for tag-name:

- it can not be null or blank
- it can not be the same as the tag name of an existing object under parent-address

Additional rules:

- Group object can not be added to Application 255
- Level object can not be added to Group 255

Examples

DBADDSAFE 254/56 Group 0 Lighting Group 0

Successful Response

200 OK.

(where no OID is used for this element), or

301 OID=*oid-value*

(where an OID is applicable to the element that has been created)

Failure response

400 Syntax Error.

401 Bad object or device ID: Can't add groups to Application 255

401 Bad object or device ID: Can't add levels to Group 255

401 Bad object or device ID: Element address in use

401 Bad object or device ID: Element address is required for this type

401 Bad object or device ID: Element address must be a number

401 Bad object or device ID: Element address must be in the range of 0 to 255

401 Bad object or device ID: Element tag name is required for this type

401 Bad object or device ID: Element tagname in use

401 Bad object or device ID: Parent address is not unique. Unique parent address required for dbaddsafes

401 Bad object or device ID: Parent element not available

401 Bad object or device ID: Unable to add element: Field <elementType> not found in <parentAddress>.

401 Bad object or device ID: Unable to add element: <exception info>

401 Bad object or device ID: <exception info>

440 There is no tag database to perform this operation on.

4.5.58 DBCOPY Command

DBCOPY

Access Level

Operate

Syntax

DBCOPY source-address destination-parent-address

source-address = the database address of the part of the database to copy

destination-parent-address = the parent element to hold the copied part of the database

Use

Copies part of a tag database tree to another place. This means a unit or network definition, for example, can be copied to create a new entry.

You can copy parts of databases between projects by using *//project-name/* as address prefixes

Notes:

The destination part of the tree has new object identifiers (OIDs) generated for all elements that have OIDs.

If the source and destination databases are the same (copying within one database) then the TagName and Address fields are not copied and are set to null in the destination. These addresses

will need to be set to allow the database contents to be addressed by Address and TagName.

However, if the source and destination databases or projects are different, then TagNames and Addresses are copied with the rest of the elements.

Examples

```
# copy the network addressed as j1 in project jim to the current project
called system
DBCOPY //jim/j1 system
```

```
# copy the unit with tagname newunit in project jim to network 1 on
project system
DBCOPY //jim/newunit //system/1
```

Successful Response

200 OK.

(where no OID is used for this element), or

301 OID=oid-value

(where an OID is applicable to the element that has been created)

Failure response

400 Syntax Error.

401 Bad object or device ID: *reason*

408 Operation failed: *reason*

440 There is no tag database to perform this operation on.

4.5.59 DBCOPYSAFE Command

DBCOPYSAFE

Access Level

Operate

Syntax

```
DBCOPYSAFE source-address destination-parent-address [destination-element-address]
[destination-tag-name]
```

source-address = the database address of the part of the database to copy

destination-parent-address = the parent element to hold the copied part of the database

destination-element-address = the address of the copied object; it is mandatory if the object has an Address field and is being copied to the same parent node

destination-tag-name = the tag name of the copied object; it is mandatory if the object has a TagName field and is being copied to the same parent node

Use

Copies part of a tag database tree to another place. This means a unit or network definition, for example, can be copied to create a new entry. The source object is not restricted to unit or network, e.g. application or group object can also be copied.

You can copy parts of databases between projects by using *//project-name/* as address prefixes

Notes:

The destination part of the tree has new object identifiers (OIDs) generated for all elements that have OIDs.

All fields of the source object are copied to the destination object, including Address and TagName fields.

There are some business rules check before copying the object and the rules are:

- the destination object's parent node must have the same type as the source object's parent node
- if element-type is Network, Unit, Application, Group or Level, destination-element-address must be the string representation of decimal number and its value must be within the range of [0, 255]
- the destination-element-address can not be the same as the address of an existing object under destination-parent-address if it is provided
- the destination-tag-name can not be the same as the tag name of an existing object under destination-parent-address if it is provided

Examples

```
# copy the group addressed as 0 in application 56 under network 254 to  
the same application as address 1 with tag name Group-1
```

```
DBCOPYSAFE 254/56/0 254/56 1 Group-1
```

```
# copy the unit at address 10 in network 254 from project jim to network  
1 on project system
```

```
DBCOPYSAFE //jim/254/p/10 //system/1
```

Successful Response

```
200 OK.
```

(where no OID is used for this element), or

```
301 OID=oid-value
```

(where an OID is applicable to the element that has been created)

Failure response

```
400 Syntax Error.
```

```
401 Bad object or device ID: Bad source address: <exception info>
```

```
401 Bad object or device ID: Bad destination address: <exception info>
```

```
401 Bad object or device ID: Can't get source or destination parent element
```

```
401 Bad object or device ID: Destination Address is not unique.
```

```
401 Bad object or device ID: Destination address is required for this object when  
copying under the same node
```

```
401 Bad object or device ID: Destination tag name is required for this object when  
copying under the same node
```

```
401 Bad object or device ID: Element address must be a number
```

401 Bad object or device ID: Element address must be in the range of 0 to 255
401 Bad object or device ID: Element address in use
401 Bad object or device ID: Element tagname in use
401 Bad object or device ID: Source Address is not unique.
408 Operation failed: *reason*
440 There is no tag database to perform this operation on.

4.5.60 DBCREATE Command

DBCREATE

Access Level

Admin

Syntax

DBCREATE

Use

Generates a tag database that represents the current connected C-Bus networks & C-Gate implementation. This replaces the current tag database.

So, to capture the state of the network and store it, perform a DBCREATE then a DBSAVE command to save the database to disk.

This command creates a new database, which fills in compulsory elements with the string [default] where no value can be determined from the network or system.

Examples

```
dbcreate  
200 OK.
```

Successful Response

```
200 OK.
```

Failure response

```
400 Syntax Error.  
443 Error creating tag database
```

See also

[DBNEW](#)

4.5.61 DBCREATENET Command

DBCREATENET

Access Level

Admin

Syntax

DBCREATENET address name type interface-address

address = address of the new network (0..255)

name = name of the new network

type = type of the network interface (Serial | Cni | Bridge)

interface-address = interface details

Use

Creates a new network in the database. This command is equivalent to the following sequence of commands:

```
dbadd Installation/Project Network
dbset <oid>/NetworkNumber "address"
dbset <oid>/Address "address"
dbset <oid>/TagName "name"
dbadd <oid> Interface
dbset <oid2>/InterfaceType "type"
dbset <oid2>/InterfaceAddress "interface-address"
```

Note that this command on its own does not persist the new object to disk. You need to follow this with a PROJECT SAVE.

Examples

```
dbcreatenet 254 "My Network 1" Serial com1
200 OK.
```

```
dbcreatenet 251 "My Network 2" Cni 192.168.0.1:14000
200 OK.
```

```
dbcreatenet 253 "My Network 3" Bridge 254/p/253
200 OK.
```

Successful Response

200 OK.

Failure response

400 Syntax Error.

440 There is no tag database to perform this operation on. 443 Error creating tag database

4.5.62 DBDELETE Command

DBDELETE

Access Level

Operate

Syntax

DBDELETE element-address

Use

Deletes an element from the current tag database.

Examples

```
# remove group 1 from lighting on network 1
DBDELETE 1/56/1
```

Successful Response

200 OK.

Failure response

```
400 Syntax Error.
401 Bad object or device ID:
440 There is no tag database to perform this operation on.
```

4.5.63 DBGET Command

DBGET

Access Level

Monitor

Syntax

DBGET parameter-address

parameter-address = Address of a tag database element or a C-Gate object address

Use

Gets the value of an element at an address. If the address points to an element with one or more sub-elements, the sub-elements are displayed.

Examples

DBGET Installation/Version

DBGET Installation/Project/Network[1]/TagName

Successful Response

342 parameter-address=value (there may be multiple responses)

Failure response

400 Syntax Error

401 Bad object or device ID:

440 There is no tag database to perform this operation on.

4.5.64 DBGETXML Command

DBGETXML

Access Level

Monitor

Syntax

DBGETXML db-address

Use

Gets a part of the current Tag Database as a snippet of XML.

The db-address given is any object in the XML database, and can be specified as either a path from the base of the tag database (ie /Installation/InstallationDetail/Installer), or a C-Bus network address, or a tag name.

This command returns:

- a 343-Begin XML snippet response
- The XML snippet requested, as one or more lines starting with "347-". This starts with a <?xml ... ?> line. Note that the XML is not pretty-printed.
- a 344 End XML snippet response

If a single element is selected by address, for example an element that simply contains a string, then an XML snippet similar to the following will be returned just giving the raw type of the field:

```
347-<?xml version="1.0"?>
```

```
347-<string>2001-11-01T13:05:34.153</string>
```

Examples

DBGETXML Installation/Version

DBGEXTXML p/1/20

Successful Response (example)

```
343-Begin XML snippet
347-<?xml version="1.0"?>
347-<Unit xmlns="http://www.clipsal.com/cis/schema/2001/cbus.xsd">
347-<TagName>n1_u20</TagName><Address>20</Address><UnitType>KEYM8</
UnitType><UnitName>NEWUNIT </UnitName><FirmwareVersion>1.3.06</
FirmwareVersion></Unit>
344 End XML snippet
```

Failure response

```
401 Bad object or device ID
440 There is no tag database to perform this operation on.
444 XML creation failed
```

See also

[DBSETXML](#)

4.5.65 DBLOAD Command

DBLOAD

Access Level

Operate

Syntax

DBLOAD filename

Use

In normal use, the [PROJECT LOAD](#) command should be used instead of DBLOAD to open project databases.

Loads a project database into C-Gate in the current project.

This makes the database the active project database for the current project.

If and only if the filename given has an extension of ".zip", C-Gate will attempt to uncompress a ZIP archive containing the XML tag database. If the extension ".gz" is given, C-Gate will attempt to use GZIP to uncompress the file before reading. If any other extension is given, or no extension is given, then the file is assumed and expected to be in an DB form, though C-Gate does *not* assume a .DB extension.

If the filename is an absolute path, then the file will be loaded from that absolute filename. If a relative path is given, then the path will be relative to the path given in the tag-base-directory

property.

Examples

```
DBLOAD myproj.db
DBLOAD project.zip
```

Successful Response

200 OK.

Failure response

```
400 Syntax Error.
442 Error reading tag database: details
```

See also

DBSAVE

4.5.66 DBNETWORKPATH Command

DBNETWORKPATH

Access Level

Admin

Syntax

```
DBNETWORKPATH start-net-address end-net-address [OID | COMPACT]
```

start-net-address = the starting network address in the database

end-net-address = the ending network address in the database

Use

Finds a network path between two networks representation in the project database, via bridge devices.

It is assumed that the standard bridge and network addresssing convention is used: the unit address of a bridge device is the same as the network address on the far side of the bridge device. A path longer than 6 bridges will not be returned.

With the OID option given, a series of OIDs are given for all the networks in the path, including the ending network, but not the source network in one or more 137 responses.

With the COMPACT option given, a compact bridge path hex string is returned in a single 136 response.

The OID option is the default if no option is given

Examples

```
#OID form
dbnetworkpath 254 252 oid
137-47951260-07e6-1027-ae52-f53240c219d9
137 4795d5b0-07e6-1027-ae56-f53240c219d9
```

```
#compact form
dbnetworkpath 254 252 compact
136 FDFC
```

Successful Response

Either:

```
"136" hex-bytes
```

or one or more lines of:

```
"137" oid
```

Failure response

```
400 Syntax Error.
401 Bad object or device ID: reason
408 Operation failed: reason
440 There is no tag database to perform this operation on.
```

4.5.67 DBNEW Command

DBNEW

Access Level

Admin

Syntax

DBNEW

Use

Clears the tag database for the current project and creates a new blank database.

This blank database contains nothing more than an Installation element, so additional information has be to added before the database is valid to the tag database specification.

Examples

```
dbnew
200 OK.
```

Successful Response

200 OK.

Failure Response

408 Project not found

**4.5.68 DBRENAMENET
Command**

DBRENAMENET

Access Level

Admin

Syntax

DBRENAMENET db-net-address new-net-address

db-net-address = the database address of a <Network> stored in the project database.

new-net-address = the parent element to hold the copied part of the database

Use

Re-addresses a <Network> in the project database, and additionally ensures that and other <Networks> that are of the bridge type have any references to the old network corrected to support the new network. Re-addressing means that the network's <Address> element is given a new value.

Warning: This command does not prevent you from re-addressing a network to be the same address as another existing network. This could be confusing if used with reckless abandon.

Examples

```
# re-address network 1 to be network 2
DBRENAMENET 1 2
```

Successful Response

200 OK.

Failure response

400 Syntax Error.

401 Bad object or device ID: *reason*

408 Operation failed: *reason*

440 There is no tag database to perform this operation on.

4.5.69 DBRENAMENETSAFE Command

DBRENAMENETSAFE

Access Level

Admin

Syntax

DBRENAMENETSAFE db-net-address new-net-address

db-net-address = the database address of a <Network> stored in the project database.

new-net-address = the parent element to hold the copied part of the database

Use

Re-addresses a <Network> in the project database, and additionally ensures that and other <Networks> that are of the bridge type have any references to the old network corrected to support the new network. Re-addressing means that the network's <Address> element is given a new value.

There are some business rules check before renaming the network and the rules are:

- db-net-address and new-net-address must be string representations of decimal number (i.e. only 0~9 is allowed)
- new-net-address can not be the same as db-net-address
- new-net-address must be within the range of [0, 255]
- new-net-address can not be the same as the address of an existing network in the same project

Examples

```
# re-address network 1 to be network 2
DBRENAMENETSAFE 1 2
```

Successful Response

200 OK.

Failure response

```
400 Syntax Error.
401 Bad object or device ID: Address is not unique
401 Bad object or device ID: Can't rename network to the same address
401 Bad object or device ID: Element address must be a number
401 Bad object or device ID: Element address must be in the range of 0 to 255
401 Bad object or device ID: Get project database entry failed
401 Bad object or device ID: Invalid network address
401 Bad object or device ID: New network address in use
401 Bad object or device ID: No network given
401 Bad object or device ID: No new address given
401 Bad object or device ID: No project in database
401 Bad object or device ID: Not a network database entry
408 Operation failed: reason
```


440 There is no tag database to perform this operation on.

4.5.70 DBSAVE Command

DBSAVE

Access Level

Admin

Syntax

DBSAVE filename

Use

Saves the current Tag database to the file or URL specified. The current tag database will be saved in the given filename.

The filename extension determines how the file will be saved:

- If the extension given is **.zip** the tag database is saved in a zip archive. The tag database is saved in the archive as an entry called 'tagdb.db' and is the only entry in the archive.
- If the extension given is **.gz** the tag database will be compressed using GZIP.

Otherwise, the file is saved in DB format. The files will only be saved with an extension of **.db** if this is given in the filename.

Examples

DBSAVE project.db
DBSAVE compressed-project.zip

Successful Response

200 OK.

Failure response

440 There is no tag database to perform this operation on.
441 Error writing tag database

See also

[DBLOAD](#), [PROJECT SAVE](#)

4.5.71 DBSET Command

DBSET

Syntax

DBSET parameter-address value

parameter-address = address of a tag database element or a C-Gate object address
value = the value to set the parameter to. This includes all characters to the end of the line.

Use

Sets a tag database field by address.

Examples

DBSET Installation/Version 2.01
200 OK.

Successful Response

200 OK.

Failure response

400 Syntax Error.
401 Bad object or device ID:
440 There is no tag database to perform this operation on.
446 Unable to set:

4.5.72 DBSETSAFE Command

DBSETSAFE

Syntax

DBSET parameter-address value

parameter-address = address of a tag database element or a C-Gate object address
value = the value to set the parameter to. This includes all characters to the end of the line.

Use

Sets a tag database field by address and apply business rules check.

The rules are:

- if the field to be set is Address and the object to be set is Network, Unit, Application, Group or Level, the value must be the string representation of decimal number (i.e. only 0~9 is allowed) and the value must be within the range of [0, 255]
- if the field to be set is Address, the value can not be the same as the address of an existing object under the same parent
- if the field to be set is TagName, the value can not be blank
- if the field to be set is TagName, the value can not be the same as the tag name of an existing object under the same parent

Examples

DBSETSAFE Installation/Version 2.01
200 OK.

Successful Response

200 OK.

Failure response

400 Syntax Error.
401 Bad object or device ID: Address is not unique. Unique address required for dbset operation
401 Bad object or device ID: Bad object address
401 Bad object or device ID: Element address must be a number
401 Bad object or device ID: Element address must be in the range of 0 to 255
401 Bad object or device ID: Element address in use
401 Bad object or device ID: Element tagname in use
401 Bad object or device ID: Field not available
401 Bad object or device ID: Field not found
401 Bad object or device ID: TagName can't be null or blank
401 Bad object or device ID: OID field can not be changed
401 Bad object or device ID: <exception info>
440 There is no tag database to perform this operation on.
446 Unable to set:

4.5.73 DBSETXML Command

DBSETXML

Access Level

Admin

Syntax

DBSETXML db-address << end-tag

Use

Sets a part of the current Tag Database as a snippet of XML given after the command issued. The db-address given is any object in the XML database, and can be specified as either a path from the base of the tag database (ie /Installation/InstallationDetail/Installer), a C-Bus network address, a tag name or OID.

The XML required for this command is provided as a here document, meaning is it provided in one or more lines following the command, with the command terminated by the end-tag given on the command line after the << characters, on a line by itself.

This command returns:

- a 301 OID= message if the XML setting succeeds. The OID returned is the OID given to the

- newly-created element that results from the XML setting operation on the database.
- one or more errors if other conditions occur.

Examples

```
DBGETXML Installation/Version << EOF
.. one or more lines of XML data ...
EOF
```

Successful Response

```
301 OID=oid-value
This response returns the OID of the created element
```

Failure response

```
401 Bad object or device ID:
440 There is no tag database to perform this operation on.
```

See also

[DBGETXML](#)

4.5.74 DBTAGLIST Command

```
DBTAGLIST
```

Access Level

Admin

Syntax

```
DBTAGLIST [pattern]
```

pattern = substring to match the tagnames against
If not given, all tag names are listed.

Use

Lists tag names and their associated objects.

Examples

```
DBTAGLIST
```

```
DBTAGLIST lighting
```

Successful Response

```
one or more lines of tag names ...
eg:
```

```
342-254/203/1/TagName=Group 1
342 254/56/TagName=Lighting
```

Failure response

```
401 Bad object or device ID:
420 Access denied.
440 There is no tag database to perform this operation on.
```

4.5.75 DBUPDATE Command

DBUPDATE

Access Level

Admin

Syntax

```
DBUPDATE network-address [unit-delete]
```

network-address = a C-Gate or Tag database address that resolves to a network or a unit

unit-delete = If this parameter is provided and set to the word "UnitDelete", then any units that are present in the database but are not present in the physical network will be deleted from the database.

Use

Updates the current database with the current physical network or unit configuration. This overwrites the current networks.

Examples

```
# update network 1
DBUPDATE 1
```

```
# update network n1
DBUPDATE n1
```

```
#update unit p/1/22
DBUPDATE p/1/22
```

Successful Response

```
200 OK.
```

Failure response

```
400 Syntax Error.
401 Bad object or device ID:
408 Operation failed:
```

440 There is no tag database to perform this operation on.

4.5.76 DBVALIDATE Command

DBVALIDATE

Access Level

Operate

Syntax

DBVALIDATE db-address

Use

Validates the section of the current tag database given by the address, checking that it conforms to the tag specification. This validation checks that required fields are filled in correctly and that there are no violations of the tag specification.

Examples

```
dbset 1/TagName Jim
200 OK.
```

Successful Response

200 OK.

Failure response

```
400 Syntax Error.
440 There is no tag database to perform this operation on.
```

4.5.77 DBVERIFY Command

DBVERIFY

Access Level

Admin

Syntax

DBVERIFY

Use

Verifies the current network against the current database and reports differences. DBVERIFY

compares the live networks that C-Gate is connected with and entries in the tag database in order to detect differences or omissions. These are reported as a series of 345 responses.

Examples

```
dbverify
200 OK.
```

Successful Response

```
200 OK.
```

Failure response

```
400 Syntax Error.
440 There is no tag database to perform this operation on.
Zero or more lines of:
345 Difference: difference-details
408 Operation failed: (Verify failed: <count> differences)
```

4.5.78 DO Command

DO

Access Level

Operate

Syntax

```
DO object-identifier method-name *(method-parameter)
```

```
method-parameter = token
```

Use

Calls the methods of an object.

A 202 Done response is returned.

The method may return one or more lines of 120 response that are information relating to this execution of the method.

Note that the [ON](#), [OFF](#) and [RAMP](#) commands are shorthand versions of DO commands with the methods on off and ramp.

Examples

```
do 1/56/1 on
202 Done: //HOME/1/56/1
```

Successful Response

"202 Done:" object-identifier

Also may include one or more lines of 120 responses giving additional method information.

Failure response

400, 401, 402, 405, 408, 420

See also

[GET or SHOW](#), [SET](#), [ON](#), [OFF](#), [RAMP](#)

4.5.79 **ENABLE Command**

ENABLE

Access Level

Operate

Syntax

ENABLE [?]

Use

Lists the ENABLE sub-commands:

ENABLE LABEL - Set labels on devices that support dynamic labels

ENABLE REMOVE - Remove any reference to this network variable

ENABLE SET - Set an enable variable to the given value

Success response

A series of lines giving help for these commands in the form: 101 Help: help-information

Failure responses

4.5.80 **ENABLE LABEL Command**

ENABLE LABEL

Access Level

Operate

Syntax

ENABLE LABEL app language group-number action-sel [variant] options


```

app = application address (normally //proj/net/$CB)
language = language code (0..255)
group-number = relevant C-Bus group number (0..255)
action-sel = action selector | "-" (dash indicates unset)
variant = F0 | F1 | F2 | F3 (sets the variant for this label)

options = ( "text" text-label ) |
          ( "icon" icon-selector ) |
          ( "dynamic" icon-selector icon-width icon-height vertical-offset dynamic-
icon ) |
          ( "set_language" ) |
          options-byte hex-bytes

```

text-label = ASCII text to end of line to be used as label
icon-selector = numeric icon selection in the range 0 through 65535
icon-width = width of the icon in pixels
icon-height = height of the icon in pixels

dynamic-icon = block of ASCII-encoded bytes representing the icon, as 16 lines of up to 62 bits, encoded as up to 16 characters of HEX, with each block separated by a colon character.

If a block is less than 16 characters, the block will be zero-filled on the right. For example:

```

AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644

```

options-byte = integer in the range of 0 through 255, used as the value of the options 1 byte in the command.

hex-bytes = a string of hex bytes to follow the command

Note: all numeric values except for dynamic-icon can be entered as decimal, 0x or \$ prefixed hex, or 0b prefixed binary.

Use

Performs label setting with devices that support dynamic labels.

Example

```

#set a text label
enable label 1/56 1 2 3 text hello
200 OK.

```

```

#set an existing icon
enable label 1/56 1 2 - icon $1234
200 OK.

```

```

#set a dynamic icon
enable label 1/56 1 2 - dynamic
0102030405060708:0102030405060708:0102030405060708:0102030405060708:010203040506070

```

```
# change language
enable label 1/56 1 2 - set_language
200 OK.

# send a raw command
enable label 1/56 1 2 - 6 3F2B0102
200 OK.
```

Success Response

200 OK.

Failure Responses

```
400 Syntax Error
401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied
```

4.5.81 ENABLE REMOVE Command

ENABLE REMOVE

Access Level

Operate

Syntax

ENABLE REMOVE enable-netvar-address

Use

Removes any memory that the server has of this enable network variable. This means that the network variable is removed, and any on-disk storage of the value of the network variable is also removed.

Example

ENABLE REMOVE 2/\$cb/1
200 OK.

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.82 ENABLE SET Command

ENABLE SET

Access Level

Operate

Syntax

ENABLE SET enable-netvar-address value ["force"]

value = integer in the range 0..255 | percentage value (0..100) followed by the % sign | level tag

Use

Sends an *enable set* message to the network to set the enable network variable given as *enable-netvar-address*, to the *value* given. This is equivalent to a LIGHTING RAMP immediate command for the lighting application.

Value is either: an integer in the range 0..255; or a percentage value (0..100) followed by the % sign; or a level tag.

Specifying the optional parameter *force*, will allow the command to be sent regardless of network state.

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.83 EREPORT Command

EREPORT

Access Level

OPERATE

Syntax

EREPORT [?]

Use

Gives an overview of the EREPORT sub-commands:

EREPORT MESSAGE - Send an error reporting event with choice of headers

Success response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure responses

A functional copy of C-Gate will always respond to this command

4.5.84 EREPORT MESSAGE Command

EREPORT MESSAGE

Access Level

Operate

Syntax

EREPORT MESSAGE app type category most-recent acknowledged most-severe severity
unit-id [<data-byte-1 [data-byte-2]]

app = application address (net/\$CE)

type = type of message (RECENT | ERROR_REPORT | ACK | CLEAR | 0..255)

category = error category number (0..1023)

most-recent = true if the error is the most recent (y | n | 1 | 0)

acknowledged = true when the error has been acknowledged (y | n | 1 | 0)

most-severe = true if the error is the most severe to date (y | n | 1 | 0)

severity = severity level (0..4)

unit-id = address of the error monitoring device (0..255)

data-byte-1 = optional byte containing category specific meaning (0..255)

data-byte-2 = optional byte containing category specific meaning (0..255)

- severity level: 0 = ALL_OK; 1 = OK; 2 = minor failure; 3 = general failure; 4 = extreme failure

Use

Sends an error report to the network using the [Error Reporting Application](#).

Note that EREPORT MESSAGE does not yet implement any of the error-reporting constraints enforced by some of the devices - it will allow you to send messages that are syntactically incorrect. This allows it to be more flexible during the transition to a new error reporting protocol.

Messages received with an \$05 or \$15 header will continue to produce the old status change message, unchanged. However, they will no longer produce a level 8 event message.

Examples

```
ereport message 254/$CE ACK 1023 n n n 7 255 255 255
```

Successful Response

```
200 OK
```

Failure responses

```
400, 401, 402, 405, 408, 420
```

4.5.85 EVENT Command

EVENT

Access Level

Monitor

Syntax

```
EVENT ON | OFF | event-mode
```

```
event-mode = "e" (IDIGIT | '+' ) "s" DIGIT "c" DIGIT
```

Use

Controls the output of events, status change and config change information to the command session that the command is executed in.

EVENT ON will turn on the output of events only with no prefix on each event. This is the same as event-mode e+s0c0.

EVENT OFF disables event output. This is equivalent to setting an event-mode of e0s0c0.

(Can also use the command EVENTS.)

Event Modes:

Giving an event-mode expression can give fine grained control of event output. An event mode is made up of a set of identifiers and an event level for each identifier.

For events, the identifier is: `e` and valid event levels are `+` and `0` through `9`. In this case, the event level indicates the maximum level of events that will be displayed. When these events are delivered, they show a prefix of `#e#`. The `+` level is a special code to indicate that no prefix is to be used and the default event level of C-Gate object are used.

For status change events, the identifier is `s` and valid events levels are `0` and `1` corresponding to disabled and enabled. When these events are delivered, they are prefixed with `#s#`.

For config change events, the identifier is `c` and valid events levels are `0` and `1` corresponding to disabled and enabled. When these events are delivered, they are prefixed with `#c#`.

Console

The console is opened in event mode `e+s0c0`. This can also be changed by command.

Examples

```
# turn on events in this session
EVENT ON

#enable scp, ccp and events at level 5
events e5s1c1

# turn off events
EVENTS OFF

# return current event mode
events
306 e5s1c1
```

Successful Response

```
200 OK.
or
306 event-mode
```

4.5.86 GET Command

GET

Access Level

Monitor

Syntax

```
GET object-identifier [parameter | ? | * | ??]
```

Use

Retrieves the values of objects. Object parameters may be viewed by naming them in the command.

The special parameter '?' returns a list of all parameters of this object. The special parameter * returns a list of all parameter values for an object. The special parameter ?? returns one line for each parameter with a description of the parameter.

(Can also use the SHOW command.)

Examples

```
show cbus networks
get //proj/level57/ units
get //proj/level57/$38/1 level
get //proj/level57/2/1 power
get //proj/level57/2 type
get //proj/level57/2 ?
```

Successful Response

One or more lines of the following 3xx series responses:

response-code [-] object-identifier ":" parameter-name=parameter-value

For example: 300 level57/56/1: level=200

Failure response

400, 401, 402, 408, 420

See also

[SET](#), [DO](#)

4.5.87 GETSTATE Command

GETSTATE

Access Level

Monitor

Syntax

GETSTATE (network-address | cgroup-name)

Use

Causes a series of events to be sent to the event stream, that gives the *current* status of a connected network. Events are returned for all devices in the network. Events are returned giving:

- status information for all units and groups
- level information for all groups and terminals

This command is an easy way for an application to get the current state of the network as a starting point for a model or for subsequent event tracking or polling.

Examples

```
GETSTATE 57
200 OK.
```

Successful Response

```
200 OK
```

Failure response

```
400, 401, 420
```

See also

[GET](#), [TREE](#)

4.5.88 HELP Command

```
HELP
```

Syntax

```
HELP [command-or-topic]
```

Access Level

Monitor

Use

Gives brief command-line help. Note that the help command will only give help for commands that are accessible by the access level of the current user.

HELP * returns a list of all the commands.

? can be used as an abbreviation of HELP.

Examples

```
help set
101-Help: syntax: SET <object-id> <param-name> <value>
101-Help: Set the parameter given in <param-name> to the value in
101-Help: <value> for the object <object-id>.
```


101-Help: <object-id> is a network, group, unit, or system entity
101-Help: <param-name> is a named parameter.
101 Help: <value> is the value to set the parameter to.

Successful Response

One or more lines in the following form:

101 [-] help-information

Failure response

400, 420

4.5.89 LIGHTING Command

LIGHTING

Access Level

Operate

Syntax

LIGHTING [?]

Use

Lists the LIGHTING sub-commands:

LIGHTING LABEL - Sets labels on devices that support dynamic labels.

LIGHTING OFF - Turns off a specified group.

LIGHTING ON - Turns on a specified group.

LIGHTING RAMP - Ramps a group to a % level using a specified ramp time.

LIGHTING TERMINATERAMP - Stops any ramping operation for a given group.

Success Response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure Responses

401 Bad object or device ID

402 Not supported by this object

405 Parameter out of range

408 Operation failed

420 Access denied

4.5.90 LIGHTING LABEL Command

LIGHTING LABEL

Access Level

Operate

Syntax

LIGHTING LABEL app language group-number action-sel [variant] options

app = application address (normally //proj/net/\$38)

language = language code (range 0 through 255)

group-number = relevant C-Bus group number (range 0 through 255)

action-sel = action selector | "-" (dash indicates unset)

variant = F0 | F1 | F2 | F3 (sets the variant for this label)

```
options = ( "text" text-label ) |
          ( "icon" icon-selector ) |
          ( "dynamic" icon-selector icon-width icon-height vertical-offset dynamic-
icon ) |
          ( "set_language" ) |
          options-byte hex-bytes
```

text-label = ASCII text to end of line to be used as label, and its maximum length is 14 characters

icon-selector = numeric icon selection in the range 0 through 65535

icon-width = width of the icon in pixels

icon-height = height of the icon in pixels

dynamic-icon = block of ASCII-encoded bytes representing the icon, as 16 lines of up to 62 bits, encoded as up to 16 characters of HEX, with each block separated by a colon character.

If a block is less than 16 characters, the block will be zero-filled on the right. For example:

```
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644
```

options-byte = ; integer in the range of 0 through 255, used as the value of the options 1 byte in the command.

hex-bytes = ; a string of hex bytes to follow the command, and its maximum length

is 14 characters (i.e. 28 hex byte characters in the command)

Note: all numeric values except for dynamic icon can be entered as decimal, 0x or \$ prefixed hex, or 0b prefixed binary.

Use

Sets labels on devices that support dynamic labels.

Note: For devices that support unicode dynamic labels, if a group/variant already has a unicode label it will not accept text labels sent with this command. To set or clear unicode labels see the LIGHTING UNICODELABEL command.

Examples

```
#set a text label
lighting label 254/56 1 2 3 text hello
200 OK.
```

```
#set an existing icon
lighting label 254/56 1 2 - icon $1234
200 OK.
```

```
#set a dynamic icon
lighting label //proj/1/56 1 2 - dynamic
0102030405060708:0102030405060708:0102030405060708:0102030405060708:010203040506070
8:0102030405060708:0102030405060708:0102030405060708:0102030405060708:0102030405060
708:0102030405060708:0102030405060708:0102030405060708:0102030405060708:01020304050
60708:0102030405060708
200 OK.
```

```
# change language
lighting label 254/56 1 2 - set_language
200 OK.
```

```
# send a raw command
lighting label 254/56 1 2 - 6 3F2B0102
200 OK.
```

Success Response

200 OK.

Failure Responses

```
400 Syntax Error
401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied
```

4.5.91 LIGHTING UNICODELABEL Command

LIGHTING UNICODELABEL

Access Level

Operate

Syntax

LIGHTING UNICODELABEL app language group-number action-sel variant option text-label|hex-bytes

app = application address (normally //proj/net/\$38)

language = language code (range 0 through 255)

group-number = relevant C-Bus group number (range 0 through 255)

action-sel = action selector | "-" (dash indicates unset) (should always be dash)

variant = F0 | F1 | F2 | F3 (sets the variant for this label)

option = ("text" text-label) |
("raw" hex bytes string encoded in UTF-8)

text-label = ASCII text to end of line to be used as label if option = text

hex-bytes = a string of hex bytes to follow the command if option = raw

Use

Sets unicode labels on devices that support unicode dynamic labels. The label string should be encoded in UTF-8.

To clear a unicode label simply provide no parameter for text-label or hex-bytes.

Note: Please use the raw option to send unicode characters to C-Gate. Using the text option to send unicode may not always work.

Examples

```
#set a text label
lighting unicodelabel 254/56 1 2 - F0 text hello
200 OK.
```

```
#set a raw hex bytes string
lighting unicodelabel 254/56 1 2 - F0 raw 414243444546
200 OK.
```

Success Response

200 OK.

Failure Responses

400 Syntax Error
401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.92 LIGHTING OFF

See [OFF](#)

4.5.93 LIGHTING ON

See [ON](#)

4.5.94 LIGHTING RAMP

See [RAMP](#)

4.5.95 LIGHTING TERMINATERAMP

See [TERMINATERAMP](#)

4.5.96 LOCK

Command

LOCK

Access Level

Operate

Syntax

LOCK object-identifier

Use

Sets an advisory lock on an object. This lock is typically used by cooperating applications that want exclusive access to a device or object for programming or configuration. A lock survives until unlocked with the UNLOCK command or the command session ends.

Note that the locking is advisory only. Applications must cooperate to make a locking system work.

Examples

LOCK p/254/2
225 //HOME/254/p/2: Locked.

Successful Response

225 <object-identifier>: Locked

Failure response

400, 401, 420, 425

See also

[UNLOCK](#)

4.5.97 LOGIN Command

LOGIN

Access Level

Connect

Syntax

LOGIN [username password]

username = user name

password = user password

Use

Changes the current access level of this command session, by providing a username and a password.

If no username and password are given, the command returns the current access level.

Usernames and passwords are case-sensitive.

Examples

LOGIN

LOGIN james aabbc

Successful Response

210 Access level: level 211

210 Access level set to: level

Failure response

400, 422

4.5.98 LOGOUT Command

LOGOUT

Access Level

Connect

Syntax

LOGOUT

Use

Logs out of the current access level of this command session.

If a user hasn't logged in, just reports the current access level.

Examples

LOGOUT

Successful Response

211 Access level set to: *level*

Failure response

(Always succeeds.)

4.5.99 MEASUREMENT Command

MEASUREMENT

Access Level

Operate

Syntax

MEASUREMENT [?]

Use

Lists the MEASUREMENT sub-commands:

MEASUREMENT DATA - Measurement data for a Channel.

Success Response

A series of 101 Help lines giving a list of the MEASUREMENT commands.

Failure Responses

4.5.100 MEASUREMENT DATA Command

MEASUREMENT DATA

Access Level

Operate

Syntax

MEASUREMENT DATA channel value multiplier units

channel = channel address (eg //proj/net/\$E4/1/1)
value = scaled measurement value (-32768..32767)
multiplier = power of ten unit multiplier (-128..127)
units = measurement units (0..\$FF)

For valid units values see the [Unit Code Table](#).

Use

Measurement Data Messages are sent by a Measurement Device either:

- (a) based on an elapsed time interval,
- (b) based on a change in the measured value, or
- (c) in response to a specific Control Trigger application request

Examples

MEASUREMENT DATA 254/\$E4/1/2 37 1 0

Successful Response

200 OK.

Failure response

400 Syntax Error
401
402
405
408
420

4.5.101 MEDIATransport Command

MEDIATransport

Access Level

Operate

Syntax

MEDIATransport [?]

Use

Lists the MEDIATransport sub-commands:

MEDIATransport CATEGORY_NAME - Sets the category name.
 MEDIATransport ENUMERATE - Request up to 16 names for a category, selection or track.
 MEDIATransport ENUMERATION_SIZE - Indicates the number of items in an enumeration.
 MEDIATransport FORWARD - Fast-forwards the current track.
 MEDIATransport NEXT_CATEGORY - Changes to the next category.
 MEDIATransport NEXT_SELECTION - Changes to the next selection.
 MEDIATransport NEXT_TRACK - Changes to the next track.
 MEDIATransport PAUSE - Pauses/unpauses playing.
 MEDIATransport PLAY - Starts playing.
 MEDIATransport REPEAT - Sets state to 'repeat'.
 MEDIATransport REWIND - Rewinds the current track.
 MEDIATransport SELECTION_NAME - Sets a selection name.
 MEDIATransport SET_CATEGORY - Chooses a category.
 MEDIATransport SET_SELECTION - Chooses a selection.
 MEDIATransport SET_TRACK - Chooses a track.
 MEDIATransport SHUFFLE - Sets state to 'shuffle'.
 MEDIATransport SOURCE_POWER - Turns on/off the output unit.
 MEDIATransport STATUS_REQUEST - Requests current status of an output unit.
 MEDIATransport STOP - Stops play.
 MEDIATransport TOTAL_TRACKS - Indicates how many tracks are available
 MEDIATransport TRACK_NAME - Sets the track name.

Success Response

A series of 101 Help lines giving a list of the MEDIATransport commands.

Failure Responses

4.5.102 MEDIATransport CATEGORY_NAME Command

MEDIATransport CATEGORY_NAME

Access Level

Operate

Syntax

MEDIATransport CATEGORY_NAME app group wni total index text

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 wni = name identification (0..7)

0 = current category
 1 = next category
 2 = next+1 category
 3,4 = reserved
 5 = previous category
 6 = previous-1 category
 7 = enumerated category
 total = total number of packets - 1 (0..3)
 index = index/sequence of this packet (0..3)
 text = all/part of the *category* name in UTF-8 (max 11 bytes per packet)

Use

Sets the name of a *category* in the output unit in the given *media link group*.
 The name is constructed from one or more packets, sent in sequence, coded as UTF-8, limited to 40 bytes total length.

Examples

```
MEDIATRANSPORT CATEGORY_NAME 254/$C0 2 1 1 0 iPod
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.103 MEDIATRANSPORT ENUMERATE Command

MEDIATRANSPORT ENUMERATE

Access Level

Operate

Syntax

```
MEDIATRANSPORT ENUMERATE app group type start
```

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 type = enumeration type where 0=category, 1=selection, 2=track
 start = enumerate from this index (0..\$FF)

Use

Requests a dump of the names of the first 16 categories, selections or tracks in the given *media link group*.

Examples

MEDIATransport Enumerate 254/\$C0 2 2 0

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.104 MEDIATransport Enumeration_Size Command

MEDIATransport Enumeration_Size

Access Level

Operate

Syntax

MEDIATransport Enumeration_Size app group type start size

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
type = enumeration type where 0=category, 1=selection, 2=track
start = enumerate from this index (0..\$FF)
size = number of items in the enumeration following (0..\$0F)

Use

Describes the number of entries the unit is about to send in response to an enumeration request on the given *media link group*.

Examples

MEDIATransport Enumeration_Size 254/\$C0 2 2 0 6

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.105 MEDIATransport Forward Command

MEDIATransport Forward

Access Level

Operate

Syntax

MEDIATRANSPORT FORWARD app group operation

app = application address (normally //proj/net/\$C0)

group = media link group (0..\$FF, where \$FF indicates unused)

operation = one of (0..\$FF):

0 = cease fast-forward, play at normal speed

2 = fast-forward at 2x speed

4 = fast-forward at 4x speed

6 = fast-forward at 8x speed

8 = fast-forward at 16x speed

10 = fast-forward at 32x speed

12 = fast-forward at 64x speed

all other values reserved

Use

Requests that the *track* currently being played in the given *media link group* be fast-forwarded.

Continues to be in effect until a PLAY, FORWARD 0 or REWIND command is received.

If the output unit is not in the *play* state, the command is ignored.

Examples

MEDIATRANSPORT FORWARD 254/\$C0 2 1 2

Successful Response

200 OK.

Failure response

400 Syntax Error

401

402

405

408

420

4.5.106 MEDIATRANSPORT NEXT_CATEGORY

Command

MEDIATRANSPORT NEXT_CATEGORY

Access Level

Operate

Syntax

MEDIATRANSPORT NEXT_CATEGORY app group operation

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 operation = one of (0..\$FF):
 0 = select previous category
 non-0 = select next category

Use

Requests that the *output unit* in the given *media link group* change to the next or previous *category*.

Examples

MEDIATRANSPORT NEXT_CATEGORY 254/\$C0 2 1

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.107 MEDIATRANSPORT NEXT_SELECTION Command

MEDIATRANSPORT NEXT_SELECTION

Access Level

Operate

Syntax

MEDIATRANSPORT NEXT_SELECTION app group operation

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 operation = one of (0..\$FF):
 0 = select previous selection
 non-0 = select next selection

Use

Requests that the *output unit* in the given *media link group* change to the next or previous *selection*.

Examples

MEDIATRANSPORT NEXT_SELECTION 254/\$C0 2 1

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.108 MEDIATRANSPORT NEXT_TRACK Command

MEDIATRANSPORT NEXT_TRACK

Access Level

Operate

Syntax

MEDIATRANSPORT NEXT_TRACK app group operation

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0..\$FF):
 0 = select previous track
 non-0 = select next track

Use

Requests that the *output unit* in the given *media link group* change to the next or previous *track*.

Examples

MEDIATRANSPORT NEXT_TRACK 254/\$C0 2 1

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.109 MEDIATRANSPORT PAUSE Command

MEDIATRANSPORT PAUSE

Access Level

Operate

Syntax

MEDIATRANSPORT PAUSE app group operation

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0,\$FF):
 0 = pause track
 255 = resume playing track
 other values reserved

Use

Requests that the *output unit* in the given *media link group* pause or resume the current *track*.
If the unit is in the stopped or paused state, a pause operation is ignored.
If the unit is in the stopped or playing state, a resume operation is ignored.

Examples

MEDIATRANSPORT PAUSE 254/\$C0 2 1

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.110 MEDIATRANSPORT PLAY Command

MEDIATRANSPORT PLAY

Access Level

Operate

Syntax

MEDIATRANSPORT PLAY app group

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)

Use

Requests that the *output unit* in the given *media link group* play tracks from the current *selection*.
If the *output unit* is in the stopped state, play begins from the first track in the *selection*. (This may be randomised if the SHUFFLE modifier is active.)
If the *output unit* is in the paused state, playing resumes from where it was paused.
If the *output unit* is in the playing state, the command has no effect other than turning off any fast-forward state activated.

Examples

MEDIATRANSPORT PLAY 254/\$C0 2

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.111 MEDIATRANSPORT REPEAT Command

MEDIATRANSPORT REPEAT

Access Level

Operate

Syntax

MEDIATRANSPORT REPEAT app group operation

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0,\$FF):
 0 = repeat is off
 1..254 = repeat current track
 255 = repeat all tracks

Use

Sets the *repeat* modifier of the given *media link group*.
The repeat modifier can be changed at any time.

Examples

MEDIATRANSPORT REPEAT 254/\$C0 2 1

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.112 MEDIATRANSPORT REWIND Command

MEDIATRANSPORT REWIND

Access Level

Operate

Syntax

MEDIATRANSPORT REWIND app group operation

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0,\$FF):
 0 = cease rewind, play at normal speed
 2 = rewind at 2x speed
 4 = rewind at 4x speed
 6 = rewind at 8x speed
 8 = rewind at 16x speed
 10 = rewind at 32x speed
 12 = rewind at 64x speed
 all other values reserved

Use

Requests that the *track* currently being played in the given *media link group* be played in reverse. Continues to be in effect until a PLAY, FORWARD 0 or another REWIND command is received. If the *output unit* is not in the play state, the command is ignored.

Examples

MEDIATRANSPORT REWIND 254/\$C0 2 0

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.113 MEDIATRANSPORT SELECTION_NAME Command

MEDIATRANSPORT SELECTION_NAME

Access Level

Operate

Syntax

MEDIATRANSPORT SELECTION_NAME app group wni total index text

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 wni = name identification (0..7)
 0 = current selection
 1 = next selection
 2 = next+1 selection
 3,4 = reserved
 5 = previous selection
 6 = previous-1 selection
 7 = enumerated selection
 total = total number of packets - 1 (0..3)
 index = index/sequence of this packet (0..3)
 text = all/part of the *selection* name in UTF-8 (max 11 bytes per packet)

Use

Sets the name of a *selection* in the output unit in the given *media link group*.
 The name is constructed from one or more packets, sent in sequence, coded as UTF-8, limited to 40 bytes total length.

Examples

```
MEDIATRANSPORT SELECTION_NAME 254/$C0 2 1 1 0 "List 2"
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.114 MEDIATRANSPORT SET_CATEGORY Command

```
MEDIATRANSPORT SET_CATEGORY
```

Access Level

Operate

Syntax

```
MEDIATRANSPORT SET_CATEGORY app group category-number
```

app = application address (normally //proj/net/\$C0)
 group = media link group (0..\$FF, where \$FF indicates unused)
 category-number = category to select (0..127)
 (the meaning of this depends on the output unit)

Use

Requests that the given *category-number* be used by the given *media link group's output unit*.

If the given *category-number* doesn't exist, no action is taken.

Examples

```
MEDIATransport SELECT_SOURCE 254/$C0 2 5
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.115 MEDIATransport SET_SELECTION Command

```
MEDIATransport SET_SELECTION
```

Access Level

Operate

Syntax

```
MEDIATransport SET_SELECTION app group selection-number
```

app = application address (normally //proj/net/\$C0)

group = media link group (0..\$FF, where \$FF indicates unused)

selection-number = selection number to select

Use

Requests that the *selection-number* used by the given *media link group* be changed to that specified.

If the given *selection-number* doesn't exist in the *active category*, then the *active selection* isn't changed.

Examples

```
MEDIATransport SET_SELECTION 254/$C0 2 0 2
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.116 MEDIATRANSPORT SET_TRACK Command

MEDIATRANSPORT SET_TRACK

Access Level

Operate

Syntax

MEDIATRANSPORT SET_TRACK app group track-number

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
track-number = track to select (0..\$FFFF)
(the meaning of this depends on the output unit)

Use

Requests that the given track-number be played by the given *media link group's output unit*.
If the given track-number doesn't exist in the *active selection*, no action is taken.

Examples

MEDIATRANSPORT SET_TRACK 254/\$C0 2 5

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.117 MEDIATRANSPORT SHUFFLE Command

MEDIATRANSPORT SHUFFLE

Access Level

Operate

Syntax

MEDIATRANSPORT SHUFFLE app group operation

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0,\$FF):
 0 = shuffle is off
 255 = shuffle is on

1..254 = reserved

Use

Sets the *shuffle* modifier of the given *media link group*.
The shuffle modifier can be changed at any time.

Examples

```
MEDIATransport SHUFFLE 254/$C0 2 0
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.118 MEDIATransport SOURCE_POWER Command

```
MEDIATransport SOURCE_POWER
```

Access Level

Operate

Syntax

```
MEDIATransport SOURCE_POWER app group state
```

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
state = one of 0 or 1..255
 0 = power off
 not 0 = power on

Use

Turns on or off, the *output unit* in the given *media link group*.
If the unit doesn't have power-control ability, this command is ignored.

Examples

```
MEDIATransport SOURCE_POWER 254/$C0 2 1
```

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.119 MEDIATransport STATUS_REQUEST Command

MEDIATransport STATUS_REQUEST

Access Level

Operate

Syntax

MEDIATransport STATUS_REQUEST app group

app = application address (normally //proj/net/\$C0)

group = media link group (0..\$FF, where \$FF indicates unused)

Use

Requests current status of the *output unit* in the given *media link group*.

Examples

MEDIATransport STATUS_REQUEST 254/\$C0 2

Successful Response

#####

Failure response

400 Syntax Error

4.5.120 MEDIATransport STOP Command

MEDIATransport STOP

Access Level

Operate

Syntax

MEDIATransport STOP app group

app = application address (normally //proj/net/\$C0)

group = media link group (0..\$FF, where \$FF indicates unused)

Use

Requests that the *output unit* in the given *media link group* stop playing. If the unit is already in the stopped state, this command is ignored.

Examples

```
MEDIATransport STOP 254/$C0 2
```

Successful Response

```
200 OK.
```

Failure response

```
400 Syntax Error
```

4.5.121 MEDIATransport TOTAL_TRACKS Command

```
MEDIATransport TOTAL_TRACKS
```

Access Level

Operate

Syntax

```
MEDIATransport TOTAL_TRACKS app group tracks
```

app = application address (normally //proj/net/\$C0)
group = media link group (0..\$FF, where \$FF indicates unused)
tracks = number of tracks (0..\$FFFF)

Use

An output unit on the Media Link Group is describing how many tracks are available in the currently selected *category* and *selection*

Examples

```
MEDIATransport TOTAL_TRACKS 254/$C0 2 17
```

Successful Response

```
200 OK.
```

Failure response

```
400 Syntax Error
```

4.5.122 MEDIATRANSPORT TRACK_NAME Command

MEDIATRANSPORT TRACK_NAME

Access Level

Operate

Syntax

MEDIATRANSPORT TRACK_NAME app group wni total index text

app = application address (normally //proj/net/\$C0)

group = media link group (0..\$FF, where \$FF indicates unused)

wni = name identification (0..7)

0 = current track

1 = next track

2 = next+1 track

3,4 = reserved

5 = previous track

6 = previous-1 track

7 = enumerated track

total = total number of packets - 1 (0..3)

index = index/sequence of this packet (0..3)

text = all/part of the *track* name in UTF-8 (max 11 bytes per packet)

Use

Sets the name of a *track* being played by the output unit in the given *media link group*.

The name is constructed from one or more packets, sent in sequence, coded as UTF-8, limited to 40 bytes total length.

Examples

MEDIATRANSPORT TRACK_NAME 254/\$C0 2 1 2 0 "Adagio for "

Successful Response

200 OK.

Failure response

400 Syntax Error

4.5.123 NET Command

NET

Access Level

Admin

Syntax

NET [?]

Use

Lists the NET sub-commands:

NET CHECK_UNRAVEL - Checks to see if a networks needs to be unravelled.

NET CLOSE - Closes a network.

NET CREATE - Defines a C-Bus network to add to the current project.

NET DELETE - Deletes a network entry by name.

NET FLUSH - Flushes or deletes the in-memory objects attached to a network.

NET LEARN - Sends learn-mode commands to a network.

NET LIST - Returns a list of available networks defined in the current project.

NET LIST_ALL - Returns a list of all networks defined in all projects.

NET LOAD - Loads network definitions for the current project.

NET OPEN - Opens a network.

NET PINGU - Performs a single MMI to get a list of units from the network.

NET PROJECT_IDENTIFY - Returns the project name of the C-Bus network attached to the specified interface.

NET RENAME - Re-addresses a network, giving it a new name and address.

NET SAVE - Saves the current network configuration.

NET SET_PROJECT_IDENTIFY - Stores a project name in a unit on a network.

NET SYNC - Initiates a network synchronisation operation.

NET SYNCNEW - Checks a network or a single unit address for any new units.

NET UNRAVELUNIT - Find all units at the given addresses and move them to unique addresses.

NET UNRAVEL - Find all units on the network and where necessary move them to unique addresses.

Success response

A series of lines giving help for these commands in the form:

101-Help: help-information

Failure responses

4.5.124 NET CHECKUNIT

Command

NET CHECKUNIT

Access Level

Admin

Syntax

NET CHECKUNIT net-address [unit-addresses]

net-address = address of the network

unit-addresses = comma delimited sequence of unit addresses, or *

Use

Interrogates unit addresses to determine whether there are single or multiple units at each address.

This command performs a single MMI followed by parallel queries to all unit addresses in question. The entire process takes about two seconds to run.

If the unit-addresses parameter is not provided or is a * wildcard, all occupied addresses in the MMI will be queried.

Each response line will begin with either "Duplicate units detected", "Single unit detected", "Single unit with error detected", "One or more units detected", "One or more units with error detected" or "No units detected", and then be followed by the text " at address: xxx".

Examples

```
net checkunit 254 0
120-completed MMI 1 of 1.
120 Duplicate units detected at address: 0
```

```
net checkunit 254 5
120-completed MMI 1 of 1.
120 Single unit detected at address: 5
```

```
net checkunit 254 0,10,15
120-completed MMI 1 of 1.
120-Duplicate units detected at address: 0
120-Single unit detected at address: 10
120 No units detected at address: 15
```

```
net checkunit 254 *
120-completed MMI 1 of 1.
120-Duplicate units detected at address: 0
120-Single unit detected at address: 5
120-Single unit detected at address: 10
120-Single unit detected at address: 11
120 Single unit detected at address: 104
```

Success response

```
120 [xxxxx] detected at address: [xxx]
```

Failure responses

```
401 Bad object or device ID
408 Operation failed: ...
420 Access denied
```

See also

NET UNRAVEL

4.5.125 NET CLOCKS

Command

NET CLOCKS

Access Level

Admin

Syntax

NET CLOCKS net-address [*target* | R]

net-address = address of the network

target = the target number of enabled clocks to achieve

R = a constant to perform an emergency recovery of clock only

Use

This command is used for managing the number of C-Bus clocks enabled on the network. Not to be confused with the [CLOCK](#) commands.

The command has three different behaviours depending on the *target* parameter.

If the *target* parameter is:

- not supplied, then the command returns a list of the unit addresses that have C-Bus clock enabled. This also includes additional information showing the clock generator and the software burdens enabled.
- an integer, then C-Gate will if needed enable or disable additional clocks in order for the network to reach the *target* number of clocks enabled. This may take several seconds.
- the letter 'R', the command will perform an emergency recovery by enabling clock on the gateway. This takes less than a second. Note that C-Gate also has an automatic [clock recovery](#) feature that achieves the same outcome.

In all cases the command requires the network to be open first.

The standard recommendation is that each C-Bus network should have three (3) clocks enabled.

WARNING: This command does not take into account network topology. The best practice is to enable clocks on units located in the physical centre of the network topology and to achieve that you will need to program units individually. However this command will always attempt to keep existing clocks enabled, so if you have enabled two clocks by hand and you specify a *target* value of '3', it will preserve the original two enabled clocks and enable one more.

Example

```
# List the clocks enabled, which includes the clock generator (clocks_active).
# Note that unravelled addresses may report more than one unit.
net clocks 254
120-address=0 output_units=1 clocks_enabled=1 clocks_active=1 burdens_enabled=0
120-address=1 output_units=1 clocks_enabled=0 clocks_active=0 burdens_enabled=0
120-address=3 output_units=1 clocks_enabled=0 clocks_active=0 burdens_enabled=1
120-address=4 output_units=1 clocks_enabled=0 clocks_active=0 burdens_enabled=0
120-address=10 output_units=1 clocks_enabled=0 clocks_active=0 burdens_enabled=0
```

```
120-address=55 output_units=0
200 OK.

# Units 0 and 3 are unplugged, we can't communicate with this network any more.
net clocks 254
408 Operation failed: MMI failed.

# Emergency recovery.
net clocks 254 r
120-address=4 output_units=1 clocks_enabled=0 clocks_active=0 burdens_enabled=0
120-Gateway clock at address 4 is now enabled.
200 OK.

# Boosting the network to the recommended number of three clocks.
net clocks 254 3
120-Keeping the master clock at address 4 enabled.
120-Clock at address 1 is now enabled.
120-Clock at address 10 is now enabled.
200 OK.

# Showing the network with three clocks.
net clocks 254
120-address=1 output_units=1 clocks_enabled=1 clocks_active=0 burdens_enabled=0
120-address=4 output_units=1 clocks_enabled=1 clocks_active=1 burdens_enabled=0
120-address=10 output_units=1 clocks_enabled=1 clocks_active=0 burdens_enabled=0
120-address=55 output_units=0
200 OK.
```

Success response

200 OK.

Failure responses

401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: ... (a range of other options here depending on particular failures)
420 Access denied

See also

[NET OPEN](#)
[C-Bus Clock Recovery](#)

4.5.126 NET CLOSE Command

NET CLOSE

Access Level

Admin

Syntax

NET CLOSE net-address

Use

Closes the network given on the command line as net-address.

Success response

A response giving the project and network closed like:

200 OK: //project/net-address

eg

200 OK: //HOME/254

Failure responses

401 Bad object or device ID

408 Operation failed: ...

420 Access denied

See also

[NET OPEN](#)

4.5.127 NET CREATE Command

NET CREATE

Access Level

Admin

Syntax

NET CREATE name type interface-address [options...]

name = the name of the network to create

type = one of the valid network types (serial, cni, etherlite, socket, bridge, modem, wiser)

interface-address = address for the PORT of this network. (This address depends on the actual type)

serial = local serial port name (as would be returned by the PORT LIST command, eg COM1)

cni = IP address with an optional port number preceeded by a ':'. With no port number given, the default CNI port is used.

socket = IP address, followed by a ':' and a port number

etherlite = IP address followed by a ':' and a serial port number, where 0 is the first port.

bridge = C-Gate object address for the near-side of the bridge that the network is

beyond.
for wiser it is the same as type cni

Use

Defines a C-Bus network to add to the current project. This creates the network in state new in the current project. This command also will create an entry for this network in the project tag database, if tag databases are being used. An event is sent on the creation of a network by this mechanism. Note: this command does not change or update a networks.txt file -- this requires use of the NET SAVE FILE command.

Success response

200 OK.

Failure responses

408 Operation failed: ...
420 Access denied

See also

[NET DELETE](#)

4.5.128 NET DELETE Command

NET DELETE

Access Level

Admin

Syntax

NET DELETE net-address

net-address = address of the network to delete

Use

Deletes a network entry by name.

This command removes the network from the C-Gate server memory.

An event is sent on the deletion of a network by this mechanism.

The network must be in InterfaceState closed in order to be deleted.

Success response

200 OK.

Failure responses

401 Bad object or device ID
408 Operation failed: ...
420 Access denied
468 Can not delete open network

See also

[NET CREATE](#), [NET LIST](#)

**4.5.129 NET FLUSH
Command**

NET FLUSH

Access Level

Admin

Syntax

NET FLUSH net-address

net-address = address of the network to flush

Use

Flushes or deletes the in-memory objects attached to this network. This means that a subsequent NET SYNC operation will start from scratch. Both Unit objects and Application objects are flushed.

Note: this command can only be executed when a sync is not in progress.

Success response

200 OK.

Failure responses

401 Bad object or device ID
408 Operation failed: ...
420 Access denied

See also

NET SYNC

**4.5.130 NET LEARN
Command**

NET LEARN

Access Level

Admin

Syntax

NET LEARN net-address app-number grade group

Use

Sends learn-mode commands to the specified network and application, using the given grade and group.

Grade values are:

init relay:	0x01
init dim:	0x02
cancel:	0x80
exit relay:	0x81
exit dim:	0x82
exit area:	0x83

Success response

200 OK.

Failure responses

One or more lines of

401 Bad object or device ID
408 Operation failed: ...
420 Access denied

See also

4.5.131 NET LIST Command

NET LIST

Access Level

Admin

Syntax

NET LIST [project]

project = the name of a project

Use

Returns a list of networks defined in the given project. If no project name is given, then the current project is used. The current project is determined by the command session that this command is issued in - see the PROJECT USE command for details of setting the current project. All networks in the project are displayed, whether the connections have been opened or not.

Example

```
net list example
131-network=254 State=new InterfaceState=closed
131 network=100 State=new InterfaceState=closed
```

Success response

One or more lines of:

```
131 network=network-address State=status InterfaceState=interface-state
```

If no networks are found, the following is returned as a response:

```
132 no networks found
```

Failure responses

```
408 Operation failed
420 Access denied
```

4.5.132 NET LIST_ALL Command

NET LIST_ALL

Access Level

Admin

Syntax

NET LIST_ALL

Use

Returns a list of networks from all open projects along with status information about the networks.

Success response

One or more lines of:

```
"135 project=" project-name "address=" net-address "OID=" net-oid "interfaceType="
interface-type "interfaceAddress=" interface-address "state=" object-state
```

```
"interfaceState=" interface-state
```

If no networks are found, the following is returned as a response:

```
132 no networks found
```

Failure responses

```
408 Operation failed
```

```
420 Access denied
```

4.5.133 NET LOAD Command

NET LOAD

Access Level

Admin

Syntax

```
NET LOAD DB | FILE [project]
```

project = the name of a project

Use

Loads network definitions from the given project into memory, from either the project tag database, or from the networks file for that project. If no project name is given, then the current project is used. The current project is determined by the command session that this command is issued in - see the PROJECT USE command for details of setting the current project. This will not over-write any existing definitions of networks held in the C-Gate server memory which have the same names as given in the networks file.

Example

```
net load db example  
200 OK.
```

```
net load db //example  
200 OK.
```

```
project use home  
200 OK.  
net load db  
200 OK.
```

Success response

```
200 OK.
```

Failure responses

One or more lines of

469 Can not create network: reason-information
401 Bad object or device ID
408 Operation failed: ...
420 Access denied

See also

[NET CREATE](#), PROJECT LOAD, PROJECT SAVE

4.5.134 NET OPEN Command

NET OPEN

Access Level

Admin

Syntax

NET OPEN net-address

Use

Opens a network. Opening a network does the following:

- Attempts to open the port address given in the NET CREATE command -- thus initiating a connection to the C-Bus network that is accessed by the port

- Performs an initial synchronisation of the attached network

Once both these steps are complete, the network State will be ok and the InterfaceState will be running.

Success response

A series of responses using:

120-NET OPEN: ... delivering progress information as the network is opened, followed by:
200 OK.

if all succeed, or by one of the failure responses if the open does not succeed.

Failure responses

401 Bad object or device ID
408 Operation failed: ...
420 Access denied

See also

[NET CLOSE](#)

4.5.135 NET PINGU Command

NET PINGU

Access Level

Admin

Syntax

NET PINGU net-address

net-address = address of the network to do a unit ping on

Use

Performs a single quick MMI operation to get a list of units from the network. A single MMI is issued, and no retries are performed.

Success response

302 Units= *comma-separated-list-of-units*

For example:

302 Units= 1,2,3,45,255

Failure responses

401 Bad object or device ID
408 Operation failed: ...
420 Access denied

See also

[NET SYNC](#)

4.5.136 NET PROJECT_IDENTIFY Command

NET PROJECT_IDENTIFY

Access Level

Admin

Syntax

NET PROJECT_IDENTIFY interface-address

interface-address = address of the interface to use, in the form type@address

Use

Returns the project name of the C-Bus network attached to the specified interface.

Example

```
net project_identify CNI@localhost:14000
305 Project=TEST UnitCount=6
```

Success response

```
305 Project=project-name UnitCount=unit-count
```

Failure responses

```
401 Bad object or device ID
408 Operation failed: ...
420 Access denied
```

See also

[NET SET PROJECT IDENTIFY](#)

4.5.137 NET RENAME Command

NET RENAME

Access Level

Admin

Syntax

```
NET RENAME net-address new-net-address ["nofixrefs"]
```

net-address = address of the network to be renamed/readdresses

new-net-address = new address for the network

"nofixrefs" prevents the command from changing bridge definitions in other networks in this project to support the new network name.

Use

Re-addresses a network, giving it a new name and address.

This takes place immediately.

This command also updates any references to the old network address in any bridge networks unless the optional `nofixrefs` is given at the end of the command. The network address of the bridge network can only be updated when the network is not open, and if this command fails to update the network address, there will be a message in the response that the network address is not

changed.

Note: This command doesn't affect or rename the database instance.

Example

```
NET RENAME 1 2
120-Interface not changed for network 2 because it is open
200 OK.
```

Success response

```
200 OK.
```

Failure responses

```
401 Bad object or device ID
408 Operation failed: ...
420 Access denied
```

See also

[DBRENAMENET](#)

4.5.138 NET SAVE Command

NET SAVE

Access Level

Admin

Syntax

```
NET SAVE DB | FILE [project]
```

project = the name of a project

Use

Saves the network configuration for the given project to either the project database, or to a networks file appropriate to the project. If no project is given, then the current project is used. The current project is determined by the command session that this command is issued in - see the PROJECT USE command for details of setting the current project. Note: this command does not save a project database to disk -- this is done with the PROJECT SAVE command.

Example

```
net save db example
200 OK.

net save db //example
```

200 OK.

project use home

200 OK.

net save db

200 OK.

Success response

200 OK.

Failure responses

408 Operation failed: ...

420 Access denied

See also

[NET CREATE](#), [PROJECT SAVE](#), [NET LOAD](#)

4.5.139 NET SET_PROJECT_IDENTIFY Command

NET SET_PROJECT_IDENTIFY

Access Level

Admin

Syntax

NET SET_PROJECT_IDENTIFY net-address project-name

net-address = address of the network to be identified

project-name = new project name, must be no more than 8 characters and conform to the sixbit packing standard

Use

Stores a project name in the appropriate unit on the given network.

Success response

200 OK.

Failure responses

401 Bad object or device ID

408 Operation failed: ...

420 Access denied

See also

[NET PROJECT IDENTIFY](#)**4.5.140 NET SYNC
Command**

NET SYNC

Access Level

Admin

Syntax

NET SYNC net-address ["fast"] [retry-count]

net-address = address of the network to be synchronised

retry-count = number of retries for commands during the sync process

Use

Initiates a network synchronisation operation, returning a number of lines of status information as the synchronisation succeeds. Applications can use this progress information to present progress graphics or show a status screen of some kind if required.

If the optional parameter 'fast' is given, the sync only completes an MMI, type, version and serial number scan before returning. This gives a quick view of a network.

If a retry-count is given, then the network's Retries parameter is set to this value for the duration of this command.

Note: if another sync of the same network is in progress, either a background sync or a sync started by 'NET SYNC' command, the sync command will attach itself to the on going sync and respond with the cached sync outputs in the beginning followed by the remaining outputs as a normal sync operation.

Note: the background sync is explained in [Starting a Project or Network](#). The background sync is controlled by network's [AutoSync](#) property which should always be set to yes as explained in [Starting a Project or Network](#). It is not necessary to turn off AutoSync for C-Bus Toolkit, Schedule Plus and other C-Gate clients to work properly. Network's [NextSyncTime](#) property indicates when the next scheduled automatic background sync will be executed, while [LastSyncTime](#) property records the time of the last network synchronisation.

Example command output

```
NET SYNC 254
120-Start of cached output.
120-Net Sync: started at Wed Oct 16 13:52:33 CST 2013
120-Net Sync: synchronising PCIs
120-Net Sync: starting network discovery
120-Net Sync: Unit count: 6
120-Net Sync: Unit error count: 0
120-Net Sync: Units at: 3, 4, 5, 8, 11, 18
120-Net Sync: Unit errors at:
```



```

120-Net Sync: address=3 type=PCINT4 version=4.6.00 serial=100385.2535
catalog=5500PC
120-Net Sync: address=4 type=RELDN12 version=2.7.00 serial=100800.2302
catalog=L5512RVF
120-Net Sync: address=5 type=PCLOCALU version=4.4.08 serial=100120.2490
catalog=5500PCU
120-End of cached output.
120-Net Sync: address=8 type=PCINTU version=5.4.00 serial=100779.1755
catalog=5500PCU
120-Net Sync: address=11 type=KEYV3 version=1.8.00 serial=100399.2339
catalog=5093NL
120-Net Sync: address=18 type=DIMDN8 version=2.7.00 serial=100800.4036
catalog=L5508D1A
120-Net Sync: synchronising bridges
120-Net Sync: synchronizing unit 1 of 6 at address 3
120-Net Sync: address 3 type PCINT4 version 4.6.00 catalog 5500PC
120-Net Sync: synchronizing unit 2 of 6 at address 4
120-Net Sync: address 4 type RELDN12 version 2.7.00 catalog L5512RVF
120-Net Sync: synchronizing unit 3 of 6 at address 5
120-Net Sync: address 5 type PCLOCALU version 4.4.08 catalog 5500PCU
120-Net Sync: synchronizing unit 4 of 6 at address 8
120-Net Sync: address 8 type PCINTU version 5.4.00 catalog 5500PCU
120-Net Sync: synchronizing unit 5 of 6 at address 11
120-Net Sync: address 11 type KEYV3 version 1.8.00 catalog 5093NL
120-Net Sync: synchronizing unit 6 of 6 at address 18
120-Net Sync: address 18 type DIMDN8 version 2.7.00 catalog L5508D1A
120-Net Sync: finished at Wed Oct 16 13:52:49 CST 2013
200 OK.

```

Success response

200 OK.

Failure responses

```

401 Bad object or device ID
408 Operation failed: ...
409 A network scan is currently in progress. Please wait for it to complete before
retrying.
420 Access denied

```

See also

[CBusNetwork Starting a Project or Network](#)

4.5.141 NET SYNCNEW Command

NET SYNCNEW

Access Level

Admin

Syntax

NET SYNCNEW net-address [unit-number]

net-address = address of the network to do a syncnew operation on
unit-number = (optional) specific unit address to check

Use

Checks a network or a single unit address to see if there are any new units.

If no unit number is given, the command checks for any new units on the network and if they are found, returns the type, version and serial number for the units.

If a unit number is given, the command check for a new unit at that address, also checking to see if any duplicate units are found at that address.

Success response

```
# command with unit address given
net syncnew 1 0
120-completed MMI 1 of 1.
120-unit found
120-duplicate test 1/3
120-duplicate test 2/3
120-duplicate test 3/3
120-no duplicate found
120-identifying unit
303 New Unit Found: address=0 type=BRIDGE2N version=4.1.00 serial=79833.98

#command without unit address given
net syncnew 1
120-completed MMI 1 of 1.
303-New Unit Found: address=2 type=PC_GIM version=4.2.00 serial=1048575.4095
303-New Unit Found: address=3 type=DIMDN8 version=1.2.18 serial=69260.99
303-New Unit Found: address=4 type=PCINT4 version=4.0.0 serial=77432.153
303-New Unit Found: address=11 type=PC_CBTI version=4.2.00 serial=84037.22
303-New Unit Found: address=18 type=GATEWLS version=4.2.00 serial=1048575.4095
303-New Unit Found: address=19 type=PC_LOCAL version=3.12 serial=69.13
303-New Unit Found: address=37 type=GATEWLS version=4.2.00 serial=1048575.4095
303 New Unit Found: address=255 type=RELDN12 version=1.2.18 serial=69923.58
```

Failure responses

```
401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: Unit at given address not found
408 Operation failed: Unit already in model
408 Operation failed: No new units found
408 Operation failed: ... (a range of other options here depending on particular failures)
420 Access denied
```

See also

[NET SYNC](#)

4.5.142 NET UNRAVELUNIT Command

NET UNRAVELUNIT

Access Level

Admin

Syntax

NET UNRAVEL net-address unit-addresses [matchdb]

net-address = address of the network

unit-addresses = comma delimited sequence of unit addresses, or *

matchdb = if specified, units will be moved to their database addresses where possible

Use

Performs a series of operations on a network to find all units at the given address or addresses and where necessary move them to unique addresses.

This is similar to the [NET UNRAVEL](#) command but only affects the addresses given. Please see the help for that command for more information about the unravelling process.

Example

```
net unravel 254 6
120-completed MMI 1 of 1.
120-Unravel: Unit count: 9
120-Unravel: Units at: 0, 1, 2, 3, 4, 5, 6, 7, 45
120-Unravel: Scanning unit 1 of 9 at address 6 (0x06).
120-Unravel: Unit at address 6 (0x06) with serial number 664712.6 left in place.
120-Unravel: Complete.
200 OK.
```

Success response

200 OK.

Failure responses

```
401 Bad object or device ID
408 Operation failed: ...
420 Access denied
```

See also

[NET UNRAVEL](#), [NET SYNC](#)

4.5.143 NET UNRAVEL Command

NET UNRAVEL

Access Level

Admin

Syntax

NET UNRAVEL net-address [matchdb]

net-address = address of the network

matchdb = if specified, units will be moved to their database addresses where possible

Use

Performs a series of operations on a network to find all units and where necessary move them to unique addresses. This is a necessary operation to commission a C-Bus network, as all units invariably come from the factory set to address 255.

The unravel process will:

- Perform an MMI to find occupied addresses.
- Interrogate the gateway unit and ensure no other units are at that address.
- If the gateway unit is a bridge, move it to the correct address matching the near side network.
- Interrogate each remaining unit address. If duplicates are found, they will be moved to unique addresses.
- If 'matchdb' is specified and a matching unit is found in the database, ensure the unit ends up at that address.
- Leave one unit at each originally occupied address. This ensures an already unravelled network will not change if it is unravelled again.
- Move all other units to vacant addresses starting at 2 and increasing.
- Cope with old units without serial numbers (though it will proceed slowly at these addresses).

Typical responses are:

```
120-completed MMI 1 of 1.
120-Unravel: Unit count: <count>
120-Unravel: Units at: <comma delimited addresses>
120-Unravel: Unravelling gateway units...
120-Unravel: Top unit was sent to <address> (<hex-address>).
120-Unravel: Restored top unit to address <address> (<hex-address>).
120-Unravel: Scanning unit <no> of <count> at <address> (<hex-address>).
120-Unravel: Unit at address <address> (<hex-address>) with serial number <serial>
moved to address <address> (<hex-address>).
120-Unravel: Unit at address <address> (<hex-address>) moved to address <address>
(<hex-address>).
120-Unravel: Last unit at address <address> (<hex-address>) with serial number
<serial> left in place.
120-Unravel: Complete.
120-Unravel: Updating network model at address <address> (<hex-address>)
```

200 OK.

Note that there is also a [NET UNRAVELUNIT](#) command that allows you to specify a subset of addresses to be unravelled.

Example

```
net unravel 254
120-completed MMI 1 of 1.
120-Unravel: Unit count: 1
120-Unravel: Units at: 255
120-Unravel: Unravelling gateway units...
120-Unravel: Unit at address 255 (0xFF) moved to address 2 (0x02).
120-Unravel: Top unit was sent to 2 (0x02).
120-Unravel: Unit at address 255 (0xFF) with serial number 100296.2545 moved to
address 3 (0x03).
120-Unravel: Unit at address 255 (0xFF) with serial number 69260.9 moved to address
4 (0x04).
120-Unravel: Unit at address 255 (0xFF) with serial number 68928.124 moved to
address 5 (0x05).
120-Unravel: Unit at address 255 (0xFF) with serial number 65504.1 moved to address
6 (0x06).
120-Unravel: Unit at address 255 (0xFF) with serial number 68742.214 moved to
address 7 (0x07).
120-Unravel: Unit at address 255 (0xFF) with serial number 68426.48 moved to
address 8 (0x08).
120-Unravel: Unit at address 255 (0xFF) moved to address 9 (0x09).
120-Unravel: Unit at address 255 (0xFF) moved to address 10 (0x0A).
120-Unravel: Unit at address 255 (0xFF) moved to address 11 (0x0B).
120-Unravel: Unit at address 255 (0xFF) moved to address 12 (0x0C).
120-Unravel: Scanning unit 1 of 1 at address 255 (0xFF).
120-Unravel: Complete.
120-Unravel: Updating network model at address 255 (0xFF)
120-Unravel: Updating network model at address 12 (0x0C)
120-Unravel: Updating network model at address 11 (0x0B)
120-Unravel: Updating network model at address 10 (0x0A)
120-Unravel: Updating network model at address 9 (0x09)
120-Unravel: Updating network model at address 8 (0x08)
120-Unravel: Updating network model at address 7 (0x07)
120-Unravel: Updating network model at address 6 (0x06)
120-Unravel: Updating network model at address 5 (0x05)
120-Unravel: Updating network model at address 4 (0x04)
120-Unravel: Updating network model at address 3 (0x03)
120-Unravel: Updating network model at address 2 (0x02)
200 OK.
```

Success response

200 OK.

Failure responses

```
401 Bad object or device ID
408 Operation failed: ...
420 Access denied
```

See also

[NET UNRAVELUNIT](#), [NET SYNC](#)

**4.5.144 NETWORK
Command**

NETWORK

Access Level

Admin

Syntax

NETWORK [?]

Use

Lists the NETWORK sub-commands:

NETWORK LOCATE - Sends a locate-yourself message to the matched units.

Success response

A series of lines giving help for these commands in the form:

101-Help: help-information

Failure responses**4.5.145 NETWORK LOCATE
Command**

NETWORK LOCATE

Access Level

Admin

Syntax

NETWORK LOCATE app options mode

app = application address for this application, normally net/\$d0

options = ("unit" address) - locate unit address
 | ("app" address) - locate units in application
 | ("group" app group-address) - locate units using group
 | ("serial" manufacturer serial-number - locate using serial &
manufacturer

mode = "on" | "off" | integer in range 0-255

Use

Sends a locate-yourself message to the matched units. This puts the unit into a commissioning mode.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.146 NEW Command

NEW

Access Level

Operate

Syntax

Defining a group:

NEW object-type object-id param

object-type = "UNIT" | "GROUP" | "PHANTOM"
object-id = a network, group, unit or system entity
param = a parameter appropriate to the object type

Use

Creates a new object. The new object is created in state *new*.

Examples

NEW unit p/1/\$34 DIMMER4 1 (last param is version)
NEW group 42/56/1 area
NEW group 42/56/2 phantom
NEW group 42/56/3

Successful Response

200 OK.

Failure response

400, 401, 402, 406, 420

See also

[GET](#), [SET](#)

4.5.147 NOOP Command

NOOP

Access Level

Connect

Syntax

NOOP

Use

Results in a 200 OK reply message with no additional details if the interface is working. Designed to allow simple verification that the interface is operating correctly. A 500 Internal error can only occur when the server has a serious internal error. If no response is received to a NOOP command, then the current command session is not responding. This may be due to the fact that this command interface is still processing an existing command.

Example

NOOP

Successful Response

200 OK

Failure response

500 Internal error [or no response in the case of a broken connection or other condition]

See Also

[# or // \(comment\)](#)

4.5.148 OFF Command

OFF

Access Level

Operate

Syntax

OFF object-identifier ["force"]

Use

Turns off the specified group. This command sends a C-Bus group OFF command to the appropriate network.

Specifying the optional parameter `force` allows the command to be sent even if the group is not in an ok state.

Examples

OFF 254/56/1

Successful Response

200 OK

Failure responses

400, 401, 402, 405, 408, 420

See also

[ON](#), [RAMP](#), [DO](#)

4.5.149 OID

Command

OID

Access Level

Operate

Syntax

OID

Use

Generates an Object ID. Object IDs are unique, and are based on underlying hardware depending on OS support.

Example

```
oid
301 OID=3da71864-0e90-11d8-ba38-a385e17d8d20
```

Success Response

302 OID=oid-value

Failure Responses

None

4.5.150 ON Command

ON

Access Level

Operate

Syntax

ON object-identifier ["force"]

Use

Turns on the specified group. This command sends a C-Bus group ON command to the appropriate network.

Specifying the optional parameter *force*, allows the command to be sent even if the group is not in an ok state.

Examples

ON 254/56/1

Successful Response

200 OK

Failure responses

400, 401, 402, 405, 408, 420

See also

[OFF](#), [RAMP](#), [DO](#)

4.5.151 PORT Command

PORT

Access Level

Program

Syntax

PORT [?]

Use

Lists the PORT sub-commands:

PORT CNISCAN - Lists the CNIs found on a network.

PORT IFLIST - Lists the IP network interfaces available on this server.

PORT LIST - Lists available ports that *might* be connected to a C-Bus PCI.

PORT PROBE - Probes a port to locate responding PCIs.

PORT REFRESH - Refreshes the known local serial ports.

Success response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure responses

4.5.152 PORT CNISCAN

Command

PORT CNISCAN

Access Level

Admin

Syntax

PORT CNISCAN [ip-address] ["fast"]

ip-address = (optional) An IP range to scan for CNI devices. If not given, all networks connected to the server will be scanned for CNI devices.

If "*fast*" is given, don't attempt to connect to each CNI.

Use

Scans a network looking for CNIs. A list of found CNIs will be returned. For a CNI to response to this command and thus be detected, it must not have network configuration disabled, and must have an IP address assigned to it that matches the network being scanned.

Success response

One or more lines of: 129 found CNI ip-address=*ip-address* status=*status* port=*port-number* where:

ip-address is the address of the CNI found, and
status is either:

available -- indicating that the port is available for use

unknown -- the port can not be connected to or is in use (it is not possible to differentiate these conditions for a CNI) or, if no CNIs are found: 130 no CNIs found

port-number is the TCP port number used to connect to the CNI's C-Bus interface.

If no CNIs are detected, the response is:
 130 no CNIs found

Note: this command may detect non-CNI devices that are based on similar components to the CNI. You must use [PORT PROBE](#) to actually verify that there is a PCI attached to a device.

Failure responses

408 Operation failed: ...
 420 Access denied

See also

[PORT IFLIST](#), [PORT PROBE](#)

4.5.153 PORT CNISCAN2 Command

PORT CNISCAN2

Access Level

Admin

Syntax

PORT CNISCAN2 [interface-address] [ip-address] ["fast"]

interface-address = (optional) A network interface on the computer. If not given, defaults to '0.0.0.0' (wildcard) in which case the Java + OS subsystem will try and route packets intelligently through all available interfaces.

ip-address = (optional) An IP range to scan for CNI devices. If not given, all networks connected to the server will be scanned for CNI devices.

If "*fast*" is given, don't attempt to connect to each CNI.

Examples

```
// invalid adapter
port cni scan2 1.1.1.1
408 Operation failed: scan failed: Can not send to network: Cannot
assign requested address: Cannot bind
```

```
// selected adapter
port cni scan2 10.176.60.37
129-ip-address=10.176.148.27 status=unknown port=10001 type=CNI
```

```

129 ip-address=10.176.61.60 status=available port=10001 type=CNI 2
mac=00:17:DD:01:0E:33 serial=00100700.3526 cbus-unit-address=37
129 ip-address=192.168.0.10 status=available port=0 type=CNI AC
mac=00:17:DD:09:00:17 serial=00000000.0000 cbus-unit-address=0
129 ip-address=10.176.60.34 status=available port=0 type=CNI AC
mac=00:17:DD:09:00:0C serial=00100327.0683 cbus-unit-address=11

// selected adapter, plus device ip-address filter
port cni scan2 10.176.60.37 10.176.60.34
129 ip-address=10.176.60.34 status=available port=0 type=CNI AC
mac=00:17:DD:09:00:0C serial=00100327.0683 cbus-unit-address=11

// wildcard with device ip-address filter
port cni scan2 0.0.0.0 10.176.60.34
129 ip-address=10.176.60.34 status=available port=0 type=CNI AC
mac=00:17:DD:09:00:0C serial=00100327.0683 cbus-unit-address=11

```

Use

Scans a network looking for CNIs. A list of found CNIs will be returned. For a CNI to response to this command and thus be detected, it must not have network configuration disabled, and must have an IP address assigned to it that matches the network being scanned.

Success response

One or more lines of:

```

129 found CNI ip-address=ip-address status=status port=port-number type=type mac=mac
serial=serial cbus-unit-address=cbus-unit-address

```

where:

ip-address is the address of the CNI found, and

status is either:

available -- indicating that the port is available for use

unknown -- the port can not be connected to or is in use (it is not possible to differentiate these conditions for a CNI) or, if no CNIs are found: 130 no CNIs found

port-number is the TCP port number used to connect to the CNI's C-Bus interface.

type is the type of the device, e.g. CNI, CNI2, CNIAC.

serial is the C-Bus serial number of the device.

cbus-unit-address is the C-Bus unit address of the device.

If no CNIs are detected, the response is:

```

130 no CNIs found

```

Note: this command may detect non-CNI devices that are based on similar components to the CNI. You must use [PORT PROBE](#) to actually verify that there is a PCI attached to a device.

Failure responses

```

408 Operation failed: ...

```

```

420 Access denied

```

See also

[PORT IFLIST](#), [PORT PROBE](#)

4.5.154 PORT IFLIST Command

PORT IFLIST

Access Level

Admin

Syntax

PORT IFLIST

Use

Returns a list of IP network interfaces available on this server. Will return a list of all the IP addresses that this host responds to, so this is principally used to work out the IP interfaces on a server to make scanning for CNIs easier.

Note: the loopback interface (127.0.0.1) will not show up in the list of interfaces.

Success response

One or more lines of:

```
127 address=ip-address interface=interface-name
```

Where *interface-name* is the name of the interface and *address* is the IP address of the interface.

The message 128 no interfaces found is presented if no interfaces are located.

Note: there will be at least **5** seconds of delay between issuing the command and the display of any results. Clients should anticipate this delay.

Failure responses

```
408 Operation failed: Path not found
420 Access denied
```

4.5.155 PORT LIST Command

PORT LIST

Access Level

Admin

Syntax

PORT LIST

Use

Returns a list of available ports that *might* be connected to a C-Bus PCI. The list of ports returned is restricted to local serial ports or network ports that use port-redirectors to appear as a local serial port.

Success response

One or more lines of:

```
125 port=port-name status=status
```

Where *port-name* is the name of the port and
status is one of:

inuse -- indicating that the port is already in use and is therefore not available

available -- indicating that the port is available for use

unknown -- for some reason, port state can not be determined

error -- the port is in an error state

If no ports are found, the following is returned as a response:

```
126 no ports found
```

Failure response

```
408 Operation failed
```

```
420 Access denied
```

See also

[PORT CNISCAN](#)

4.5.156 PORT PROBE Command

PORT PROBE

Access Level

Admin

Syntax

```
PORT PROBE type address
```

type = The type of device to probe. This can be the string: `serial` or `socket`, `chni`, `wiser` or `etherlite`. (Case is not significant)

address = The address to probe.

This address depends on the type above:

for type `serial`, the address is a local serial port name (as would be returned by the PORT LIST command).

for type `cni`, the address is an IP address with an *optional* port number preceded by a ':'.
With no port number given, the default CNI port is used.
for type `wiser`, it is the same as type `cni`
for type `socket`, the address is an IP address, followed by a ':' and a port number
for type `etherlite`, the address is an IP address followed by a ':' and a serial port number,
where 0 is the first port.

Note: the probe will fail if a `cni` device is configured to use a password on connection or is configured to not accept connections from the server.

Use

Probes a port attempting to locate an interface to a C-Bus network (a PCI) that is responding.
This command does the following:

1. Attempts to opens a connection of the specified type of port at the given address. If a connection can not be opened, the command fails with an error
2. If possible, sets appropriate HW flow control pins (necessary for C-Touch in PCI mode)
3. Probes the port for a connected and operating PCI.

The tests used will attempt to identify all versions of PCI by tests like: Sending `~~<CR>` and then checking for an echo response Sending a CAL command to PCI and check for valid response

Success response

230 Probe succeeded: C-Bus network detected (PCISerial=<serial_no>)

Note serial number returned if it can be located for the unit in question. If no serial number is present, then the bracketed PCISerial= will not be included in the response.

Failure responses

431 Probe failed: Port in use
432 Probe failed: No C-Bus network detected
408 Operation failed:
420 Access denied

See also

[PORT LIST](#), [PORT CNISCAN](#), [PORT IFLIST](#)

4.5.157 PORT REFRESH

Command

PORT REFRESH

Access Level

Admin

Syntax

PORT REFRESH

Use

Refreshes the known local serial ports.

Success response

200 OK.

Failure responses

408 Error <message>

420 Access denied

4.5.158 PROJECT Command

PROJECT or PROJECT?

Access Level

Admin

Syntax

PROJECT [?]

Use

Lists the PROJECT sub-commands:

PROJECT ARCHIVE - Archives a project from the existing project directory to a given filename.

PROJECT CLOSE - Closes a project.

PROJECT COPY - Copies a project in the project directory on the server disk.

PROJECT DELETE - Deletes a project from the project directory on the server disk or at another path.

PROJECT DIR - Returns a list of available projects.

PROJECT LIST - Returns a list of all open projects in this C-Gate server.

PROJECT LOAD - Loads a project from disk into the C-Gate server.

PROJECT NEW - Creates a new project.

PROJECT RENAME - Renames a project in the project directory on the server disk.

PROJECT RESTORE - Restores a project from a given archive file.

PROJECT SAVE - Saves a project to disk from the C-Gate server.

PROJECT START - Starts the named or current project.

PROJECT STOP - Stops the named or current project.

PROJECT USE - Sets the context of the current command session to the given project name.

Success response

A series of lines listing these commands in the form:

101-Help: help-information

4.5.159 PROJECT CLOSE Command

PROJECT CLOSE

Access Level

Admin

Syntax

PROJECT CLOSE [project-name]

project-name = (optional) Name of the project to close. If not given, then the current project name is used.

Use

Closes a project. Closing a project means that it is no longer held in C-Gate server memory, and is thus can no longer be referenced. This command should be used to close projects that are not actively being used to conserve server resources. You can not close a running project, it must be stopped first.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied
467 Can not close running project

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT LOAD](#)

4.5.160 PROJECT COPY Command

PROJECT COPY

Access Level

Admin

Syntax

PROJECT COPY project-name new-project-name

project-name = Name of the project that will be copied

new-project-name = Name that the project will be copied to. *new-project-name* must be a valid project name and use characters allowed in the server's filesystem.

Use

Copies a project in the project directory on the server disk. **Note:** *This action only copies the on-disk project. A currently loaded and running project cannot be copied.*

Success response

200 OK.

Failure responses

401 Bad Object or Device ID

408 Operation failed: ...

420 Access denied

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT LOAD](#), [PROJECT DELETE](#), [PROJECT RENAME](#)

4.5.161 PROJECT DELETE

Command

PROJECT DELETE

Access Level

Admin

Syntax

PROJECT DELETE [project-name]

Parameters

project-name = (optional) Name of the project to delete its directory. If not given, the current project directory is deleted.

Use

Deletes a project directory on the server disk or at another path. So that project directory will no longer appear in the list given by the PROJECT DIR command. **Note:** that a currently loaded and running project cannot be deleted.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT LOAD](#)

4.5.162 PROJECT DIR Command

PROJECT DIR

Access Level

Admin

Syntax

PROJECT DIR ["all"]

"all"(optional) Here we can pass parameter as all. It will list all project list from C-Gate server's base directory. It includes XML projects, SQL project and converting project.

Example:

command PROJECT DIR

123-project=CONVERT
123-project=TEST
123-project=EXAMPLE
123 project=HOME

Example:

command: PROJECT DIR all

Response: 123 project=HOME repository=XML description= creationDate=31/12/2099

Response: 123 project=EXAMPLE repository=SQL description=C-Bus project generated from live network by C-Gate creationDate=31/12/2099

Response: 123 project=OFFICE repository=Converting description= creationDate=31/12/2099

Response: 123 project=TEST repository=Failed description= creationDate=31/12/2099 (here *Failed* indicates that the conversion was unsuccessful)

Use

Returns a list of available projects, either from the [default project directory](#), or from the specified path on the **C-Gate server computer** if an optional path name is given.

Success response

One or more lines of:

123 project=*project-name* Where *project-name* is the name of the project file, or

124 no projects found if there are no projects in the project directory.

Failure responses

408 Operation failed: Path not found

420 Access denied

4.5.163 PROJECT LIST Command

PROJECT LIST

Access Level

Admin

Syntax

PROJECT LIST

Use

Returns a list of all open projects in this C-Gate server. An open project is one that has been loaded into server memory either at startup or via the [PROJECT LOAD](#) command.

Success response

One or more lines of:

123 project=*project-name* state=*project-state*

Where *project-name* is the name of the project file and *project-state* is the state of the project (one of *started*, *stopped*, *starting* or *stopping*), or

124 no projects found

If there are no open projects.

Failure responses

408 Operation failed: ..
420 Access denied

See also

[PROJECT DIR](#), [PROJECT LOAD](#)

4.5.164 PROJECT LOAD Command

PROJECT LOAD

AccessLevel

Admin

Syntax

PROJECT LOAD [project-name [file-name]]

project-name = (optional) name of the project to load. If not specified, the [current project](#) is used.
file-name = (optional) Name of the file the project can be found in. This can only be specified if the **project-name** is given.

Use

Loads a project from disk into the C-Gate server. This makes the project available to be started, or accesses with DB and other commands.

Examples

Normal Opening

```
project load cis  
200 OK.
```

Loading a project of the wrong DBVersion will fail:

```
project load cis  
408 Operation failed: Project load failed: Incompatible project: DBVersion must be  
2.1 - perform a TRANSFORM PROJECT command to fix this
```

Success response

```
200 OK.
```

Failure responses

```
401 Bad Object or Device ID  
408 Operation failed: ...  
420 Access denied  
464 Project already loaded  
465 Project not found
```

408 Operation failed: Project load failed: Incompatible project: DBVersion must be 2.1 - perform a TRANSFORM PROJECT command to fix this

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT CLOSE](#)

4.5.165 PROJECT NEW Command

PROJECT NEW

Access Level

Admin

Syntax

PROJECT NEW project-name

project-name = Name of the project to create.

Use

Creates a new project with the given name.

The project is created under the given name and is initialized in the following way:

- There are no network entries associated with the project
- A blank tag database is created and attached to the project.
- The project is in state stopped.

The new project exists in server memory only. [PROJECT SAVE](#) must be used to save the project to a project file on the server disk.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT LOAD](#)

4.5.166 PROJECT RENAME Command

PROJECT RENAME

Access Level

Admin

Syntax

PROJECT RENAME project-name new-project-name [remove-existing-project-directory ? yes|no]

project-name = Name of the existing project to be rename.

new-project-name = New name of this project. The new name must be a valid project name and use characters allowed in the server's filesystem.

remove-existing-project-directory(optional) = **yes**: If a project with the new name already exists, it will be deleted before renaming.

no (default): If a project with the new name exists, the operation will fail with a "project exists" error.

Use

Renames a project in the project directory on the server disk. **Note:** *This action only renames the on-disk project. A currently loaded and running project cannot be renamed.*

Success response

200 OK.

Failure responses

401 Bad Object or Device ID

408 Operation failed: ...

420 Access denied

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT LOAD](#), [PROJECT DELETE](#)

4.5.167 PROJECT REPAIR

Command

PROJECT REPAIR

Access Level

Admin

Syntax

PROJECT REPAIR project-name

project-name = Name of the project to repair.

Use

Attempts to repair a project that is corrupt or contains invalid data.

This command will:

1. Manually process the file and fix common corruptions in the XML.
2. Execute the transform 'repair.xslt'.
3. Execute the transform 'tidyduplicategroups.xslt'.

A backup copy of the project will be placed in the tag directory.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

PROJECT ARCHIVE, PROJECT DIR, PROJECT SAVE, PROJECT LOAD, PROJECT DELETE

4.5.168 PROJECT RESTORE Command

PROJECT RESTORE

Access Level

Admin

Syntax

PROJECT RESTORE project-name archive-file

project-name = Name of the project that will change name.

archive-file = File to restore from. The filename extension will determine the behaviour of the restore process. If the extension is ".zip", the project will be restored from a ZIP archive (where the tag database is stored as the only entry in the archive, called tagdb.db). If the extension is ".gz", then file will be unzipped. If any other extension is given (including .db), the file will be assumed to be uncompressed DB.

Use

Restores a project from the given archive file to the project name given. The extension of the archive file determines if decompression of the archive file is attempted by this command. See parameters above for details of file extensions and behaviour.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

PROJECT ARCHIVE, PROJECT DIR, PROJECT SAVE, PROJECT LOAD, PROJECT DELETE

**4.5.169 PROJECT SAVE
Command**

PROJECT SAVE

Access Level

Admin

Syntax

PROJECT SAVE [project-name]

project-name = (optional) Name of the project to save. If not specified, the [current project](#) is saved.

file-name = (optional) Name of the file the project will be saved to. This can only be specified if the project-name is given. If not specified, then the filename used for the loading of the project will be used.

Use

Saves a project to disk from the C-Gate server, in a form that can be loaded later.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied
466 Can not save project

See also

[PROJECT DIR](#), [PROJECT SAVE](#), [PROJECT CLOSE](#)

4.5.170 PROJECT START Command

PROJECT START

Access Level

Admin

Syntax

PROJECT START [project-name]

project-name = (optional) Name of the project. Project names are as returned by the [PROJECT DIR](#) command.

Use

Starts a named project (or the current one). This means that C-Gate will open and commence operating with the networks specified in the definition of this project.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

[PROJECT STOP](#)

4.5.171 PROJECT STOP Command

PROJECT STOP

Access Level

Admin

Syntax

PROJECT STOP [project-name]

project-name = (optional) Name of the project to stop. Project names are as returned by the PROJECT DIR command.

Use

Stops the given or current project.

Success response

200 OK.

Failure responses

401 Bad Object or Device ID
408 Operation failed: ..
420 Access denied

See also

[PROJECT START](#)

4.5.172 PROJECT USE Command

PROJECT USE

Access Level

Admin

Syntax

PROJECT USE [project-name]

project-name = (optional) Name of the project. Project names are as returned by the PROJECT DIR command.

Use

Sets the context of the current command session to the project name given. This means that all further addressing of networks, groups etc can be performed without using absolute addressing by using the *//project-name/* prefix. Note that the project must have already been loaded before it can be set to be the default project with the PROJECT USE command. If no *project-name* is given the command returns a line: 123 project=*project-name*

Success response

200 OK.

or, if no project name is given:

123 project=*project-name*

Failure responses

401 Bad Object or Device ID
408 Operation failed: ...
420 Access denied

See also

[PROJECT LIST](#)

4.5.173 QUIT Command

QUIT

Access Level

Connect

Syntax

QUIT | EXIT

Use

Closes the connection to the C-Gate server command interface after sending a final response. Note that this command does not stop the C-Gate server operations or stop operation of the event interface.

Examples

QUIT

EXIT

Successful Response

204 Closing Connection

Failure response

400 Syntax Error

4.5.174 RAMP Command

RAMP

Access Level

Operate

Syntax

RAMP object-identifier ramp-level [ramp-time ["force"]]

ramp-level = 1*3DIGIT [%]; in the range 0-255, or 0-100 if "%" is used

ramp-time = 1*2DIGIT ["s" | "m"]; s for seconds, m for minutes

Use

Ramps the specified group or area to the percentage level using the specified ramp time. Ramp time means the time need to perform this particular ramp operation.

Ramp Level Parameter

The ramp level can be given in an absolute (0..255) or percentage (0%-100%) range.

Ramp Time Parameter

Supported ramp times are between 0 seconds and 17 minutes. This command sends the C-Bus Group RAMP command to the specified network.

If there is no symbol following the ramp time, the time is assumed to be in seconds. If followed by an 's', then seconds are used. Following the time by 'm' means the time is assumed to be in minutes.

The ramp time parameter is optional. If not given, then the ramp time is set to 0 seconds and the ramp is immediate.

Specifying the optional parameter *force*, will allow the command to be sent even if the group is not in an ok state.

Examples

```
RAMP 57/$38/15 0 4s
```

```
RAMP Lobby/$38/12 253 5m
```

```
ramp 57/56/1 255 0
```

```
ramp 1/56/1 100%
```

Successful Response

```
200 OK
```

Failure responses

```
400, 401, 402, 405, 408, 420
```

See also

[ON](#), [OFF](#), [DO](#)

4.5.175 REPORT Command

REPORT

Access Level

Monitor

Syntax

REPORT network-address

Deprecated. Use TREE.

Use

See [TREE](#).

4.5.176 RUN Command

RUN

Access Level

Operate

Syntax

RUN filename [QUIET]

Use

Executes a series of commands in the file specified.

Specifying QUIET means no output or response events are shown unless an error occurs.

Examples

RUN all-lights-off-slowly

Successful Response

203 Run complete.

Failure response

4xx syntax and client errors

410 Macro command error

411 Enable command not allowed in macros

412 Macro loop detected

See also

cgate parameter macro-path

4.5.177 SCENE Command

SCENE

Access Level

Operate

Syntax

SCENE command scene-set scene

command = ("PLAY" | "RECORD")

scene-set = NAME

scene = NAME

Use

Performs functions for scenes created with the Scene Module – see the section on the Scene Module in the C-Gate manual.

Use this command to record or play a scene from the command line.

Examples

SCENE play conf1 allon

Successful Response

200 OK.

Failure response

401, 402, 407, 420

4.5.178 SECURITY Command

SECURITY

Access Level

Operate

Syntax

SECURITY [?]

Use

Lists the SECURITY sub-commands:

SECURITY ARM - Arms a security device.

SECURITY DISPLAY_MESSAGE - Displays a message on the security panel display.

SECURITY EMULATE_KEYPAD - Sends a key press message to the Security system.

SECURITY RAISE_ALARM - Raises an alarm condition with the Security system.

SECURITY REQUEST_ZONE_NAME - Sends a request zone name message to the security system.

SECURITY STATUS_REQUEST - Sends a Security Status Request message to the Security device.

SECURITY TAMPER - Sends a Tamper message to the Security device.

Success Response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure Responses

4.5.179 SECURITY ARM Command

SECURITY ARM

Access Level

Operate

Syntax

SECURITY ARM app arm-mode

app = application address for this application, normally net/\$d0

```
arm-mode =    "away" - arm to away mode
              | "night" - arm to night (home) mode
              | "day" - arm to day mode
              | "vacation" - arm to vacation mode
              | "highest" - arm to highest level of protection
```

Use

Arms a security device.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

**4.5.180 SECURITY_DISPLAY_MESSAGE
Command**

SECURITY_DISPLAY_MESSAGE

Access Level

Operate

Syntax

SECURITY_DISPLAY_MESSAGE app [message]

app = application address for this application, normally net/\$d0
message = ASCII message to display, up to 17 characters long

Use

Displays a message on the security panel display. The message is given as the last argument to the command, and includes all characters to the end of the line. If the message is missing, then a display message will be generated with no display information.

This message may not be supported by some security systems.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

**4.5.181 SECURITY_EMULATE_KEYPAD
Command**

SECURITY_EMULATE_KEYPAD

Access Level

Operate

Syntax

SECURITY EMULATE_KEYPAD app key

app = application address for this application, normally net/\$d0

key = value from 0 to 127 representing an ASCII character, or a value above 127 representing a custom keypad value

Use

Sends a key press message to the Security system. This is equivalent to pressing the given key on the security system's keypad.

This message may not be supported by some security systems.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID

402 Not supported by this object

405 Parameter out of range

408 Operation failed

420 Access denied

4.5.182 SECURITY RAISE_ALARM Command

SECURITY ARM

Access Level

Operate

Syntax

SECURITY RAISE_ALARM app

app = application address for this application, normally net/\$d0

Use

Raises an alarm condition with the Security system.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.183 SECURITY REQUEST_ZONE_NAME Command

SECURITY REQUEST_ZONE_NAME

Access Level

Operate

Syntax

SECURITY REQUEST_ZONE_NAME zone-address

zone-address = address for the zone, normally: net/\$d0 zone-number
zone-number = decimal number, or HEX number (preceded by '\$')

Use

Sends a *request zone name* message to the security system. The security system will report the zone name using a [Zone Name](#) event.

This message may not be supported by some security systems.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.184 SECURITY STATUS_REQUEST Command

SECURITY STATUS_REQUEST

Access Level

Operate

Syntax

SECURITY STATUS_REQUEST app ("1"|"2")

app = application address for this application, normally net/\$d0

Use

Sends a Security Status Request message to the Security device. There are two possible request messages, Status Request 1 and Status Request 2. Select the appropriate number as the argument following the application address.

This command simply performs the request. Applications connected to the C-Gate server that wish to access the results of the status request must receive and decode the [Status Report 1](#) and [Status Report 2](#) event.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.185 SECURITY TAMPER Command

SECURITY TAMPER

Access Level

Operate

Syntax

SECURITY TAMPER app RAISE | DROP

app = application address for this application, normally net/\$d0

Use

Sends a Tamper Raise or Tamper Drop message to the Security device.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID

402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.186 SESSION_ID Command

SESSION_ID

Access Level

Operate

Syntax

SESSION_ID

Use

When a client application (like C-Bus Toolkit or Schedule Plus) connects to C-Gate it is assigned a sequential command session id to identify its connection with C-Gate. This command returns the ID of the client application's command session.

Example

```
session_id  
300 sessionID=cmd3
```

Success Response

```
"300 sessionID="ID
```

ID: The command session id.

Failure Responses

400 Syntax Error

4.5.187 SESSION_ID ALL Command

SESSION_ID ALL

Access Level

Operate

Syntax

SESSION_ID ALL

Use

Returns the ID with extended information of all open command sessions.

Example

```

session_id all
300-sessionID=cmd1 origin=internal from=20130715-135940 tag=Console
300-sessionID=cmd5 origin=MyLaptop/127.0.0.1:63459 from=20130715-142035 tag=C-Bus
Toolkit v1.12.6 build 985
300 sessionID=cmd7 origin=/127.0.0.1:64588 from=20130715-153547

```

Success Response

One or more lines of:

```
"300 sessionID="ID "origin="Origin "from="From ["tag="Tag]
```

ID: The command session id.

Origin: The combination of the host name, ip address and port number of the client connection and the format is 'Host Name'/'IP Address':'Port Number'. In some cases the Host Name field can be blank. For C-Gate's internal console command session it is 'internal'.

From: The time when the client application establishes the connection with C-Gate. The time format is 'YYYYMMDD-HHMMSS'.

Tag: The name and version information about the client application which is reported by the client application using 'SESSION_ID TAG' command. It will not be included in the response if the client application has not reported the information to C-Gate.

Failure Responses

```

400 Syntax Error
500 no session found

```

4.5.188 SESSION_ID TAG Command

SESSION_ID TAG

Access Level

Operate

Syntax

SESSION_ID TAG tag_information

Use

Reports the client application's name and version information to C-Gate which is shown in 'SESSION_ID ALL' command's responses. This command can only be used once per connection.

Example

```
session_id tag C-Bus Toolkit v1.12.6 build 985
200 OK
```

Success Response

```
200 OK
```

Failure Responses

```
400 Syntax Error
408 Operation failed
```

4.5.189 SET Command

SET

Access Level

Operate

Syntax

```
SET object-identifier parameter parameter-value
```

```
parameter = token parameter-value = token
```

Use

Sets parameters in objects. Objects may be part of connected controlled networks, or may be internal to the C-Gate server. Object parameters are described in detail in the guide for the particular object.

Many parameters can be set with the set command. Major general parameters for most or all objects are described here. Specific detail parameters for objects are dealt with in the specific guides for the relevant objects.

Example

```
set #57.1.1 load-power 125

set level42.1 name lobby
```

Successful Response

```
200 OK
```


Failure response

400, 401, 402, 405, 408, 420

See also

[GET or SHOW](#)

4.5.190 SHOW Command

SHOW

Deprecated. Use GET.

Syntax

See [GET](#).

4.5.191 SHUTDOWN Command

SHUTDOWN

Access Level

Admin

Syntax

SHUTDOWN

Use

Shuts down the C-Gate server. The server will **not** restart, so this command must be used with care. Once the command has been confirmed with the [CONFIRM](#) command, the C-Gate server will shut down, closing the command connection.

Examples

SHUTDOWN

Success Response

600 Critical operation. Type CONFIRM to continue.

CONFIRM
206 Shutdown confirmed.

Failure response

420 Access denied.

See also

[CONFIRM](#)

4.5.192 SHORTMESSAGE Command

SHORTMESSAGE

Access Level

Operate

Syntax

SHORTMESSAGE [?]

Use

Lists the SHORTMESSAGE sub-commands:

SHORTMESSAGE REFRESH - Transmit the information message of the given type as soon as possible.

SHORTMESSAGE SEND - Sends a short message.

Success Response

A series of 101 Help lines giving a list of the SHORTMESSAGE commands.

Failure Responses

4.5.193 SHORTMESSAGE REFRESH Command

SHORTMESSAGE REFRESH

Access Level

Operate

Syntax

SHORTMESSAGE REFRESH app info-type

app = application address for this application, normally net/\$AD

info-type = the type of the message (0..\$3F)

- 0 = generic text
- 1 = weather forecast
- 2 = surf report
- 3 = tide times
- 4 = unread emails

5 = voice-mail count
6 = stock prices ticker
7 = news headline
8 = other headlines
9 = air-quality forecast
10 = UV radiation forecast
11 = pollen count
12 = traffic report
13 = sports result
14 = horoscope
15 = electrical brownout warning
16 = irrigation notice

Use

Requests that an information message of the given type be transmitted as soon as possible.

Examples

```
SHORTMESSAGE REFRESH 254/$AD 4
```

Success Response

```
200 OK.
```

Failure Responses

```
401 Bad object or device ID  
402 Not supported by this object  
405 Parameter out of range  
408 Operation failed  
420 Access denied
```

4.5.194 SHORTMESSAGE SEND Command

```
SHORTMESSAGE SEND
```

Access Level

Operate

Syntax

```
SHORTMESSAGE SEND app total index info-type number symbol text
```

app = application address for this application, normally net/\$AD

total = total number of messages (0..7)

index = the number for this message (0..7)

info-type = the type of the message (0..\$3F)

0 = generic text
1 = weather forecast
2 = surf report
3 = tide times
4 = unread emails
5 = voice-mail count
6 = stock prices ticker
7 = news headline
8 = other headlines
9 = air-quality forecast
10 = UV radiation forecast
11 = pollen count
12 = traffic report
13 = sports result
14 = horoscope
15 = electrical brownout warning
16 = irrigation notice

number = a value in the range 0..\$FFFF or a dash (-) if no number portion

symbol = a value in the range 0..\$FF or a dash (-) if no symbol portion

text = a text string of up to 14 bytes coded as UTF-8, which can be quoted or escaped

Use

Sends a short message.

Examples

```
SHORTMESSAGE SEND 254/$AD 7 1 4 - - "test"
```

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.195 STOP Command

STOP

Access Level

Operate

Syntax

STOP command-id

Use

Stops a command that is running in the background of this command session. Commands placed in the background are given a unique ID with special command syntax. See the section on [Unique Commands IDs](#) for details.

Stop only works with commands that specially support it, and all stops will be performed in a safe way.

Example

```
# example of stopping a command
[33] net sync myNet
stop 33
[33] 207 Stopped.
```

Success Response

A response is returned if the command is stopped, using the id of the stopped command.

```
[id] 207 Stopped.
```

Failure Responses

```
400 Syntax Error
401
408 Operation failed: Stop failed: message
420 Access denied
```

4.5.196 TELEPHONY Command

TELEPHONY

Access Level

OPERATE

Syntax

TELEPHONY [?]

Use

Lists the TELEPHONY sub-commands:

TELEPHONY CLEAR_DIVERSION - Instructs the telephony device to clear any diversion currently in place.

TELEPHONY DIVERT - Sends a message to the telephony device requesting a diversion be set to

the given number.

TELEPHONY ISOLATE_SECONDARY_OUTLET - Sends a message to change the isolation behaviour of the secondary outlet of the Clipsal 5500TAU Telephone Interface.

TELEPHONY RECALL_LAST_NUMBER_REQUEST - Asks a telephony device to return the last number called or received.

TELEPHONY REJECT_INCOMING_CALL - Sends a message to the telephony device to reject the current incoming call.

Success response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure responses

A functional copy of C-Gate will always respond to this command

4.5.197 TELEPHONY CLEAR_DIVERSION Command

TELEPHONY CLEAR_DIVERSION

Access Level

Operate

Syntax

TELEPHONY CLEAR_DIVERSION app

Use

Instructs the telephony device to clear any diversion currently in place.

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.198 TELEPHONY DIVERT Command

TELEPHONY DIVERT

Access Level

Operate

Syntax

TELEPHONY DIVERT app number

app = address of the telephony application

number = up to 16 characters that can be interpreted as a telephone number

Use

Sends a message to the telephony device requesting a diversion be set to the given number.

Success response

200 OK

Failure responses

401 Bad object or device ID

402 Not supported by this object

405 Parameter out of range

408 Operation failed

420 Access denied

4.5.199 TELEPHONY ISOLATE_SECONDARY_OUTLET

Command

TELEPHONY ISOLATE_SECONDARY_OUTLET

Access Level

Operate

Syntax

TELEPHONY ISOLATE_SECONDARY_OUTLET app mode

app = address of the telephony application

mode = "normal" | "isolate"

Use

Sends a message to change the isolation behaviour of the secondary outlet of the Clipsal 5500TAU Telephone Interface.

Setting mode to normal will make the secondary outlet behave as normal.

Setting mode to isolate isolates, or switches out, the secondary outlet

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

**4.5.200 TELEPHONY RECALL_LAST_NUMBER_REQUEST
Command**

TELEPHONY RECALL_LAST_NUMBER_REQUEST

Access Level

Operate

Syntax

TELEPHONY RECALL_LAST_NUMBER_REQUEST app type

app = address of the telephony application
direction = "out" | "in"

Use

Requests that a telephony device return the last number called or received. The information is returned in a RECALL_LAST_NUMBER_RESPONSE message, which will be delivered to the Status Change Port and the event list if appropriately configured.

Setting direction to *out* will return the number of the last outgoing call made.

Setting direction to *in* will return the number of the last incoming call received

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.201 TELEPHONY REJECT_INCOMING_CALL Command

TELEPHONY REJECT_INCOMING_CALL

Access Level

Operate

Syntax

TELEPHONY REJECT_INCOMING_CALL app

app = address of the telephony application

Use

Sends a message to the telephony device to reject the current incoming call.

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.202 TEMPERATURE Command

TEMPERATURE

Access Level

Operate

Syntax

TEMPERATURE [?]

Use

Lists the TEMPERATURE sub-commands:
TEMPERATURE BROADCAST - Broadcasts a temperature value on the given Temperature Group.

Success response

A series of lines giving help for these commands in the form:

101-Help: help-information

Failure responses

4.5.203 TEMPERATURE BROADCAST Command

TEMPERATURE BROADCAST

Access Level

Operate

Syntax

TEMPERATURE BROADCAST temp-group-address celsius-temperature ["force"]

temp-group-address = the address of the temperature group to send the temperature for as a C-Gate address, for example //proj/net1/\$19/1

celsius-temperature = temperature in degrees celsius with a maximum of one decimal place, with no units given

Specifying the optional parameter *force* will allow the command to be sent even if the group is not in an ok state.

Use

Broadcasts a temperature value on the given Temperature Group.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.204 TERMINATERAMP Command

TERMINATERAMP

Access Level

Operate

Syntax

TERMINATERAMP object-identifier [force]

Use

Sends a TERMINATERAMP message on the specified C-Bus Group in the Lighting application. This stops any ramping operation currently occurring for the given group.

Specifying the optional parameter force will allow the command to be sent even if the group is not in an ok state.

Example

```
# stop ramping of group 1 on lighting application
terminateramp 1/56/1
```

Success Response

200 OK.

Failure Responses

400 Syntax Error
402
405
408 Operation failed: *message*
420 Access denied

See Also

[RAMP](#)

4.5.205 TEST_SPAM Command

TEST_SPAM

Access Level

PROGRAM

Syntax

TEST_SPAM [?]

Use

Lists the TEST_SPAM sub-commands:

TEST_SPAM EREPORT - Generate random error reporting messages on a network

TEST_SPAM LIGHTING - Generate random lighting messages on a network

TEST_SPAM LIST - List active generation sessions.

TEST_SPAM STOP - Stop an active generation session.

Success response

A series of lines giving help for these commands in the form:

```
101 Help: help-information
```

Failure responses

A functional copy of C-Gate will always respond to this command

4.5.206 TEST_SPAM EREPORT Command

```
TEST_SPAM EREPORT
```

Access Level

```
PROGRAM
```

Syntax

```
TEST_SPAM EREPORT net-address interval [max]
```

interval = number of milliseconds between messages

max = the maximum number of messages to generate

Use

Generate random [error reporting application](#) messages on a C-Bus network.

The messages are completely random and not guaranteed to use the network's actual groups, thus it is expected that some of the messages will have no impact.

Example

```
# Send five messages one second apart
test_spam ereport 254 1000 5
301-id=5
200 OK.
```

Success response

```
200 OK.
```

Failure responses

```
401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: ... (a range of other options here depending on particular
failures)
420 Access denied
```

4.5.207 TEST_SPAM LIGHTING

Command

TEST_SPAM LIGHTING

Access Level

PROGRAM

Syntax

TEST_SPAM LIGHTING net-address interval [max]

interval = number of milliseconds between messages

max = the maximum number of messages to generate

Use

Generate random lighting application messages (on application address 56) on a C-Bus network.

The messages are completely random and not guaranteed to use the network's actual groups, thus it is expected that some of the messages will have no impact.

Example

```
# Send messages 100 milliseconds apart indefinitely until stopped
test_spam lighting 254 100
301-id=4
200 OK.
```

Success response

200 OK.

Failure responses

```
401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: ... (a range of other options here depending on particular
failures)
420 Access denied
```

See also

4.5.208 TEST_SPAM LIST Command

TEST_SPAM LIST

Access Level

PROGRAM

Syntax

TEST_SPAM LIST

Use

Display the currently running spam sessions.

Each row provides the spam session id, the type (lighting or ereport), the network it's running on, the time it was started, the interval in milliseconds, the maximum number of messages (if specified) and the current number of messages already sent.

Example

```
test_spam list
300-id=1 type=lighting network=254 started=20131203-172002 interval=1000 max=-1
current=9
300 id=2 type=ereport network=254 started=20131203-172007 interval=500 max=-1
current=7
```

Success response

300

Failure responses

```
401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: ... (a range of other options here depending on particular
failures)
420 Access denied
```

4.5.209 TEST_SPAM STOP Command

TEST_SPAM STOP

Access Level

PROGRAM

Syntax

```
TEST_SPAM STOP [id]
```

Use

Stop the spam session with the given id.

If no id is given, all spam sessions are stopped.

Example

```
test_spam stop
207-stopped session 1
207-stopped session 2
200 OK.
```

Success response

```
200 OK.
```

Failure responses

```
401 Bad object or device ID: Network not found
401 Bad object or device ID: addressed object is not a network
408 Operation failed: ... (a range of other options here depending on particular
failures)
420 Access denied
```

4.5.210 TOPOLOGY

Command

```
TOPOLOGY
```

Access Level

```
PROGRAM
```

Syntax

```
TOPOLOGY [?]
```

Use

Lists the TOPOLOGY sub-commands:

TOPOLOGY EXPLORE - Instructs the telephony device to clear any diversion currently in place.

Success response

A series of lines giving help for these commands in the form:

```
101 Help: help-information
```

Failure responses

A functional copy of C-Gate will always respond to this command

4.5.211 TOPOLOGY EXPLORE Command

TOPOLOGY EXPLORE

Access Level

PROGRAM

Syntax

TOPOLOGY EXPLORE addresses

addresses = list of space-separated addresses in the form:

```
<type>@address;option=value;option=value"
```

Some address examples are serial@COM1, cni@192.168.1.40:10001,
modem@COM2;number=85431111;username=222;password=2345

Note: A port serial must not be in use, to run this command on it.

Use

Explores the topology of a set of interfaces.

Example

```
TOPOLOGY EXPLORE serial@COM1
```

Success response

A series of lines giving help for these commands in the form:

```
323-Network Found NET0 COM1
321-Network Serial NET0 1048575.4095
324 Bridges Found NET0 0
```

Failure responses

```
472 Can not open network:
```

4.5.212 TREE Command

TREE

Access Level

Monitor

Syntax

TREE network-address

Use

Returns a tree representation of a network.

Examples

TREE 254

Successful Response

Response lines starting with 320 that give human-readable information about the network:

```
320- Network name:253 type=cni address=192.168.1.223:10001 state=ok
320- Unit count=6
320- Units:
320-//HOME/254/p/0 ($0) type=KEYH4A app=56($38), 255($ff), 255($ff), 255($ff)
state=ok groups=
320-//HOME/254/p/7 ($7) type=KEYGL5 app=56($38), 255($ff) state=ok groups=
320-//HOME/254/p/8 ($8) type=RELDN8A app=56($38), 127($7f), 171($ab), 220($dc)
state=ok groups=0,1,2,3,4,5,6,7,8,9,2,3,4,5,6,15
320-//HOME/254/p/9 ($9) type=DIMDN8 app=56($38), 255($ff) state=ok
groups=0,36,250,250,36,36
320-//HOME/254/p/14 ($e) type=PC_CNIEI app=255($ff), 255($ff) state=ok groups=
320-//HOME/254/p/66 ($42) type=PC_NAC2 app=255($ff), 255($ff) state=ok groups=
320-Applications:
320- Application 3 ($3) [application]
320- Groups:
320- Application 56 ($38) [lighting]
320- Groups:
320- //HOME/254/56/0 ($0) level=0 state=ok units=8,9
320- //HOME/254/56/1 ($1) level=0 state=ok units=8
320- //HOME/254/56/2 ($2) level=0 state=ok units=8
320- //HOME/254/56/3 ($3) level=0 state=ok units=8
320- //HOME/254/56/4 ($4) level=0 state=ok units=8
320- //HOME/254/56/5 ($5) level=0 state=ok units=8
320- //HOME/254/56/6 ($6) level=0 state=ok units=8
320- //HOME/254/56/7 ($7) level=0 state=ok units=8
320- //HOME/254/56/8 ($8) level=0 state=ok units=8
320- //HOME/254/56/9 ($9) level=0 state=ok units=8
320- //HOME/254/56/15 ($f) level=0 state=ok units=8
320- //HOME/254/56/36 ($24) level=255 state=ok units=9
320- //HOME/254/56/250 ($fa) level=255 state=ok units=9
320- Application 127 ($7f) [application]
320- Groups:
320- Application 171 ($ab) [application]
```

```
320- Groups:
320- Application 220 ($dc) [application]
320- Groups:
320 -end-
```

Failure response

```
401 Bad object or device ID
402 Not supported by this object
408 Operation failed
420 Access denied
```

See also

[GETSTATE](#), [GET](#), [TREEXML](#)

4.5.213 TREEXML Command

TREEXML

Access Level

Monitor

Syntax

TREEXML network-address ["withsync"] ["withpsync"] ["withqsync"]

Use

Returns an XML-formatted view of an open network and basic information about unit parameters. The *withsync*, *withpsync* and *withqsync* options can be used to initiate the appropriate methods on the unit objects prior to producing the XML output.

Alternatively [TREEXMLDETAIL](#) command can be used to get [<State>](#) and [<OnlineStatus>](#) of units in addition to all the parameters returned for TREEXML command.

Examples

TREEXML 254

Successful Response

A starting line in the form:

```
343-Begin XML snippet
followed by zero or more lines of XML data preceeded by "347-" in the treexml format, followed by:
344 End XML snippet
```

Example

```
343-Begin XML Snippet
```

```
347- <Network>
347- <Unit>
347- <Error>DUPLICATE</Error>
347- <Address>3</Address>
347- </Unit>
347- <Type>KEYH4A</Type>
347- <CatalogueNumber>R5044ZW</CatalogueNumber>
347- <Version>1.3.0</Version>
347- <SerialNo>101273.892</SerialNo>
347- <Address>0</Address>
347- <PartName></PartName>
347- <Application>56, 255, 255, 255</Application>
347- <Groups></Groups>
347- <Voltage>32.9</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- </Unit>
347- <Unit>
347- <Type>RELDN8A</Type>
347- <CatalogueNumber>5508RVF</CatalogueNumber>
347- <Version>1.0.0</Version>
347- <SerialNo>123457.3794</SerialNo>
347- <Address>8</Address>
347- <PartName>NEWUNIT </PartName>
347- <Application>56, 127, 171, 220</Application>
347- <Groups>0,1,2,3,4,5,6,7,8,9,2,3,4,5,6,15</Groups>
347- <Voltage>33.1</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- </Unit>
347- <Unit>
347- <Type>PC_LOCAL</Type>
347- <CatalogueNumber>SLC5500PC</CatalogueNumber>
347- <Version>3.12</Version>
347- <SerialNo>69.13</SerialNo>
347- <Address>19</Address>
347- <PartName>NEWUNIT </PartName>
347- <Application>255, 255</Application>
347- <Groups></Groups>
347- <Voltage>33.6</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- </Unit>
347- <Unit>
347- <Type>GATEWLSN</Type>
347- <CatalogueNumber>SLC5800WCGD</CatalogueNumber>
347- <Version>4.2.00</Version>
347- <SerialNo>1048575.4095</SerialNo>
347- <Address>37</Address>
347- <PartName>W2 </PartName>
347- <Application>56, 255</Application>
347- <Groups></Groups>
347- <Voltage>31.0</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- </Unit>
347- <Unit>
347- <Type>RELDN12</Type>
```

```

347- <CatalogueNumber>SLC5512HRVFP</CatalogueNumber>
347- <Version>1.2.18</Version>
347- <SerialNo>69923.58</SerialNo>
347- <Address>255</Address>
347- <PartName>NEWUNIT </PartName>
347- <Application>56, 255</Application>
347- <Groups>17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32</Groups>
347- <Voltage>31.5</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- </Unit>
347-</Network>
344 End XML Snippet

```

Failure response

```

401 Bad object or device ID
402 Not supported by this object
408 Operation failed
420 Access denied

```

See also

[TREE TREEXMLDETAIL](#)

4.5.214 TREEXMLDETAIL Command

TREEXMLDETAIL

Access Level

Monitor

Syntax

TREEXMLDETAIL network-address ["withsync"] ["withpsync"] ["withqsync"]

Use

Returns an XML-formatted view of an open network and basic information about unit parameters. Comparing to the TREEXML command, TREEXMLDETAIL returns two additional parameters for the unit information, e.g. [<State>](#) and [<OnlineStatus>](#).

The *withsync*, *withpsync* and *withqsync* options can be used to initiate the appropriate methods on the unit objects prior to producing the XML output.

Possible values for [<OnlineStatus>](#) are:

- new: as defined in the network property [<DBUnitAddressesNew>](#)
- missing: as defined in the network property [<DBUnitAddressesMissing>](#)
- online: as defined in the network property [<DBUnitAddressesOnline>](#)
- error: as defined in the network property [<DBUnitAddressesError>](#)
- duplicate: as defined in the network property [<DBUnitAddressesDuplicate>](#)
- unknown: this status is not defined and one possible reason is that the unit address is not in the project database or the MMI report but it has been seen by C-Gate at least once since the project

is opened in C-Gate, e.g. a unit is plugged into the network then unplugged while it is not in the project database

Examples

TREEXMLDETAIL 254

Successful Response

A starting line in the form:

343-Begin XML snippet

followed by zero or more lines of XML data preceded by "347-" in the treexml format, followed by:

344 End XML snippet

Example

343-Begin XML Snippet

347-<Network>

347- <Unit>

347- <Type>KEYH4A</Type>

347- <CatalogueNumber>R5044ZW</CatalogueNumber>

347- <Version>1.3.0</Version>

347- <SerialNo>101273.892</SerialNo>

347- <Address>0</Address>

347- <PartName></PartName>

347- <Application>56, 255, 255, 255</Application>

347- <Groups></Groups>

347- <Voltage>32.9</Voltage>

347- <Burden>no</Burden>

347- <Clock>no</Clock>

347- <State>ok</State>

347- <OnlineStatus>online</OnlineStatus>

347- </Unit>

347- <Unit>

347- <Type>PCINT4</Type>

347- <CatalogueNumber>5500PC</CatalogueNumber>

347- <Version>4.6.00</Version>

347- <SerialNo>100385.2535</SerialNo>

347- <Address>1</Address>

347- <PartName>PCI1 </PartName>

347- <Application>255, 255</Application>

347- <Groups></Groups>

347- <Voltage>30.6</Voltage>

347- <Burden>yes</Burden>

347- <Clock>yes</Clock>

347- <State>ok</State>

347- <OnlineStatus>online</OnlineStatus>

347- </Unit>

347- <Unit>

347- <Error>DUPLICATE</Error>

347- <Address>3</Address>

347- </Unit>

347- <Unit>

347- <Type>PCLOCALU</Type>

```
347- <CatalogueNumber>5500PCU</CatalogueNumber>
347- <Version>4.4.08</Version>
347- <SerialNo>100120.2490</SerialNo>
347- <Address>5</Address>
347- <PartName>PCIU </PartName>
347- <Application>255, 255</Application>
347- <Groups></Groups>
347- <Voltage>31.7</Voltage>
347- <Burden>yes</Burden>
347- <Clock>no</Clock>
347- <State>ok</State>
347- <OnlineStatus>online</OnlineStatus>
347- </Unit>
347- <Unit>
347- <Type>PCINTU</Type>
347- <CatalogueNumber>5500PCU</CatalogueNumber>
347- <Version>5.4.00</Version>
347- <SerialNo>100779.1755</SerialNo>
347- <Address>6</Address>
347- <PartName>TVE212 </PartName>
347- <Application>255, 255</Application>
347- <Groups></Groups>
347- <Voltage>31.8</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- <State>ok</State>
347- <OnlineStatus>new</OnlineStatus>
347- </Unit>
347- <Unit>
347- <Type>RELDN8A</Type>
347- <CatalogueNumber>5508RVF</CatalogueNumber>
347- <Version>1.0.0</Version>
347- <SerialNo>123457.3794</SerialNo>
347- <Address>8</Address>
347- <PartName>NEWUNIT </PartName>
347- <Application>56, 127, 171, 220</Application>
347- <Groups>0,1,2,3,4,5,6,7,8,9,2,3,4,5,6,15</Groups>
347- <Voltage>33.1</Voltage>
347- <Burden>no</Burden>
347- <Clock>no</Clock>
347- <State>ok</State>
347- <OnlineStatus>new</OnlineStatus>
347- </Unit>
347- </Network>
344 End XML Snippet
```

Failure response

401 Bad object or device ID
402 Not supported by this object
408 Operation failed
420 Access denied

See also

[TREE](#) [TREEXML](#)

4.5.215 TRIGGER Command

TRIGGER

Access Level

Operate

Syntax

TRIGGER [?]

Use

Returns help information for the TRIGGER commands

Success response

A series of lines giving help for these commands in the form: 101 Help: help-information

Failure responses

4.5.216 TRIGGER EVENT Command

TRIGGER EVENT

Access Level

Operate

Syntax

TRIGGER EVENT trigger-group-address action-selector

Use

Sends a trigger event message to the network to set the trigger-group given as an address to the value of the given action-selector. This is equivalent to a LIGHTING RAMP immediate command for the lighting application. *action-selector* is either: an integer in the range 0 through 255; or a percentage value (0 - 100) followed by the % sign.

Success response

200 OK

Failure responses

401 Bad object or device ID
402 Not supported by this object

405 Parameter out of range
 408 Operation failed
 420 Access denied

4.5.217 TRIGGER LABEL Command

TRIGGER LABEL

Access Level

Operate

Syntax

TRIGGER LABEL app language group-number action-sel [variant] options

app = application address (normally //proj/net/\$CA)
 language = language code (range 0 through 255)
 group-number = relevant C-Bus group number (range 0 through 255)
 action-sel = action selector | "-" (dash indicates unset)
 variant = F0 | F1 | F2 | F3 (sets the variant for this label)
 options = ("text" text-label) |
 ("icon" icon-selector) |
 ("dynamic" icon-selector icon-width icon-height vertical-offset dynamic-
 icon) |
 ("set_language") |
 options-byte hex-bytes
 text-label = ASCII text to end of line to be used as label, and its maximum length
 is 14 characters
 icon-selector = numeric icon selection in the range 0 through 65535
 icon-width = width of the icon in pixels
 icon-height = height of the icon in pixels
 dynamic-icon = block of ASCII-encoded bytes representing the icon, as 16 lines of
 up to 62 bits, encoded as up to
 16 characters of HEX, with each block separated by a colon character.

If a block is less than 16 characters, the block will be zero-filled on the right.
 For example:

```
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:
AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644:AABBCCDDEEFF6644
```

- options-byte = integer in the range 0-255, used as the value of the options 1 byte in the command.
- hex-bytes = a string of hex bytes to follow the command, and its maximum length is 14 characters (i.e. 28 hex byte characters in the command)

Note: all numeric values except for dynamic icon can be entered as decimal, 0x or \$ prefixed hex, or 0b prefixed binary.

Use

Performs label setting on devices that support dynamic labels.

Note: For devices that support unicode dynamic labels, if a group/variant already has a unicode label it will not accept text labels sent with this command. To set or clear unicode labels see the TRIGGER UNICODELABEL command.

Example

```
#set a text label
trigger label 1/56 1 2 3 text hello
200 OK.
```

```
#set an existing icon
trigger label 1/56 1 2 - icon $1234
200 OK.
```

```
#set a dynamic icon
trigger label 1/56 1 2 - dynamic
0102030405060708:0102030405060708:0102030405060708:0102030405060708:010203040506070
8:0102030405060708:0102030405060708:0102030405060708:0102030405060708:0102030405060
708:0102030405060708:0102030405060708:0102030405060708:0102030405060708:01020304050
60708:0102030405060708
200 OK.
```

```
# change language
trigger label 1/56 1 2 - set_language
200 OK.
```

```
# send a raw command
trigger label 1/56 1 2 - 6 3F2B0102
200 OK.
```

Success Response

200 OK.

Failure Responses

```
400 Syntax Error
401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied
```

4.5.218 TRIGGER UNICODELABEL Command

TRIGGER UNICODELABEL

Access Level

Operate

Syntax

TRIGGER UNICODELABEL app language group-number action-sel variant option text-label|hex-bytes

app = application address (normally //proj/net/\$CA)

language = language code (range 0 through 255)

group-number = relevant C-Bus group number (range 0 through 255)

action-sel = action selector | "-" (dash indicates unset) (should always be dash)

variant = F0 | F1 | F2 | F3 (sets the variant for this label)

option = ("text" text-label) |
("raw" hex bytes string encoded in UTF-8)

text-label = ASCII text to end of line to be used as label if option = text

hex-bytes = a string of hex bytes to follow the command if option = raw

Use

Sets unicode labels on devices that support unicode dynamic labels. The label string should be encoded in UTF-8.

To clear a unicode label simply provide no parameter for text-label or hex-bytes.

Note: Please use the raw option to send unicode characters to C-Gate. Using the text option to send unicode may not always work.

Examples

```
#set a text label
trigger unicodelabel 254/56 1 2 - F0 text hello
200 OK.
```

```
#set a raw hex bytes string
trigger unicodelabel 254/56 1 2 - F0 raw 414243444546
200 OK.
```

Success Response

200 OK.

Failure Responses

400 Syntax Error
401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.5.219 UNLOCK Command

UNLOCK

Access Level

Operate

Syntax

UNLOCK object-identifier

Use

Releases an advisory lock on an object. Such a lock would have been set with the [LOCK](#) command.

Note that the locking is advisory only. Applications must cooperate to make a locking system work.

Examples

UNLOCK p/1/11

Successful Response

226 <object-identifier>: Unlocked

Failure response

400, 401, 420, 425

See also

[LOCK](#)

4.5.220 CONVERTUNIT CHECK Command

CONVERTUNIT CHECK

Access Level

Clipsal

Syntax

"syntax: CONVERTUNIT CHECK <mode> <old address> <unit type> <unit cat>",

"Old device in database. Along with old unit address the target device unit type

and catalog number should be passed".

Example: CONVERTUNIT CHECK 1 254/p/1 "DIMDD8" "5508D1D"

"**syntax:** CONVERTUNIT CHECK <mode> <old address DB> <new address DB>",

"It is assumed that before running the command, the new device has already been created in the database. The address is different from the old one"

Example: CONVERTUNIT CHECK 2 "254/p/1" "254/p/2"

"syntax: CONVERTUNIT CHECK <mode> <old address DB and Network>"

"Physical device is connected to the network, in the same address as the old device in database"

Example: CONVERTUNIT CHECK 3 "254/p/1"

Use

Check whether unit is convertible/compatible.

Examples

Success responses

Response summary:

301 no
200 yes

Failure responses

301 no

See Also

To be linked

4.5.221 CONVERTUNIT CONVERT Command

CONVERTUNIT CONVERT

Access Level

Clipsal

Syntax

Syntax 1: CONVERTUNIT CONVERT <mode> <old address> <unit type> <unit cat>
[Mapping rules set]

Old device in database. Along with old unit address the target device unit type and catalog number should be passed.

Example: CONVERTUNIT CONVERT 1 254/p/1 "DIMDD8" "5508D1D"

Example: CONVERTUNIT CONVERT 1 254/p/1 "RELDN16A" "5516RVF"
"DirectReplacement"

Syntax 2: CONVERTUNIT CONVERT <mode> <old address DB> <new address DB>

It is assumed that before running the command, the new device has already been created in the database. The address is different from the old one

Example: CONVERTUNIT CONVERT 2 "254/p/1" "254/p/2"

Syntax 3: CONVERTUNIT CONVERT <mode> <old address DB and Network>

Physical device is connected to the network, in the same address as the old device in database

Example: CONVERTUNIT CONVERT 3 "254/p/1"

Use

Convert when the unit is compatible or convertible; otherwise, return response as no.

Success responses

Response summary:

301 no
200 ok

Failure responses

401 Bad object or device ID
301 Unable to create lock
301 Unit conversion rules set is not found.
408 Operation failed: No matching unit specification.
408 Operation failed: No Valid parameters.
401 Unable to save data to DB

See Also

To be linked

4.5.222 NAC ROUTING

Command

DBGETJSON NAC_ROUTING_TABLE

Access Level

Admin

Syntax

DBGETJSON NAC_ROUTING_TABLE unit address

Use

Gets the routing table response only for the network bridged from local network.

This command returns:

- a 345-Begin of JSON
- The JSON response
- a 346 End of JSON

Examples

DBGETJSON NAC_ROUTING_TABLE 254/p/5

254 : Network address

5 : SYS_NAC address

Successful Response (example)

345-Begin of JSON

```
346-[
  {
    "destination": 253,
    "route": "FD"
  },
  {
    "destination": 252,
    "route": "FDFC"
  },
  {
    "destination": 251,
    "route": "FDFCFB"
  }
]
347 End of JSON
```

Failure response

401 Invalid unit identifier.
401 More than one units are matched
500 Element not found

**4.5.223 DBGETJSON
Command**

DBGETJSON

Access Level

Admin

Syntax

DBGETJSON [?]

sub-commands:

DBGETJSON NAC_ROUTING_TABLE
DBGETJSON NAC_TAGMAP
DBGETJSON NAC_OBJECTS_LIST

**4.5.224 NAC TAGMAP
Command**

DBGETJSON NAC_TAGMAP

Access Level

Admin

Syntax

DBGETJSON NAC_TAGMAP unit address

Use

It will return JSON response for the object which is included in controller of commission app.

This command returns:

- a 345-Begin of JSON
- The JSON response
- a 346 End of JSON

Examples

DBGETJSON NAC_ROUTING_TABLE 254/p/5

254 : Network address
5 : SYS_NAC address

Successful Response (example)

345-Begin of JSON

```
346-[
  {
    "address": "253",
    "cbusTagMap": {
      "network": "A2"
    }
  },
  {
    "address": "253/202",
    "cbusTagMap": {
      "network": "A2",
      "application": "Trigger Control"
    }
  },
  {
    "address": "253/202/1",
    "cbusTagMap": {
      "network": "A2",
      "application": "Trigger Control",
      "levels": [
        {
          "value": "Action Selector 0",
          "text": "0"
        }
      ]
    }
  }
]
```

347 End of JSON

Failure response

401 Invalid unit identifier.
401 More than one units are matched
500 Element not found

4.5.225 NAC OBJECTS LIST

Command

DBGETJSON NAC_OBJECTS_LIST

Access Level

Admin

Syntax

DBGETJSON NAC_OBJECTS_LIST unit address

Use

It will return JSON response for the object which is included in controller of commission app.

This command returns:

- a 345-Begin of JSON
- The JSON response
- a 346 End of JSON

Examples

DBGETJSON NAC_ROUTING_TABLE 254/p/5

254 : Network address

5 : SYS_NAC address

Successful Response (example)

345-Begin of JSON

```
346-[
{
  "export": true,
  "log": true,
  "address": "0/56/23",
  "datahex": "00FF03FC",
  "cbustagmap": {
    "network": "Locals",
    "application": "Lighting1",
    "levels": [
      {
        "value": 14,
        "text": "Level 14"
      },
      {
        "value": 12,
        "text": "12312"
      }
    ]
  }
}
```

```
    }  
  ],  
  "group": "Group 23"  
},  
"comment": "",  
"highpriolog": true,  
"tags": []  
}  
]  
347 End of  JSON
```

Failure response

401 Invalid unit identifier.
401 More than one units are matched
500 Element not found

4.5.226 LOG EXTRACT Command

LOG EXTRACT

Access Level

Admin

Syntax

LOG EXTRACT number_of_days destination_path

Use

Extracts the server logs into a specified destination folder.

Example

LOG EXTRACT 5 C:\Users\Documents\C-Bus Commission
200 OK.

Success Response

200 OK.

Failure Responses

400 Syntax Error
420 Access denied

301 Invalid directory path
301 Failed to create destination folder

4.6 Configuration

The behaviour of C-Gate is controlled by Configuration Parameters that are found in different places in a hierarchy. Parameters defined lower down in the hierarchy will override the same parameters defined higher up.

The configuration hierarchy is as follows:

1. Internal Configuration

C-Gate has an internal default configuration that is hard-coded. All [Configuration Parameters](#) and their default values are defined here.

2. Global Configuration

This is retrieved from an external [configuration file](#) (e.g. C-GateConfig.txt). These values override the Internal Configuration.

3. Project Configuration

These parameters are stored in the project database and will override the Global Configuration. For more information see [Scope and Scope Objects](#).

4. Network Configuration

These parameters are stored in the project database and will override the Project Configuration. For more information see [Scope and Scope Objects](#).

4.6.1 Configuration File

The Configuration File is the customary method for the user to manipulate Configuration Parameters.

Default File

When C-Gate starts it looks for a Global Configuration from a file called C-GateConfig.txt located in the /config directory.

The first time C-Gate is started it will use its default configuration to generate this file from scratch.

The configuration file can also be re-loaded again with the [CONFIG LOAD](#) command.

Alternate File

The configuration filename can be altered by supplying a new configuration file as the first parameter when C-Gate is run from the command line. For example, where C-Gate might have been run using:

```
java -jar cgate.jar
```

the following command:

```
java -jar cgate.jar c:\alternate\alternateconfig.txt
```

would load an alternate configuration file instead of the normal C-GateConfig.txt.

4.6.2 Scope and Scope Objects

Scope

Configuration parameters belong to one of three levels of scope.

Global parameters are defined only in the [Configuration File](#). They are specific to the C-Gate instance and cannot be defined at the Project or Network level.

Project parameters are copied from the configuration file to a newly created project. After this point the parameters no longer depend on or use the configuration file. Any changes to project parameters are saved to the project file.

Network parameters are also copied to a newly created project. Initially they behave like project parameters and are only stored once in the file. However different values for the parameter can also be stored against individual networks.

To find out a parameter's scope you can look at its page in [Configuration Parameters](#) or read its generated comments in the [Global Configuration](#) file.

Scope Objects

In order to manipulate these three levels of scope C-Gate recognises the following four types of scope objects:

global	This is a literal constant that refers to the global scope.
project	This is a literal constant that refers to the current project in use.
<project>	This is a reference to the project address. See Objects and Addresses .
<network>	This is a reference to the network address. See Objects and Addresses .

C-Gate provides the following commands: [CONFIG OBGET](#) and [CONFIG OBSET](#).

In order to discard a property value, resetting it to a value higher in scope, use the [CONFIG OBRESET](#) command.

Use of the legacy commands [CONFIG GET](#) and [CONFIG SET](#) is discouraged.

4.6.3 Configuration Commands

The values of configuration parameters can be viewed and set using the following commands in the CONFIG command set. Refer to the descriptions in the Command Descriptions section for more details of using these commands:

- [CONFIG](#) - Gives an overview of the CONFIG commands
- [CONFIG GET](#) - Get the value of a configuration parameter
- [CONFIG SET](#) - Set the value of a configuration parameter
- [CONFIG INFO](#) - Get information, including description, scope, default value about a configuration parameter
- [CONFIG LOAD](#) - Load configuration parameters from file or project

- [CONFIG SAVE](#) - Save configuration parameters to file or project
- [CONFIG OBGET](#) - Get the value of a configuration parameter for a scope object
- [CONFIG OBSET](#) - Set the value of a configuration parameter for a scope object
- [CONFIG OBRESET](#) - Reset the value of a configuration parameter for a scope object to the global value.

4.6.4 Configuration Parameters

This sections lists the configuration parameters and explains their use.

4.6.4.1 **accept-connections-from**

Parameter Name

accept-connections-from

Description

This parameter gives a space-separated list of host names or IP addresses that C-Gate will accept connections from. The special value `all` will accept connections from all hostnames and addresses.

This is a separate mechanism to Access Control, and it is recommended that the Access Control features be used for security instead of setting this parameter.

Default Value

`all`

Scope

Global

Effective

Restart

4.6.4.2 **access-control-file**

Parameter Name

access-control-file

Description

This parameter sets the name of the access control file. The file is relative to the

Default Value

`access.txt`

Scope

Global

Effective

Restart

4.6.4.3 allow-fast-start**Parameter Name**

allow-fast-start

Description

Setting this parameter to yes will allow C-Bus Networks to fast start. Fast starting means that when the network is opened, an initial set of units, groups and other network entities are recovered from the project database, and the details are then verified by checking the network, replacing units in the model that don't physically exist. This means that starting C-Gate for a known network can be performed much faster.

In general, turn this on for stable systems, and turn it off when the networks a C-Gate server connects to frequently change.

Default Value

no

Scope

Network

Effective

CloseOpen

4.6.4.4 allow-v3-pci**Parameter Name**

allow-v3-pci

Description

If this parameter is set to yes, C-Gate will take advantage of advanced features in version 3 and later of the C-Bus PCI and CNI modules for network connections. The new features make network operations more reliable.

Leave this set to yes. Setting it to no will decrease C-Bus network reliability.

Default Value

yes

Scope

Global

Effective

CloseOpen

4.6.4.5 application.catalog.filename**Parameter Name**

application.catalog.filename

Description

The filename of the application catalog.

Default Value

applications.xml

Scope

Global

Effective

Restart

4.6.4.6 application.catalog.directory**Parameter Name**

application.catalog.directory

Description

The directory name where the application catalog is found.

Default Value

unitspec

Scope

Global

Effective

Restart

4.6.4.7 auto-reopen**Parameter Name**

auto-reopen

Description

This parameter, if set to yes, ensures that C-Gate will attempt to re-open a connection to a C-Bus network that has failed due to network errors or other problems. If set to no, the network will close as a result of an error and will not reopen without user intervention.

Note for re-sync of the network after auto-reopen: if the network is closed due to network errors or other problems but not 'NET CLOSE' command, after the network is automatically reconnected, a background is scheduled immediately after the reconnection.

Note for bridge network: when the routing network is closed by 'NET CLOSE' command or due to network errors, the bridge network is closed automatically. In either case, C-Gate will try to re-open the connection to the bridge network if auto-reopen is set to yes, thus after the routing network is reconnected by automatic re-open or by 'NET OPEN' command, the bridge network will be reconnected if there is no other issue with the bridge connection.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.8 cbus-application**Parameter Name**

cbus-application

Description

This parameter gives the default application number to be used when a C-Bus Network is asked about defaults. It is given in decimal. The default of 56 is the normal Lighting Application.

Default Value

56

Scope

Network

Effective

CloseOpen

4.6.4.9 cbus-tx-delay**Parameter Name**

cbus-tx-delay

Description

This parameter sets the delay in milliseconds between commands sent to a C-Bus Network if there is no other flow control mechanism in use (like flow control or sync pci). The default value of 250 is the optimal setting for a network where there is only one PC Interface or CNI connected to it. This value should be increased if other PCIs are connected and sending packets into the network.

Default Value

250

Scope

Network

Effective

CloseOpen

4.6.4.10 cbus.tx-cache**Parameter Name**

cbus.tx-cache

Description

If this parameter is set to yes, C-Gate will cache outgoing C-Bus commands if they are identical to a previously sent command that C-Gate is still waiting for a response to. When the original command completes C-Gate will apply the same response to the cached commands. In short the redundant commands are never issued onto C-Bus. This behavior is transparent to C-Gate clients however it can be observed in the C-Gate log.

Note that disabling the cache can result in intermittent but unpredictable behavior due to different

threads executing identical commands.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.11 ccp.display-oids**Parameter Name**

ccp.display-oids

Description

If set to yes, then the [Config Change Port](#) will display Object Identifiers (OIDs) along with config change information.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.12 ccp.display-state**Parameter Name**

ccp.display-state

Description

If set to yes, then the [Config Change Port](#) will send object state change information out to the port.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.13 cgate-name**Parameter Name**

cgate-name

Description

This parameter allows this particular C-Gate server to be given a name that is reported when connections are made to the server.

Default Value

Clipsal C-Gate

Scope

Global

Effective

Restart

4.6.4.14 clock.master**Parameter Name**

clock.master

Description

If set to yes, then the [Clock and Timekeeping Application](#) will operate in master mode.

Default Value

no

Scope

Network

Effective

CloseOpen

4.6.4.15 clock.mastermode**Parameter Name**

clock.mastermode

Description

This parameter sets the type of master mode used by the [Clock and Timekeeping Application](#).

Valid values are primary and secondary. In primary master mode, C-Gate is much more assertive about setting the time for the C-Bus Networks it is connected to.

Default Value

secondary

Scope

Network

Effective

CloseOpen

4.6.4.16 clock.update-interval**Parameter Name**

clock.update-interval

Description

This parameter sets the time in minutes between normal clock and timekeeping updates being sent to the C-Bus Network. *Leave this set at the default value unless you are very sure you want to make it different. Use combinations of [clock.master](#) and [clock.mastermode](#) to change [Clock and Timekeeping Application](#) behaviour.*

Default Value

30

Scope

Network

Effective

CloseOpen

4.6.4.17 cgroups-file**Parameter Name**

cgroups-file

Description

This parameter gives the filename used to hold C-Groups information, relative to the config directory set with the config-path parameter.

Default Value

Cgroups.txt

Scope

Project

Effective

CloseOpen

4.6.4.18 command-local-address**Parameter Name**

command-local-address

Description

If set to an IP address or hostname, this is the address that [Command Interface](#) listens on for connections.

Default Value*The default value is the empty string***Scope**

Global

Effective

Restart

4.6.4.19 command-port**Parameter Name**

command-port

Description

This parameter gives the TCP/IP port that the [Command Interface](#) listens on waiting for connections.

Default Value

20023

Scope

Global

Effective

Restart

4.6.4.20 command.encoding**Parameter Name**

command.encoding

Description

This parameter gives the character encoding to be used on the command, even, Status Change and Config Change Port. Encodings can be chosen to suit devices connected to C-Gate, though operating with utf-8 is likely to give the best integration when working with project databases.

Default Value

utf-8

Scope

Global

Effective

Restart

4.6.4.21 comms-debug**Parameter Name**

comms-debug

Description

If set to yes, this parameter enables logging of all network traffic to disk in a file called comm-debug-<network-name> in the C-Gate home directory.

Leave this set at it's default value of no unless you are asked to change it by Clipsal Staff.

Do not turn this on for fun, as it will cause the server to work hard and it will fill a disk quickly, as it verbosely logs everything going in and out of all network ports.

Default Value

no

Scope

Network

Effective

CloseOpen

4.6.4.22 config-change-port

Parameter Name

config-change-port

Description

This parameter gives the TCP/IP port number for the [Config Change Port](#).

Default Value

20026

Scope

Global

Effective

Restart

4.6.4.23 config-path

Parameter Name

config-path

Description

This parameter sets the path that C-Gate uses to look for configuration files, like C-GateConfig.txt and access.txt. This is given relative to the C-Gate home directory.

Default Value

config

Scope

Global

Effective

Restart

4.6.4.24 console.enable-commands**Parameter Name**

console.enable-commands

Description

If set to yes, allows commands to be entered on the C-Gate console.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.25 default-tag-db**Parameter Name**

default-tag-db

Description

This parameter is deprecated.

Default Value

there is no default value.

Scope

Global

Effective

Restart

4.6.4.26 enable.save-state**Parameter Name**

enable.save-state

Description

If set to yes, this causes the Enable Control Application to save the state of network variables persistantly to disk when they change.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.27 enable-xml-to-sql-background-job**Parameter Name**

enable-xml-to-sql-background-job

Description

If set to yes, this parameter enables the background job of XML projects to SQL projects transformation.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.28 event-filename**Parameter Name**

event-filename

Description

This parameter gives the filename for the an on-disk copy of the event log. This filename is relative to the C-Gate home directory. The given file will have events added to the end of it if parameter [use-event-file](#) is enabled.

Default Value

event . log

Scope

Global

Effective

Restart

4.6.4.29 event-file.event-level**Parameter Name**

event-file.event-level

Description

This parameter controls the event level of the event file independently of [global-event-level](#). The default value is 9. If adverse performance is experienced this value may be decreased to 5.

Default Value

9

Scope

Global

Effective

Restart

4.6.4.30 event-file.keep-days**Parameter Name**

event-file.keep-days

Description

This parameter determines the number of days worth of event-logs to keep. The default value is seven days. Note that there may be several log files for each day.

Note that zero is not a valid value, ie. it is not possible to keep all logs indefinitely.

Default Value

7

Scope

Global

Effective

Restart

4.6.4.31 event-file.split**Parameter Name**

event-file.split

Description

This parameter determines whether the event-file will be split.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.32 event-file.split-count**Parameter Name**

event-file.split-count

Description

This parameter determines the maximum number of split event-files.

Default Value

50

Scope

Global

Effective

Restart

4.6.4.33 event-file.split-size**Parameter Name**

event-file.split-size

Description

This parameter determines the size in bytes at which to split the event-file. The default value is 5,000,000 bytes (4.768 MB)

Default Value

5000000

Scope

Global

Effective

Restart

4.6.4.34 event-host**Parameter Name**

event-host

Description

This parameter sets the name of the host to send events to. This is used when the [event-mode](#) parameter is set to socket.

Default Value

localhost

Scope

Global

Effective

Restart

4.6.4.35 event-millis**Parameter Name**

event-millis

Description

If set to yes, this parameter causes millisecond timing to be added to events.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.36 event-mode**Parameter Name**

event-mode

Description

This parameter selects the mode for serving or sending events. There are two valid values:

server, which will run an event server on event-port and wait for connections, and
socket, which will send the list of events to the host specified in [event-host](#) at port event-port.

Default Value**Scope**

Global

Effective

Restart

4.6.4.37 event-port**Parameter Name**

event-port

Description

This parameter gives the TCP/IP port number that events will be provided on or delivered to. (See [event-mode](#) for more information about this).

Default Value

20024

Scope

Global

Effective

Restart

4.6.4.38 event-printer**Parameter Name**

event-printer

Description

If given a value, this parameter gives the name of the printer to send events to. As such, the parameter is a filename, which can be set to values like LPT1: to redirect output to the printer. If you mis-name the printer, it is likely you will end up with a local file of the name given in the parameter that will continue to grow as C-Gate runs.

Default Value

The default value of this parameter is the empty string.

Scope

Global

Effective

Restart

4.6.4.39 event.display-oids**Parameter Name**

event.display-oids

Description

If set to yes, this parameter causes the OID of the referenced object to be added to event output.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.40 file.base**Parameter Name**

file.base

Description

This parameter sets the base or root directory for the FILE commands. The FILE commands allow the upload and download of files from C-Gate via a command session. The FILE command allow access to any files below this directory.

There are security implications with the settings of this parameter. Be careful when setting it.

Default Value

The default value is . or the directory from which C-Gate was started.

Scope

Global

Effective

Immediate

4.6.4.41 global-event-level**Parameter Name**

global-event-level

Description

This parameter sets the level of events sent to the event port. Valid values are between 0 (no events issued) and 9 (all events, including lots of debugging).

Default Value

5

Scope

Global

Effective

Restart

4.6.4.42 heartbeat-time**Parameter Name**

heartbeat-time

Description

This parameter sets the delay in seconds between heartbeat events being issued. This can be used as a keepalive mechanism or a way of ensuring that C-Gate is still operating. A value of 0 disables heartbeat events.

Default Value

0

Scope

Global

Effective

Restart

4.6.4.43 **hide-project-names**

Parameter Name

hide-project-names

Description

If set to yes, this parameter will hide all project names from event and command output. This is provided to work with legacy environments.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.44 **instance.lock-file**

Parameter Name

instance.lock-file

Description

Filename, relative to the C-Gate base directory, that give the lock file used to ensure that only a single instance of C-Gate is run. To run multiple instances of C-Gate on one computer, make a different configuration file for each instance and start C-Gate specifying the configuration file on the command line.

Default Value

cgate.lock

Scope

Global

Effective

Restart

4.6.4.45 instance.lock-timeout**Parameter Name**

instance.lock-timeout

Description

Time in seconds that an instance lock file (see [instance.lock-file](#)) is still valid. This timeout is kept short, so after a reboot, any old locks will be invalid and won't prevent C-Gate starting.

Default Value

20

Scope

Global

Effective

Restart

4.6.4.46 lighting.learn-update**Parameter Name**

lighting.learn-update

Description

If set to yes, this parameter will allow the Lighting Application to perform database updates as a result of learn mode on the C-Bus Network.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.47 load-change-port**Parameter Name**

load-change-port

Description

This parameter sets the TCP/IP port number used for the [Status Change Port](#).

Default Value

20025

Scope

Global

Effective

Restart

4.6.4.48 local-flow-control

Parameter Name

local-flow-control

Description

If set to yes, flow control will be performed in C-Gate rather than by the serial port. This applies to serial network connections only.

Default Value

no

Scope

Network

Effective

CloseOpen

4.6.4.49 macro-path

Parameter Name

macro-path

Description

This parameter sets the pathname, relative to the C-Gate home directory, where macros are stored.

Default Value

macros

Scope

Global

Effective

Immediate

4.6.4.50 memory-report**Parameter Name**

memory-report

Description

If set to yes, causes memory usage reports to be issued along with heartbeat events. heartbeat-time must be non-zero for memory reports to be issued.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.51 network.application-connect**Parameter Name**

network.application-connect

Description

If set to yes, models bridge Application Connect features by ensuring that any SAL messages sent to a network are relayed to networks via CBus Bridges with Application Connect or Forwarding enabled.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.52 network.error.commands-failed**Parameter Name**

network.error.commands-failed

Description

This parameter defines the number of consecutive commands that need to fail before the network is deemed to be in error.

Default Value

3

Scope

Network

Effective

OpenClose

4.6.4.53 network.error.units-failed**Parameter Name**

network.error.units-failed

Description

This parameter defines the threshold beyond which the number of units that fail during a sync will place the network into error.

Thus for a value of 3, the network will go into error if 4 units fail to sync.

Default Value

3

Scope

Network

Effective

OpenClose

4.6.4.54 network.error.units-failed-hysteresis**Parameter Name**

network.error.units-failed-hysteresis

Description

This parameter controls the number of units that need to recover in the subsequent sync before the network can be taken out of error.

It applies only to networks that are already in error, and is subtracted from the value of *network.error.units-failed* to determine a new threshold for failed units.

For example, if *network.error.units-failed*=3 and *network.error.units-failed-hysteresis*=1, and a network sync failed with four units in error, then the next sync must report two or fewer units in error in order to return to an ok state.

This is useful in cases where an intermittent faulty unit is repeatedly sending a network in and out of error. Using a hysteresis will keep the network in error until the problem is properly fixed.

Default Value

0

Scope

Network

Effective

OpenClose

4.6.4.55 network.pci.poll-interval**Parameter Name**

network.pci.poll-interval

Description

This parameter defines the interval in seconds to poll gateway to check the network connection status and unit online statuses.

Default Value

60

Scope

Network

Effective

OpenClose

4.6.4.56 network.retries**Parameter Name**

network.retries

Description

This parameter gives the default number of retries to be performed by each command sent to a network for a response or for processing. This sets the value of the Retries parameter for all networks when they are opened.

Default Value

2

Scope

Network

Effective

OpenClose

4.6.4.57 network.retries.pci-check**Parameter Name**

network.retries.pci-check

Description

This parameter indicates when to perform a PCI connection check when retrying a failed command. A value of 1 means that the command will be retried once, then a PCI connection check will take place, then the remainder of the retries will take place up to the number specified by *network.retries*. A value of zero means that no PCI connection check will be performed.

Default Value

1

Scope

Network

Effective

OpenClose

4.6.4.58 network.source**Parameter Name**

network.source

Description

This parameter gives the default source for network definitions. Valid values are `db`, for project database derived network definitions, or `file`, for `network.txt` file network definitions.

Default Value

`db`

Scope

Project

Effective

Restart

4.6.4.59 network.state-interval**Parameter Name**

network.state-interval

Description

if non-zero, this parameter gives the interval in seconds between reporting of network state information as an [event 704](#).. A value of 0 disables the reporting.

Default Value

0

Scope

Global

Effective

Restart

4.6.4.60 networks-file**Parameter Name**

networks-file

Description

This parameter gives the name of the file used to hold network definitions. This is prefixed by the project name to make the network filename.

Default Value

networks.txt

Scope

Global

Effective

Restart

4.6.4.61 patch.archive-file**Parameter Name**

patchset.file

Description

This parameter sets the filename for the patch set.

Default Value

patchset.zip

Scope

Global

Effective

Restart

4.6.4.62 pci-flow-control**Parameter Name**

pci-flow-control

Description

If set to yes, enables XON/XOFF flow control in version 3 or later PC Interfaces.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.63 pci.local-sal**Parameter Name**

pci.local-sal

Description

If set to yes, places version 3 or later PCI or CNi devices into Local_SAL mode, which makes application messaging compatible with bridge Application Connect messaging.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.64 pp.spec-base-directory**Parameter Name**

pp.spec-base-directory

Description

This parameter sets the base directory for Unit Specifications.

Default Value

unitspec/

Scope

Global

Effective

Restart

4.6.4.65 project.default

Parameter Name

project.default

Description

This parameter sets the name of the default project.

Default Value

system

Scope

Global

Effective

Restart

4.6.4.66 project.default.archive-dir

Parameter Name

project.default.archive-dir

Description

This parameter sets the default directory used for archiving project using the PROJECT ARCHIVE and PROJECT RESTORE commands.

Default Value

tag/archived

Scope

Global

Effective

Restart

4.6.4.67 project.default.dir**Parameter Name**

project.default.dir

Description

This parameter gives the default directory to look for project database files in. This is relative to the C-Gate home directory.

Default Value

tag/

Scope

Global

Effective

Restart

4.6.4.68 project.start**Parameter Name**

project.start

Description

This parameter gives a space-separated list of projects to start when C-Gate is started. If there are no projects listed, then no projects will be started.

Default Value

system

Scope

Global

Effective

Restart

4.6.4.69 reopen-delay**Parameter Name**

reopen-delay

Description

This parameter gives the delay time in milliseconds between attempts to open a network that has been closed or failed during opening.

Default Value

15000

Scope

Network

Effective

CloseOpen

4.6.4.70 report-new-objects**Parameter Name**

report-new-objects

Description

If set to yes, this parameter will cause events to be issued indicating when new units are located on the C-Bus Networks that C-Gate has connected to.

Default Value

no

Scope

Network

Effective

CloseOpen

4.6.4.71 response-delay**Parameter Name**

response-delay

Description

This parameter gives the time in milliseconds to wait for a response from a C-Bus Network. If no response is received in this time, then retries will be initiated.

Default Value

3000

Scope

Network

Effective

CloseOpen

4.6.4.72 scene-base**Parameter Name**

scene-base

Description

This parameter gives the base directory for holding Scenes. This is relative to the C-Bus home directory.

Default Value

scene

Scope

Global

Effective

Restart

4.6.4.73 secure.bind-address**Parameter Name**

secure.bind-address

Description

This parameter gives the local address to bind [SSL Connections](#) to. This can be given as a hostname or IP address. The default value is the empty string, meaning bind to all local connections.

Default Value

The default value is the empty string.

Scope

Global

Effective

Restart

4.6.4.74 secure.port-base**Parameter Name**

secure.port-base

Description

This parameter gives the base TCP/IP port number for the SSL Connection. The SSL Connections are all indexed from this base. So, if secure.port-base=20123, the default, then:

- command port is on base+0 or 20123
- event port is on base+1 or 20124
- status change port (SCP) is on base+2 or 20125
- config change port (CCP) is on base+3 or 20126

Default Value

20123

Scope

Global

Effective

Restart

4.6.4.75 serial.fixbaud**Parameter Name**

serial.fixbaud

Description

This parameter, valid for serial network types, allows C-Gate to attempt to perform baud-rate scanning and changing the baud rate of a serial connection if there is no response at the standard 9600bps.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.76 speed-write**Parameter Name**

speed-write

Description

If set to yes, this parameter enables fast writing of lighting and other application commands to the connected C-Bus networks, by performing write caching and command concatenation. This results in much higher throughput to the C-Bus Networks, though commands are queued and OK responses are given immediately, meaning that individual errors will go undetected if the client does not watch the output from the event port.

Use speed-write=yes in situations where a client application wants to send a large block of commands to C-Gate and requires quick responses.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.77 startup-delay**Parameter Name**

startup-delay

Description

This parameter is DEPRECATED. Do not use.

Default Value

0

Scope

Network

Effective

CloseOpen

4.6.4.78 sweep-timeout**Parameter Name**

sweep-timeout

Description

This parameter gives the delay in milliseconds between executions of the queue sweeper.

Default Value

2000

Scope

Network

Effective

CloseOpen

4.6.4.79 sync-fast-pci**Parameter Name**

sync-fast-pci

Description

If set to yes, this parameter causes network connections to be operated synchronously when supported by a PC Interface or CNI device. In general, this should be left set to yes to ensure connection integrity.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.80 sync-time**Parameter Name**

sync-time

Description

This parameter gives the time in seconds over which a network will be synchronised. As network synchronisation is spaced out to perform continuous sync operations, this is the maximum time that will occur before a unit is re-synchronised.

Note: This option

Note: C-Bus Toolkit provides a "C-Gate Config Options" dialog that can directly modify this value.

Default Value

3600

Scope

Network

Effective

CloseOpen

4.6.4.81 sync.free-periods**Parameter Name**

sync.free-periods

Description

This parameter defines a set of periods during which background syncs are avoided. This is done by means of a comma-delimited string which represents one or more schedules.

Only weekly schedules are supported at this time. A weekly schedule takes the format:

W[:<days-of-week>]:<period>

<days-of-week> is in one of two formats:

<dow-int> a decimal integer [1..127] representing a bitmask of days from Sunday to Saturday.

<dow-bit> seven binary digits representing days from Sunday to Saturday, eg. 1010101.

These formats are equivalent, for example:

```
0000001 (bin) == 1 (dec) == Saturday
0111110 (bin) == 62 (dec) == Monday to Friday
1000001 (bin) == 65 (dec) == Saturday and Sunday
1111111 (bin) == 127 (dec) == Every day of the week
```

If the value is zero, then no days are active ie. the schedule is 'deactivated'.

If this field is not provided, then every day is assumed.

<period> is in one of two formats:

```
<start-time>-<end-time>           where all times
and durations are hours and minutes
<start-time>+<duration>           in military time,
eg. 0800, 1030, 0200.
```

These formats are equivalent and are both supported for convenience.

Some useful examples are:

W:0111110:0800-1000	No background syncing between
8am and 10am on weekdays.	
W:62:0800+0200	Equivalent to the above.
W:0:0800+0200	Schedule is deactivated.
W:0800+0200	No background syncing between
8am and 10am every day.	
W:0111110:0800-1000,W:2:1600+0200	No background syncing between
8am and 10am on weekdays,	
	or between 4pm-6pm Fridays.

For more information on how this parameter is used by C-Gate, see [Sync-Free Periods](#).

Default Value

<blank>

Scope

Network

Effective

CloseOpen

4.6.4.82 sync.global-pool-size

Parameter Name

sync.global-pool-size

Description

This parameter controls the maximum number of background network syncs that may be in progress at any one time. Too much network sync traffic can saturate the network infrastructure between C-Gate and the C-Bus networks or it can affect the performance of the C-Gate software, its clients, or the machine C-Gate is running on.

Default Value

25

Scope

Global

Effective

Restart

4.6.4.83 sync.padding.enabled**Parameter Name**

sync.padding.enabled

Description

This parameter enables the [Sync Padding](#) feature.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.84 sync.padding.sync-time-factor**Parameter Name**

sync.padding.sync-time-factor

Description

When [sync.padding.enabled](#) is yes, this parameter provides a multiplier that is applied to [sync-time](#)

to determine the amount of time over which to space syncs.

It should not be less than zero or exceed 1.0.

Default Value

0.75

Scope

Network

Effective

CloseOpen

4.6.4.85 sync.gateway-pool-size**Parameter Name**

sync.gateway-pool-size

Description

This parameter controls the maximum number of background network syncs that may be in progress at any one time through a single C-Gate gateway device (PCI, CNI, etc). This only applies when network bridges are in use to connect to more than one network behind a gateway device. Too much network sync traffic can overwhelm the limited processing capability in the gateway device.

Default Value

3

Scope

Global

Effective

Restart

4.6.4.86 tag-autosave**Parameter Name**

tag-autosave

Description

If set to yes, this parameter causes a project database modified by a learn update to be automatically saved to disk.

Default Value

no

Scope

Global

Effective

Immediate

4.6.4.87 tag-name-output**Parameter Name**

tag-name-output

Description

If set to yes, this parameter causes tag names to be output in place of addresses in events, and the config and status ports.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.88 tag-use-zip**Parameter Name**

tag-use-zip

Description

If set to yes, this parameter causes project databases to be stored in ZIP format.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.89 tag-validate-db

Parameter Name

tag-validate-db

Description

If set to yes, this parameter causes XML validation to always be performed on project databases.

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.90 unit-auto-update-db

Parameter Name

unit-auto-update-db

Description

If set to yes, this parameter causes units to be automatically updated to the tag database as they are synchronised.

Default Value

no

Scope

Network

Effective

Immediate

4.6.4.91 unit-auto-delete**Parameter Name**

unit-auto-delete

Description

If set to yes, this parameter causes units to be automatically deleted from the in-memory network model of the unit if no matching unit is found on the corresponding C-Bus Network.

Default Value

yes

Scope

Network

Effective

Immediate

4.6.4.92 unitcatalog.directory**Parameter Name**

unitcatalog.directory

Description

This parameter gives the directory where the Unit Catalog is stored. This is relative to the C-Gate home directory)

Default Value

unitspec

Scope

Global

Effective

Restart

4.6.4.93 unitcatalog.filename**Parameter Name**

unitcatalog.filename

Description

This parameter gives the filename of the Unit Catalog. This is in the directory given by [unitcatalog.directory](#).

Default Value

cbusunits.xml

Scope

Global

Effective

Restart

4.6.4.94 use-1.0-addressing**Parameter Name**

use-1.0-addressing

Description

If set to yes, this parameter causes the server to use addresses in the form of C-Gate 1.0 in all output.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.95 use-cgroups**Parameter Name**

use-cgroups

Description

If set to yes, this parameter enables C-Groups.

Default Value

no

Scope

Project

Effective

CloseOpen

4.6.4.96 use-config-change-port**Parameter Name**

use-config-change-port

Description

If set to yes, this parameter enables the [Config Change Port](#).

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.97 use-event-file**Parameter Name**

use-event-file

Description

If set to yes, this parameter enables the storage of events as a file in the local filesystem. This filename is determined by [event-filename](#).

Default Value

yes

Scope

Global

Effective

Restart

4.6.4.98 use-load-change-port

Parameter Name

use-load-change-port

Description

If set to yes, this parameter enables the Status Change Port.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.99 use-queue-sweeper

Parameter Name

use-queue-sweeper

Description

If set to yes, this parameter enables the queue sweeper, which clears up networking issues and packet failures. This should be left set to yes, or problems may result.

Default Value

yes

Scope

Network

Effective

CloseOpen

4.6.4.100 use-scenes**Parameter Name**

use-scenes

Description

If set to yes, this parameter enables the scene functions in C-Gate.

Default Value

no

Scope

Global

Effective

Restart

4.6.4.101 use-tags**Parameter Name**

use-tags

Description

If set to yes, this parameter enables the use of project databases. Leave this set to yes for all normal operation.

Default Value

yes

Scope

Global

Effective

Restart

4.7 C-Bus Applications

4.7.1 Specifying Applications Application Addresses

All C-Bus applications are identified in C-Bus messages by an application address, an address in the range 0 (\$0) through 255 (\$FF). The application address is used to associate the particular message with an application on the network.

Application Catalog

The C-Gate server associates Application classes (that is, code to process application messages and commands) with actual applications using the application catalog, normally called `applications.xml`. This file is normally located in the same place as unit specifications, the `unitspec` folder.

The application catalog is loaded into the C-Gate server the first time any application is referred to.

For the format and usage of the catalog, see the [applications.xml](#) section in this manual.

Extension mechanism (`application.spe`)

A file called `application.spe` can be placed in the `config` directory to allow application definitions to be altered at a specific site only. C-Gate does not ship with this file in the `config` directory.

This file is scanned for application definitions and these take precedence over the definitions in the `applications.xml` file.

See the section on `application.spe` override for details of the file format and extension mechanism.

Best Practice Guide

*Use the **extension mechanism provided by the `applications.spe` override**, rather than editing the `applications` catalog itself to make changes to the way C-Gate maps applications to application classes.*

4.7.1.1 `applications.xml`

The application catalog, `applications.xml`, is the resource that C-Gate uses to resolve network application numbers to decode and encode application messages.

Configuration variables

The name of the application catalog is determined by the value of the `application.catalog.filename` parameter (by default, `applications.xml`) contained in the `application.catalog.directory` directory (default `unitspec`). These values can be changed for special applications if needed. However, general useage and best practice suggest leaving these set to their defaults and using the `application.spe` override mechanism for specific intergrated applications.

File Format

This is a short but complete `applications.xml` file. Refer to the actual example in the `config` directory.

```

<Applications>
  <Application>
    <Name>Network</Name>
    <Description>C-Bus Network Management and Control</Description>
    <AddressRangeStart>$FF</AddressRangeStart>
    <AddressRangeEnd>$FF</AddressRangeEnd>
    <DefaultAddress>$FF</DefaultAddress>
    <CGateClass>CBusNetManagementApplication</CGateClass>
    <AllowMMIs>No</AllowMMIs>
    <AutoCreate>No</AutoCreate>
    <Editable>No</Editable>
    <HasGroups>No</HasGroups>
    <LightingCompatibility>None</LightingCompatibility>
    <NagIfUsed>No</NagIfUsed>
    <RenameAllowed>No</RenameAllowed>
  </Application>
  <Application>
    <Name>lighting</Name>
    <Description>C-Bus Lighting Application</Description>
    <AddressRangeStart>$30</AddressRangeStart>
    <AddressRangeEnd>$5F</AddressRangeEnd>
    <DefaultAddress>$38</DefaultAddress>
    <CGateClass>CBusLightingApplication</CGateClass>
    <AllowMMIs>Yes</AllowMMIs>
    <AutoCreate>Yes</AutoCreate>
    <Editable>No</Editable>
    <HasGroups>No</HasGroups>
    <LightingCompatibility>Complete</LightingCompatibility>
    <NagIfUsed>No</NagIfUsed>
    <RenameAllowed>No</RenameAllowed>
    <KeyMacrofunctions>
      <KeyMacrofunction>
        <Name>On/Off</Name>
        <Microfunction>
          <Code>SP</Code>
          <Function>TOGGLE</Function>
        </Microfunction>
        <Microfunction>
          <Code>SR</Code>
          <Function>TOGGLE</Function>
        </Microfunction>
        <Microfunction>
          <Code>LP</Code>
          <Function>TOGGLE</Function>
        </Microfunction>
        <Microfunction>
          <Name>LR</Name>
          <Function>TOGGLE</Function>
        </Microfunction>
      </KeyMacrofunction>
    </KeyMacrofunctions>
  </Application>
</Applications>

```

Element Descriptions

An <Application> element contains a definition of an application. It contains the following elements:

<Name> gives the short name used by C-Gate to define the applications. This short name is used to prefix commands and it used to prefix events shown in the event stream and status change stream. This name should be short, have no whitespace, and be in lowercase. It needs to be unique.

<Description> gives a one-sentence description of the application. This is not currently used by C-Gate. This can contain whitespace.

<AddressRangeStart> gives the lowest application address that this application uses. This is in the range 0-255 or \$0 - \$FF.

<AddressRangeEnd> gives the highest application address that this application uses. This is in the range 0-255 or \$0 - \$FF.

<DefaultAddress> gives the normal, default address for this application. This is in the range 0-255 or \$0 - \$FF.

<CGateClass> gives the java classname that C-Gate will load to handle this application. If a fully qualified class path name is given, that is used. Otherwise the class is interpreted relative to com.clipsal.cgate.cbus.app.

The remaining elements are used internally by C-Bus Toolkit and other Clipsal software. These elements are optional.

<AllowMMIs>, if set to yes, allows MMIs to be used on this application.

<AutoCreate>, if set to yes, indicates that the application is automatically created in C-Bus Toolkit projects.

<Editable>No</Editable>, if set to yes, allows the application details to be edited in C-Bus Toolkit.

<HasGroups>No</HasGroups>, if set to yes, indicates that this application has groups.

<LightingCompatibility> indicates the level of lighting application support this application can use. Valid values are: complete, partial, or none.

<NagIfUsed>, if set to yes, ensures that C-Bus Toolkit will verify use of this application during commissioning.

<RenameAllowed>, if set to yes, allows this application to be renamed in projects.

The <KeyMacrofunctions> block gives hints to C-Bus Toolkit for the default definitions used in key input devices. It contains zero or more <Macrofunction> blocks.

Each <Macrofunction> block contains a <Name> and one or more <Microfunction> blocks.

Each Microfunction block has a <Code> and <Function> Element.

Failure modes

C-Gate will fall back to looking for the application.spe file if the applications.spe file is not found.

Historical Note

The application catalog replaces the application.spe file used internally in C-Gate as of version 2.5. The application.spe override is still supported, however.

4.7.1.2 applications.spe override

The application.spe file, contained in the config directory, provides a mechanism to allow individual installations to override the definitions of the application catalog. This is a plain text file with lines that define the behaviour of C-Gate for particular application addresses.

Warning: *by creating overly general overrides in this file, you can stop C-Gate working properly with normal applications.*

Lines starting with # are comments, and entries define the following fields, with fields separated by whitespace:

name gives the short name of the application -- this is used as the command prefix and for event prefixes as well.

low and **high** give the range of application numbers that will be recognised as this application. You can see that lighting has a range of application numbers attached to it.

default is the default application number, which will be used when the application is created other than from a network event.

class to load gives the name of the application class (must be a subclass of CBusApplication). If no full package is given on the class, it is assumed to be in the com.clipsal.cgate.cbust.app package. Alternatively if a full package and class is given, that is loaded.

description gives a description of the application.

Some sample application.spe entries are given below:

#name	low	high	default	class to load	description
network	\$FF	\$FF	\$FF	CBusNetManagementApplication	C-Bus Network
Management and Control					
lighting	\$30	\$5F	\$38	CBusLightingApplication	*C-Bus Lighting
Application					
heating	\$88	\$88	\$88	CBusLightingApplication	*C-Bus Heating
Application (Clipsal 5 Star)					
rsc	\$26	\$26	\$26	CBusLightingApplication	*Room Control System
(Clipsal 5 Star)					
security	\$D0	\$D0	\$D0		
com.clipsal.cgate.cbust.app.security.CBusSecurityApplication					Security

If no overrides are required, then this file can be empty or missing.

Best practice guide

Only ever use this mechanism if there is no way to renumber C-Bus units to fit into the regular application framework. Renumbering the units will allow other application-aware applications to

easily work with the network. Redefining standard applications can cause serious C-Bus malfunctions.

4.7.2 Airconditioning Application

The Airconditioning Application is used to manage airconditioned zones.

A number of events are provided by HVAC devices and a number of commands are available to control HVAC devices via this application.

The Airconditioning Application has a C-Bus application address of \$AC, or 172 decimal.

4.7.2.1 Airconditioning Application Overview

This gives a short overview of the Airconditioning Application implementation in the C-Gate server

Application name

Airconditioning

Application short name

(used as first part of commands or events to refer to this application)

aircon

C-Bus Application range

\$AC (172)

Commands

(these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

set_humidity_setback_limit
set_humidity_lower_guard_limit
set_humidity_upper_guard_limit
set_hvac_lower_guard_limit
set_hvac_setback_limit
set_hvac_upper_guard_limit
set_ward_off
set_ward_on
set_zone_humidity_mode
set_zone_hvac_mode

Events

(these are detected by the C-Gate server and will be shown in the event list and status port but are not available as commands)

humidity_schedule_entry
hvac_schedule_entry
zone_humidity
zone_humidity_plant_status
zone_hvac_plant_status
zone_temperature

Notes and implementation issues

4.7.2.2 Objects

One new object is provided for the implementation of this application:

Airconditioning Application

This is the application object created in the server when an airconditioning command is detected on the C-Bus network, or if an airconditioning command is sent from the command line. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
ShortName	The short name for this application. This is set to aircon when the application is loaded	Monitor	Read Only

4.7.2.3 Events

The events in this section are recognised by the Airconditioning application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP)

The events can be divided into two main groups: Status Events, which emit information about things happening on the application, and Command Events, which indicate that an Airconditioning command has been executed on the application.

4.7.2.3.1 Status Events

Airconditioning Status Events emit information about things happening on the application.

4.7.2.3.1.1 Humidity Schedule Entry

Event

Humidity Schedule Entry

Condition causing event

Setting a humidity schedule entry.

Event text

humidity_schedule_entry app ward zone-list entry format mode raw-flag setback-enabled guard-enabled use-aux-level start-time level

app = air-conditioning application address
 ward = air-conditioning ward number (0-255)
 zone-list = comma-separated list of zone numbers (max 7)
 entry = entry number (0..255)
 format = format (1..7)
 mode = humidity mode (0..3)

raw-flag = 0 if level is a humidity percentage, 1 if level is a raw value
 setback-enabled = 1 for setback enabled
 guard-enabled = 1 for guard enabled
 use-aux-level = 1 if aux-level is used, else 0 for automatic operation
 start-time = time this schedule starts
 level = humidity % value or raw level

- format:

- 1 = 4 periods/day, all days same
- 2 = 4 periods/day, week/weekend
- 3 = 4 periods/day, each day different
- 4 = 6 periods/day, all days same
- 5 = 6 periods/day, week/weekend
- 6 = 6 periods/day, each day different
- 7 = no fixed number of periods

- mode:

- 0 = off
- 1 = humidify only
- 2 = dehumidify only
- 3 = humidity control

- start-time: number of minutes from Sunday 00:00 (ie Sunday 00:00 = 0)

- humidity level is 0% to 100%

- raw humidity level can be a fraction of plant capacity eg 50% or -10%

Sample Event output

```
20081022-084833 702 sys [aircon] humidity_schedule_entry ward=1 zones=0,1,2 entry=0
format=1 mode=3 rawlevel=0 setbackenabled=0 guardenabled=0 useauxlevel=0
starttime=750 setlevel=13107 sourceUnit=0
```

Sample SCP output

```
# aircon humidity_schedule_entry //BOARD/254/172 1 0,1,2 0 1 3 0 0 0 0 750 13107
#sourceunit=0 OID=...
```

4.7.2.3.1.2 HVAC Schedule Entry

Event

HVAC Schedule Entry

Condition causing event

Setting a HVAC schedule entry.

Event text

```
hvac_schedule_entry app ward zone-list entry format mode raw-flag setback-enabled
guard-enabled use-aux-level start-time level
```

app = air-conditioning application address
 ward = air-conditioning ward number (0-255)
 zone-list = comma-separated list of zone numbers (max 7)
 entry = entry number (0..255)
 format = format (1..7)
 mode = HVAC mode (0..4)
 raw-flag = 0 if level is a temperature, 1 if level is a raw value
 setback-enabled = 1 for setback enabled
 guard-enabled = 1 for guard enabled
 use-aux-level = 1 if aux-level is used, else 0 for automatic operation
 start-time = time this schedule starts
 level = temperature or raw level

- format:

- 1 = 4 periods/day, all days same
- 2 = 4 periods/day, week/weekend
- 3 = 4 periods/day, each day different
- 4 = 6 periods/day, all days same
- 5 = 6 periods/day, week/weekend
- 6 = 6 periods/day, each day different
- 7 = no fixed number of periods

- mode:

- 0 = off
- 1 = heating only
- 2 = cooling only
- 3 = heating & cooling
- 4 = fan only

- start-time: number of minutes from Sunday 00:00 (ie Sunday 00:00 = 0)
- temperature is in degrees centigrade * 256
- raw temperature level can be a fraction of plant capacity eg 50% or -10%

Sample Event output

```

20081022-093613 702 sys [aircon] hvac_schedule_entry ward=1 zones=0,1,2 entry=0
format=7 mode=3 rawlevel=0 setbackenabled=0 guardenabled=0 useauxlevel=0
starttime=1500 setlevel=5120 sourceUnit=0
  
```

Sample SCP output

```

# aircon hvac_schedule_entry //BOARD/254/172 1 0,1,2 0 7 3 0 0 0 0 1500 5120
#sourceunit=0 OID=...
  
```

4.7.2.3.1.3 Zone Humidity Plant Status

Event

Zone Humidity Plant Status

Condition causing event

Reporting the status of a humidity plant.

Event text

zone_humidity_plant_status app ward zone-list type status error

app = air-conditioning application address
 ward = air-conditioning ward number (0-255)
 zone-list = comma-separated list of zone numbers (max 7)
 type = type of humidity plant (0..3)
 status = humidity plant status (0..255)
 error = humidity plant error (0..11)

- type:
 - 0 = none
 - 1 = evaporative
 - 2 = refrigerative
 - 3 = both
- status: a cumulative bitmask of Humidity Plant Status (0..255)
 - 1 = humidifier active
 - 2 = dehumidifier active
 - 4 = fan active
 - 8 = damper open
 - 16 = (unused)
 - 32 = busy
 - 64 = error
 - 128 = expansion
- error:
 - 0 = none
 - 1 = humidifier failure
 - 2 = dehumidifier failure
 - 3 = fan failure
 - 4 = sensor failure
 - 5 = humidifier problem
 - 6 = dehumidifier problem
 - 7 = fan problem
 - 8 = humidifier service required
 - 9 = dehumidifier service required
 - 10 = fan service required
 - 11 = filter replacement required

Sample Event output

```
20081021-152001 702 sys [aircon] zone_humidity_plant_status ward=1 zones=1,2 type=3
status=1 error=0 sourceUnit=0
```

Sample SCP output

```
# aircon zone_humidity_plant_status //BOARD/254/172 1 1,2 3 1 0 #sourceunit=0
OID=...
```

4.7.2.3.1.4 Zone HVAC Plant Status

Event

Zone HVAC Plant Status

Condition causing event

Reporting the status of a HVAC plant.

Event text

zone_hvac_plant_status app ward zone-list type status error

app = air-conditioning application address

ward = air-conditioning ward number (0-255)

zone-list = comma-separated list of zone numbers (max 7)

type = type of HVAC plant (0..10,255)

status = HVAC plant status (0..255)

error = HVAC plant error (0..11)

- type:
 - 0 = none
 - 1 = furnace
 - 2 = evaporative
 - 3 = reverse cycle
 - 4 = heat-pump heating
 - 5 = heat-pump cooling
 - 6 = furnace/evap
 - 7 = furnace/heat-pump cooling
 - 8 = hydronic
 - 9 = hydronic/heat-pump cooling
 - 10 = hydronic/evap
 - 255 = any
- status: a cumulative bitmask of HVAC plant status (0..255):
 - 1 = cooler active
 - 2 = heater active
 - 4 = fan active
 - 8 = damper open
 - 16 = (unused)
 - 32 = busy
 - 64 = error
 - 128 = expansion
- error:
 - 0 = none
 - 1 = heater failure
 - 2 = cooler failure
 - 3 = fan failure
 - 4 = sensor failure
 - 5 = heater problem
 - 6 = cooler problem
 - 7 = fan problem
 - 8 = heater service required
 - 9 = cooler service required
 - 10 = fan service required

11 = filter replacement required

Sample Event output

```
20081022-101347 702 sys [aircon] zone_hvac_plant_status ward=1 zones=0,1,2 type=3
status=1 error=0 sourceUnit=0
```

Sample SCP output

```
# aircon zone_hvac_plant_status //BOARD/254/172 1 0,1,2 3 1 0 #sourceunit=0 OID=...
```

4.7.2.3.1.5 Zone Humidity

Event

Zone Humidity

Condition causing event

Reporting the humidity of HVAC zones.

Event text

zone_humidity app ward zone-list level sensor-status

app = air-conditioning application address
 ward = air-conditioning ward number (0-255)
 zone-list = comma-separated list of zone numbers (max 7)
 level = humidity in % * 655.35 (ie 100% = 65535)
 sensor-status = humidity plant status (0..3)

- sensor-status:
 - 0 = no error
 - 1 = sensor operating in relaxed accuracy
 - 2 = sensor out of calibration
 - 3 = sensor total failure (humidity reading is meaningless)

Sample Event output

```
20081022-124135 702 sys [aircon] zone_humidity ward=1 zones=0,1,2 level=10485
sensorstatus=0 sourceUnit=0
```

Sample SCP output

```
# aircon zone_humidity //BOARD/254/172 1 0,1,2 10485 0 #sourceunit=0 OID=
```

4.7.2.3.1.6 Zone Temperature

Event

Zone Temperature

Condition causing event

Reporting the temperature of HVAC zones.

Event text

zone_temperature app ward zone-list level sensor-status

app = air-conditioning application address
 ward = air-conditioning ward number (0-255)
 zone-list = comma-separated list of zone numbers (max 7)
 level = temperature in degrees Centigrade * 256
 sensor-status = HVAC plant status (0..3)

- temperature is a two-byte signed number, so 0deg = \$0000, 0.1deg = \$0019, 1deg = \$0100, -1deg = \$FF00
- sensor-status:
 - 0 = no error
 - 1 = sensor operating in relaxed accuracy
 - 2 = sensor out of calibration
 - 3 = sensor total failure (temp reading is meaningless)

Sample Event output

```
20081022-113542 702 sys [aircon] zone_temperature ward=1 zones=0,1,2 level=5120
sensorstatus=0 sourceUnit=0
```

Sample SCP output

```
# aircon zone_temperature //BOARD/254/172 1 0,1,2 5120 0 #sourceunit=0 OID=...
```

4.7.2.3.2 Command Events

Airconditioning Command Events indicate that the equivalent Airconditioning command has been executed on the application.

4.7.2.3.2.1 Refresh

Event

Refresh

Condition causing event

The equivalent command has been invoked on the application.

Event text

refresh app ward

For description of parameters see the [AIRCON_REFRESH](#) command.

Sample Event output


```
20081022-084833 702 sys [aircon] refresh ward=1 sourceUnit=0
```

Sample SCP output

```
# aircon refresh //BOARD/254/172 1 #sourceunit=0 OID=...
```

4.7.2.3.2.2 Set Ward On

Event

Set Ward On

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_ward_on app ward

For description of parameters see the [AIRCON SET_WARD_ON](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_ward_on ward=1 sourceUnit=0
```

Sample SCP output

```
# aircon set_ward_on //BOARD/254/172 1 #sourceunit=0 OID=...
```

4.7.2.3.2.3 Set Ward Off

Event

Set Ward Off

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_ward_off app ward

For description of parameters see the [AIRCON SET_WARD_OFF](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_ward_off ward=1 sourceUnit=0
```

Sample SCP output

```
# aircon set_ward_off //BOARD/254/172 1 #sourceunit=0 OID=...
```

4.7.2.3.2.4 Set Zone HVAC Mode

Event

Set Zone HVAC Mode

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_zone_hvac_mode app ward zone-list mode raw-flag setback-enabled guard-enabled use-aux-level type level auxlevel

For description of parameters see the [AIRCON SET_ZONE_HVAC_MODE](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_zone_hvac_mode ward=255 zone-list=1,2,3,4,5,6
mode=7 rawlevel=1 setbackenabled=1 guardenabled=1 useauxlevel=1 type=255
level=65535 auxlevel=255
```

Sample SCP output

```
# aircon set_zone_hvac_mode 254/172 255 1,2,3,4,5,6 7 1 1 1 1 255 65535 255
```

4.7.2.3.2.5 Set HVAC Upper Guard Limit

Event

Set HVAC Upper Guard Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_hvac_upper_guard_limit app ward zone-list limit mode raw-level setback-enabled guard-enabled use-aux-level

For description of parameters see the [AIRCON SET_HVAC_UPPER_GUARD_LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_hvac_upper_guard_limit ward=255 zone-  
list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 guardenabled=1  
useauxlevel=1
```

Sample SCP output

```
# aircon set_hvac_upper_guard_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1  
#sourceunit=0 OID=...
```

4.7.2.3.2.6 Set HVAC Lower Guard Limit

Event

Set HVAC Lower Guard Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

**set_hvac_lower_guard_limit app ward zone-list limit mode raw-level setback-enabled
guard-enabled use-aux-level**

For description of parameters see the [AIRCON SET HVAC LOWER GUARD LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_hvac_lower_guard_limit ward=255 zone-  
list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 guardenabled=1  
useauxlevel=1
```

Sample SCP output

```
# aircon set_hvac_lower_guard_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1  
#sourceunit=0 OID=...
```

4.7.2.3.2.7 Set HVAC Setback Limit

Event

Set HVAC Setback Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

**set_hvac_setback_limit app ward zone-list limit mode raw-level setback-enabled guard-
enabled use-aux-level**

For description of parameters see the [AIRCON SET_HVAC_SETBACK_LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_hvac_setback_limit ward=255 zone-
  list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 guardenabled=1
  useauxlevel=1
```

Sample SCP output

```
# aircon set_hvac_setback_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1
#sourceunit=0 OID=...
```

4.7.2.3.2.8 Set Zone Humidity Mode

Event

Set Zone Humidity Mode

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_zone_humidity_mode app ward zone-list mode raw-flag setback-enabled guard-enabled use-aux-level type level auxlevel

For description of parameters see the [AIRCON SET_ZONE_HUMIDITY_MODE](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_zone_humidity_mode ward=255 zone-
  list=1,2,3,4,5,6 mode=7 rawlevel=1 setbackenabled=1 guardenabled=1
  useauxlevel=1 type=255 level=65535 auxlevel=255
```

Sample SCP output

```
# aircon set_zone_humidity_mode 254/172 255 1,2,3,4,5,6 7 1 1 1 1 255 65535 255
```

4.7.2.3.2.9 Set Humidity Upper Guard Limit

Event

Set Humidity Upper Guard Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_humidity_upper_guard_limit app ward zone-list limit mode raw-level setback-enabled guard-enabled use-aux-level

For description of parameters see the [AIRCON SET HUMIDITY UPPER GUARD LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_humidity_upper_guard_limit ward=255 zone-
list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 gardenabled=1
useauxlevel=1
```

Sample SCP output

```
# aircon set_humidity_upper_guard_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1
#sourceunit=0 OID=...
```

4.7.2.3.2.10 Set Humidity Lower Guard Limit

Event

Set Humidity Lower Guard Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_humidity_lower_guard_limit app ward zone-list limit mode raw-level setback-enabled guard-enabled use-aux-level

For description of parameters see the [AIRCON SET HUMIDITY LOWER GUARD LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_humidity_lower_guard_limit ward=255 zone-
list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 gardenabled=1
useauxlevel=1
```

Sample SCP output

```
# aircon set_humidity_lower_guard_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1
#sourceunit=0 OID=...
```

4.7.2.3.2.11 Set Humidity Setback Limit

Event

Set Humidity Setback Limit

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_humidity_setback_limit app ward zone-list limit mode raw-level setback-enabled guard-enabled use-aux-level

For description of parameters see the [AIRCON SET HUMIDITY SETBACK LIMIT](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_humidity_setback_limit ward=255 zone-  
list=1,2,3,4,5,6 limit=65535 mode=7 rawlevel=1 setbackenabled=1 gardenabled=1  
useauxlevel=1
```

Sample SCP output

```
# aircon set_humidity_setback_limit 254/172 255 1,2,3,4,5,6 65535 7 1 1 1 1  
#sourceunit=0 OID=...
```

4.7.2.4 Commands

The following commands are supported by this application:

- [AIRCON REFRESH](#) - Sends a refresh request to an air-conditioning ward.
- [AIRCON SET HUMIDITY SETBACK LIMIT](#) - Sets the error allowed in the set humidity for zones.
- [AIRCON SET HUMIDITY LOWER GUARD LIMIT](#) - Sets the absolute minimum humidity allowed in zones.
- [AIRCON SET HUMIDITY UPPER GUARD LIMIT](#) - Sets the absolute maximum humidity allowed in zones.
- [AIRCON SET HVAC LOWER GUARD LIMIT](#) - Sets the absolute minimum temperature allowed in zones.
- [AIRCON SET HVAC SETBACK LIMIT](#) - Sets the error allowed in the set temperature for zones.
- [AIRCON SET HVAC UPPER GUARD LIMIT](#) - Sets the absolute maximum temperature allowed in zones.
- [AIRCON SET WARD OFF](#) - Switches off all plant in all the zones in the specified ward.
- [AIRCON SET WARD ON](#) - Returns an air-conditioning ward to its previous operational state.
- [AIRCON SET_ZONE HUMIDITY MODE](#) - Broadcasts Humidity mode and level required for zones.
- [AIRCON SET_ZONE HVAC MODE](#) - Broadcasts HVAC mode and level required for zones.

4.7.2.5 Specification

For details regarding the function of the Airconditioning Application on the C-Bus network, refer to "C-Bus Application Messages and Behaviour -- Chapter 25: Air Conditioning (CBUS-APP/25), Issue 1.12, 26 August 2008".

4.7.3 Audio Application

The Audio Application is used to manage Multi Room Audio activity.

A number of events are provided by MRA devices and a number of commands are available to control MRA devices via this application.

The Audio Application has a C-Bus application address of \$CD, or 205 decimal.

4.7.3.1 Audio Application Overview

This gives a short overview of the Audio Application implementation in the C-Gate server

Application name

Audio

Application short name

(used as first part of commands or events to refer to this application)

audio

C-Bus Application range

\$CD (205)

Commands

(these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

off
on
ramp
terminateramp
next_feed
previous_feed
mute
dynamic_1
dynamic_2
next_language
high_priority
zone_descriptor_request
zone_feed_label_request
output_device_status_request
output_common_control
output_error_code
request_current_feed

current_feed
set_feed

Events

(these are detected by the C-Gate server and will be shown in the event list and status port but are not available as commands)

label
load_icon

Notes and implementation issues

4.7.3.2 Objects

One new object is provided for the implementation of this application:

Audio Application

This is the application object created in the server when an audio command is detected on the C-Bus network, or if an audio command is sent from the command line. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
ShortName	The short name for this application. This is set to audio when the application is loaded	Monitor	Read Only

4.7.3.3 Events

The events in this section are recognised by the Audio application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP).

4.7.3.3.1 Current Feed

Event

Current Feed

Condition causing event

Reporting the current feed for a zone.

Event text

current_feed app (multiplexer zone feed) | (Z function) gain

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
feed = feed number (0..7)


```
function = zone function (0..255)
gain = amount of gain (0..4)
```

Sample Event output

```
20081022-084833 702 sys [audio] current_feed multiplexer=1 zone=2 feed=4 gain=0
```

Sample SCP output

```
# audio current_feed //BOARD/254/205 1 2 4 0 #sourceunit=0 OID=...
```

4.7.3.3.2 Dynamic 1

Event

Dynamic 1

Condition causing event

Setting the dynamic 1 operation for a zone.

Event text

```
dynamic_1 app ( multiplexer zone ) | ( Z function )
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] dynamic_1 multiplexer=1 zone=2
```

Sample SCP output

```
# audio dynamic_1 //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.3.3 Dynamic 2

Event

Dynamic 2

Condition causing event

Setting the dynamic 2 operation for a zone.

Event text

```
dynamic_2 app ( multiplexer zone ) | ( Z function )
```

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

Sample Event output

20081022-084833 702 sys [audio] dynamic_2 multiplexer=1 zone=2

Sample SCP output

audio dynamic_2 //BOARD/254/205 1 2 #sourceunit=0 OID=...

4.7.3.3.4 High Priority

Event

High Priority

Condition causing event

Sending a high priority operation for a zone.

Event text

high_priority app multiplexer level feed

app = audio application address (net/\$CD)
multiplexer = multiplexer number (0..2)
level = level number (0..7)
feed = feed number (0..7)

Sample Event output

20081022-084833 702 sys [audio] high_priority multiplexer=1 level=2 feed=4

Sample SCP output

audio high_priority //BOARD/254/205 1 2 4 #sourceunit=0 OID=...

4.7.3.3.5 Mute

Event

Mute

Condition causing event

Setting the mute operation for a zone.

Event text

mute app (multiplexer zone) | (Z function) mode

app = audio application address (net/\$CD)
 multiplexer = multiplexer number (0..2)
 zone = zone number (0..7)
 function = zone function (0..255)
 mode = mode (0, 2, 5, 7, 255)
 0 = turn amplifier off
 2 = amplifier on, volume normal, speakers off
 5 = amplifier on, volume preset, speakers off
 7 = amplifier on, volume preset, speakers on
 255 = amplifier on, volume normal, speakers on

Sample Event output

20081022-084833 702 sys [audio] mute multiplexer=1 zone=2 mode=0

Sample SCP output

audio mute //BOARD/254/205 1 2 0 #sourceunit=0 OID=...

4.7.3.3.6 Next Feed

Event

Next Feed

Condition causing event

The zone is being set to the next feed.

Event text

next_feed app (multiplexer zone) | (Z function)

app = audio application address (net/\$CD)
 multiplexer = multiplexer number (0..2)
 zone = zone number (0..7)
 function = zone function (0..255)

Sample Event output

20081022-084833 702 sys [audio] next_feed multiplexer=1 zone=2

Sample SCP output

audio next_feed //BOARD/254/205 1 2 #sourceunit=0 OID=...

4.7.3.3.7 Next Language

Event

Next Language

Condition causing event

The zone is being set to the next language.

Event text

```
next_language app ( multiplexer zone ) | ( Z function )
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] next_language multiplexer=1 zone=2
```

Sample SCP output

```
# audio next_language //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.3.8 Off

Event

Off

Condition causing event

An 'off' operation is performed on a zone.

Event text

```
off app ( multiplexer zone code ) | ( Z function )
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = function code (0..7)
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] off multiplexer=1 zone=2 code=3
```

Sample SCP output

```
# audio off //BOARD/254/205 1 2 3 #sourceunit=0 OID=...
```

4.7.3.3.9 On

Event

On

Condition causing event

An 'on' operation is performed on a zone.

Event text

```
on app ( multiplexer zone code ) | ( Z function )
```

```
app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
code = function code (0..7)
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] on multiplexer=1 zone=2 code=3
```

Sample SCP output

```
# audio on //BOARD/254/205 1 2 3 #sourceunit=0 OID=...
```

4.7.3.3.10 Output Common Control

Event

Output Common Control

Condition causing event

A control command is sent to all output devices.

Event text

```
output_common_control control-code
```

```
control-code = control code (0)
0 = turn off
```

Sample Event output

```
20081022-084833 702 sys [audio] output_common_control control-code=0
```

Sample SCP output

```
# audio output_common_control //BOARD/254/205 0 #sourceunit=0 OID=...
```

4.7.3.3.11 Output Device Status Request

Event

Output Device Status Request

Condition causing event

A request is sent for all output devices to return their status.

Event text

output_device_status_request parameter

parameter = reserved value (0)

Sample Event output

```
20081022-084833 702 sys [audio] output_device_status_request parameter=0
```

Sample SCP output

```
# audio output_device_status_request //BOARD/254/205 0 #sourceunit=0 OID=...
```

4.7.3.3.12 Output Error Code

Event

Output Error Code

Condition causing event

A device advertises an error condition.

Event text

output_error_code app multiplexer zone code

app = audio application address (net/\$CD)

multiplexer = multiplexer number (0..2)

zone = zone number (0..7)

code = error code (0..1)

0 = overheating

1 = low voltage

Sample Event output

```
20081022-084833 702 sys [audio] output_error_code multiplexer=1 level=2 code=1
```

Sample SCP output

```
# audio output_error_code //BOARD/254/205 1 2 1 #sourceunit=0 OID=...
```

4.7.3.3.13 Previous Feed

Event

Previous Feed

Condition causing event

The zone is being set to the previous feed.

Event text

```
previous_feed app ( multiplexer zone ) | ( Z function )
```

```
app = audio application address (net/$CD)  
multiplexer = multiplexer number (0..2)  
zone = zone number (0..7)  
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] previous_feed multiplexer=1 zone=2
```

Sample SCP output

```
# audio previous_feed //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.3.14 Ramp

Event

Current Feed

Condition causing event

A 'ramp' operation is in progress on a zone.

Event text

```
current_feed app ( multiplexer zone code ) | ( Z function ) level rate
```

```
app = audio application address (net/$CD)  
multiplexer = multiplexer number (0..2)  
zone = zone number (0..7)  
code = zone code (0..7)  
function = zone function (0..255)
```

```

level = level to ramp to (0..255)
rate = time taken to perform the ramp (0..15)
    0 = instantaneous
    1 = 4 seconds
    2 = 8 seconds
    3 = 12 seconds
    4 = 20 seconds
    5 = 30 seconds
    6 = 40 seconds
    7 = 1 minute
    8 = 90 seconds
    9 = 2 minutes
    10 = 3 minutes
    11 = 5 minutes
    12 = 7 minutes
    13 = 10 minutes
    14 = 15 minutes
    15 = 17 minutes

```

Sample Event output

```
20081022-084833 702 sys [audio] ramp multiplexer=1 zone=2 code=4 level=255 rate=1
```

Sample SCP output

```
# audio ramp //BOARD/254/205 1 2 4 255 1 #sourceunit=0 OID=...
```

4.7.3.3.15 Request Current Feed

Event

Request Current Feed

Condition causing event

The current feed for a zone is requested.

Event text

```
request_current_feed app ( multiplexer zone ) | ( Z function )
```

```

app = audio application address (net/$CD)
multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

```

Sample Event output

```
20081022-084833 702 sys [audio] request_current_feed multiplexer=1 zone=2
```

Sample SCP output

```
# audio request_current_feed //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.3.16 Set Feed

Event

Set Feed

Condition causing event

The current feed is being set for a zone.

Event text

```
set_feed app ( multiplexer zone feed ) | ( Z function ) option
```

```
app = audio application address (net/$CD)
```

```
multiplexer = multiplexer number (0..2)
```

```
zone = zone number (0..7)
```

```
feed = feed number (0..7)
```

```
function = zone function (0..255)
```

```
option = value (0..1)
```

```
0 = allow labels and annunciation on feed change
```

```
1 = disable labels and annunciation on feed change
```

Sample Event output

```
20081022-084833 702 sys [audio] set_feed multiplexer=1 zone=2 feed=4 option=0
```

Sample SCP output

```
# audio set_feed //BOARD/254/205 1 2 4 0 #sourceunit=0 OID=...
```

4.7.3.3.17 Terminate Ramp

Event

Terminate Ramp

Condition causing event

A ramp in progress is being terminated on a zone.

Event text

```
terminateramp app ( multiplexer zone code ) | ( Z function )
```

```
app = audio application address (net/$CD)
```

```
multiplexer = multiplexer number (0..2)
```

```
zone = zone number (0..7)
```

```
code = function code (0..7)
```

```
function = zone function (0..255)
```

Sample Event output

```
20081022-084833 702 sys [audio] terminatoramp multiplexer=1 zone=2 code=3
```

Sample SCP output

```
# audio terminatoramp //BOARD/254/205 1 2 3 #sourceunit=0 OID=...
```

4.7.3.3.18 Zone Descriptor Request

Event

Zone Descriptor Request

Condition causing event

DLT devices are requested to report their zone descriptor labels.

Event text

```
zone_descriptor_request app ( multiplexer zone ) | ( Z function )
```

app = audio application address (net/\$CD)

multiplexer = multiplexer number (0..2)

zone = zone number (0..7)

function = zone function (0..255)

Sample Event output

```
20081022-084833 702 sys [audio] zone_descriptor_request multiplexer=1 zone=2
```

Sample SCP output

```
# audio zone_descriptor_request //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.3.19 Zone Feed Label Request

Event

Zone Feed Label Request

Condition causing event

DLT devices are requested to report their feed description and dynamic labels.

Event text

```
zone_feed_label_request app ( multiplexer zone ) | ( Z function )
```

app = audio application address (net/\$CD)

```

multiplexer = multiplexer number (0..2)
zone = zone number (0..7)
function = zone function (0..255)

```

Sample Event output

```
20081022-084833 702 sys [audio] zone_feed_label_request multiplexer=1 zone=2
```

Sample SCP output

```
# audio zone_feed_label_request //BOARD/254/205 1 2 #sourceunit=0 OID=...
```

4.7.3.4 Commands

The following commands are supported by this application:

- [AUDIO CURRENT FEED](#) - Reports the current feed for the given zone.
- [AUDIO DYNAMIC 1](#) - Request a matrix switcher to send a Dynamic 1 operation in the given zone.
- [AUDIO DYNAMIC 2](#) - Request a matrix switcher to send a Dynamic 2 operation in the given zone.
- [AUDIO HIGH PRIORITY](#) - Request an output device to turn on and go to a set output level and feed.
- [AUDIO MUTE](#) - Set the mute mode of an amplifier.
- [AUDIO NEXT FEED](#) - Set the next feed for the given zone.
- [AUDIO NEXT LANGUAGE](#) - Set the next language for the given zone.
- [AUDIO OFF](#) - Send an off operation.
- [AUDIO ON](#) - Send an on operation.
- [AUDIO OUTPUT COMMON CONTROL](#) - Request all devices to perform a function.
- [AUDIO OUTPUT DEVICE STATUS REQUEST](#) - Request status of all devices.
- [AUDIO OUTPUT ERROR CODE](#) - Send an error code notification.
- [AUDIO PREVIOUS FEED](#) - Set the previous feed for the given zone.
- [AUDIO RAMP](#) - Send a ramp operation.
- [AUDIO REQUEST CURRENT FEED](#) - Request the current feed of a zone.
- [AUDIO SET FEED](#) - Set the current feed of a zone.
- [AUDIO TERMINATERAMP](#) - Terminate a ramp in progress.
- [AUDIO ZONE DESCRIPTOR REQUEST](#) - Send feed description to DLT labelling devices.
- [AUDIO ZONE FEED LABEL REQUEST](#) - Send feed description and dynamic labels to DLT labelling devices.

4.7.3.5 Specification

For details regarding the function of the Audio Application on the C-Bus network, refer to "C-Bus Application Messages and Behaviour -- Chapter 14: Air Conditioning (CBUS-APP/14)".

4.7.4 Clock and Timekeeping Application

The Clock and Timekeeping Application allows C-Gate to interact with devices that use clocks on the C-Bus Network, and also to set and get the network time as a slave or primary or secondary master clock.

4.7.4.1 Application Overview

The table gives a short overview of the Clock & Timekeeping Application implementation in the C-Gate server.

Application name

Clock and Timekeeping

Application short name (used as first part of commands or events to refer to this application)

clock

C-Bus Application range

\$DF (only one)

Commands (These can be entered from C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

date
request_refresh
time

Events (these are detected by C-Gate and will be shown in the event list and status port but are not available as commands)

Notes and implementation issues

4.7.4.2 Objects supported

One new object is provided for the implementation of this application:

Clock and Timekeeping Application

This is the application object created in the server when a clock and timekeeping command is detected on the C-Bus network or if a clock and timekeeping command is sent from the command line. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
ShortName	The short name for this application. This is set to clock when the application is loaded	Monitor	Read Only

4.7.4.3 Methods

This application support the following methods, which mirror the commands described later:

Method Name	Description	Access Control
ClockDate	See CLOCK DATE	Operate
ClockTime	See CLOCK TIME	Operate
ClockRequestRefresh	See CLOCK REQUEST_REFRESH	Operate

4.7.4.4 Commands

The following commands are supported by this application:

- [CLOCK](#)
- [CLOCK DATE](#)
- [CLOCK TIME](#)
- [CLOCK REQUEST_REFRESH](#)

4.7.4.5 Events issued

Events are issued for the clock messages detected on a C-Bus network. They are in the form shown below:

```
20081217-100512 702 sys [clock] time 10:05:12 255 sourceUnit=0 sessionId=cmd1
commandId={none}
20081217-100512 702 sys [clock] date 2008-12-17 2 sourceUnit=0 sessionId=cmd1
commandId={none}
20081217-100512 702 sys [clock] request_refresh, sourceUnit=3 sessionId=cmd3
commandId={none}
```

4.7.4.6 Status Change Port

This application places CLOCK commands into the Status Change Port (SCP) in the same format as if they were issued as CLOCK commands.

Samples are shown below:

```
#s# clock time //BOARD/254/223 10:08:25 0 #sourceunit=3 OID=...
#s# clock date //BOARD/254/223 2008-12-17 2 #sourceunit=3 OID=...
#s# clock request_refresh //BOARD/254/223 #sourceunit=3 OID=...
```

4.7.4.7 Network time

This applications is implemented so that the time indicated on the network is kept separately from the system time maintained by the platform that the C-Gate server is running on.

When a date or time update is received from the network, C-Gate will calculate the offset from the system clock and use this offset to calculate the network time if requested. The offset will be recalculated if a command is sent from the command line, or if a command is received from the network.

#masterIn master mode (configuration property [clock.master](#)=yes), the time and date will be sent to the network every `clock.update-interval` minutes, subject to the protocol required in the C-Bus Application Messages and Behaviour -- Chapter 23: Clock & Timekeeping (CBUS-APP/23).

4.7.4.8 Configuration properties

The following configuration properties are used by this application:

- [clock.master](#) - turn on/off clock master behaviour
- [clock.mastermode](#) - select primary or secondary master mode
- [clock.update-interval](#) - set the update interval

4.7.4.9 Usage Notes

There are no network variables available for this application.

4.7.4.10 Specification

For detailed knowledge regarding the function of the Clock and Timekeeping Application, refer to "C-Bus Application Messages and Behaviour -- Chapter 23: Clock & Timekeeping (CBUS-APP/23)".

4.7.5 Enable Control Application

This summarizes the Enable Control Application as used by C-Gate

4.7.5.1 Enable Introduction

This document covers the implementation of the C-Bus Enable Control Application within the C-Gate server.

4.7.5.2 Enable Application Overview

The table gives a short overview of the Enable Control Application's implementation in C-Gate.

Application name Enable Control

Application short name (used as first part of commands or events to refer to this application)
enable

C-Bus Application range \$CB

Commands (these can be entered from C-Gate command interface preceded by the short name of this application. All commands are also recognised as events) set remove (see notes below)

Events (these are detected by C-Gate and will be shown in the event list and status port but are not available as commands)

Notes and implementation issues C-Gate to keep an on-disk copy of any enable control variables that are in use for a particular project/network/application. This to be turned on/off by the enable.save-state configuration parameter. **remove** command is used to clear the C-Gate server's memory of a particular variable. This will erase any value held (delete the network variable?) and remove any disk entry for it.

4.7.5.3 Persistent storage

If the [enable.save-state](#) configuration parameter is set (set to yes, the default value), then this application will store the value of all network variables in this application to disk, and will re-set these values when the server is restarted (eg. after power failure or manual restart).

One file is stored for each network variable value. This file contains a single number represented as a series of ASCII decimal digits, either terminated by whitespace (including CR or LF, space or other whitespace) or the end of the file.

The files are contained in a new directory structure under the projects directory, in a subdirectory named app/enable. Files are named as follows <project>-<net>-<application>-<netvar> (the elements of the normal address of the network variable, with the '/' path symbols replaced by the '-' dash character.)

For example, if the projects directory is C:\Users\<username>\Documents\C-Bus Commission\Projects and the network variable is //system/1/203/25 then the filename used to hold the state of the variable would be C:\Users\<username>\Documents\C-Bus Commission\Projects\app\enable\system-1-203-25 .

4.7.5.4 Objects supported

Two new objects are provided for the implementation of this application:

4.7.5.4.1 Enable Control Application Object

This is the application object created in the server when an enable control application is detected on the C-Bus network. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
----------------	-------------	---------------------	----------------------

ShortName	The short name for this application. This is set to enable when the application is loaded	Monitor	Read Only
-----------	--	---------	-----------

4.7.5.4.2 Enable Network Variables

4.7.5.5 Commands supported

The following commands are supported by this application:

- [ENABLE](#)
- [ENABLE SET](#)
- [ENABLE REMOVE](#)

4.7.5.6 Events issued

Events are issued for the enable set messages detected on a C-Bus network. They are in the form show below:

```
20020910-172843.078 702 //system/1/202/144 [enable] set value=2 sourceUnit=131
```

In the example shown, the address given is the enable network variable that was activated, the command given is `event`, the `action=` parameter gives the action selector value, and the `sourceUnit=` gives the source unit id on the network where the command was sent.

The ENABLE REMOVE command does not cause an event to be issued.

4.7.5.7 Status Change Port Reporting

This application places ENABLE SET commands into the Status Change Port (SCP).

A sample are shown below.

```
enable set //system/1/203/144 2 #sourceunit=131
```

The enable REMOVE command does not cause a SCP entry.

4.7.5.8 Project database

This application does **not** support the <Group> element under <Application>. Instead, it supports the <NetVar> element which can contain zero or more <Level> elements containing Level Tag information.

To use level tags:

Add one or more level entries to the <NetVar> element in the tag database. For each of the NetVar elements, add a <Value> element with the appropriate application selector level in the range 0 through 255.

So, an except would look like:

```
<NetVar>
  <TagName>Switch1</TagName>
  <Level>
    <TagName>AllOff</TagName>
    <Value>0</Value>
  </Level>
```

```

    <Level>
      <TagName>Level1</TagName>
      <Value>2</Value>
    </Level>
  </NetVar>

```

This allows the ENABLE SET command to use the tags AllOff and Level1 as action selector values, for example:

```
enable set Switch1 AllOff
```

4.7.5.9 Usage Notes

Network Variables

Network variables are created automatically when events for them are received, or they are detected in other ways.

4.7.5.10 Enable Specification

For detailed knowledge regarding the function of the Enable Control Application, refer to "C-Bus Application Messages and Behaviour -- Chapter 8: Enable Control (CBUS-APP/08)".

4.7.6 Error Reporting Application

This section of the manual covers the implementation of the Error Reporting Application (\$CE) in the C-Gate server.

4.7.6.1 Application Overview

Application name

Error Reporting

Application short name (used as first part of commands or events to refer to this application)

ereport

C-Bus Application range

\$CE (206)

Commands/Events (these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

[message](#)

Events (these are detected by the C-Gate server and will be shown in the event list and status port but are not available as commands)

Notes and implementation issues

See the full documentation of this application to cover operation and use of this application.

4.7.6.2 Objects Supported

One new object is provided for the implementation of this application:

Error Reporting Application

This is the application object created in the server when an ereport command is detected on the C-Bus network or if an ereport command is sent from the command line. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
ShortName	The short name for this application. This is set to ereport when the application is loaded	Monitor	Read Only

4.7.6.3 Send Event Event

Send Event

Condition causing event

An Error Report has been received.

This event can be generated by the [EREPOR MESSAGE](#) command.

Event Text

```
send app category most-recent acknowledged most-severe severity unit-id [<data-byte-1 [data-byte-2]]
```

Sample Event output

```
20060111-160154 702 //cis/1/206 37be1ec0-0d69-1028-a7c5-e25002b8b622 [ereport]
ereport message 254/$CE ACK 1023 n n n 7 255 255 255
```

Sample SCP output

```
ereport message 254/$CE ACK 1023 n n n 7 255 255 255
```

4.7.7 Measurement Application

This document covers the implementation of the C-Bus Measurement Application within the C-Gate server.

The Measurement Application is used by C-Bus devices that report analog measurements. Each measurement comes from a [Channel](#), with Channels grouped together in [Measurement Devices](#).

4.7.7.1 Specification

For detailed knowledge regarding the function of the Measurement Application on the C-Bus network, refer to "C-Bus Application Messages and Behavior -- Chapter 28: Measurement (CBUS-APP/28), Issue 1, 7 August 2002".

4.7.7.2 Application Overview

This gives a short overview of the Media Transport Control Application implementation in the C-Gate server

Application name

Measurement

Application short name (used as first part of commands or events to refer to this application)

measurement

C-Bus Application range

\$E4 (228)

Commands/Events (these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

[data](#)

Events (these are detected by the C-Gate server and will be shown in the event list and status port but are not available as commands)

data

Notes and implementation issues

See the full documentation of this application to cover operation and use of this application.

4.7.7.3 Objects Supported

There are three new objects supported by the Measurement Application implementation.

The Measurement Application Object models the application as a whole. The Measurement Application Object is addressed using the application address for the Measurement Application (\$E4 or 228) on the C-Bus network that the measurement devices are connected to.

The Measurement Device Object models a measurement device. The measurement device is addressed as a sub-address of the Measurement Application that it is associated with.

Each measurement device can have one or more Channels, up to a total of 256 Channels per Measurement Device.

4.7.7.3.1 Measurement Application Object

The Measurement Application Object models a C-Bus network's Measurement Application.

The Measurement Application Object supports up to 256 [Measurement Device Objects](#).

Addressing

The Measurement Application Object is addressed using the standard application address for the

Measurement Application (\$E4 or decimal 228) on the relevant C-Bus network.

For example, for a hypothetical network called net1, the Measurement Application's address will be net1/\$E4 or net1/228

4.7.7.3.1.1 Commands Supported

The following command line commands are accepted by the Measurement Application Object.

Command

MEASUREMENT ?

Syntax

"MEASUREMENT" ["?"]

Use

Use this command to get help information about the MEASUREMENT commands. Help information is returned as a number of lines of help information.

Success Response

A series of lines giving help for these commands in the form:

101 Help: help-information

Failure Responses

401 Bad object or device ID
402 Not supported by this object
405 Parameter out of range
408 Operation failed
420 Access denied

4.7.7.3.1.2 Events Supported

There are no events supported by this object.

4.7.7.3.1.3 Object Parameters

The following parameters are supported by the Measurement Application Object.

They can be accessed with the GET command acting on the object address of the relevant Measurement Application object.

Parameter Name	Type & Description	Can set with SET command?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronize with the	No	

	C-Bus network "sync" = the object is synchronising with the C-Bus network "error" = an error has occurred.		
Devices	Returns a list of device numbers that are known in this application. The list items are separated by commas	No	

4.7.7.3.2 Measurement Device Object

The Measurement Device object models a device that is capable of making measurements. The measurement device can have 0 or more Channels, where each channel can provide measurement data.

Addressing Measurement Device Objects

A [Measurement Application](#) can support up to 256 separate Measurement Devices.

Measurement Device Objects are addressed as subelements of the Measurement Application Object. Valid Measurement Device addresses range from 0 through 255 (or in hexadecimal, \$00 through \$FF). No other values are allowed or valid.

So, on an example network called net1, the Measurement Application would be addressed as net1/\$E4 or net1/228. Measurement device number 1 would be addressed as net1/228/1, and Measurement Device \$25 would be addressed as net1/\$E4/\$25 or net1/228/\$25.

4.7.7.3.2.1 Commands Supported

There are no commands supported by the Measurement Device Object.

4.7.7.3.2.2 Events Supported

There are no events supported by this object.

4.7.7.3.2.3 Object Parameters

The following parameters are supported by the Measurement Device Object.

They can be accessed with the GET command acting on the object address of the relevant Measurement Device object.

Parameter Name	Type & Description	Can be set with SET command?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronise with the C-Bus network "sync" = the object is attempting to synchronise with the C-Bus network "error" = an error has occurred.	No	

Channels	A comma-separated list of channel numbers that are known on this Measurement Device.	No	
----------	--	----	--

4.7.7.3.3 Channel Object

The Channel Object models a single measurement channel on a [Measurement Device](#).

Addressing Channel Objects

Channel addresses are in the range of 0 through 255.

Channel Objects are addressed as subelements of the Measurement Device Object. Valid Channel addresses range from 0 through 255 (or in hexadecimal, \$00 through \$FF). No other values are allowed or valid.

So, on an example network called net1, the Measurement Application would be addressed as net1/\$E4 or net1/228. Measurement device number 1 would be addressed as net1/228/1. Channel 2 on Measurement Device 1 would be addressed as: net1/\$E4/1/2 or net1/228/1/2 .

4.7.7.3.3.1 Commands Supported

The following commands are supported by this object

Command

MEASUREMENT DATA

Syntax

"MEASUREMENT" "DATA" channel-address measurement-value exponent units

channel-address = ;the address of the measurement channel to send the data for
;as a C-Gate address, for example net1/228/1/1

measurement-value = ; integer part of the measurement (mantissa)

exponent = power of ten to be applied to the measurement value

units = an integer giving the units of the measurement value.

See the [unit code table](#) for conversion details.

Use

Use this command to send measurement data to the C-Bus network.

Success Response

200 OK.

Failure Responses

401 Bad object or device ID

402 Not supported by this object

405 Parameter out of range
 408 Operation failed
 420 Access denied

4.7.7.3.3.2 Events Supported

The following events are supported by the Channel Object.

Event

Measurement Data

Condition causing event

Measurement data has been received. This event can be generated by the [MEASUREMENT DATA](#) command.

Event Text

"data" measurement-data exponent units

measurement-data = ; integer measurement value

exponent = power of ten to be applied to the measurement value

units = an integer giving the units of the measurement value. See the [unit code table](#) for conversion details.

Sample Event output

```
20020910-172843 702 //system/1/228/1/1 [measurement] data 10234 -2 2 sourceUnit=100
```

Sample SCP output

```
measurement data //system/1/228/1/1 10234 -2 2 #sourceunit=100
```

4.7.7.3.3.3 Object Parameters

The following parameters are supported by the Channel Object.

They can be accessed with the GET command acting on the object address of the relevant Channel object.

Parameter Name	Type & Description	Can be set with SET command?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronise with the C-Bus network "sync" = the object is attempting to synchronise with the C-Bus network	No	

	"error" = an error has occurred.		
Data	Four numbers, separated by commas that give the following information from the last measurement received for this channel. The four numbers are: 1. the integer portion of the measurement 2. the exponent (power of 10) to be applied to the integer portion of the measurement 3. the unit code for this measurement 4. the number of milliseconds since this measurement was taken (if set to -1, indicates that no measurement has been received for this channel)	No	If no measurement has been received, the this parameter will be set to "0,0,0,-1"

4.7.7.4 Events Issued

The events in this section are generated by the Measurement application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP)

4.7.7.4.1 Data

Event

Data

Condition causing event

Receiving a MEASUREMENT DATA command sequence.

Event text

[measurement] data device channel units multiplier value sourceUnit

device = measurement device (0..\$FF)
channel = measurement channel (0..\$FF)
units = measurement units (0..\$FF)
multiplier = power of ten unit multiplier (-128..127)
value = scaled measurement value (-32768..32767)
sourceUnit = source of the command

For meanings of units values see the [Unit Code Table](#).

Sample Event output

```
20090220-173207 702 //BC/254/228 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[measurement] data device=1 channel=2 units=0 multiplier=1 value=37
sourceUnit=3
```

Sample SCP output

```
#s# measurement data //BC/254/228 1 2 0 1 37 #sourceunit=3 OID=...
```

4.7.7.5 Unit Code Table

The following table gives the relation between unit codes (from the [MEASUREMENT DATA](#) command and [Measurement Data](#) event), unit names and typical uses.

Unit Code	Units	Typical Use
\$00	°C	Temperature
\$01	Amps	Current
\$02	Angle (degrees)	Angular displacement
\$03	Coulomb	(Electric) charge
\$04	False = 0 True otherwise	Boolean stuff
\$05	Farads	Capacitance
\$06	Henrys	Inductance
\$07	Hertz	Frequency
\$08	Joules	Energy
\$09	Katal	Rate of catalytic activity
\$0A	Kg / m ³	Density
\$0B	Kilograms	Mass
\$0C	Litres	Volume
\$0D	Litres per hour	Very slow flow rates
\$0E	Litres per minute	Slow flow rate
\$0F	Litres per second	Flow rate
\$10	Lux	Light level
\$11	Metres	Distance
\$12	Metres per minute	Slow speed
\$13	Metres per second	Speed
\$14	Metres/s ²	Acceleration
\$15	Mole	Quantity of substance
\$16	Newton metre	Torque
\$17	Newtons	Force
\$18	Ohms	Resistance
\$19	Pascal	Pressure
\$1A	Percent	Humidity, generic percentages & linear ratios
\$1B	Decibels	Logarithmic ratio
\$1C	PPM	Concentrations
\$1D	RPM	Angular speed
\$1E	Second	Elapsed Time
\$1F	Minutes	Elapsed Time
\$20	Hours	Elapsed Time
\$21	Sieverts	Radiation
\$22	Steradian	Units of solid angle
\$23	Tesla	Magnetic field strength
\$24	Volts	Voltage
\$25	Watt hours	Power consumption
\$26	Watts	Power
\$27	Webers	Magnetic Flux
\$FE	No units	Unitless quantities

\$FF	Custom	User defined
------	--------	--------------

4.7.8 Media Transport Control Application

The Media Transport Control Application is used to control audio and video equipment attached to, or used with, C-Bus equipment.

This can include recorders, players, tuners, and other audio and video media equipment.

The Media Transport Control Application has a C-Bus application address of \$C0, or 192 decimal.

4.7.8.1 Application Overview

This gives a short overview of the Media Transport Control Application implementation in the C-Gate server

Application name

Airconditioning

Application short name

(used as first part of commands or events to refer to this application)

mediatransport

C-Bus Application range

\$C0 (192)

Commands

(these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

category_name
 enumerate
 enumeration_size
 forward
 next_category
 next_selection
 next_track
 pause
 play
 repeat
 rewind
 selection_name
 set_category
 set_selection
 set_track
 shuffle
 source_power
 status_request
 stop
 track_name

Events

(these are detected by the C-Gate server and will be shown in the event list and status port)

category_name
 enumerate
 enumeration_size
 forward
 next_category
 next_selection
 next_track
 pause
 play
 repeat
 rewind
 selection_name
 set_category
 set_selection
 set_track
 shuffle
 source_power
 status_request
 stop
 track_name

Notes and implementation issues

4.7.8.2 Objects Supported

An object is provided for the implementation of this application:

Media Transport Control Application

This is the application object created in the server when a Media Transport Control command is detected on the C-Bus network, or if a Media Transport Control command is sent from the command line. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
Name	The name for this application.	Monitor	Read/Write
ShortName	The short name for this application. This is set to mediatransport when the application is loaded	Monitor	Read Only

4.7.8.3 Events Issued

The events in this section are generated by the Media Transport Control application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP)

4.7.8.3.1 Category Name

Event

Category Name

Condition causing event

Receiving a MEDIATRANSPORT CATEGORY_NAME command sequence.

Event text**[mediatransport] category_name group wni total index text sourceUnit**

group = media link group (0..\$FF, where \$FF indicates unused)
 wni = name identification (0..7)
 0 = current category
 1 = next category
 2 = next+1 category
 3,4 = reserved
 5 = previous category
 6 = previous-1 category
 2 = enumerated category
 total = total number of packets - 1 (0..3)
 index = index of this packet in the sequence (0..3)
 text = all/part of the category name in UTF-8 (max 11 bytes per packet)
 sourceUnit = source of the command

Sample Event output

```
20090220-173207 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] category_name group=2 wni=1 total=1 sequence=0 text=list
\x204 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport category_name //BC/254/192 2 1 1 0 list\x204
#sourceunit=3 OID=...
```

4.7.8.3.2 Enumerate

Event

Enumerate

Condition causing event

Receiving a MEDIATRANSPORT ENUMERATE command sequence.

Event text**[mediatransport] enumerate group type start sourceUnit**

group = media link group (0..\$FF, where \$FF indicates unused)
 type = enumeration type where 0=category, 1=selection, 2=track
 start = enumerate from this index (0..\$FF)
 sourceUnit = source of the command

Sample Event output

```
20090220-164033 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] enumerate group=2 type=1 start=0 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport enumerate //BC/254/192 2 1 0 #sourceunit=3 OID=...
```

4.7.8.3.3 Enumeration Size

Event

Enumeration Size

Condition causing event

Receiving a MEDIATRANSPORT ENUMERATION_SIZE command sequence.

Event text

```
[mediatransport] enumeration_size group type start size sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
 type = enumeration type where 0=category, 1=selection, 2=track
 start = enumerate from this index (0..\$FF)
 size = number of items in the enumeration following (0..\$0F)
 sourceUnit = source of the command

Sample Event output

```
20090220-164033 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] enumeration_size group=2 type=1 start=0 size=4
sourceUnit=3
```

Sample SCP output

```
#s# mediatransport enumeration_size //BC/254/192 2 1 0 4 #sourceunit=3
OID=...
```

4.7.8.3.4 Forward

Event

Forward

Condition causing event

Receiving a MEDIATRANSPORT FORWARD command sequence.

Event text

[mediatransport] forward group operation sourceUnit

group = media link group (0..\$FF, where \$FF indicates unused)
 operation = one of (0..\$FF):
 0 = cease fast-forward, play at normal speed
 2 = fast-forward at 2x speed
 4 = fast-forward at 4x speed
 6 = fast-forward at 8x speed
 8 = fast-forward at 16x speed
 10 = fast-forward at 32x speed
 12 = fast-forward at 64x speed
 all other values reserved
 sourceUnit = source of the command

Sample Event output

```
20090220-100958 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] forward group=0 operation=0 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport forward //BC/254/192 0 0 #sourceunit=3 OID=...
```

4.7.8.3.5 Next Selection

Event

Next Selection

Condition causing event

Receiving a MEDIATRANSPORT NEXT_SELECTION command sequence.

Event text

[mediatransport] next_selection group operation sourceUnit

group = media link group (0..\$FF, where \$FF indicates unused)
 operation = one of (0..\$FF):
 0 = select previous selection
 non-0 = select next selection
 sourceUnit = source of the command

Sample Event output

```
20090220-164033 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] next_selection group=2 operation=1 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport next_selection //BC/254/192 2 1 #sourceunit=3 OID=...
```

4.7.8.3.6 Next Category

Event

Next Category

Condition causing event

Receiving a MEDIATRANSPORT NEXT_CATEGORY command sequence.

Event text

[mediatransport] next_category group operation sourceUnit

group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0..\$FF):
 0 = select previous category
 non-0 = select next category
sourceUnit = source of the command

Sample Event output

```
20090223-085323 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] next_category group=2 operation=1 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport next_category //BC/254/192 2 1 #sourceunit=3 OID=...
```

4.7.8.3.7 Next Track

Event

Next Track

Condition causing event

Receiving a MEDIATRANSPORT NEXT_TRACK command sequence.

Event text

[mediatransport] next_track group operation sourceUnit

group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0..\$FF):
 0 = select previous track
 non-0 = select next track
sourceUnit = source of the command

Sample Event output

```
20090220-171413 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] next_track group=2 operation=1 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport next_track //BC/254/192 2 1 #sourceunit=3 OID=...
```

4.7.8.3.8 Pause

Event

Pause

Condition causing event

Receiving a MEDIATRANSPORT PAUSE command sequence.

Event text

```
[mediatransport] pause group operation sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
operation = one of (0..\$FF):
 0 = pause track
 255 = resume playing track
sourceUnit = source of the command

Sample Event output

```
20090220-171841 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] pause group=2 operation=0 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport pause //BC/254/192 2 0 #sourceunit=3 OID=...
```

4.7.8.3.9 Play

Event

Play

Condition causing event

Receiving a MEDIATRANSPORT PLAY command sequence.

Event text

```
[mediatransport] play group sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
sourceUnit = source of the command

Sample Event output

```
20090220-172356 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] play group=2 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport play //BC/254/192 2 #sourceunit=3 OID=...
```

4.7.8.3.10 Repeat

Event

Repeat

Condition causing event

Receiving a MEDIATRANSPORT REPEAT command sequence.

Event text

```
[mediatransport] repeat group operation sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)

operation = one of (0,\$FF):

0 = repeat is off

1..254 = repeat current track

255 = repeat all tracks

sourceUnit = source of the command

Sample Event output

```
20090220-174549 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] repeat group=2 operation=1 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport repeat //BC/254/192 2 1 #sourceunit=3 OID=...
```

4.7.8.3.11 Rewind

Event

Rewind

Condition causing event

Receiving a MEDIATRANSPORT REWIND command sequence.

Event text

```
[mediatransport] rewind group operation sourceUnit
```


group = media link group (0..\$FF, where \$FF indicates unused)
 operation = one of (0,\$FF):
 0 = cease rewind, play at normal speed
 2 = rewind at 2x speed
 4 = rewind at 4x speed
 6 = rewind at 8x speed
 8 = rewind at 16x speed
 10 = rewind at 32x speed
 12 = rewind at 64x speed
 sourceUnit = source of the command

Sample Event output

```
20090220-175000 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] rewind group=2 operation=0 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport rewind //BC/254/192 2 0 #sourceunit=3 OID=...
```

4.7.8.3.12 Selection Name

Event

Selection Name

Condition causing event

Receiving a MEDIATRANSPORT SELECTION_NAME command sequence.

Event text

```
[mediatransport] selection_name group wni total index text sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
 wni = name identification (0..7)
 0 = current selection
 1 = next selection
 2 = next+1 selection
 3,4 = reserved
 5 = previous selection
 6 = previous-1 selection
 2 = enumerated selection
 total = total number of packets - 1 (0..3)
 index = index of this packet in the sequence (0..3)
 text = all/part of the selection name in UTF-8 (max 11 bytes per packet)
 sourceUnit = source of the command

Sample Event output

```
20090220-172714 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] selection_name group=2 wni=1 total=1 sequence=0
text=list\x204 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport selection_name //BC/254/192 2 1 1 0 list\x204  
#sourceunit=3 OID=...
```

4.7.8.3.13 Set Selection

Event

Set Selection

Condition causing event

Receiving a MEDIATRANSPORT SET_SELECTION command sequence.

Event text

```
[mediatransport] set_selection group selection sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
selection = number of selection selected
sourceUnit = source of the command

Sample Event output

```
20090220-175233 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] select_selection group=2 selection=1 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport select_selection //BC/254/192 2 1 #sourceunit=3  
OID=...
```

4.7.8.3.14 Set Category

Event

Set Category

Condition causing event

Receiving a MEDIATRANSPORT SET_CATEGORY command sequence.

Event text

```
[mediatransport] set_category group category sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
category = number of category selected
sourceUnit = source of the command

Sample Event output

```
20090220-175448 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] select_category group=2 category=5 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport select_category //BC/254/192 2 5 #sourceunit=3 OID=...
```

4.7.8.3.15 Set Track

Event

Set Track

Condition causing event

Receiving a MEDIATRANSPORT SET_TRACK command sequence.

Event text

```
[mediatransport] set_track group track sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)

track = number of track selected

sourceUnit = source of the command

Sample Event output

```
20090220-175822 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] set_track group=2 track=7 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport set_track //BC/254/192 2 7 #sourceunit=3 OID=...
```

4.7.8.3.16 Shuffle

Event

Shuffle

Condition causing event

Receiving a MEDIATRANSPORT SHUFFLE command sequence.

Event text

```
[mediatransport] shuffle group operation sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)

operation = one of (0,\$FF):

0 = shuffle is off

255 = shuffle is on

sourceUnit = source of the command

Sample Event output

```
20090220-175956 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] shuffle group=2 operation=255 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport shuffle //BC/254/192 2 255 #sourceunit=3 OID=...
```

4.7.8.3.17 Source Power Control

Event

Source Power Control

Condition causing event

Receiving a MEDIATRANSPORT SOURCE_POWER command sequence.

Event text

```
[mediatransport] source_power group operation sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)

operation = one of 0 or 1..255

0 = power off

not 0 = power on

sourceUnit = source of the command

Sample Event output

```
20090220-173847 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] source_power group=2 operation=255 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport source_power //BC/254/192 2 255 #sourceunit=3 OID=...
```

4.7.8.3.18 Status Request

Event

Status Request

Condition causing event

Receiving a MEDIATRANSPORT STATUS_REQUEST command sequence.

Event text

```
[mediatransport] status_request group sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
sourceUnit = source of the command

Sample Event output

```
20090220-174237 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] status_request group=2 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport status_request //BC/254/192 2 #sourceunit=3 OID=...
```

4.7.8.3.19 Stop

Event

Stop

Condition causing event

Receiving a MEDIATRANSPORT STOP command sequence.

Event text

```
[mediatransport] stop group sourceUnit
```

group = media link group (0..\$FF, where \$FF indicates unused)
sourceUnit = source of the command

Sample Event output

```
20090220-174149 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5  
[mediatransport] stop group=2 sourceUnit=3
```

Sample SCP output

```
#s# mediatransport stop //BC/254/192 2 #sourceunit=3 OID=...
```

4.7.8.3.20 Track Name

Event

Track Name

Condition causing event

Receiving a MEDIATRANSPORT TRACK_NAME command sequence.

Event text

```
[mediatransport] track_name group which total index text
```

group = media link group (0..\$FF, where \$FF indicates unused)
 wni = name identification (0..7)
 0 = current track
 1 = next track
 2 = next+1 track
 3,4 = reserved
 5 = previous track
 6 = previous-1 track
 2 = enumerated track
 total = total number of packets - 1 (0..3)
 index = index of this packet in the sequence (0..3)
 text = all/part of the *track* name in UTF-8 (max 11 bytes per packet)
 sourceUnit = source of the command

Sample Event output

```

20090220-144729 702 //BC/254/192 fb95a8a0-e046-102b-a64e-ce2810e2f0a5
[mediatransport] track_name group=2 wni=0 total=0 sequence=0 text=track
\x203 sourceUnit=3
  
```

Sample SCP output

```

#s# mediatransport track_name //BC/254/192 2 0 0 0 track\x203
#sourceunit=3 OID=...
  
```

4.7.9 Security Application

The Security Application is used by security devices, such as security panels, that are connected to a C-Bus network.

A number of events are provided by security devices and a number of commands are available to control security devices via the Security Application.

The Security Application has a C-Bus application address of \$D0, or 208 decimal.

4.7.9.1 Security Application Overview

The table gives a short overview of the Security Application implementation in the C-Gate server

Application name	Security
Application short name (used as first part of commands or events to refer to this application)	security
C-Bus Application range	\$D0
Commands (these can be entered from C-Gate command interface preceded by the short name)	status_request arm tamper [raise drop] raise_alarm

of this application. All commands are also recognised as events)	emulate_keypad display_message request_zone_name
Events (these are detected by C-Gate and will be shown in the event list and status port but are not available as commands)	system_armed system_disarmed exit_delay_started alarm_on alarm_off tamper_on tamper_off panic_cleared panic_activated zone_unsealed zone_sealed zone_open zone_short zone_isolated low_battery_detected low_battery_corrected battery_charging zone_name status_report_1 status_report_2 password_entry_status mains_failure mains_restored arm_ready arm_not_ready
Notes and implementation issues	C-Gate is not capable of behaving as a security panel.

4.7.9.2 Objects Supported

There are two new object supported by the Security Application implementation.

The Security Application Object models security and a security panel as a whole. The security application object is addressed using the application address for the Security Application (\$D0 or 208) on the C-Bus network that the security system is connected to.

The Security Zone Object models one security zone. There can, and generally will be more than one security zone. The security zone is addressed as a sub-address of the Security application that it is associated with.

4.7.9.2.1 Security Application Object

The Security Application Object models a C-Bus network's Security Application as a whole. Generally, one or more security devices or panels are connecte to a C-Bus network and they communicate using the Security Application messaging.

The Security Applicatin Object receives and models Security Application messages ([events](#)) and provides the ability to send Security Application messages from the C-Gate server using a number of [commands](#).

The Security Application Object also supports a number of [parameters](#) that can be accessed with the GET and SET commands, quoting the Security Application Object's address.

Addressing

The Security Application Object is addressed using the standard application address for the Security Application (\$D0 or 208) on the C-Bus network that the security system is connected to.

For example, for a hypothetical network called net1, the Security Application's address will be net1/\$D0 or net1/208.

4.7.9.2.1.1 Commands Supported

The following commands are supported by the Security Application object:

- [SECURITY](#)
- [SECURITY ARM](#)
- [SECURITY DISPLAY MESSAGE](#)
- [SECURITY EMULATE KEYPAD](#)
- [SECURITY RAISE ALARM](#)
- [SECURITY STATUS REQUEST](#)
- [SECURITY TAMPER](#)

4.7.9.2.1.2 Events Supported

Event

System Armed/Disarmed

Condition causing event

The security system has just become armed. The type of arming is indicated in the type parameter.

Event Text

"system_arm" type

```
type = 1*3DIGIT      ; in the range of 1 - 127.  values 128-255 are reserved.
                      ; 0 = disarmed, 1=fully armed, 2=partially armed
                      ; 3 - 127 are manufacturer dependent
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] system_armed 1 sourceUnit=100
```

Sample SCP output

```
#security system_armed //system/1/224 1 #sourceunit=100
```

Event

System Disarmed

Condition causing event

The security system has just become disarmed. This is equivalent to a [System Armed/Disarmed](#) condition with an argument of 0.

Event Text

"system_disarmed"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] system_disarmed sourceUnit=100
```

Sample SCP output

```
#security system_disarmed //system/1/224 #sourceunit=100
```

Event

Exit Delay Started

Condition causing event

The security system has commenced its exit delay processing.

Event Text

"exit_delay_started"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] exit_delay_started sourceUnit=100
```

Sample SCP output

```
#security exit_delay_started //system/1/224 #sourceunit=100
```

Event

Entry Delay Started

Condition causing event

The security system has commenced its entry delay processing.

Entry delay processing will normally commence in a security system when it detects a zone becoming unsealed in some defined entry path. The Entry Delay allows time for disarming the system. If the system is not disarmed during the Entry Delay period, the system will normally raise an alarm condition.

Event Text

"entry_delay_started"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] entry_delay_started sourceUnit=100
```

Sample SCP output

```
#security entry_delay_started //system/1/224 #sourceunit=100
```

Event

Alarm On

Condition causing event

The security system has commenced some alarm or notification activity. This message may optionally be followed by an Alarm Type message, to convey additional information about the alarm condition.

Event Text

```
"alarm_on"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] alarm_on sourceUnit=100
```

Sample SCP output

```
#security alarm_on //system/1/224 #sourceunit=100
```

Event

Alarm Off

Condition causing event

The security system alarm has been switched off. An Alarm Off message implies that any alarm condition, tamper condition or panic condition has been cleared

Event Text

```
"alarm_off"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] alarm_off sourceUnit=100
```

Sample SCP output

```
#security alarm_off //system/1/224 #sourceunit=100
```

Event

Tamper Off

Condition causing event

The security system has detected clearing of the tampering. A security system is expected to continue an alarm condition if tampering is removed. Some security systems only allow a tamper condition to be cleared by Disarming.

This event may not be implemented by some security systems.

Event Text`"tamper_off"`**Sample Event output**`20020910-172843 702 //system/1/224 [security] tamper_off sourceUnit=100`**Sample SCP output**`#security tamper_off //system/1/224 #sourceunit=100`**Event**

Tamper On

Condition causing event

The security system alarm has detected tampering becoming active. The security system may, at the discretion of the manufacturer or installer, cause an alarm condition if tampering is detected. In this case, an Alarm On would be used.

Event Text`"tamper_on"`**Sample Event output**`20020910-172843 702 //system/1/224 [security] tamper_on sourceUnit=100`**Sample SCP output**`#security tamper_on //system/1/224 #sourceunit=100`**Event**

Current Alarm Type

Condition causing event

The security system uses this event to describe the type of the active alarm. This alarm will have been presented in the previous message.

Event Text

```
"current_alarm_type" type
type = 1*3DIGIT
    ; 0 = reserved
    ; 1 = intruder
    ; 2 = line cut
    ; 3 = arm failed
    ; 4 = fire
    ; 5 = gas
    ; 6 - 127 are reserved
    ; 128 - 254 used at alarm manufacturer's discretion
    ; 255 - reserved
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] current_alarm_type 1 sourceUnit=100
```

Sample SCP output

```
#security alarm_off //system/1/224 1 #sourceunit=100
```

Event

Panic Cleared

Condition causing event

The security system has detected cancellation of the panic condition. A security system may cancel an alarm condition if panic is cancelled. (Note: Some security systems only allow a panic condition to be cleared by Disarming)

Event Text

```
"panic_cleared"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] panic_cleared sourceUnit=100
```

Sample SCP output

```
#security panic_cleared //system/1/224 #sourceunit=100
```

Event

Panic Activated

Condition causing event

The security system has detected operation of the panic button.

Event Text

"panic_activated"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] panic_activated sourceUnit=100
```

Sample SCP output

```
#security panic_activated //system/1/224 #sourceunit=100
```

Event

Low Battery Corrected

Condition causing event

The security system has detected that its backup battery **was** running low, and is now acceptable

This event may not be supported by some security systems.

If supported, the security system will emit this message when the battery was previously running low (with less than 1 hour capacity left), but due to some corrective action the battery is now acceptable again.

Event Text

"low_battery_corrected"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] low_battery_corrected sourceUnit=100
```

Sample SCP output

```
#security low_battery_corrected //system/1/224 #sourceunit=100
```

Event

Low Battery Detected

Condition causing event

The security system has detected that its backup battery is running low.

This event may not be supported by some security systems.

If supported, the security system will emit this message when the battery has less than one hour of capacity left.

Event Text

"low_battery_detected"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] low_battery_detected sourceUnit=100
```

Sample SCP output

```
#security low_battery_detected //system/1/224 #sourceunit=100
```

Event

Battery Charging

Condition causing event

The security system has started or stopped charging its battery. The argument indicates if the battery charging has stopped or started.

This event may not be supported by some security systems.

Event Text

"battery_charging" charge-mode

charge-mode = "stopped" | "started"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] battery_charging started  
sourceUnit=100
```

Sample SCP output

```
#security battery_charging //system/1/224 started #sourceunit=100
```

Event

Status Report 1

Condition causing event

The security system reports its current state using this and the [Status Report 2](#) event. This may result from the security system receiving a [SECURITY STATUS REQUEST](#) command for Status Report 1.

Event Text

"status_report_1" armed-state tamper-state panic-state zone-report

armed-state = 1*3DIGIT ; in the range of 1 - 127. values 1, 128-255 are reserved.

; 0 = disarmed, 1=fully armed, 2=partially armed
; 3 - 127 are manufacturer dependent

tamper-state = 1*3DIGIT ; 0 = no tamper active
; 255 = tamper currently active
; 1-254 = reserved

panic-state = 1*3DIGIT ; 0 = no panic active
; 255 = panic currently active
; 1-254 = reserved

zone-report = 32*(zone-state)
; 32 zone-state digits for zones 1 through 32,
separated by spaces
; zones not present in the system will report as zone
sealed (0)

zone-state = 1*DIGIT ; the state of the zone
; 0 = zone sealed
; 1 = zone unsealed
; 2 = zone open
; 3 = zone short

Sample Event output

```
20020910-172843 702 //system/1/224 [security] status_report_1 1 0 0 0 0 0 0 0 0 0 0
1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

Sample SCP output

```
#security status_report_1 //system/1/224 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 #sourceunit=100
```

Event

Status Report 2

Condition causing event

The security system reports its current state using this and the [Status Report 1](#) event. This may result from the security system receiving a [SECURITY STATUS REQUEST](#) command for Status Report 2.

Event Text

"status_report_2" extended-zone-report

```

extended-zone-report = 48*(zone-state SP)
                        ; 48 zone-state digits for zones 33 through 80,
separated by spaces    ; zones not present in the system will report as zone
                        ; sealed (0)

zone-state = 1*DIGIT ; the state of the zone
                ; 0 = zone sealed
                ; 1 = zone unsealed
                ; 2 = zone open
                ; 3 = zone short

```

Sample Event output

```

20020910-172843 702 //system/1/224 [security] status_report_2 0 0 0 0 0 0 0 0 0 0 0 1
1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

```

Sample SCP output

```

#security status_report_2 //system/1/224 0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
#sourceunit=100

```

Event

Password Entry Status

Condition causing event

The security system reports the state of password entry, including successful attempts, failed attempts, and when entry is barred for a time after too many failed attempts.

Event Text

"password_entry_status" password-entry-status

```

password-entry-status = 1*3DIGIT ; the state of the zone
                        ; 1 = password entry succeeded
                        ; 2 = password entry failed
                        ; 3 = password entry disabled
                        ; 4 = password entry re-enabled (after previous
disable)
                        ; 0, 5-255 reserved

```

Sample Event output

```

20020910-172843 702 //system/1/224 [security] password_entry_status 1

```


Sample SCP output

```
#security password_entry_status //system/1/224 1 #sourceunit=100
```

Event

Mains Failure

Condition causing event

The security system has detected a failure of its mains power.

Event Text

"mains_failure"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] mains_failure
```

Sample SCP output

```
#security mains_failure //system/1/224 #sourceunit=100
```

Event

Mains Restored

Condition causing event

The security system has detected restoration of its mains power.

Event Text

"mains_restored"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] mains_restored
```

Sample SCP output

```
#security mains_restored //system/1/224 #sourceunit=100
```

Event

Arm Ready

Condition causing event

The security system has armed correctly (all zones are ready)

Event Text

"arm_ready"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] arm_ready
```

Sample SCP output

```
#security arm_ready //system/1/224 #sourceunit=100
```

Event

Line Cut Alarm Raised

Condition causing event

The security system has detected the attached phone line being cut.

Event Text

"line_cut_alarm line_cut_alarm_raised"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] line_cut_alarm line_cut_alarm_raised  
sourceUnit=100
```

Sample SCP output

```
#security line_cut_alarm line_cut_alarm_raised //system/1/224 #sourceunit=100
```

Event

Line Cut Alarm Cleared

Condition causing event

The security system has detected the attached phone line being reconnected.

Event Text

"line_cut_alarm line_cut_alarm_cleared"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] line_cut_alarm line_cut_alarm_cleared  
sourceUnit=100
```

Sample SCP output

```
#security line_cut_alarm line_cut_alarm_cleared //system/1/224 #sourceunit=100
```

Event

Fire Alarm Raised

Condition causing event

The security system has detected fire.

Event Text

"fire_alarm fire_alarm_raised"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] fire_alarm fire_alarm_raised  
sourceUnit=100
```

Sample SCP output

```
#security fire_alarm fire_alarm_raised //system/1/224 #sourceunit=100
```

Event

Fire Alarm Cleared

Condition causing event

The security system has detected that a fire condition has ceased.

Event Text

"fire_alarm fire_alarm_cleared"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] fire_alarm fire_alarm_cleared  
sourceUnit=100
```

Sample SCP output

```
#security fire_alarm fire_alarm_cleared //system/1/224 #sourceunit=100
```

Event

Arm Failed Raised

Condition causing event

The security system has failed to arm.

Event Text

"arm_failed arm_failed_raised"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] arm_failed arm_failed_raised  
sourceUnit=100
```

Sample SCP output

```
#security arm_failed arm_failed_raised //system/1/224 #sourceunit=100
```

Event

Arm Failed Cleared

Condition causing event

The security system has been able to arm after having previously failed.

Event Text

"arm_failed arm_failed_cleared"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] arm_failed arm_failed_cleared  
sourceUnit=100
```

Sample SCP output

```
#security arm_failed arm_failed_cleared //system/1/224 #sourceunit=100
```

Event

Gas Alarm Raised

Condition causing event

The security system has detected the presence of gas.

Event Text

"gas_alarm gas_alarm_raised"

Sample Event output

```
20020910-172843 702 //system/1/224 [security] gas_alarm gas_alarm_raised  
sourceUnit=100
```

Sample SCP output

```
#security gas_alarm gas_alarm_raised //system/1/224 #sourceunit=100
```

Event

Gas Alarm Cleared

Condition causing event

The security system has detected that the presence of gas has cleared.

Event Text

```
"gas_alarm gas_alarm_cleared"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] gas_alarm gas_alarm_cleared  
sourceUnit=100
```

Sample SCP output

```
#security gas_alarm gas_alarm_cleared //system/1/224 #sourceunit=100
```

Event

Other Alarm Raised

Condition causing event

The security system has detected special alarm condition.

Event Text

```
"other_alarm other_alarm_raised"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] other_alarm other_alarm_raised  
sourceUnit=100
```

Sample SCP output

```
#security other_alarm other_alarm_raised //system/1/224 #sourceunit=100
```

Event

Other Alarm Cleared

Condition causing event

The security system has detected the removal of a special alarm condition.

Event Text

```
"other_alarm other_alarm_cleared"
```

Sample Event output

```
20020910-172843 702 //system/1/224 [security] other_alarm other_alarm_cleared
sourceUnit=100
```

Sample SCP output

```
#security other_alarm other_alarm_cleared //system/1/224 #sourceunit=100
```

4.7.9.2.1.3 Object Parameters

The following parameters are supported by the Security Application Object.

They can be accessed with the [GET](#) command acting on the object address of the relevant Security Application object.

Parameter Name	Type & Description	Can set with SET command?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronize with the C-Bus network and security system "sync" = the object is synchronising with the C-Bus network and security system "error" = an error has occurred.		In any state except for "ok", all parameters will return "unknown" or -1 results until synchronisation has been achieved. Synchronisation may be initiated by using the SECURITY STATUS REQUEST command which will attempt to refresh the variables below from the results. Note that some variables can not be refreshed in this way. These will be refreshed when the appropriate events are received.
ArmState	Integer. Returns the armed state of the security system. Valid values are:	no	

	0 = disarmed 1 = fully armed 2 = partially armed 3 - 127 = manufacturer dependent -1 = arm state unknown		
AlarmState	String. Returns the state of alarm of the security system. Valid values are: "on" = alarm is on/in progress "off" = no alarm in progress "unknown" = alarm state is unknown	no	
TamperState	String. Returns the state of tamper of the security system. Valid values are: "on" = tampering detected "off" = no tampering detected "unknown" = tamper state is unknown	no	
PanicState	String. Returns the state of panic of the security system. Valid values are: "on" = panic detected "off" = no panic detected "unknown" = panic state is unknown	no	
LowBatteryState	String. Returns the low battery state of the security system: "detected" = low battery charge detected "corrected" = low battery charge corrected "unknown" = battery state is unknown	no	
ChargingState	String. Returns the charging state of the security system: "on" = battery is being charged "off" = battery is not being charged "unknown" = charging state is unknown	no	
PassEntryState	Integer. Returns the current password entry state of the security system. Valid values are: 1 = password entry succeeded 2 = password entry failed 3 = password entry disabled 4 = password entry re-enabled (after previous disable) -1 = unknown	no	
MainsState	String: Return the mains state of the security system. Valid values are:	no	

	"on" = Mains supply on "off" = Mains supply off "unknown" = Mains supply state unknown		
--	---	--	--

4.7.9.2.2 Security Zone Object

The Security Zone object models a security zone which is controlled by a security panel or other security device that interfaces to a C-Bus network.

A Security Application may have 0 or more security zones associated with them.

Addressing Security Zone Objects

Security Zone Objects are addressed as subelements of the Security Application object. Valid security zone addresses range from 1 through 127 (or in hexadecimal, \$01 through \$7f). No other values for security zones are allowed or valid.

So, on an example network called net1, the security application would be addressed as net1/\$D0 or net1/208. Security Zone number 1 would be addressed as net1/208/1, and zone \$25 would be addressed as net1/\$D0/\$25.

4.7.9.2.2.1 Commands Supported

The following commands are supported by the Security Zone object.

- [SECURITY REQUEST_ZONE_NAME](#)

4.7.9.2.2.2 Events Supported

The following events are supported by the Security Zone object.

Event

Zone Unsealed

Condition causing event

The security system alarm has detected a zone becoming unsealed that was previously sealed.

Event Text

"zone_unsealed"

Sample Event output

```
20020910-172843 702 //system/1/224/1 [security] zone_unsealed sourceUnit=100
```

Sample SCP output

```
#security zone_unsealed //system/1/224/1 #sourceunit=100
```


Event

Zone Sealed

Condition causing event

The security system alarm has detected a zone becoming sealed that was previously unsealed.

Event Text

"zone_sealed" zone-number

Sample Event output

```
20020910-172843 702 //system/1/224/9 [security] zone_sealed sourceUnit=100
```

Sample SCP output

```
#security zone_sealed //system/1/224/9 #sourceunit=100
```

Event

Zone Open

Condition causing event

The security system alarm has detected a protected loop zone becoming open circuit.

Event Text

"zone_open"

Sample Event output

```
20020910-172843 702 //system/1/224/5 [security] zone_open sourceUnit=100
```

Sample SCP output

```
#security zone_open //system/1/224/5 #sourceunit=100
```

Event

Zone Short

Condition causing event

The security system alarm has detected a protected loop zone becoming short circuit.

Event Text

"zone_short"

Sample Event output

```
20020910-172843 702 //system/1/224/2 [security] zone_short sourceUnit=100
```

Sample SCP output

```
#security zone_short //system/1/224/2 #sourceunit=100
```

Event

Zone Isolated

Condition causing event

The security system alarm has isolated, bypassed or shunted a zone.

Event Text

"zone_isolated"

Sample Event output

```
20020910-172843 702 //system/1/224/3 [security] zone_isolated sourceUnit=100
```

Sample SCP output

```
#security zone_isolated //system/1/224/3 #sourceunit=100
```

Event

Zone Name

Condition causing event

The security system has emitted a text name for a zone. This may result from the security system receiving a [SECURITY REQUEST ZONE NAME](#) command.

Event Text

"zone_name" zone-name

Sample Event output

```
20020910-172843 702 //system/1/224 [security] zone_name KITCHEN1 sourceUnit=100
```

Sample SCP output

```
#security zone_name //system/1/224 KITCHEN1 #sourceunit=100
```

Event

Arm Not Ready

Condition causing event

This event is sent during Arming if any security zones do not seal correctly. One message will be sent for each zone that does not correctly seal.

Event Text

"arm_not_ready"

Sample Event output

```
20020910-172843 702 //system/1/224/9 [security] arm_not_ready sourceUnit=100
```

Sample SCP output

```
#security arm_notReady //system/1/224/9 #sourceunit=100
```

4.7.9.2.2.3 Object Parameters

The following parameters are supported by the Security Zone Object.

They can be accessed with the GET command acting on the object address of the relevant Security Zone object.

Parameter Name	Type & Description	Can be set with SET command ?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronise with the C-Bus network and security system "sync" = the object is synchronising with the C-Bus network and security system "error" = an error has occurred.	No	In any state except for "ok", all parameters will return "unknown" or -1 results until synchronisation has been achieved. Synchronisation may be initiated by using the SECURITY STATUS_REQUEST command which will attempt to refresh the variables below from the results. Note that some variables can not be refreshed in this way. These

			will be refreshed when the appropriate events are received.
ZoneState	The state of this zone. Valid values are: 0 = sealed 1 = unsealed 2 = open 3 = short -1 = unknown	No	
ZoneName	String. The name of this zone, if supplied by the security system, or "unknown" otherwise.	No	

4.7.9.3 Specification

For detailed knowledge regarding the function of the Security Application on the C-Bus network, refer to "C-Bus Application Messages and Behavior -- Chapter 5: Security (CBUS-APP/5), Issue 3, 13 December 2002".

4.7.10 Short Message Application

The Short Message Application is used to transfer small messages around a C-Bus network. The messages comprise text, and optionally a single number and/or reference to a predefined graphical icon.

The Short Message Application has a C-Bus application address of \$AD, or 173 decimal.

4.7.10.1 Short Message Overview

This gives a short overview of the Media Transport Control Application implementation in the C-Gate server

Application name

Short Message

Application short name

(used as first part of commands or events to refer to this application)

shortmessage

C-Bus Application range

\$AD (173)

Commands

(these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

refresh
send

Events

(these are detected by the C-Gate server and will be shown in the event list and status port)

refresh
send

Notes and implementation issues

4.7.10.2 Events Issued

The events in this section are generated by the Short Message application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP)

4.7.10.2.1 Short Message Refresh

Event

Short Message Refresh

Condition causing event

Receiving a SHORTMESSAGE REFRESH command sequence.

Event text

[shortmessage] refresh info-type sourceUnit

info-type:

- 0 = generic text
- 1 = weather forecast
- 2 = surf report
- 3 = tide times
- 4 = unread emails in inbox
- 5 = voice mail messages
- 6 = stock prices ticker
- 7 = news headline
- 8 = other headline
- 9 = air quality forecast
- 10 = UV radiation forecast
- 11 = pollen count
- 12 = traffic report
- 13 = sports result
- 14 = horoscope
- 15 = joke of the day
- 16 = trivia of the day
- 17 = todays appointments
- 18 = things to do today
- 19 = birthdays today
- 20 = management notice
- 21 = electrical power brownout warning
- 22 = irrigation

sourceUnit = source of the command

Sample Event output

```
20090310-171721 702 //PROJ/254/173 fdf4fde0-e392-102b-a572-829dbedeadeab  
[shortmessage] refresh info-type=1 sourceUnit=3
```

Sample SCP output

```
#s# shortmessage refresh //PROJ/254/173 1 #sourceunit=3 OID=fdf4fde0-  
e392-102b-a572-829dbedeadeab
```

4.7.10.2.2 Short Message Send

Event

Short Message Send

Condition causing event

Receiving a SHORTMESSAGE SEND command sequence.

Event text

```
[shortmessage] send info-type sourceUnit
```

info-type:

- 0 = generic text
- 1 = weather forecast
- 2 = surf report
- 3 = tide times
- 4 = unread emails in inbox
- 5 = voice mail messages
- 6 = stock prices ticker
- 7 = news headline
- 8 = other headline
- 9 = air quality forecast
- 10 = UV radiation forecast
- 11 = pollen count
- 12 = traffic report
- 13 = sports result
- 14 = horoscope
- 15 = joke of the day
- 16 = trivia of the day
- 17 = todays appointments
- 18 = things to do today
- 19 = birthdays today
- 20 = management notice
- 21 = electrical power brownout warning
- 22 = irrigation

sourceUnit = source of the command

Sample Event output

```
20090310-171721 702 //PROJ/254/173 fdf4fde0-e392-102b-a572-829dbedeadeab  
[shortmessage] refresh info-type=1 sourceUnit=3
```

Sample SCP output

```
#s# shortmessage refresh //PROJ/254/173 1 #sourceunit=3 OID=fd4fde0-  
e392-102b-a572-829dbedeadeab
```

4.7.11 Telephony Application

The Telephony Application gives C-Gate the ability to interact with C-Bus capable telephone devices such as the C-Bus Telephone Interface.

4.7.11.1 Telephony Application Overview

Application name

Telephony

Application short name (used as first part of commands or events to refer to this application)

telephony

C-Bus Application range

\$E0

Commands

(these can be entered from C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

isolate_secondary_outlet
recall_last_number_request
reject_incoming_call
divert
clear_diversion

Events

(these are detected by C-Gate and will be shown in the event list and status port but are not available as commands)

line_on_hook
line_off_hook
dial_out_failure
dial_in_failure
ringing
recall_last_number_response
internet_connection_request_made

4.7.11.2 Objects supported

One new object is provided for the implementation of this application, the [Telephony Application Object](#) which is described in the sections which follow.

4.7.11.2.1 Telephony Application Object

This is the application object created in the server when a telephony command is detected on the C-Bus network or if a telephony command is sent from the command line. This object provides the

following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access	Write Access
ShortName	The short name for this application. This is set to telephony when the application is loaded	Monitor Only	Read Only

4.7.11.3 Commands supported

The following commands are available in the Telephony application:

- [TELEPHONY](#)
- [TELEPHONY CLEAR DIVERSION](#)
- [TELEPHONY DIVERT](#)
- [TELEPHONY ISOLATE SECONDARY OUTLET](#)
- [TELEPHONY RECALL LAST NUMBER REQUEST](#)
- [TELEPHONY REJECT INCOMING CALL](#)

4.7.11.4 Events supported

The events in this section are recognised by the Telephony Application and are made available as events in the C-Gate server event port, or via the Status Change Port (SCP)

4.7.11.4.1 Line On Hook

Event Line On Hook

Condition causing event The telephony device has detected that the line has gone on hook.

Event text line_on_hook

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
line_on_hook sourceUnit=131

Sample SCP output # telephony line_on_hook //system/1/224 #sourceunit=131

4.7.11.4.2 Line Off Hook

Event Line Off Hook

Condition causing event The telephony device has detected that the line has gone off hook.

Event text line_off_hook <direction> <type> [<number>]

<direction> = in for incoming calls, out for outgoing calls <reason> = voice when the telephony device is handling a voice call, data when the telephony device is handling

a data call, or other when another device is handling the call.

<number> = (optional) the rightmost 16 bytes of the calling number identification string (numbers or characters)

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
line_off_hook in other 5551212 sourceUnit=131

Sample SCP output # telephony line_off_hook //system/1/224 in other
5551212 #sourceunit=131

4.7.11.4.3 Dial Out Failure

Event Dial Out Failure

Condition causing event The telephony device has indicated that the previous dial-out attempt has failed.

Event text dial_out_failure <reason>

<reason> = no_dialtone Or no_answer Or no_ack_prompt Or unobtainable Or busy

no_dialtone indicates that no dialtone was detected

no_answer indicates that the call was not answered

no_ack_prompt indicates that there was no valid acknowledgement of prompts

unobtainable indicates that the number was unobtainable or does not exist

busy indicates that the number was busy

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
dial_out_failure busy sourceUnit=131

Sample SCP output # telephony dial_out_failure //system/1/224 busy
#sourceunit=131

4.7.11.4.4 Dial In Failure

Event Dial In Failure

Condition causing event The telephony device has indicated that the previous dial-in attempt has failed.

Event text dial_in_failure <reason>

<reason> = no_answer no_answer indicates that the call was not answered

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
dial_in_failure no_answer sourceUnit=131

Sample SCP output # telephony dial_in_failure //system/1/224 no_answer
#sourceunit=131

4.7.11.4.5 Ringing

Event Ringing

Condition causing event The telephony device has detected that the telephone line is ringing

Event text ringing [<number>]

<number> = (optional) the rightmost 16 bytes of the calling number identification string (numbers or characters)

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
ringing 5551212 sourceUnit=131

Sample SCP output # telephony ringing //system/1/224 5551212
#sourceunit=131

4.7.11.4.6 Recall Last Number Response

Event Recall Last Number Response

Condition causing event The telephony device's response to a Recall Last Number Request.

Event text last_number <direction> <number>

<direction> = in for incoming calls, out for outgoing calls

<number> = (optional) the rightmost 16 bytes of the calling number identification string (numbers or characters)

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
last_number in 5551212 sourceUnit=131

Sample SCP output # telephony last_number //system/1/224 in 5551212
#sourceunit=131

4.7.11.4.7 Internet Connection Request Made

Event Internet Connection Request Made

Condition causing event The telephony device indicates that it has received a request to initiate an Internet connection.

Event text internet_connection_request_made

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
internet_connection_request_made sourceUnit=131

Sample SCP output # telephony internet_connection_request_made //system/1/224
#sourceunit=131

4.7.11.4.8 Isolate Secondary Outlet

Event Isolate Secondary Outlet

Condition causing event A network device has requested that the telephony device set the isolation for the secondary outlet

Event text isolate_secondary_outlet <mode>

<mode> = set to normal to request normal operation of the outlet, or set to isolate to request isolation of the outlet.

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
isolate_secondary_outlet normal sourceUnit=131

Sample SCP output telephony isolate_secondary_outlet //system/1/224
normal #sourceunit=131

4.7.11.4.9 Recall Last Number Request

Event Recall Last Number Request

Condition causing event A network device has requested that the telephony device recall the last number.

Event text recall_last_number_request <direction>

<direction> = in to get the last incoming call number, out for the last outgoing call number

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
recall_last_number_request in sourceUnit=131

Sample SCP output telephony recall_last_number_request //system/1/224 in
#sourceunit=131

4.7.11.4.10 Reject Incoming Call

Event Reject Incoming Call

Condition causing event A network device has requested that the telephony device reject the incoming call.

Event text reject_incoming_call

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
reject_incoming_call sourceUnit=131

Sample SCP output telephony reject_incoming_call //system/1/224 #sourceunit=131

4.7.11.4.11 Divert

Event Divert

Condition causing event A network device has requested that the telephony device divert calls.

Event text divert [<number>]

<number> = (optional) the number to perform the divert to

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
divert 5551212 sourceUnit=131

Sample SCP output telephony divert //system/1/224 5551212 #sourceunit=131

4.7.11.4.12 Clear Diversion

Event Clear Diversion

Condition causing event A network device has requested that the telephony device clear any current diversion

Event text clear_diversion

Sample Event output 20020910-172843 702 //system/1/224/144 [telephony]
clear_diversion sourceUnit=131

Sample SCP output telephony clear_diversion //system/1/224 #sourceunit=131

4.7.11.4.13 Configuration properties

There are no configuration properties for this application.

4.7.11.5 Usage Notes

There are no network variables available for this application.

4.7.11.6 Specification

For detailed knowledge regarding the function of the Telephony Application on the C-Bus network, refer to "C-Bus Application Messages and Behaviour -- Chapter 24: Telephony Status & Control (CBUS-APP/24), Issue 1, 7 August 2002".

4.7.12 Temperature Broadcast Application

The Temperature Broadcast Application is used by C-Bus devices that are performing temperature measurements to broadcast temperature measurements to the C-Bus network.

The C-Gate server can receive temperature broadcast events and send temperature broadcasts from the command line.

4.7.12.1 Application Overview

The table gives a short overview of the Temperature Broadcast Application implementation in the C-Gate server

Application name	Temperature Broadcast
Application short name (used as first part of	temperature

commands or events to refer to this application)	
C-Bus Application range	\$19 (decimal 25)
Commands (these can be entered from C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)	broadcast
Events (these are detected by C-Gate and will be shown in the event list and status port but are not available as commands)	
Notes and implementation issues	

4.7.12.2 Objects Supported

There are two new objects supported by the Temperature Broadcast Application implementation.

The Temperature Broadcast Application Object models the application as a whole. The Temperature Broadcast Application Object is addressed using the application address for the Temperature Broadcast Application (\$19 or 25) on the C-Bus network that the temperature sensing devices are connected to.

The Temperature Group Object models one group of temperature readings. The temperature group is addressed as a sub-address of the Temperature Broadcast Application that it is associated with.

4.7.12.2.1 Temperature Broadcast Application Object

The Temperature Broadcast Application Object models a C-Bus network's Temperature Broadcast Application.

The Temperature Broadcast Application Object supports up to 256 [Temperature Group Objects](#).

Addressing

The Temperature Broadcast Application Object is addressed using the standard application address for the Temperature Broadcast Application (\$19 or decimal 25) on the relevant C-Bus network.

For example, for a hypothetical network called net1, the Temperature Broadcast Application's address will be net1/\$1E or net1/25.

4.7.12.2.1.1 Commands Supported

The following command line commands are accepted by the Temperature Broadcast Application Object:

- [TEMPERATURE BROADCAST](#)

4.7.12.2.1.2 Events Supported

There are no events supported by this object.

4.7.12.2.1.3 Object Parameters

The following parameters are supported by the Temperature Broadcast Application Object.

They can be accessed with the GET command acting on the object address of the relevant Temperature Broadcast Application object.

Parameter Name	Type & Description	Can set with SET comm and?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronize with the C-Bus network "sync" = the object is synchronising with the C-Bus network "error" = an error has occurred.	No	

4.7.12.2.2 Temperature Group Object

The Temperature Group object models a group of one or more temperature sensing devices. The Temperature group used by a C-Bus sensor will normally be set when the temperature sensor device is programmed.

Addressing Temperature Group Objects

A Temperature Broadcast Application can typically have 256 temperature groups associated with the application.

Temperature Group Objects are addressed as subelements of the Temperature Broadcast Application Object. Valid Temperature Group addresses range from 0 through 255 (or in hexadecimal, \$00 through \$FF). No other values are allowed or valid.

So, on an example network called net1, the temperature broadcast application would be addressed as net1/\$1E or net1/208. Temperature group number 1 would be addressed as net1/208/1, and temperature group \$25 would be addressed as net1/\$D0/\$25.

4.7.12.2.2.1 Commands Supported

The following commands are supported by the Temperature Group Object.

- [TEMPERATURE BROADCAST](#)

4.7.12.2.2.2 Events Supported

The following events are supported by the Temperature Broadcast object.

Event

Temperature Broadcast

Condition causing event

A temperature broadcast has been received.

This event can be generated by the [TEMPERATURE BROADCAST](#) command.

Event Text

"broadcast" celsius-temperature

Sample Event output

```
20020910-172843 702 //system/1/224/1 [temperature] broadcast 25.5 sourceUnit=100
```

Sample SCP output

```
temperature broadcast //system/1/224/1 #sourceunit=100
```

4.7.12.2.2.3 Object Parameters

The following parameters are supported by the Temperature Group Object.

They can be accessed with the GET command acting on the object address of the relevant Temperature Group object.

Parameter Name	Type & Description	Can be set with SET command ?	Notes
State	String. The overall state of this object. Valid states are: "ok" = normal operation "new" = the object has been created and has yet to synchronise with the C-Bus network "sync" = the object is attempting to synchronise with the C-Bus network "error" = an error has occurred.	No	

Temperature	The last temperature update received for this temperature group.	No	
-------------	--	----	--

4.7.12.3 Specification

For detailed knowledge regarding the function of the Temperature Broadcast Application on the C-Bus network, refer to "C-Bus Application Messages and Behavior -- Chapter 9: Temperature Broadcast (CBUS-APP/09), Issue 1, 7 August 2002".

4.7.13 Trigger Control Application

This document covers the implementation of the C-Bus Trigger Control Application within C-Gate.

4.7.13.1 Trigger Overview

This is a short overview of the Trigger Control Application's implementation in C-Gate

Application name

Trigger Control

Application short name (used as first part of commands or events to refer to this application)

trigger

C-Bus Application range

\$CA (202)

Commands/Events (these can be entered from the C-Gate command interface preceded by the short name of this application. All commands are also recognised as events)

event

Events (these are detected by the C-Gate server and will be shown in the event list and status port but are not available as commands)

Notes and implementation issues

MMIs supported Units not directly supported Trigger groups are automatically created if named on the command line or if event is received from the network.

4.7.13.2 Trigger Objects Supported

Two new objects are provided for the implementation of this application:

4.7.13.2.1 Trigger Application Object

This is the application object created in the server when a trigger control application is detected on the C-Bus network. This object provides the following parameters in addition to those supported by the standard Application object.

Parameter Name	Description	Read Access Control	Write Access Control
ShortName	The short name for this application. This is set to trigger when the application is loaded	Monitor	Read Only

4.7.13.2.2 Trigger Group Object

The groups of this application support the following methods, which mirror the command described later:

Method Name	Description	Access Control
TriggerEvent	See TRIGGER EVENT command below	Operate

4.7.13.3 Trigger Events

Events are issued for the Event messages detected on a C-Bus network. They are in the form show below:

```
20020910-172843.078 702 //system/1/202/144 [trigger] event action=2 sourceUnit=131
```

In the example shown, the address given is the trigger group that was activated, the command given is **event**, the **action=** paramter gives the action selector value, and the **sourceUnit=** gives the source unit id on the network where the command was sent.

4.7.13.3.1 Trigger Commands

The following commands are supported by this application:

[TRIGGER](#) and
[TRIGGER](#) EVENT

4.7.13.4 Status Change Port

This application places TRIGGER EVENT commands into the Status Change Port (SCP). Samples are shown below.

```
trigger event //system/1/202/144 2 #sourceunit=131
```

4.7.13.5 Configuration

There are no applicable parameters.

4.7.13.6 Project Database Issues

This application does **not** support the <Group> element under <Application>. Instead, it supports the <NetVar> element which can contain zero or more <Level> elements containing Level Tag information.

To use level tags:

Add one or more level entries to the <NetVar> element in the tag database. For each of the NetVar elements, add a <Value> element with the appropriate application selector level in the range 0 through 255.

So, an except would look like:

```
<NetVar>
  <TagName>Scene1</TagName>
    <Level Value="0">
      <TagName>One</TagName>
    </Level>
    <Level Value="76">
      <TagName>Two</TagName>
    </Level>
</NetVar>
```

This allows the Trigger Event command to use the tags AllOff and Level1 as action selector values, for example:

```
trigger event Switch1 AllOff
```

4.7.13.7 Usage Notes

Groups are created automatically when events for them are received, or they are detected in other ways.

4.7.13.8 Trigger Specification

For detailed knowledge regarding the function of the Trigger Control Application, refer to "C-Bus Application Messages and Behaviour -- Chapter 7: Trigger Control (CBUS-APP/07)".

4.8 Projects

The project command set supersedes the tag management commands and is used for management of entire projects which consist of multiple C-Bus networks.

4.8.1 Concept of Project

A **project** is a set of C-Bus networks that are operated together or work together. A project is normally programmed as one entity by the C-Bus Toolkit commissioning software.

Information about a project is stored in a Project Database, which is also known as a Tag Database. A Project Database holds the specification of a project in SQLite format.

4.8.2 Projects And Tags

A project is contained in a single project or tag database. The tag database has a <Project> element that contains the specification for the networks that are to be used in the project. The name of the project is the same as the project database file, excepting extensions (such as .db, .db or .zip).

To remain compatible with C-Bus Toolkit, all project names should be formatted as described in [Project Names](#) below.

4.8.3 Project Names

Project names should be constructed using the following rules to work smoothly across the full set of C-Bus software:

- All uppercase letters only
- First character must not be a number
- Maximum of 8 characters in length
- **Not** be one of the reserved words:
 - P, CBUS, VM, CMDINT, CGATE, TAG, CMDn (*where n is a number*)

4.8.4 Projects And Networks

There are one or more C-Bus networks in a Project.

4.8.5 Addressing

The introduction of projects makes some fundamental changes to the absolute addressing used in addressing objects in C-Gate, although all addressing is backward compatible apart from the addressing required to refer to any project that is not the default project, or the current project referred to by a command session.

An absolute project address starts with the characters // followed by the project name. Following this, the address has a / character, and then either a p character or a network name. An address without the //project/ is a relative address — it is relative to the current project for the context of the command. This means the setting of the project use command. Here are some example addresses using absolute and relative addressing:

Address	Meaning
//BUILDING/1/56/22	Group 22, application 56, network 1, project BUILDING
1/56/22	Group 22, application 56, network 1 in the current project

//PROJ/p/6/20	Unit 20, network 6 in the project called PROJ.
//MYLIGHTS	This is the address of the project called MYLIGHTS

Note: There is no way to combine pre-1.5 addressing (dot addressing) with project addresses. To use C-Gate 2.0 and later, do not use dot addressing.

4.8.6 Default Project

C-Gate has one default project. This default project is defined at startup by the [project.default](#) configuration parameter and can be changed by changing this parameter. Note that this is a global parameter and changing it may impact all sessions connected to C-Gate.

4.8.7 Current Project

Each command session has a current project that it uses when relative addresses are given.

The current project is a property of that command session only.

When a command session starts, it sets its command session to the value of C-Gate's default project. This project value can be viewed and changed using the [PROJECT USE](#) command.

4.8.8 Project Properties

The following properties are used with the PROJECT commands:

[project.default.dir](#) - The default directory where projects tag databases are stored

[project.default](#) - The default project for this C-Gate server.

[project.start](#) - The list of projects to be started on C-Gate start-up. The list is a list of project names separated by spaces. The projects will be started in the specified order.

4.8.9 Access Control Issues

All of the commands required an access level of **admin** to perform the commands.

The response 420 Access denied will be returned if **admin** or higher access level is not held by the command session that is performing the commands.

4.9 Networks

A **network** is a model of a C-Bus network that is normally programmed by the C-Bus Toolkit commissioning software.

C-Gate maintains two different models of a network, the database model and the live model.

Database model of a network

This is a static copy that is maintained in the Tag Database. It is accessed and manipulated using commands such as DBGET and DBSET.

Live model of a network

This is a live model that is maintained in C-Gate only while it is running. It is accessed and

manipulated using commands such as GET and SET.

C-Gate keeps this model synchronised with the C-Bus network through [Network Syncing](#) and by listening for C-Bus messages.

There are also commands to copy the live model to the database.

4.9.1 Opening a Network

In order to communicate with a network, C-Gate first needs to *open* it.

When a network is opened, C-Gate will:

1. Set the network object's InterfaceState to *opening*.
2. Identify the gateway unit.
3. Establish an active connection to the gateway unit.
4. If successful:
 - a. Set the network object's InterfaceState to *running*.
 - b. Place the network into the background sync queue. In most cases it will start syncing almost immediately.
5. If unsuccessful C-Gate will return the network object's InterfaceState back to *closed*.

For more information on InterfaceState see the following section: [Network Interface State Diagram](#).

Opening a network

If you have one network in your project, or if you just want to start one network, then use the NET OPEN command:

```
net open 254
120-initializing
120-opening port
120-starting network threads
120-pci reset
120-open complete
200 OK: //HOME/254
```

See that the net open command returns a number of lines of information as the network is being opened.

Starting a project

To open all the networks in your project, use the PROJECT START command.

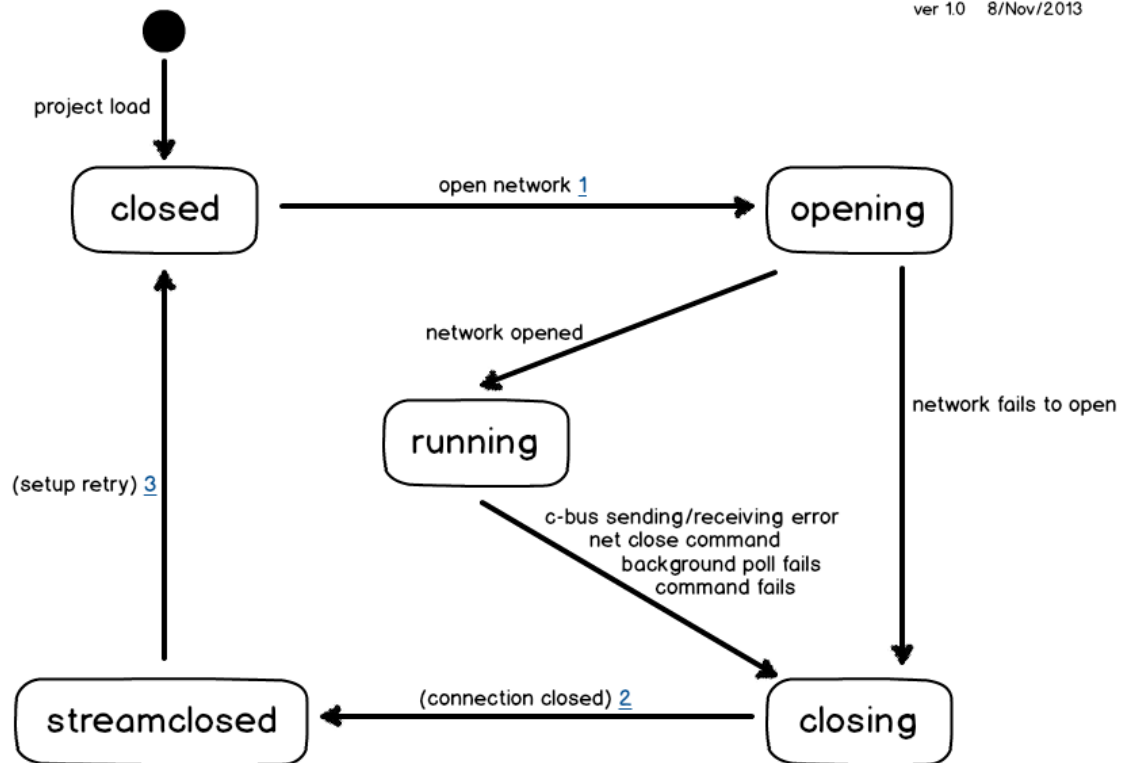
```
project start
200 OK.
```

This will open all networks in the project.

4.9.1.1 Network Interface State Diagram

C-Gate Network Interface State Diagram

ver 1.0 8/Nov/2013

**open network 1**

- open network can be triggered by
- net open command
- retry after network is closed because of polling failure
- retry after network is closed because of c-bus sending or receiving errors
- retry after bridge network is closed because routing network is closed
- retry after network fails to open
- (note: retry only happens when auto-reopen is set to yes which is the default value)

(connection closed) 2

- transition from closing to streamclosed is handled internally

(setup retry) 3

- transition from streamclosed closed is handled internally
- during the transition TargetInterfaceState is set to running if a retry is needed, then after a retry delay the network is reattempted to be opened

4.9.2 Network State

Each network has a state which indicates whether the network has been synchronised or not. The states are:

new: no sync has been performed
sync: network synchronisation is in progress
error: an error occurred while attempting to sync
ok: the network is synchronised and operating normally.

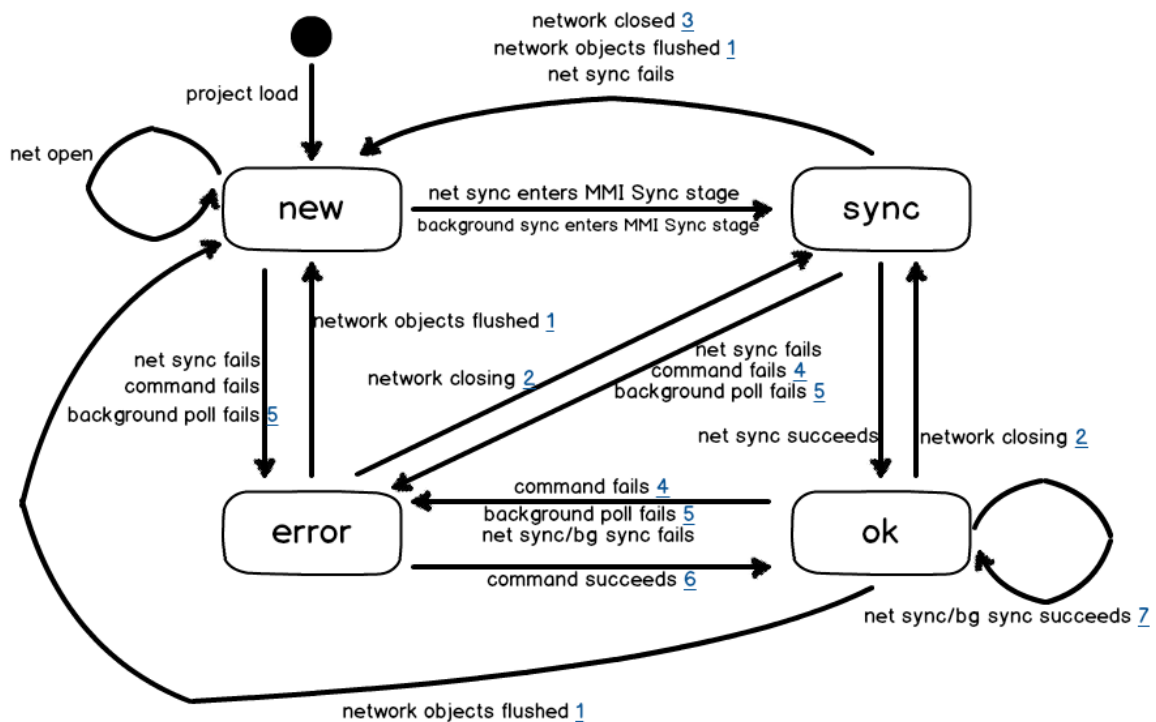
The goal of the sync process is for the network to end up in the ok state. Once the ok state has been reached, then C-Gate has a full model of the network and it is easy to work out how to control and monitor the network.

See the following [Network State Diagram](#) for a diagram of these states.

4.9.2.1 Network State Diagram

C-Gate Network State Diagram

ver 1.0 8/Nov/2013

**network objects flushed 1**

(note: a background sync starts shortly after objects are flushed)

- net flush command(not allowed during sync)
- net load command after changing network interface in db
- change Interface/InterfaceAddress using set command
(only allowed when interface state is closed)
- change bridge network address using net rename command
(only allowed when interface state is closed)

network closing 2 network closed 3

- network close can be triggered by net close command, polling failure, command retry failure, routing network close, etc
- when network is being closed, the interface state transitions from closing to streamclosed then closed
- when interface state becomes closing, network state is set to sync
- when interface state becomes closed, network state is set to new

command fails 4

- network state is set to error if the number of failed commands exceeds threshold (3 by default)

background poll fails 5

- if background poll fails, the network state will be set to error then the network will be closed shortly after

command succeeds 6

- network state is set to ok if a command is sent and responded

net sync/bg sync succeeds 7

- when network is in the ok state, 'net sync' command or background sync does not change the network state to sync and the network state will remain ok if the sync succeeds

other notes: CG-1589

4.9.3 Networks In Error

This section describes how C-Gate responds to error conditions in the network's state or interface state.

State

If C-Gate is having difficulties communicating effectively with a network, it will place the network into an *error* state (see [Network State Diagram](#)).

The following will cause C-Gate to place the network in an *error* state:

- A crucial C-Bus command fails during a [network sync](#), for example:
 - the gateway unit fails to initialise,
 - an Installation MMI fails, or
 - an Application MMI fails.
- Too many units fail to sync during a network sync (see [network.error.units-failed](#)).
- Too many consecutive failed C-Bus commands to the network, during a network sync or otherwise (see [network.error.commands-failed](#)).

Note that for all of the above C-Gate's [command retry](#) mechanism is in effect so it would take more than a single transmission failure to place a network into error.

C-Gate attempts to recover a network in *error* by setting the next sync time for the network to be 5 minutes later. Once the network successfully syncs and returns to an *ok* state the sync intervals will also return to normal (as defined by [sync-time](#)).

InterfaceState

If C-Gate cannot communicate with a network at all, it will close the network (see [Network Interface State Diagram](#)).

Things that will cause C-Gate to close the network:

- A failure while opening the network (eg. port does not exist or is in use).
- A failed [PCI Polling](#) operation.
- A failed [PCI Connection Check](#) operation.
- A C-Bus Send Error.

If [auto-reopen](#) is set to true C-Gate will attempt to re-open the closed network again in 15 seconds. The next sync time is also set to be in 15 seconds which means that when the network successfully re-opens it will also sync again.

4.9.4 Network Syncing

A network sync reads data from the C-Bus Devices on the C-Bus Network in order to build a model of the network in C-Gate's memory.

Once the first network sync is complete, C-Gate clients are able to understand and manipulate the network through C-Gate.

There are two types of syncing: background syncing and demand syncing.

Background Syncing

C-Gate performs a background network sync when a network is first opened and again at regular intervals as determined by the [sync-time](#) option.

Because syncing generates a lot of traffic, C-Bus manages the background syncs with a priority queue as follows:

1. Only 25 networks can be synced at any one time (see [sync.global-pool-size](#) option).
2. Only 3 networks can be synced at any one time through a single gateway (see [sync.gateway-pool-size](#) option).
3. Networks that have not yet been synced are synced first.
4. Networks that have recently failed to sync or that have gone into error are re-attempted after 5 minutes.
5. Finally, networks that are due for another background sync.

As a result large sites may see some networks not being synced as soon as they are opened, however they will sync eventually.

The [CBusNetwork](#) property *NextSyncTime* will indicate when the next background sync is estimated to take place. *LastSyncTime* records the time of the last synchronisation.

WARNING: Reducing *sync-time* or increasing the *pool-size* options may cause performance issues.

Demand Syncing

Demand syncs are initiated by a C-Gate client using commands such as [NET SYNC](#). For example: in C-Bus Toolkit when the user clicks "Scan".

In order to provide responsiveness to clients C-Gate allows demand syncs to bypass the traffic management criteria used for background syncs.

If a demand sync is initiated when there is a background sync already underway for a network, C-Gate will return a cached result of the sync so far and then return the remainder of the sync in real-time.

WARNING: Avoid using demand syncs where possible as they may cause performance issues.

Sync Duration

The duration of a network sync is primarily determined by the number of units on the network.

As a guideline: approximately 25 units are synced per minute. Thus a network of 75 units will take about 3 minutes to sync.

Bear in mind that other factors influence the time taken to sync a network:

- the number of bridges to the network (longer round-trip times)
- the number of older units pre-2006 (commands are not supported and C-Gate falls back to legacy commands)
- unusual gateway units (e.g. B&W C-Touch requires a special initialisation)
- units without mains power (e.g. some commands to MRA units will time out)
- other concurrent syncs (C-Bus traffic bottlenecks)

- high operational traffic (C-Bus traffic bottlenecks, C-Gate PC bottlenecks)
- reverting to the original serial library in use before Toolkit 1.20.0 (new library is faster)

Sync Stages

A sync proceeds through several stages. These are indicated in the log as SyncSubState values.

As a guideline:

- | | | |
|----------|------------------------|----------------------------|
| • pci | Gateway initialisation | (1 second) |
| • mmi | Unit presence; and | (1 second) |
| • | Unit duplicate check | (15 seconds per 25 units) |
| • tvs | Unit Identification | (15 seconds per 25 units) |
| • bridge | Bridge sync | (1 second) |
| • units | Unit sync | (30 seconds per 25 units) |
| • app | Application sync | (1 second per application) |

History

- C-Gate [v2.9.0](#) (along with C-Bus Toolkit 1.12.0) introduced permanent background syncing, ie. the *autosync* property can and should be left enabled. This makes it possible to run C-Bus Toolkit, Schedule Plus and other clients side by side. Before this release these clients would fight over this property.

4.9.4.1 Calculation of NextSyncTime

When C-Gate completes a network sync it calculates a new value for the network's NextSyncTime property.

- If the sync FAILED, the NextSyncTime is set to a time five minutes in the future.
- If the sync was a SUCCESS, the NextSyncTime is set to a time in the future as determined by:
 - o the [sync-time](#) config option,
 - o the [Sync Padding](#) feature, and
 - o any [Sync-Free Periods](#) defined for the network.

4.9.4.2 Sync Padding

The Sync Padding feature aims to prevent two or more networks from beginning their sync at the same time by spacing them out at intervals.

It is used only when C-Gate [calculates a new NextSyncTime](#) value upon successful completion of a sync.

Sync Padding does not affect:

- the first sync of any network,
- any sync requested by a C-Gate client (eg [NET SYNC](#)),
- the value of NextSyncTime manually set by a C-Gate client, or
- the NextSyncTime value that C-Gate calculates following a failed sync.

C-Gate calculates the padding interval between syncs, as follows:

$$\text{Padding Interval} = \frac{\text{<project sync-time> * <factor>}}{\text{<number of networks>}}$$

Minimum value : 5 seconds

Maximum value: 300 seconds (5 minutes)

When C-Gate calculates a new NextSyncTime value it will check to see if any other networks already have a NextSyncTime set within the padding interval. If so then the new NextSyncTime will be increased until the desired padding interval is achieved.

Configuration

Next Sync Time Padding is enabled by this option, which has a default value of yes.

[sync.padding.enabled](#)=yes

The following option provides a factor which is applied to the *sync-time* to determine a period of time within which to distribute syncs.

[sync.padding.sync-time-factor](#)=0.75

Any factor of less than 1.0 has the effect of providing a "free" buffer zone at the end of the *sync-time* which provides some allowance for newly created networks to enter the sync rotation easily.

Note that if networks have been configured with individual *sync-time* values, those values are ignored for this calculation. Only the project's sync-time value is used. But the networks will continue to sync according to their configured *sync-time* value.

Example

If:

- the project's sync-time is the default value of 3600 seconds (1 hour),
- the factor is the default value of 0.75 (75%), and
- there are 10 networks in the project

Then:

- the padding interval will be 270 seconds (4.5 minutes), and
- the networks will be distributed evenly across a 45 minute period leaving 15 minutes of allowance.

History

- C-Gate [v2.11.0](#) implemented the Next Sync Time Padding feature.

4.9.4.3 Sync-Free Periods

The Sync-Free Periods feature enables C-Gate to postpone background syncs during times known for high levels of ambient C-Bus network traffic.

It is used only when C-Gate [calculates a new NextSyncTime](#) value upon successful completion of a sync.

Sync-Free Periods are not obeyed for:

- the first sync of any network,
- any sync requested by a C-Gate client (eg [NET SYNC](#)),
- the value of NextSyncTime manually set by a C-Gate client, or
- the NextSyncTime value that C-Gate calculates following a failed sync.

When C-Gate calculates a new NextSyncTime value it will check to see if it falls into a configured Sync-Free Period. If so then the new NextSyncTime will be increased until it falls outside of the Sync-Free Period. If [Sync Padding](#) is enabled it will also be applied so that multiple network syncs don't cluster at the end of the Sync-Free Period.

Configuration

The Sync-Free Periods are controlled by this option:

[sync-free.periods](#)=

By default, this option is blank meaning that there are no sync-free periods defined.

See the linked page for the config option for information on how to define the periods.

Example

To prevent background syncs in an office when people are starting work and triggering lots of groups and scenes:

```
sync-free.periods=W:0111110:0800-0930
```

History

- C-Gate [v2.11.0](#) implemented the Sync-Free Periods feature.

4.9.5

Unit Syncing

About Unit Syncing

Unit syncing ensures that the C-Gate model of units reflects the units on the network. It is:

- done automatically for every unit as part of a [network sync](#).
- initiated by clients using DO <unit> SYNC | PSYNC | QSYNC commands.

Unit sync states

Each unit has a state which indicates whether the unit has been synchronised or not. The states are:

```
new: no sync has been performed
sync: unit synchronisation is in progress
error: an error occurred while attempting to sync
ok: the unit is synchronised and operating normally.
```

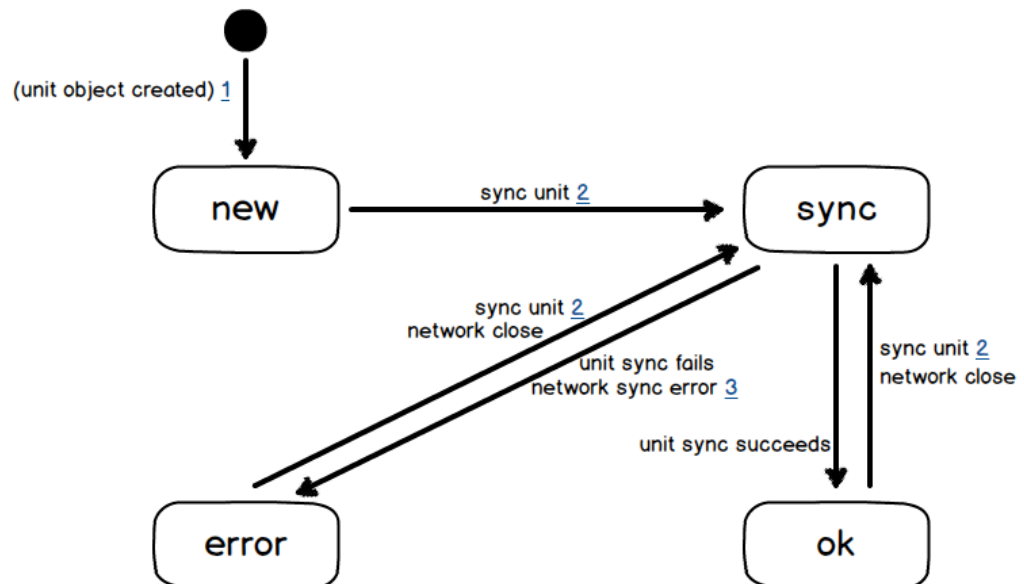
The goal of the sync process is for the unit to end up in the ok state.

See the following [Unit State Diagram](#) for a diagram of these states.

4.9.5.1 Unit State Diagram

C-Gate Unit State Diagram

ver 1.0 11/Nov/2013



(unit object created) 1

- unit object is created automatically by C-Gate when the unit is first seen during network sync
- unit state never goes back to new

sync unit 2

unit is synced

- for 'do unit_address psync' command (only allowed after the unit object is created during a network sync)
- after the local PCI unit's parameter is changed the PCI unit is re-synced
- as a part of the network sync (note: unit is not re-synced during a network sync if its last sync time is less than PSyncTime ago. PSyncTime can be changed for each unit using set command and it is 5 minutes by default)

network sync error 3

- network sync error means that the network sync completes and there are errors during the sync but the error count does not exceed the threshold to fail the entire network sync

4.9.6 Command Retries

This section describes how C-Gate retries C-Bus commands.

C-Gate Retries

The majority of C-Bus commands sent by C-Gate to a C-Bus network are subject to a time-out period defined by the [response-delay](#) option. If a command fails to receive a response in this time it will be retried.

The maximum number of retries is defined by the [network.retries](#) option. After all retries are exhausted, the command is deemed to be failed.

While retrying commands, C-Gate may execute a [PCI Connection Check](#) to check that the gateway unit is working.

PCI Retries

The C-Bus gateway unit has an internal retry mechanism where it will retry commands that have failed at some point on the C-Bus network between the gateway and the destination unit. Typically each command is attempted up to three times by the gateway. This retry mechanism is not visible to C-Gate nor its clients but may be seen when viewing raw C-Bus messages. It is only mentioned in this document in order to clearly distinguish those retries from C-Gate's retries.

4.9.7 C-Bus Clock Recovery

The C-Bus protocol requires one unit to act as the clock generator (not to be confused with the [Clock and Timekeeping Application](#)).

When C-Gate sends a C-Bus command to the network and receives a packet confirmation code terminator (%) that indicates a missing C-Bus clock, it will:

1. Attempt to enable the C-Bus clock on the gateway unit,
2. Wait one second for the network to perform an automatic clock contention process, and
3. Retry the C-Bus command it originally sent.

Upon completion of this recovery process the gateway should now be the clock generator.

To achieve redundancy consider enabling clock on multiple units. The [NET CLOCKS](#) command provides a convenient method to do this.

Configuration

The C-Bus Clock Recovery feature is always enabled.

History

- C-Gate [v2.11.0](#) implemented the C-Bus Clock Recovery feature.

4.9.8 PCI Connection Check

When C-Gate is retrying a failed command, it may first perform a PCI Connection Check.

A PCI Connection check is a single, fast command that is sent directly to the local gateway device to confirm that it is still present and working. This provides C-Gate with a diagnostic point of difference compared to a command which may be failing due to long or slow network connections, excessive traffic, multiple C-Bus network bridges, or a destination unit that is missing.

If a PCI connection check fails, C-Gate places the network immediately into error and abandons the

command retry process.

Configuration

The PCI Connection Check is controlled by this option:

[`network.retries.pci-check=1`](#)

With the default value of 1, C-Gate will perform a PCI connection check before every retry.

With a value of 2, C-Gate will perform the first retry, then perform a PCI connection check before the second and subsequent retries.

Setting this value to zero will turn off the PCI connection check entirely. Note that doing this will reinstate the legacy behaviour where C-Gate would spend a long time retrying commands before realising that a network has gone offline.

History

- C-Gate 2.8.0 build 2309 implemented the PCI Connection Check feature.

4.9.9 PCI Polling

PCI Polling is a feature where C-Gate sends a command to the gateway device at regular intervals. This achieves two things:

1. It confirms that the gateway device is still present, and by assumption so is the C-Bus network.

If the command sent to the PCI receives no response then C-Gate assumes that the network has gone offline and closes the network, placing it in a state where C-Gate will regularly try to re-open it.

This solves the historical problem that if a C-Bus network goes offline and there are no outgoing commands C-Gate will not notice the network's absence until the next scheduled sync which could be up to an hour later.

2. It updates the [Unit Online Status](#) record.

This is a bonus effect which indicates to clients whether units have been added or removed without having to wait for the next sync to take place.

Configuration

PCI Polling is controlled by these options:

<code>network.pci.poll-interval=60</code>	# poll every 60 seconds
<code>reopen-delay=15000</code>	# attempt to reopen a closed network after 15 seconds

With this default configuration it will take up to 1 minute for C-Gate to close a missing network, and up to 1 minute to re-open it once it comes back.

Setting the poll-interval to zero will turn off PCI polling entirely. Note that this will reinstate the legacy behaviour where offline networks may not be detected for up to an hour.

Troubleshooting

1. If using a CNI2 through a firewall C-Gate may take up to 15 minutes to recognise that it is back online. This is because the CNI2 is still trying to "keep-alive" the original C-Gate connection and is preventing C-Gate from re-connecting. Some firewalls are known to block the XON/XOFF messages that the CNI2 relies on to detect the dropped connection. This problem doesn't exist with the original CNI because it uses a simpler networking stack.
2. If PCI Polling commands fail regularly it may be due to excessive traffic (especially if you have lots of networks). Try increasing the poll interval to 120 or 300, or set it to zero to turn it off entirely.

History

- C-Gate [v2.9.5](#) introduces the PCI Polling feature (using Identity 0x01 during times of C-Bus inactivity).
- C-Gate [v2.10.0](#) changes the polling command to use an Installation MMI, executed every 60 seconds regardless of C-Bus activity.
- C-Gate [v2.10.4](#) suspends the PCI polling during syncs and prevents it from falling back onto a PCI connection check.
- C-Gate [v2.11.0](#) will fall back to an Identity command if the Installation MMI fails (which sometimes happens on bridged networks).
- C-Gate [v2.11.5](#) suspends the PCI polling during any live traffic and further improves stability.

4.9.10 Unit Online Status

Unit Online Status is a feature where C-Gate tries to provide timely information about which unit addresses are occupied, and where discrepancies exist between the network and the database.

It does this by providing several properties on the network object:

```
DBUnitAddressesNew=31
DBUnitAddressesOnline=1, 2, 3, 6, 8
DBUnitAddressesDuplicate=20
DBUnitAddressesError=5
DBUnitAddressesMissing=55
```

These properties are described in detail in the [CBusNetwork](#) section.

These properties are updated by:

- a network sync operation.
- a [PCI Polling](#) response. If polling changes one of these properties C-Gate will also emit an event message describing the change.

Calculation Method

These properties are determined using a complex algorithm. See the following section [Unit Online Status Diagram](#) for a graphical representation.

In technical terms C-Gate uses any responses that it receives to Installation MMIs. C-Gate keeps track of the last five MMI responses in order to increase confidence in its duplicate detection - as a result some properties may not change until several PCI pollings have taken place.

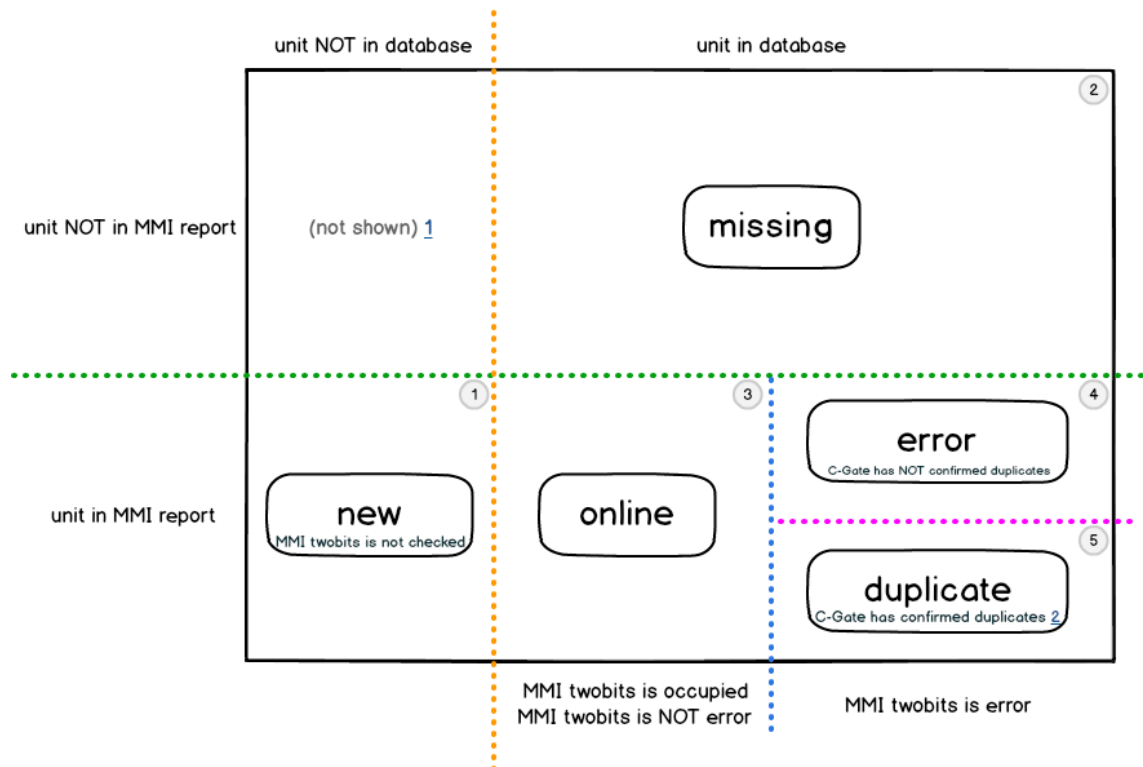
History

- C-Gate [v2.10.0](#) introduces the Unit Online Status properties.

4.9.10.1 Unit Online Status Diagram

C-Gate Unit Online Status Diagram

ver 1.0 12/Nov/2013

**IMPORTANT NOTE**

- unit online status is only valid when the network interface state is running
- unit online status can be retrieved using TREEXMLDETAIL command or from the below network properties
 - DBUnitAddressessNew
 - DBUnitAddressessOnline
 - DBUnitAddressessMissing
 - DBUnitAddressessError
 - DBUnitAddressessDuplicate

(not shown) [1](#)

- if a unit is not in database or MMI report it won't be shown in the TREEXMLDETAIL command response or the DBUnitAddresses* network properties

C-Gate has confirmed duplicates [2](#)

- C-Gate confirms duplicates during network sync or unravel by checking the number of replies from the same address

Reference:

C-Gate Manual >> Reference >> Object Overview >> CBusNetwork

Hints for reading diagram: read from 1 to 5

4.10 Access Control

The C-Gate server provides access control which applies to C-Gate's [Command and Monitoring Interfaces](#). Access control is defined by an in-server-memory access control table that is loaded from the access control file in the config directory.

4.10.1 Access Control Basics

The access control file is normally called `access.txt`, but can be named differently if the 'access-control-file' value in the C-Gate configuration file is altered.

If the file doesn't exist when it is first needed (eg on a login command), a default version is automatically written to disk.

The access control list is a list of lines based on three keywords:

1. `interface` defines an access level for a TCP/IP interface on the C-Gate server
2. `remote` defines an access level for a remote connecting IP address
3. `user` defines a user name & password and access level which can be set with the `LOGIN` command

When read in, any faulty lines in the file are ignored.

The access control list in memory is checked every time a connection is made to C-Gate command, event, load or status port, or a login command is received.

IP addresses are resolved on the fly at that time, not when the file is read.

At these times, the list of rules in the access control list is reviewed, and the access level granted is the result of the most **favourable** rule. However, using the login command, it is possible to set your access level to a lower level than would normally be configured by the interface and remote rules.

Access levels are as follows, with each later level incorporating the functions of the previous level.

`none` - no access at all. Use this to refuse connections.

`connect` - allow a connection to be established (to the command interface only) and execute the `LOGIN` command or the license challenge & response commands.

`monitor` - allow monitoring and query of the status of objects and C-Bus, but do not allow any changes. This is the minimum level to allow connection to the event, load or configuration change ports.

`operate` - allow set, on, off, ramp operations – allow changes to be made to the system

`admin` - allow C-Gate shutdown and administration functions

`program` - allow C-Bus networks to be programmed

`debug` - allow debugging functions to be performed

Possible entries in the access control list are as follows:

`interface <IP address> <access level>`

`remote <IP address> <access level>`

`user <userid> <password> <access level>`

Where the fields are:

<IP address> a full IP address, either for the interface that the connection comes in on, or a remote address. This can be a hostname, or it can be a dotted quad (ie. 10.234.2.23). If a dotted quad is used, then any part of the address that is set to 255 will match any address, allowing whole networks to be allowed access.

<userid> a username that is used in the login command along with the password.

<password> the password that goes with the username.

Sample access control file

A sample access control file would be:

```
#sample access control file
interface 127.0.0.1 admin
interface 10.50.1.2 monitor
remote 10.50.1.255 operate
remote microsoft.com none
user debugger xxAA1 debug
```

Blank lines and those beginning with # are ignored.

This file would set access control in the following way:

1. All sessions from the local computer on the *localhost* interface get *admin* level access.
2. All sessions on the ethernet interface 10.50.1.2 (ie all accesses from the external network) get *monitor* level
3. All sessions on the 10.50.1.* subnetwork get *operate* access
4. All connect attempts from the URL *microsoft.com* are denied.
5. A user that types the command:
 login debugger xxAA1
is granted *debug* access level.

4.11 CGL Format

CGL is a data exchange format, nominally a JSON string, that allows a subset of data, chiefly tagnames, to be exported from a C-Gate project, and imported back into a C-Gate project.

The current specification is [v1.0](#).

This format is used by the [CGL](#) command set.

4.11.1 CGL Specification v1.0

4.11.1.1 CGL Specification v1.0 Example

```
{
  "cglVersion": "1.0",
  "createdBy": "C-Gate v2.11.0 (build 9999)",
  "createdTime": "2016-07-22T00:18:17.408+0000",
  "networks": [
```

```

{
  "address": 254,
  "name": "n254",
  "routes": [
    {
      "address": 253,
      "path": [ 253 ]
    },
    {
      "address": 252,
      "path": [ 253, 252 ]
    }
  ],
  "applications": [
    {
      "address": 56,
      "type": 56,
      "name": "C-Bus Lighting Application",
      "groups": [
        {
          "address": 1,
          "name": "Kitchen",
          "levels": [
            {
              "address": 255,
              "name": "On"
            }
          ]
        }
      ]
    }
  ]
}

```

4.11.1.2 CGL Specification v1.0 Constraints

Property	Sending System Constraints	Receiving System Validations
<all>	Names are in Lower Camel Case (ie. lowerCamelCase).	Properties can be accepted in any order, e.g. "address" may come before or after "name". If a property is missing it is assumed to have a value of <i>null</i> .
cglVersion	A string in "x.y" format where x and y are positive integers. Must be in a known subset, currently { 1.0 }. {0.y} can be used freely for development.	ERROR if major value (x) is unsupported. WARNING or INFO if minor value (y) is unsupported.
createdBy	An arbitrary free-text string describing the system or	None. For informational purposes only.

	software that generated the CGL. Include useful information such as version.	
createdTime	A timestamp in ISO-8601 basic format (no punctuation) to a resolution of seconds and including the timezone offset of the sending system. The inclusion of milliseconds is optional. Example: 20160301T163344+1030.	ERROR if invalid timestamp. WARNING if the timestamp is in the future.
Network instances	Up to 255 instances.	
Network "address"	Integer in range 0..254. Must be unique within the "networks" collection.	ERROR if invalid syntax. WARNING if non-unique. Duplicate values are ignored.
Network "name"	A free-text string. Must be unique within the "networks" collection.	
Route instances	Up to 255 instances.	
Route "address"	Integer in range 0..255. Must be unique within the "routes" collection. This is the destination network address.	ERROR if invalid syntax. WARNING if non-unique, and duplicate entries are ignored.
Route "path"	Array of integers, each in the range 0..255. Each integer is an interim network address on the route towards the destination network. The order is important. The first integer is the first and nearest network, and the last integer must always be the destination network itself. No part of the route is allowed to form a loop.	ERROR if the route contains an invalid address. ERROR if the route contains a loop.

Application instances	Up to 256 instances.	
Application “address”	Integer in range 0..255. Must be unique within the “applications” collection.	ERROR if invalid syntax. WARNING if non-unique, and duplicate entries are ignored.
Application “type”	Integer in range 0..255. This defines the known “behaviour” of the Application and is effectively the default address from the standard list of C-Bus Applications. For example, a second Lighting application with address 57 would have a type of 56. However in most cases it would be the same value as “address”.	
Application “name”	A free-text string. Must be unique within the “applications” collection.	
Group instances	Up to 256 instances.	
Group “address”	Integer in range 0..255. Must be unique within the “groups” collection.	ERROR if invalid syntax. WARNING if non-unique, and duplicate entries are ignored.
Group “name”	A free-text string. Must be unique within the “groups” collection.	
Level instances	Up to 256 instances.	
Level “address”	Integer in range 0..255. Must be unique within the “levels” collection.	ERROR if invalid syntax. WARNING if non-unique, and duplicate entries are ignored.
Level “name”	A free-text string. Must be unique within the “levels” collection.	

4.12 Object Overview

This section gives an overview of the parameters and methods of C-Gate objects, including C-Bus objects that represent networks and devices in connected and controlled C-Bus networks.

In general, all object parameters can be retrieved from the command interface using the [GET](#) or [SHOW](#) command. Some parameters can be set with the [SET](#) command.

Methods given here can be invoked from the command interface using the [DO](#) command.

4.12.1 Working with Objects

Any addressable object in C-Gate contains one or more parameters and zero or more methods.

Parameters

Parameters are values that can be viewed and possibly set from the command interface. In many cases, where the objects in C-Gate are modelling or mirroring the physical or logical devices on a C-Bus network, the parameters represent the latest information retrieved from the actual physical or logical devices.

The [GET](#) command makes getting parameters easy.

The [SET](#) command allows some parameters to be set to new values from the command line. In the following tables, if the SET access level is shown as *None*, then the set operation will fail.

Methods

Methods allow the command line to be used to cause objects to do something.

The [DO](#) command is provided to run object methods.

Exploring Objects

You can explore the parameters provided in any object by using a special form of the GET command.

get <obj-add> *

which returns a list of parameter names

For example, for a C-Bus Network:

get 254 ?

300 //

HOME/254:Parameters=Interface,AutoSync,Retries,AutoUpdate,Options,Type,State,InterfaceState,AutoUnravel,DefaultApplication,SyncState,QuickDetect,LSP,RxQ,ShortSync,SyncTime,TxQ,FastResponse,ResponseDelay,TxEnable,EventLevel,Units,FreeUnit,FreeApplication,InterfaceAddress,Groups,TargetInterfaceState,Stats,Applications,XState,Name

To get a little more information, try:

get <obj-add> ??

returns a list of parameter names with short explanations

For example, again for a C-Bus Network:

get 254 ??

102-//HOME/254: Interface - Address of the interface for this network
102-//HOME/254: AutoSync - If set to yes, will perform automatic network synchronisations.
102-//HOME/254: Retries - The number of retries made before a command fails
102-//HOME/254: AutoUpdate - If set to yes, automatically update the tag database for a new unit.
102-//HOME/254: Options - The interface options for this interface
102-//HOME/254: Type - Type of network interface
102-//HOME/254: State - The state of connection of this object
102-//HOME/254: InterfaceState - The current state of this interface
102-//HOME/254: AutoUnravel - If set to yes, will perform automatic unravels when errors are detected in sync operations.
102-//HOME/254: DefaultApplication - The default application for this network
102-//HOME/254: SyncState - Get the sync state for this network
102-//HOME/254: QuickDetect - If set to yes, perform continuous install MMIs looking for new units.
102-//HOME/254: LSP - The count of Lost Sync Packets
102-//HOME/254: RxQ - List of the commands waiting for responses
102-//HOME/254: ShortSync - If set to yes, perform only sync to type-version-serial.
102-//HOME/254: SyncTime - The time between full network sync operations
102-//HOME/254: TxQ - List of the commands in the transmit queue
102-//HOME/254: FastResponse - If set to yes, retries and timeouts are shortened to give quick responses.
102-//HOME/254: ResponseDelay - Time in ms to delay before a command is retried.
102-//HOME/254: TxEnable - True if the transmitter is enabled
102-//HOME/254: EventLevel - The level of displayed events for this object
102-//HOME/254: Units - The list of units that are in this network
102-//HOME/254: FreeUnit - The next free unit address on this network
102-//HOME/254: FreeApplication - The next free application address on this network
102-//HOME/254: InterfaceAddress - Address of the interface for this network
102-//HOME/254: Groups - The list of groups in the DEFAULT application
102-//HOME/254: TargetInterfaceState - The current state of this interface
102-//HOME/254: Stats - List of statistics for this interface
102-//HOME/254: Applications - List of applications on this network
102-//HOME/254: XState - Get the extended state for this network
102 //HOME/254: Name - Name of this network

To quickly get a list of all parameters and their values, try:

get <obj-add> *

For example:

```
get 254 *
300-//HOME/254: Applications=56,228
300-//HOME/254: AutoSync=yes
300-//HOME/254: AutoUnravel=no
300-//HOME/254: AutoUpdate=no
300-//HOME/254: DefaultApplication=56
300-//HOME/254: EventLevel=9
300-//HOME/254: FastResponse=no
300-//HOME/254: FreeApplication=0
300-//HOME/254: FreeUnit=0
300-//HOME/254: Groups=1,2,10,44
300-//HOME/254: Interface=localhost:14000
300-//HOME/254: InterfaceAddress=localhost:14000
300-//HOME/254: InterfaceState=running
300-//HOME/254: LSP=0
300-//HOME/254: Name=1
300-//HOME/254: Options=
300-//HOME/254: QuickDetect=no
300-//HOME/254: ResponseDelay=3000
300-//HOME/254: Retries=2
300-//HOME/254: RxQ=now=1138600828749
300-//HOME/254: ShortSync=no
300-//HOME/254: State=ok
300-//HOME/254: Stats=:SendCommand=114
300-//HOME/254: SyncState=units
300-//HOME/254: SyncTime=300
300-//HOME/254: TargetInterfaceState=running
300-//HOME/254: TxEnable=yes
300-//HOME/254: TxQ=now=1138600828749
300-//HOME/254: Type=cni
300-//HOME/254: Units=2,10,14,15,19,249
300 //HOME/254: XState=ok+units
```

4.12.2 All Objects

All C-Gate objects support the following object parameters. These can be retrieved with the GET and if allowed, can be set with the SET command.

All-Objects Parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
EventLevel	Values range from 0 (no events) to 9 (all events). Events that are at a level lower to or equal to the value of event-level will be sent.	The value of the cgate global event-level parameter at the time this	monitor	operate

		object is created. This is the parameter that is provided in the global configuration.		
State	Represents the state of an object in C-Gate. The following states are possible: unknown : The state of this object is not known. new : This is a new object that is not synchronised with a real device. This state occurs when an object has been created with the NEW command but is yet to synchronise with a real device. sync : This object is in the process of synchronising with a real device. It may not contain valid data in this state. ok : Object in state of normal operation, synchronised to external controlled objects. error : This object is in an error state and its data may not reflect actual status. deleted : This object has been deleted.	A new object is created in state new .		

All-Objects Methods:

Method name	Access Level	Parameters	Description
Gc	operate	None	Performs a manual global garbage collection for C-Gate. This operation is provided for debugging purposes only.

4.12.3 CGate Object

The C-Gate system object **cgate**, hold the version of this C-Gate server. The **get cgate version** command is often used to check version compatibility, though the **apiver** command gives greater detail.

cgate object parameters

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>

Version	The current version of this C-Gate	None	monitor	None
----------------	------------------------------------	------	---------	------

4.12.4 Projects

The C-Gate system object **projects**, hold some properties common to all open projects.

The `get projects *` command will return the following parameters in addition to those available for [All Objects](#).

projects object parameters

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
NetStateInterval	The interval in seconds between state events for all networks. A value of zero disables state events.	0	monitor	operate

4.12.5 CBusNetworkManager

The C-Bus network manager, system object **cbus**, controls the definition, opening and operation of attached C-Bus networks.

cbus object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Networks	A list of the current know C-Bus networks, comma separated.	empty if no networks are defined	monitor	None

4.12.6 CBusNetwork

Inherits all the parameters and methods from: [All Objects](#)

Network object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Applications	Returns a comma-separated list of the C-Bus application numbers in use on this network	empty if no applications are defined	monitor	None

Type	The type of C-Bus network interface this is. Possible values are: serial : local serial port interface to C-Bus PCI socket : TCP/IP socket interface to C-Bus PCI cni : Clipsal C-Bus Network Interface (provides direct TCP/IP connection to C-Bus). wiser : same as type cni . etherlite : Interface to C-Bus PCI via Etherlite module.	No default, defined at interface startup time in networks file.	monitor	None
Interface	A string that defines the interface address and port name. For serial type, the string is a COM port identifier, for socket and etherlite or cni types, the string is in the form: ip-address:port-number, such as: 10.1.1.66:2001	No default	monitor	None
InterfaceState	A string that defines the current state of this interface to the C-Bus network. Possible values and their meanings are: closed : The interface has not been opened or has closed. opening : The interface has been opened (but is not yet fully operating). running : The interface is operating normally. closing : The interface is in the process of closing. streamsclosed : The interface is in the process of closing.	closed	monitor	None
OriginateInProject	Returns 'yes' if the network is loaded from project, otherwise 'no'.	no	monitor	none

Network object parameters (continued):

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Name	The name of this interface. This is equivalent to the network name specified when the C-Bus network is defined.	No default	monitor	None
Units	A list of the C-Bus unit numbers that are known on this network.	No default	monitor	None
Groups	A comma-separated list of the C-Bus group numbers for the default application of this network.	No default	monitor	None
TxQ	[For debugging only] A list of commands that are queued in the transmit queue waiting to be sent	No default	debug	None
RxQ	[For debugging only] A list of the commands that have been sent and are waiting for a response or confirmation.	No default	debug	None
Stats	Basic statistics for this C-Bus interface	No default	debug	None
FreeUnit	Returns the unit address of the next free unit address.	No default	monitor	None
FreeApplication	Returns the application address of the next free application address	No default	monitor	None
SyncTime	Time (in seconds) between synchronising the local objects and the actual C-Bus networks. (NB, the check for sync-time is only performed at the interval of scan-time .)	300	monitor	operate

Network object parameters (continued):

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
ScanTime	Time (in seconds) between checks of psync and sync times. Every scan-time checks are made if the sync-time has been exceeded, if so, a sync is performed for this network. Also, the psync-time is checked for each unit in this network and psyncs are performed on those objects if required.	30	monitor	operate
LSP	Returns the count of lost sync packets. That is, the number of responses that are returned that are not matched to a sent command waiting for a response.	No default	monitor	None
DefaultApplication	The default application number for this network. The default application can be addresses with shorthand and version 1.0 addressing.	56, or \$38 (The generic lighting application)	monitor	operate
TxEnable	Returns 'yes' if the transmitter is currently enabled for this network. This is generally used for testing purposes only.	no	monitor	None
SyncState	Returns the state of synchronisation of the network. Valid values are: idle - no sync in progress; init - sync started; mmi - initial network poll done; tvs - type, version and serial numbers gathered; pci - pci scan complete;	idle	monitor	None

	bridge - bridge scan complete; units - full unit sync complete; app - application scan complete.			
--	--	--	--	--

Network object parameters (continued):

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
XState	Returns a combination of network and sync states separated by a + character.		monitor	None
Retries	Set the number of command retries before giving up the command as failed.	2	monitor	operate
ResponseDelay	Time in milliseconds to wait for a response to the command from the network	3000 (3 seconds)	monitor	operate
FastResponse	On setting to "yes", sets Retries to 0. On setting to "no" sets Retries to 0.	no	monitor	operate
AutoSync	If set to yes, will perform automatic network synchronisations.	yes	program	program
NextSyncTime	The next scheduled sync time for this network	No default	monitor	program
LastSyncTime	The last sync time of this network	No default	monitor	None

Network object parameters (continued):

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
DBUnitAddressesNew	Only valid when InterfaceState is running; List of unit addresses which are not in project database but are occupied in MMI report, e.g.	No default	monitor	readonly

	DBUnitAddressesNew=31			
DBUnitAddressesOnline	Only valid when InterfaceState is running; List of unit addresses which are in project database and are occupied in MMI report and the MMI statuses are not error, e.g. DBUnitAddressesOnline=1, 2, 3, 4, 5	No default	monitor	readonly
DBUnitAddressesMissing	Only valid when InterfaceState is running; List of unit addresses which are in project database but are not occupied in MMI report, e.g. DBUnitAddressesMissing=10, 21	No default	monitor	readonly
DBUnitAddressesError	Only valid when InterfaceState is running; List of unit addresses which are in project database and are occupied in MMI report and the MMI statuses are error and C-Gate has not confirmed there are duplicates at the addresses, e.g. DBUnitAddressesError=9	No default	monitor	readonly
DBUnitAddressesDuplicate	Only valid when InterfaceState is running; List of unit addresses which are in project database and are occupied in MMI report and the MMI statuses are error and C-Gate has confirmed there are duplicates at the addresses, e.g. DBUnitAddressesDuplicate=18, 19	No default	monitor	readonly

Note for DBUnitAddress* parameters:

- C-Gate periodically polls the network gateway using MMI when its InterfaceState is running to check the network connection and unit online statuses.
The poll interval is defined by [network.pci.poll-interval](#). During each poll C-Gate updates the DBUnitAddress* parameters and if any of these parameters change, it will also send out an 744 event as described in [Network Information Event Table](#).
- Possible transitions during MMI Polling

- o the unit online status can become Error from New, Online, Missing (note: since the MMI two bits is a random number from the unit, possibly you need to wait for several polls to see the status change)
- o the unit online status can become New, Online, Missing from Error/Duplicate (note: there is a confidence check before the error status is reset and the threshold is 5, that means after you remove a duplicate unit physically from the network and do not perform a network scan, you need to wait 5 polls before the error status disappear)
- Possible transitions during Network Scan/Unravelling
 - o the unit online status can transition among all 5 statuses

Network object methods

<u>Method name</u>	<u>Access Level</u>	<u>Parameters</u>	<u>Description</u>
Sync	admin	None	Perform a network synchronisation immediately. This pre-empts the usual background synchronisation operation that performs sync operations every sync-time .
Open	admin	None	Opens the network interface if it is not already open.
Close	admin	None	Close this network interface.
AllOn	operate	None	Sets the state of all unprotected groups in the default application to on .
AllOff	operate	None	Sets the state of all unprotected groups in the default application to off .
AllRamp	operate	ramp-level ramp-time (see the RAMP command for the definition of ramp-level and ramp-time)	Ramp all of the unprotected groups in the default application to the specified level in the specified ramp time. See the RAMP command for additional details.
Unravel	program	[unit-address] If unit address is given (optional) then the unravel is performed at that unit address only.	Scan, detect and re-address any C-Bus units that have duplicate unit addresses. Progress is reported as a series of 120 responses.

4.12.7 CBusUnit

The C-Bus Unit is a model of a physical C-Bus device connected to a C-Bus network.

Inherits all the parameters and methods from: [All Objects](#)

Unit object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Name	The name of this object. (This is not the same as the address and may be set independently of the address)	"unknown"	monitor	operate
Address	The numeric address of this unit. Setting this value will change the C-BusUnit's address on the network. Use this with care.	No default	monitor	program
Application	The numeric value of the primary application for this C-Bus unit.	255 (The value indicating the application is unassigned)	monitor	None
Application2	The numeric value of the secondary application for this C-Bus unit.	255 (The value indicating the application is unassigned)	monitor	None
Type	The C-Bus unit device type string	No default	monitor	None
Version	The version string for this C-Bus unit.	No default	monitor	None
ClassName	The name of the java class that is used by C-Gate to model this unit.	com.clipsal.cgate.cbust.CBusUnit	debug	None
PsyncTime	The time between parameter sync operations for this unit. This is measured in seconds. (Note, parameter syncs can not occur more often than the scan-time global parameter unless invoked by the psync method (see below))	The default value is set by the cgate global parameter <i>default-psync-time</i> when the unit is created.	monitor	operate

PatchVersion	The version of patch used in this unit	\$FF means not patch used in unit. -1 means not synchronised yet.	program	None
SlotGroups	The contents of the unit's group slots	\$FF is the uninitialized value	monitor	None

Unit object methods

<u>Method name</u>	<u>Access Level</u>	<u>Parameters</u>	<u>Description</u>
Sync	admin	None	Perform a network synchronisation immediately. This pre-emptes the usual background synchronisation operation that performs sync operations every sync-time .
Psync	admin	None	Perform a parameter synchronisation on this unit. Psync usually fetches operational parameters that need to be sampled more often than syncs are performed.
Qsync	admin	None	Perform a quick synchronisation of selected unit parameters to support commissioning software.
ResetErrorFlags	admin	None	Immediately reset the physical C-Bus Unit's error flag.

4.12.8 CBus Output Units

Inherits all the parameters and methods from: [CBusUnit](#)

Output Unit object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Area	The C-Bus Area address that is set for this unit.	255 (this is the unassigned value)	monitor	None
Groups	A comma-separated list of C-Bus groups used by this unit.	None	monitor	None
Terminals	The comma-separated list of terminals supported by this unit	None	monitor	None

TerminalCount	The number of terminals supported by this unit	0	monitor	None
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4.12.9 CBus Output Terminal

This object represents an individual output terminal on a C-Bus output unit. Individual terminals represent an individual dimmer or relay contact set.

Inherits all the parameters and methods from: [All Objects](#)

Output Terminal object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Level	The current level of this terminal – in the range 0 through 255.	0	monitor	None
Power	The current output power calculated for this terminal based on the value of the Level and the Load parameters. Note: this is not an accurate power measurement nor is based upon actual current flowing.	0	monitor	None
Load	The load power of this terminal. This value defaults to 0 and must be set in order for the Power parameter to give any value other than 0.	0	monitor	operate
Groups	A comma-separated list of C-Bus groups used by this terminal	None	monitor	None
Logic	The type of logic used on the groups supported by this terminal. Values are: greater : The greater of the C-Bus groups will be used to set the output level. lesser : The lesser of the C-Bus groups will be used to set the output level.	None	monitor	None
Name	The name of this terminal	None	monitor	operate

4.12.10 DIN and Pro Dimmers and Relays

C-Bus DIN Rail and Professional Series relays have enhanced features over the standard Relay and Dimmer output devices.

Inherits parameters and methods from: [CBusOutputUnit](#)

Dimmer object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
NetVoltage	The C-Bus network voltage at this unit. This parameter is read when parameter sync (psync) is performed. Execute the psync method on this object to update this value.	None	monitor	None
Serial	The unique serial number of this device	None	monitor	None

4.12.11 CBus Din/Pro Terminal

Inherits from: [CBus Terminal](#)

Terminal object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Current	The measured current for this terminal. -1 is returned if no current sense information is available, otherwise, a value is returned from 0 through 255, representing current as a fraction of the terminal's maximum current rating. This parameter is read from the device using the parameter sync (PSync) operation. Execute the PSync method on the unit this terminal is attached to in order to update current ratings.	None	monitor	None

4.12.12 CBus Temperature Sensor

Inherits parameters and methods from: [C-BusUnit](#)

Sensor object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Temperature	The temperature recorded by this sensor when the last parameter sync (PSync) operation was performed. The temperature is given in degrees Celsius to 2 decimal places. A typical value might be 20.44.	None	monitor	None

Target	The target temperature value for this unit. This value is given in whole degrees celsius . Setting this parameter will change the target value set in the unit.	None	monitor	operate
Margin	The margin value for this unit in whole degrees celsius. Setting this parameter will change the margin value stored in the unit.	None	monitor	operate
Offset	The offset value for this unit in whole degrees. Setting this parameter will change the offset value stored in the unit.	None	monitor	operate
Mode	The heating or cooling mode for this unit. Valid values are heating or cooling . Setting this value will change the value stored in the unit.	None	monitor	operate
High	The high-side offset for this sensor	None	monitor	None
Low	The low-side offset for this sensor	None	monitor	None

4.12.13 CBus PE Cell / Light Level Sensor

Inherits from: [CBusUnit](#)

Sensor object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
LightLevel	The light level recorded by this sensor when the last parameter sync (psync) was performed. The light level is given in lux and has an approximate range of between 0 and 1600.	None	monitor	None
Target	The target value for this unit. This value is given in lux. Setting this parameter will change the target value set in the unit. Range: 0 – 1600	None	monitor	operate
Margin	The margin value for this unit. Setting this parameter will change the margin value stored in the unit. Range 0 – 1600	None	monitor	operate
RawTarget	The raw, unscaled value of the Target parameter	None	monitor	None

RawMargin	The raw, unscaled value of the Margin parameter.	None	monitor	None
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4.12.14 CBusApplication

Inherits parameters and methods from: [All Objects](#)

Application object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Name	The name of this object. (This is not the same as the address and may be set independently of the address)	"unknown"	monitor	operate
Address	The numeric address of this application.	None	monitor	None
Groups	A comma-separated list of the C-Bus group numbers for this application.	None	monitor	None
FreeGroup	The address of the first unassigned group address in this application.	None	monitor	None

Application object methods:

<u>Method name</u>	<u>Access Level</u>	<u>Parameters</u>	<u>Description</u>
AllOn	operate	None	Sets the state of all unprotected groups in this application to on .
AllOff	operate	None	Sets the state of all unprotected groups in this application to off .
AllRamp	operate	ramp-level ramp-time (see the RAMP command for the definition of ramp-level and ramp-time)	Ramp all of the unprotected groups in this application to the specified level in the specified ramp time. See the RAMP command for additional details.

4.12.15 CBusGroup

Inherits from: [All Objects](#)

Group object parameters:

<u>Parameter</u>	<u>Values and impact</u>	<u>Default value</u>	<u>GET Access Level</u>	<u>SET Access Level</u>
Name	The name of this object. (This is not the same as the address and may be set independently of the address)	"unknown"	monitor	operate
Address	The numeric address of this group.	None	monitor	None
Units	A comma-separated list of units that this group can control <i>on the local network</i> .	None	monitor	None
Protected	If set to yes, this group will not be included in AllOn, AllOff and AllRamp methods executed on this group's Application.	None	monitor	None
Level	The current level of this C-Bus group. Level can be in the range of 0 through 255. 0 is considered off. Level 255 in C-Bus is considered on. Setting the level causes a ramp command to be sent with a ramp time of 0 to the level to be set.	None	monitor	operate
Type	The type of C-Bus group. There are two possible values: group and area .	"group"	monitor	operate
RampTime	The stored ramp time to be used by the CI command when setting the group level. If a group is set with the CI commands, then the ramp time in seconds stored here is used as the ramp time for the level change.	0	monitor	operate

Group object methods:

<u>Method name</u>	<u>Access Level</u>	<u>Parameters</u>	<u>Description</u>
Off	operate	None	Sets the level of this group to 0 by executing a C-Bus off command.
On	operate	None	Sets the level of this group to 255 by executing a C-Bus on command.

Ramp	operate	ramp-level ramp-time (see the RAMP command (section 3.3.5) for the definition of ramp-level and ramp-time)	Sets the level of this group to ramp-level using a ramp time of ramp-time.
Protect	operate	None	Set the protected parameter for this group.
UnProtect	operate	None	Clear the protected parameter for this group.

4.13 Building Clients for C-Gate

This section has helper tips and documentation references useful when building client software to work with C-Gate.

4.13.1 Communicating with C-Gate

Client software can communicate with C-Gate a number of ways. This is described in detail in [Command and Monitoring Interfaces](#).

To summarise here:

Standard Input, Output, Error on local Host

For a client running on the same host as the server, the client can start C-Gate as a subprocess, and use the standard input, output and standard error to communicate with C-Gate.

TCP/IP sockets

TCP/IP sockets can be used to access the server. Separate connections are available for Commands, Event, Configuration Changes, and Status changes. The standard ports for these are 20023 - 20026. You must make sure that [access control](#) will allow remote connections from your client's location.

TCP/IP sockets with SSL

C-Gate also supports connections over SSL when it is desirable that the information communicated between C-Gate and the client is encrypted. Client authentication via SSL client certificates can also be enabled if necessary. The port range used here is 20123-20126.

See [Command and Monitoring Interfaces](#) for more information about these communication options.

Multiple client connections using any or all of the above interfaces is supported by C-Gate

4.13.2 Controlling Communications with C-Gate

C-Gate gives clients a pretty rich set of ways to control line-by-line commands with C-Gate.

Simple commands can be issued and a response waited for. See [Commands and Replies](#) for basic details of using communications.

In more complex situations, commands can be issued with a unique identifier, and all responses to that command will then be returned with the unique identifier. See [Unique Command Ids](#) for details of this.

And in addition to this, command issued with a unique identifier can be placed into the background so a new command can be issued. This allows multiple commands to be issued and performed at once if the client can use the unique identifiers to match responses to the commands.

4.13.3 Multiple Clients

Because multiple clients can work with C-Gate, it may be useful for clients to communicate with each other.

C-Gate provides a simple mechanism to perform this, the [BROADCAST_EVENT](#) command. With this command, you can share and communicate between client connected to the same C-Gate.

4.13.4 Logging with C-Gate

C-Gate logs event information to disk. This can be a very useful debugging technique.

For more detail, refer to the [Logging](#) section in the the reference.

5 Version History

This section covers the significant changes in each new version.

5.1 v2.11.6

C-Gate v2.11.6 was released in April 2021 with C-Bus Toolkit v1.15.8.

Changes:

- Cybersecurity patches.
- Fixed issue causing eDLT units to ignore certain bus messages.

5.2 v2.11.5

C-Gate v2.11.5 was released in June 2019 with C-Bus Toolkit v1.15.6.

Changes:

- Support for 40 Series Modules.
- Improvements to the PCI Polling feature.

5.3 v2.11.4

C-Gate v2.11.4 was released in July 2018 with C-Bus Toolkit v1.15.5.

Changes:

- Upgrade Silabs USB drivers.

5.4 v2.11.3

C-Gate v2.11.3 was released in June 2018 with C-Bus Toolkit v1.15.4.

Changes:

- Fix for bridge routing when NAC/SHAC is used as a gateway.

5.5 v2.11.2

C-Gate v2.11.2 was released in February 2018 with C-Bus Toolkit v1.15.3.

Changes:

- New default values for light level target and margin in sensor units.
- Now produces an event for "[trigger] indicatorkill" message.
- New properties for Clock Application.
- Now correctly sets date on bus if C-Gate is configured as the clock master.
- Better handling of corrupted messages on the bus.

5.6 v2.11.1

C-Gate v2.11.1 was released in June 2017 with C-Bus Toolkit v1.15.1.

Changes:

- Fixed issue with eDLT dialogs after Windows 10 Creator's Update.

5.7 v2.11.0

C-Gate v2.11.0 build 3232 was released on 15th May 2017 with C-Bus Toolkit v1.15.0.

Changes:

- [Java runtime](#) upgraded from v7 to v8.

- Silabs USB Drivers upgraded from v6.6.1 to v6.7.3 (improved Windows 10 support).
- Support for the Automation Controller (5500NAC, LSS5500NAC, 5500SHAC, LSS5500SHAC).
- [New C-Gate Launcher](#).
- [Sync Padding](#) feature.
- [Sync-Free Periods](#) feature.
- [Automatic Clock Recovery](#) capability.
- [NET CLOCKS](#) command.
- [PORT CNISCAN2](#) command.
- [Array Filtering](#) syntax.
- [PCI Polling](#) improved for bridged networks.
- Command caching fixes.

5.8 v2.10.6_3169

C-Gate v2.10.6 build 3169 was released on 19th April 2015 with C-Bus Toolkit v1.14.5. Also included with C-Bus Toolkit v1.14.6 and v1.14.7.

Changes:

- Added support for EN_UNIV
- Fixed installation issue on Windows 10

5.9 v2.10.6

C-Gate v2.10.6 build 3145 was released on 7th April 2015 with C-Bus Toolkit v1.14.2. Also included with C-Bus Toolkit v1.14.3 and v1.14.4.

Changes:

- Updated unit catalog and unit specifications.

5.10 v2.10.5

C-Gate v2.10.5 build 3140 was released on 9th December 2014 with C-Bus Toolkit v1.14.0. Also included with C-Bus Toolkit v1.14.1.

Changes:

- Java runtime upgraded from v6 to v7.
- [Transmit Cache](#) feature.
- Config options to reduce sync traffic (see separate release notes).
- [CONFIG OBRESET](#) command.
- [PCI Polling](#) fixes.
- Support for Saturn Zen range, including unit types KEYH1, KEYH2, KEYH3, and KEYH4.

5.11 v2.10.2

C-Gate v2.10.2 build 3106 was released on 28th May 2014 with C-Bus Toolkit v1.13.2. Also included with C-Bus Toolkit v1.13.3.

Changes:

- Fixed phantom burdens reported by eDLT and SENTEMP4.
- Fixed Logging performance issues.

5.12 v2.10.0

C-Gate v2.10.0 build 3087 was released on 9th December 2013 with C-Bus Toolkit v1.13.0.

Changes:

- New [Logging](#) subsystem.
- [PCI Polling](#) now uses an Installation MMI and executes every minute regardless of traffic.
- [TEST SPAM](#) commands.
- [DBADDSAFE](#), [DBSETSAFE](#), [DBCOPYSAFE](#) and [DBRENAMENETSAFE](#) commands.
- [TREEXMLDETAIL](#) command.
- [sync.global-pool-size](#) and [sync.gateway-pool.size](#) config options.
- USB driver updated to v6.6.
- Support for Windows 8 & 8.1.
- Sync/unravel now handles the edge case where the PCI has moved unexpectedly to another address.
- Modifying the network name or interface now flushes the network model. This solves several problems where clients were retrieving outdated data from the model.
- Fixed a bug where messages from a C-Touch or PAC on a local network with bridges were not replicated by C-Gate to the far-side network model.

This and future releases also include a standalone zip distribution of C-Gate which contains some cross-platform support for Mac and Linux operating systems.

5.13 v2.9.8

C-Gate v2.9.8 build 3077 was released on 18th November 2013 with C-Bus Toolkit v1.12.8.

Changes:

- Support for Occupancy Sensors.
- Updated support for eDLT units.

5.14 v2.9.7

C-Gate v2.9.7 build 2569 was released on 17th September 2013 with C-Bus Toolkit v1.12.7.

Changes:

- [PCI Polling](#) feature.
- [PROJECT REPAIR](#) command can now remove duplicate groups.
- Support for eDLT units.
- Upgraded the USB driver to v6.4.0.2.
- Closing the network on the near side of a bridge now also closes the network on the far side.
- Reconnecting a disconnected network on the near side of a bridge now allows the network on the far side to reopen automatically.
-

5.15 v2.9.5

C-Gate v2.9.5 build 2460 was released on 27th November 2012 with C-Bus Toolkit v1.12.5.

Changes:

- Fixed a bug introduced in v2.9.2 that would cause large scenes executed in Schedule Plus to fail.

5.16 v2.9.4

C-Gate v2.9.4 build 2458 was released on 13th November 2012 with C-Bus Toolkit v1.12.4.

Changes:

- Fixed a bug introduced in v2.9.2 that caused FILE UPLOAD to fail.

The Partial installer (without a JRE) will no longer be released. For this and future releases only the Full installer will be distributed.

5.17 v2.9.2

C-Gate v2.9.2 build 2455 was released on 25th October 2012 with C-Bus Toolkit v1.12.2.

Changes:

- Installer now installs a private Java run-time for C-Gate instead of installing a system JRE.
- cgate.exe now supports the -restart parameter (new entry for this added to Start Menu).
- Sync now detects and copes with units that have worn-out EEPROMs.
- Fixed exception caused by an invalid C-Bus message produced by a DALI Gateway.
- Fixed a bug in FILE DOWNLOAD that causes a subsequent [PROJECT SAVE](#) to fail.
- Fixed a bug introduced in 2.9.0 where [DBGET](#) no longer returns a single property.
- Fixed performance issues with CNI v2.

C-Gate v2.9.2 build 2456 was released on 30th October 2012 with C-Bus Toolkit v1.12.3.

Changes:

- Fixed bug causing C-Gate service to shut down after 2 minutes.

5.18 v2.9.0

C-Gate v2.9.0 build 2449 was released on 16th June 2012 with C-Bus Toolkit v1.12.1.

Changes:

- Installer will now default to installing [C-Gate as a Windows Service](#).
- Permanent background [syncing](#).
- Sync now detects and reports duplicate units.
- When a network sync is cancelled by the user the network will now not go into error.
- Unravelling is completely overhauled.
- Client reference counting for projects. This prevents one client from closing a project that another client is using.
- Config options are now saved in the project file.
- [DBGET](#) now supports wildcards and responses include the array index.
- Support for Windows 7.
- Fixed a bug when upgrading project files that contain unicode characters.
- ... and more.

5.19 Upgrade from 1.5 to 2.5

This section describes major changes from version 1.5 of C-Gate and how to make C-Gate 2.5 operate in a similar way to version 1.5 to allow existing client applications to still function.

5.19.1 Major changes from 1.5 to 2.x

The following are major changes from C-Gate 1.5 to 2.x:

Projects

C-Gate 2 introduced the concept of projects into C-Gate. Projects group several networks together to

operate as a group. C-Gate can run several projects simultaneously, each with a separate set of networks.

By default, C-Gate does not open networks immediately on opening. This can be configured, however.

Projects also introduced an additional addressing element, the project name (eg. //HOME) that appears at the front of fully qualified C-Gate addresses. See [Objects and Addresses](#) for details of addressing changes.

The [PROJECT](#) command set is available for manipulating projects.

In addition, some project and network specific configuration variables can be stored along with a project file. These are detailed below.

See the [Projects](#) section of the reference for details.

Project Database

As of C-Gate 2.x, projects and project information are now held in a Project Database. The contents of the project database are managed by the DB commands. See command starting with DB in the manual.

Network file changes

In C-Gate 1.5, networks were defined in the networks.txt file, and these were the set of networks that operated with C-Gate. To change this, C-Gate had to be restarted after the file contents were changed. As of 2.x, networks are defined in the Project Database, or in a Network file named for the project.

The [NET](#) and [PORT](#) command sets were introduced to work with networks. Most network operations can now be performed without restarting C-Gate, allowing on-the-fly definition and use of networks and projects.

Configuration changes

Configuration variables are now most easily managed from the command line with the [CONFIG](#) command set. See the [Configuration](#) section of the guide for details.

Access Control Commands

Access control can now be managed with the ACCESS control commands.

5.19.2 Setting 1.5 compatibility

It is possible to set C-Gate 2.x into a mode where it is basically compatible with 1.5. To do that, you'll need to do the following:

Set the following configuration variables to the values given here. Do this in the C-GateConfig.txt file.

```
hide-project-names=yes  
event.display-oids=no
```



```
ccp.display-oids=no  
tag-name-output=no  
use-1.0-addressing=no  
network.source=file
```

Then proceed as for C-Gate 1.5. There are some additional things to be aware of:

1. The output of addresses from C-Gate has changed slightly so that a leading slash '/' is now added to all addresses output. This will required changes to address parsing.
2. On output of addresses, C-Gate now uses the form /<network>/p/<unit> instead of the 1.5 style: p/<network>/<unit>
3. Many more events are now issued, especially at higher global event levels.
4. Version 1.0 addressing is no longer fully supported.

6 Not Used

Enter topic text here.

6.1 Set Plant HVAC Level

[Not implemented yet]

Event

Set Plant HVAC Level

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_ward_on app ward zone-list mode raw-flag setback-enabled guard-enabled use-aux-level type level auxlevel

For description of parameters see the [AIRCON SET_PLANT_HVAC_LEVEL](#) command.

Sample Event output

```
20081022-084833 702 sys [aircon] set_plant_hvac_level ward=255 zone-  
list=1,2,3,4,5,6 mode=7 rawlevel=1 setbackenabled=1 guardenabled=1  
useauxlevel=1 type=255 level=65535 auxlevel=255
```

Sample SCP output

```
# aircon set_ward_off //BOARD/254/172 1 #sourceunit=0 OID=...
```

6.2 Set Plant Humidity Level

[Not implemented yet]

Event

Set Plant Humidity Level

Condition causing event

The equivalent command has been invoked on the application.

Event text

set_ward_on app ward

app = air-conditioning application address
ward = air-conditioning ward number (0-255)

Sample Event output

```
20081022-084833 702 sys [aircon] set_ward_off ward=1 sourceUnit=0
```

Sample SCP output

```
# aircon set_ward_off //BOARD/254/172 1 #sourceunit=0 OID=...
```

Index

- A -

Access 64

Alarm 61

- B -

Backus-Naur Form 41

Bridge 21

- C -

C-Bus Bridge 21

C-Bus Network 21, 22

C-Bus Unit 21

CNI 22

COM Port 22

Command 57, 59

Command Message 59

Connect 62

Connected 62

Critical 61

- D -

Debug 59

- E -

Energy 56

- F -

File 63

- I -

Information 56

Interface 58, 62

- L -

License 58, 63

- N -

Network 21, 22, 57, 60, 62

Network Error 60

Network Status 57

- P -

PC Interface 58

PCI 58

Power 56

Project 21

- R -

RS-232 22

- S -

Scene 58, 64

Scene Error 64

Security 60

Serial 22

Spec 63

Specification 63

- U -

Unit 59, 63

Unit Spec 63